MAGNETOM

Installation Instructions System

Installation Instructions

Symphony, A Tim System

Document Version

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1.1 Please read this first

The following chapters are used as instructions during MR installations. The individual installation steps are described in detail in the respective chapters of this manual.

1.1.1 Installation personnel

> Only trained personnel are authorized to install the system.

Four persons are required for moving in the magnet, the electronic cabinets, the filter plate, etc.

Two people are required for the remainder of the installation procedure.

1.1.2 Installation table

The table below provides an overview of the entire installation process.

Start of installation
General
Magnet Installation
Steps after positioning
Installation in the magnet room
Installation in the equipment room
Cabling
Starting the magnet cooling
Installing the Cover Panels

1.1.3 Safety information

> Installation personnel and system engineers should refer to the list of operating notes for the magnet. It has been included in the bag on the patient table.

> Super-conducting magnets are highly sensitive components. Proceed with extreme caution during transport and delivery to avoid hazardous conditions for both the system and the personnel involved. Only specially trained personnel may install and service the magnet. The safety instructions have to be read prior to working on the magnet.

To ensure the safety of personnel and system, the chains, belts, measurement devices, tools, lifting devices, hydraulic lifts and transport rollers have to be in good condition.

Please note the general safety information in the MR - Allgemeine Sicherheitsinformationen section of this manual when you are executing the work described in this chapter.

Wearing of the following protective clothing:

- Safety shoes
- Face mask (when handling cryogens)
- Special gloves (when handling cryogens)
- Leather gloves (RF-room)
- Hard hat

Always:

- Wear protective clothing (leather gloves, safety shoes, hard hat)
- Check all devices and tools for defects prior to use.
- Check tools as well as auxiliary tools to ensure they are in good condition. Ensure that they meet manufacturer's specifications and comply with legal requirements.
- Use of specific transporting devices.
- Inform the workforce reporting to Siemens employees about possible hazards and the safety requirements to be met when handling heavy loads.
- Secure the transport devices and loads prior to moving the load.
- In case of defects, immediately inform the person responsible.

Never:

- Stand underneath heavy loads.
- Use damaged tools, defective devices, or uncalibrated measuring equipment.
- Keep personnel in the vicinity of moving loads.
- Exceed the permissible load for devices.
- Route or pull ropes, chains, or belts across sharp edges.
- Knot ropes or chains together.
- Twist ropes or chains.
- Block emergency exits.
- Block escape routes.

1.2 Handling of packaging

1.2.1 In the EU

- **Transport packaging** must be returned.
- **Sales and resale packaging** can be disposed of according to national regulations.

1.2.2 Overseas

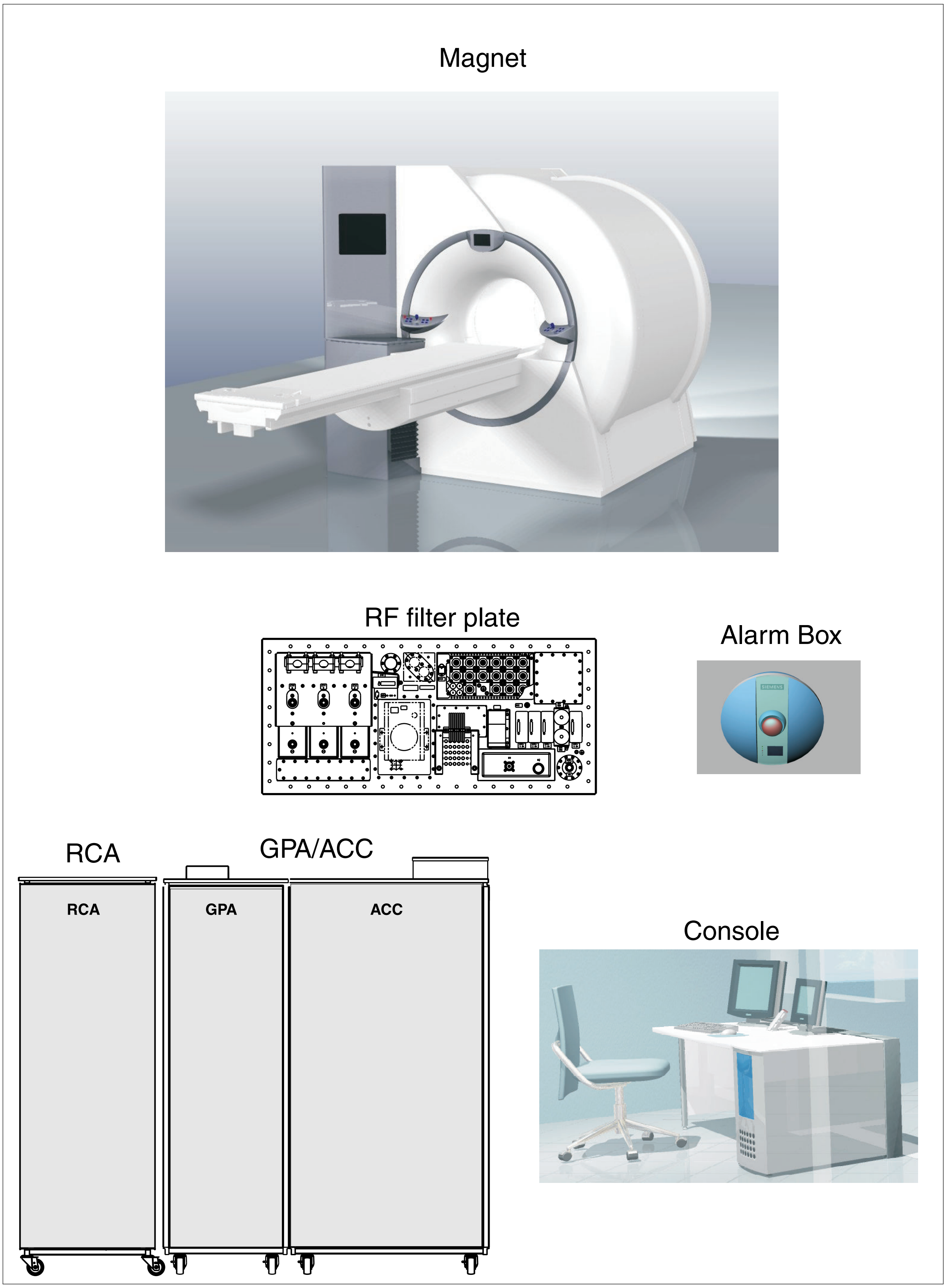
- **Transport packaging** must be returned.
  USA: Allport pallets are returned
  Japan: Wooden crate is shipped back
- **Sales and resale packaging** can be disposed of according to national regulations.

1.3 Overview of the components

The system consists of the following components:

- Magnet Type OR 70: MAGNETOM Symphony, A Tim System including the patient table and covers.
- MR electronics cabinet (consisting of GPA/ACC/CCS)
- RF filter plate
- Alarm box
- Helium compressor
- RCA
- Host PC and console components

Fig. 1: Symphony, A Tim System configuration (RCA cabinet remains on site!)



1.4 Shipping dimensions of the components

1.4.1 Magnet

Fig. 2: Symphony, A Tim System; front view

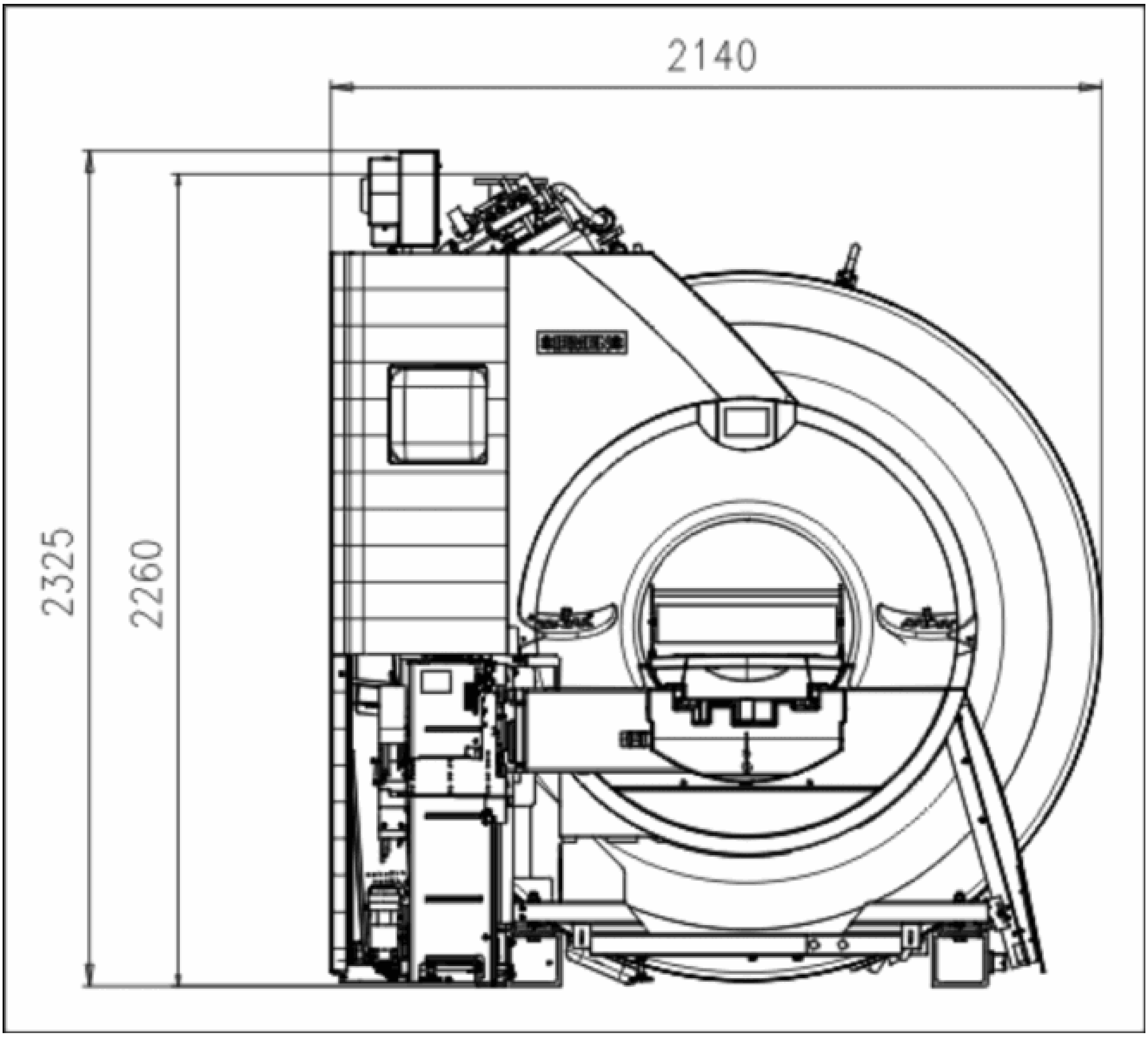
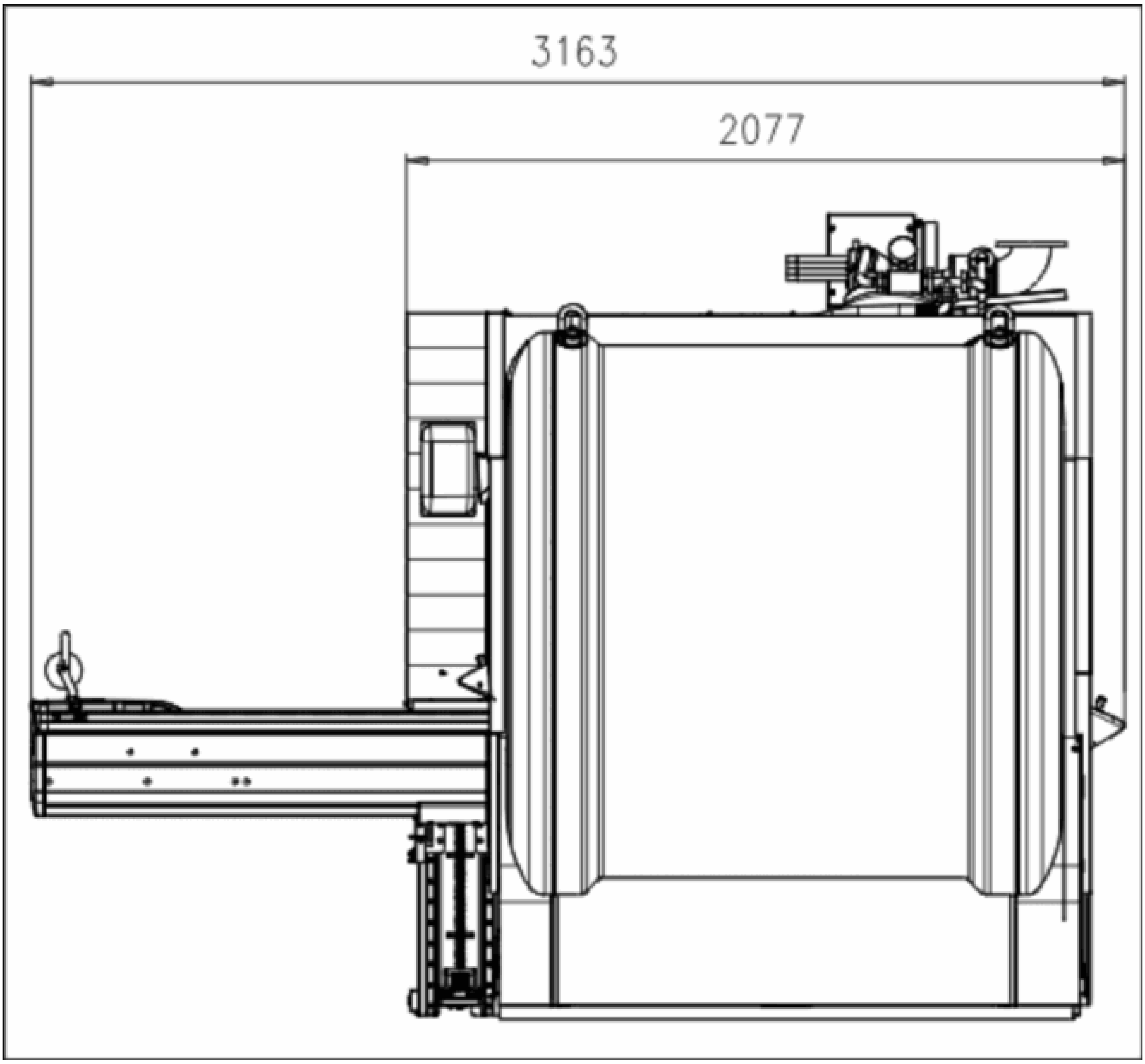


Fig. 3: Symphony, A Tim System; side view



Overall height	2325 mm.
Overall width	2140 mm.
Total length	3163 mm.
Total weight (with cryogens and gradient coil)	Approx. 5500 kg.

Gradient coil weight	800 kg.
Minimum ceiling height for transport	2325 mm[1]

1. Excluding transport rollers

1.4.2 GPA/ACC cabinet

Overall cabinet height + power supply box	1870 mm. + 100 mm
Overall width	650 mm.
Total length	1560 mm.
Total weight	1250 kg.

1.4.3 Filter plate

Overall height	560 mm.
Overall width	800 mm.
Total length	1180 mm.
Total weight	130 kg.

1.4.4 RCA cabinet with helium compressor

Overall height	1920 mm.
Overall width	650 mm.
Total length	650 mm.
Total weight	330 kg.

1.4.5 Helium compressor

Overall height	600 mm.
Overall width	550 mm.

Total length	460 mm.
Total weight	120 kg.

1.5 Explanation of the abbreviations used

The installation instructions contain certain abbreviations which are explained below:

ACC	⇒	Advanced Control Cabinet
GPA	⇒	Gradient Power Amplifier
CCS	⇒	Cabinet Cooling System
RCA	⇒	Refrigeration and Cooling Assembly
MAG	⇒	Magnet
BR	⇒	Image processor
BC	⇒	Body Coil
GC	⇒	Gradient Coil

1.6 Information for preparatory steps

1.6.1 Transport

- Only qualified carriers may handle transport activities, including delivery in the RF-room.
- SIEMENS employees are prohibited from transporting the magnet or other heavy components.
- Prior to presenting the carrier with orders, the project manager as well as the architect have to determine the load capacity of the transport route and temporary storage areas. The architect has to confirm in writing that the transport route can handle the necessary load.
- It is the responsibility of the carrier to obtain the information regarding the various transport routes prior to the actual transport. The carrier has to be provided with copies and a confirmation of the transport plans at the time of the incoming order.

1.6.2 Load capacity of the transport route

i In case of doubt with respect to the transport route, request that the project manager obtain the necessay advice from construction engineers, stress analysts, or the architect.

1.7 Damage during transport

- Notify the project manager and carrier immediately of any damage to the packaging or product.
- After the project manager and the carrier have been informed, send a fax to Siemens Magnet Technology (Fax: 0044-1865-850176) and to CS PS MR (Fax: 0049-9131-843593) in Erlangen. Describe the kind and extent of damage in the fax. If possible, enclose photos of the damaged parts.

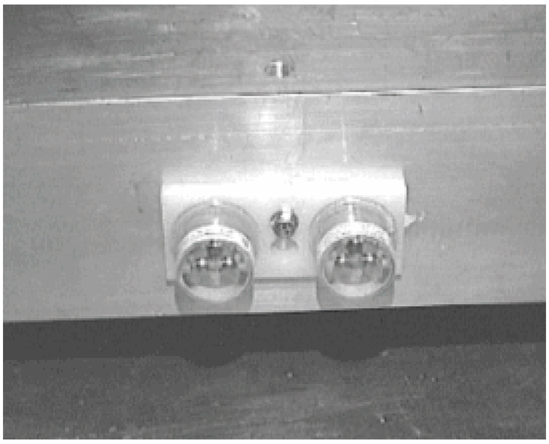
Compare all parts to the packing list included with each system and notify the project manager and MRL in Erlangen of missing or damaged parts by fax.

- Checking the shock indicators **or** the ShockBug.

1.7.1 Checking the shock indicators

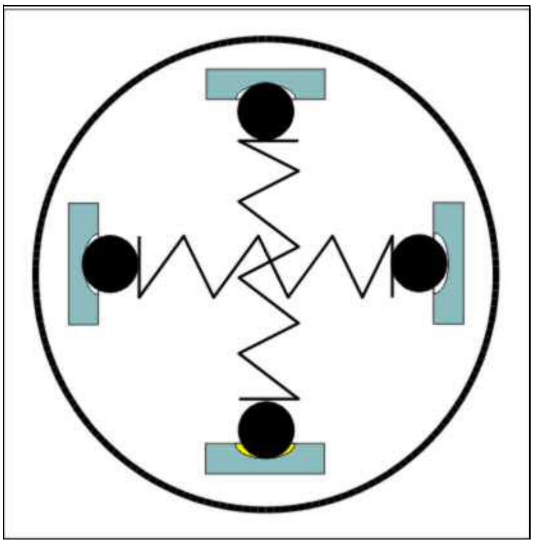
As seen from the patient table, the shock indicators are located in the right magnet box section. One of the displays is configured for 2 g horizontal, the other for 5-6 g vertical.

Fig. 4: Location of the shock indicator



- Check the shock indicators (1) while the magnet is still on the truck when it arrives at the installation site.
- Check the status of the shock indicators again after the magnet is in its final position in the magnet room.
- Remove the shock indicators after the magnet is in its final position.

Fig. 5: Shock indicator



- The shock indicators are in good condition if they resemble the figure on the left.
- The status of the shock indicator must be entered in the installation protocol.

> If the status of the shock indicator is not okay, the project manager must be informed immediately.

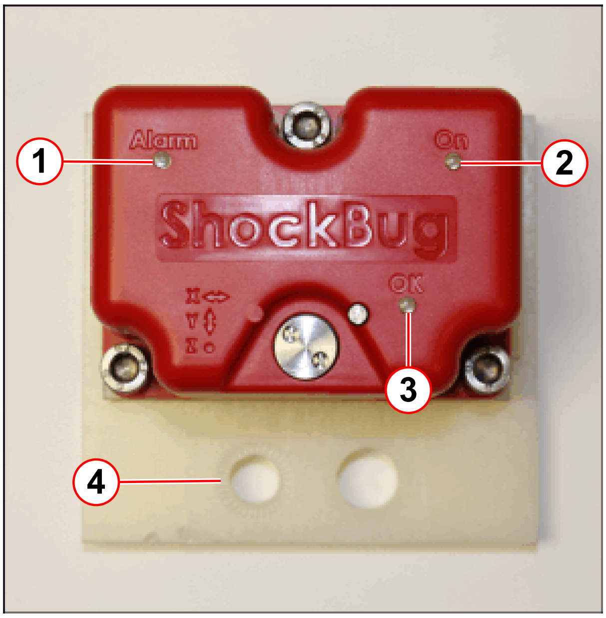
1.7.2 ShockBug

1.7.2.1 Unpacking

Checking the ShockBug indicators

A ShockBug indicator is fitted to all magnets during transportation.

Fig. 6: ShockBug indicators

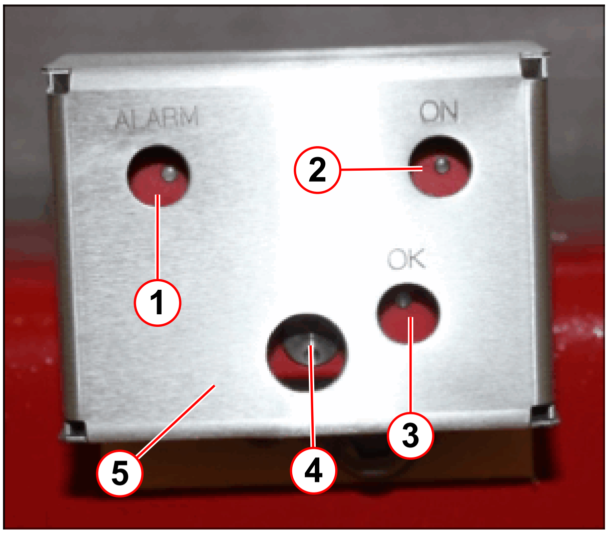


(1) Alarm LED
(2) 'ON' LED
(3) OK setup LED
(4) Mounting holes

The assembly incorporates 2 levels of shock indicators.

- **2G** indicator measures **horizontal** impact.
- **5G** indicator measures **vertical** impact.

Fig. 7: ShockBug with cover



(1) Alarm LED
(2) ON LED
(3) OK setup LED
(4) I Button
(5) Protective metal cover

The LED indicators are visible through the holes provided in the aluminium protection box.

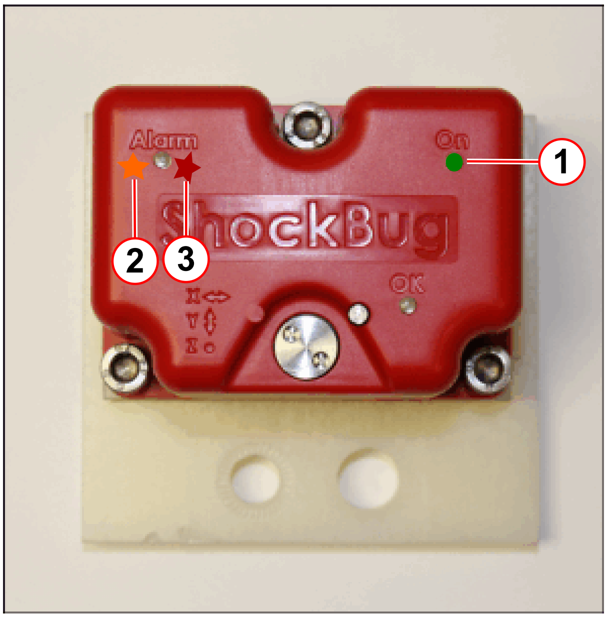
The iButton is accessible through a slot in the protective box.

The ‘ON’ LED should be flashing Green if the ShockBug is running.

The ‘OK’ LED is only lit during the download process.

- Check the ShockBug indicator after the magnet system is moved to the final installation position.

Fig. 8: ShockBug warning LED display



(1) ON = Green
(2) Alarm ORANGE = 2G
(3) Alarm RED = 5G

- The ShockBug alarm LED will flash:
  - **Orange** when the **2G** indicator has recorded an event.
  - **Red** when the **5G** indicator has recorded an event.
- The status of the shock indicator must be entered in the Installation Protocol.

> If the shock indicator has been activated, inform the Project Manager immediately.

- If there are no alarms, remove the ShockBug, when the magnet is secured in its final position.
- **Disposal** - The ShockBug (and internal battery) is 'single-use' unit.
  - » Remove from the magnet.
  - » Dispose of locally.

1.7.2.2 Transport Damage

> Notify the Project Manager and Transport carrier immediately of any damage to the Magnet system, or its component packaging.

ShockBug alarm LED:-

- Flashes **ORANGE** when triggered at the **2G** threshold.
- Flashes **RED** when triggered at the **5G** threshold.

Arrival at customer site, data from the ShockBug will require downloading if:

- The alarm LED is flashing **RED**.
- The alarm LED is flashing **ORANGE**
- Download kit (material number, **10615924)**

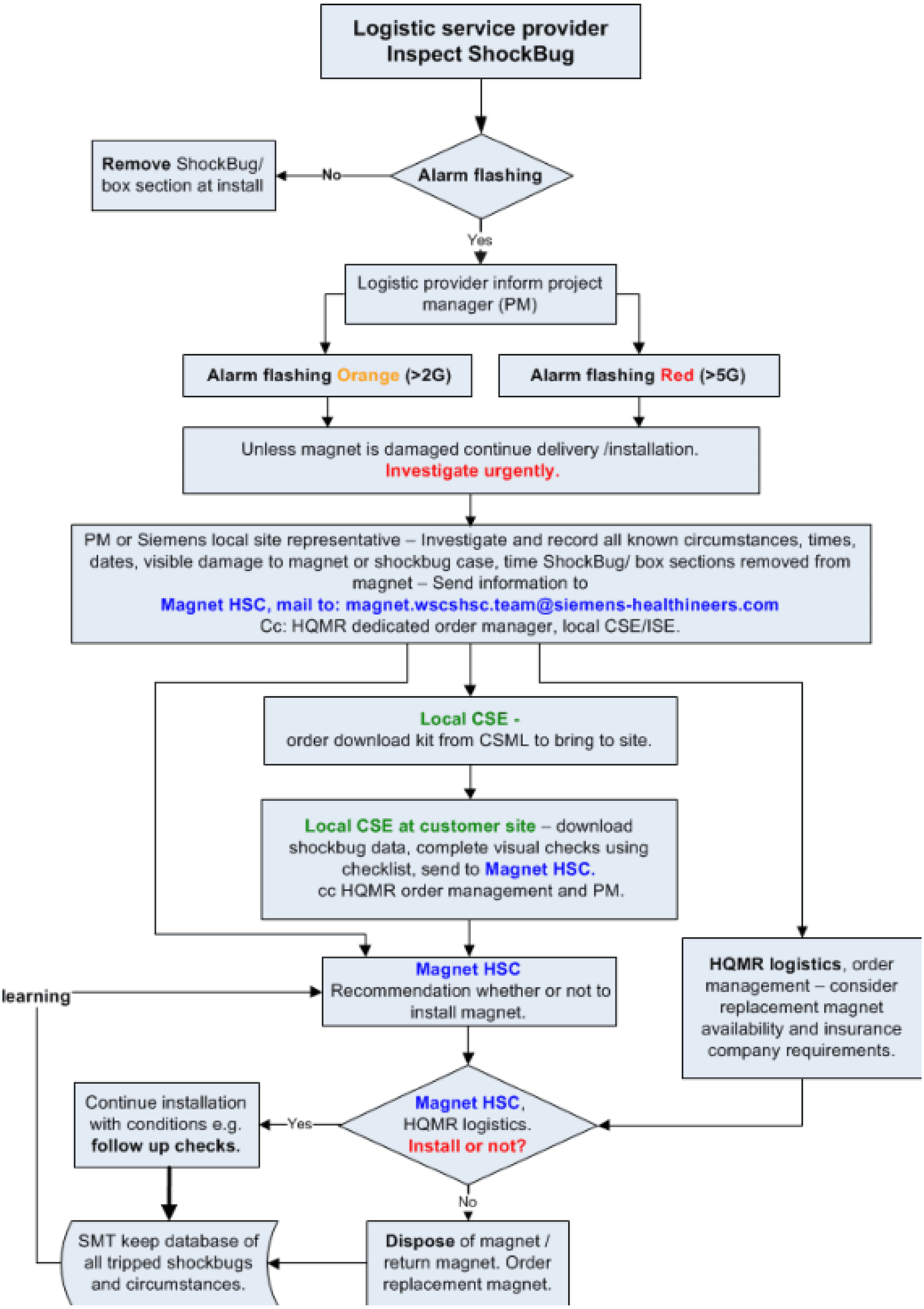
Follow the ShockBug Transport Review Process (➔ Fig. 9 Page 23)

- Email: Siemens Magnet Technology (➔ , Siemens Healthineers Magnet WS CS HSC <magnet.wscshsc.team@siemens-healthineers.com>)
- Email: MR logistics dedicated order manager,

  klaus.walter@siemens.com, karlheinz.ideler@siemens.com.
- Record the extent of damage.
- Photograph any damage to the magnet Shockbug, and components.
- Inspect the ShockBug cover for damage.
- Check the integrity of the tamper proof paint on the Shockbug screw, prior to removal.

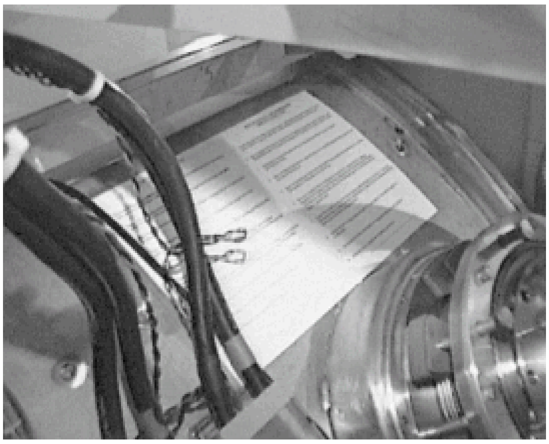
Compare all parts to the packing list included with each system and notify the Project Manager and MRL in Erlangen by email about damaged parts.

Fig. 9: ShockBug - Transport review process



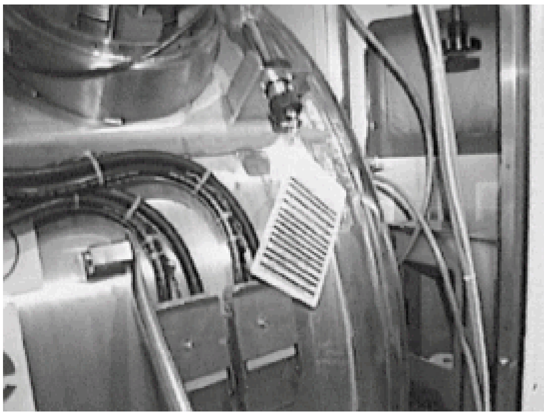
1.7.3 Important notes for the operation of the magnet

Fig. 10: Important notes



- The **Important Notes** are located on the left side of the magnet as seen from the patient table(→ Fig. 10 Page 24)
- These notes are very important for the operation of the magnet.

Fig. 11: Smart syphon operating instructions



- The operating instructions for handling the smart syphon are attached to the LHe transfer line(→ Fig. 11 Page 24)

- The **Important Notes** are located on the left side of the magnet as seen from the patient table
- These notes are very important for the operation of the magnet.
- The operating instructions for handling the smart syphon are attached to the LHe transfer line.

1.7.3.1 Table of important notes

This table is located at the magnet.

1	It is essential that the installation/operators manual be read and understood before commencing any operation on the system.
2	**Always** wear protective clothing when handling cryogens and heed all warnings and safety notices associated with cryogens and magnetic fields.
3	**Never** allow recommended pressure levels to be exceeded in cryogen dewars or vessels inside the magnet.

4	Before energizing the magnet, ensure that the ERDU is connected to the magnet and in full working order.
5	**Always** ensure that the helium vessel is fully depressurized before removing any blanking plug, baffle set, or fitting from the turret.
6	**Always** check that the current lead (DCL) is connected for the correct magnet polarity before running current into or out of the magnet, and ensure that gas is venting through the lead (DCL) with adequate flow.
7	**Never** leave a running magnet power supply unattended while it is in operation. Before removing the current lead, ensure the power supply is in standby and no fault lights are illuminated.
8	**Always** keep a log of any operation performed and record cryogen levels on a daily basis.
9	**Never** allow the helium level to fall below the recommended minimum.
10	**Always** keep a stock of essential spares on-site.

1.8 Prerequisites for installing the system

1.8.1 Installation-related prerequisites

- Check to ensure that the magnet room is complete.
- Check to ensure that the openings of the RF-shielding are large enough (height and width in accordance with the Planning Guide).
- There is enough room for installing the magnet.
- Take into account narrow passages due to suspended ceilings, carriers, and columns.
- Determine the final location of the magnet in the magnet room.
- Ensure that the primary cold water supply is ready for operation.
- Block off the unloading zone to ensure that spectators are not harmed.
- Avoid blocking escape routes or essential access routes to the hospital.
- The rooms are completed. This means that the RF-shielding including the opening for moving in the magnet has been installed. In addition, the magnet room is equipped with sufficient lighting. Also the flooring is clean to facilitate transporting the magnet and the system components.
- The four corners of the magnet are drawn on the floor of the magnet room to determine its location. Refer to the project plan.
- The floor from the unloading zone up to the final location has been covered with metal plates (at least 5 mm thick plates).
- The tools required for the installation have to be on-site (steel rollers, hydraulic lifts, etc.)

1.8.2 Tools required

- Magnet Transport Device

Fig. 12: Magnet transport device

- Forklift/lift truck for transporting the cabinets
- four hydraulic lifts (optional in case of earthquake-specific mounting)

- A sufficient number of steel (aluminium) plates (5 mm thick to cover the transport route).
- Hot air dryer
- Standard tool kit
- Drill
- Torque wrench, 20 - 50 Nm.
- Torque wrench, 5 - 20 Nm.
- Accessories for torque wrench.
- Long and short (32 - 40 cm) levels
- Ruler, measuring tape
- Multimeter for measuring V/A/Ohm
- Protective conductor meter (recommendation: SECUTEST S3)
- Four chains/ropes (each of them approx. **3 m** having a load bearing capacity of 3000 kg each)
- MSUP Display Unit
- Manual water pump (Optional for filling the secondary water circuit, part no.: 81 15 136)

Fig. 13: Manual water pump

- Table transport device (optional: removal of the patient table)

Fig. 14: Table transport device

Fig. 15: Patient table lifting element (Part no. 103 99 521)

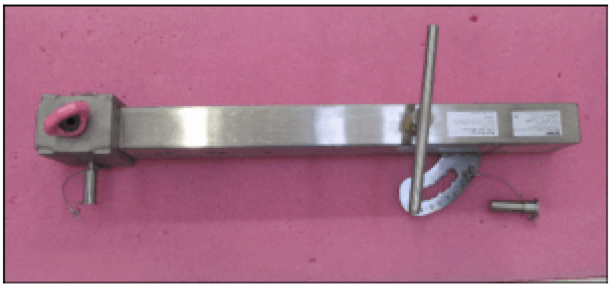
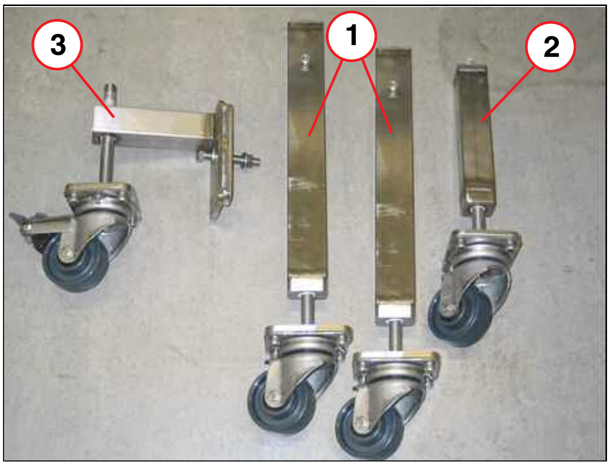


Fig. 16: Table transport device

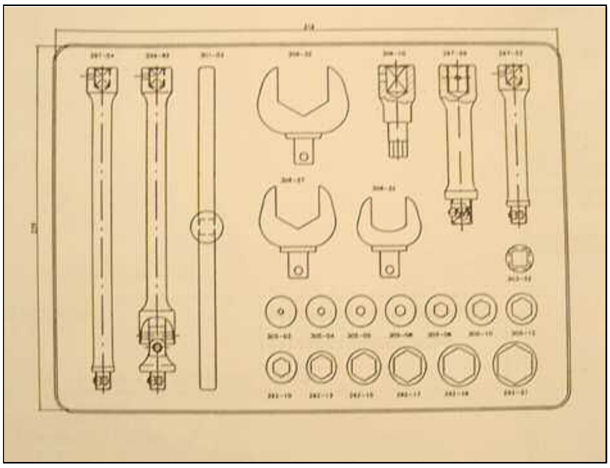


(1) Transport roller 1
(2) Transport roller 2
(3) Transport roller 3

1.8.2.1 Torque values

Fig. 17: Torque wrench

Fig. 18: Torque wrench accessories



Designation	Torque:	Wrench size
Gradient cable	22 Nm / 35 Nm	19 mm
Ramp cable	20 Nm	17 mm
7/16[1]	25 Nm	27 and 32 mm
HN[2]	6 Nm	18 mm
Water connections	50 Nm	60 mm
Water connections	25 Nm	36 and 42 mm

1. Desgination for the transmit cable connections
2. Designation for transmit and/or receive cable connections

2.1 Overview

Magnet installation	
⇒	Unloading the magnet ▪ Lifting the magnet from the Kilian pallet **or** ▪ Lifting the magnet from the truck
⇒	Moving the magnet
⇒	Remove the lifting eyes
⇒	Areas prone to earthquakes (optional)
Steps after positioning	

2.2 Unloading the magnet

WARNING

Lifting of the magnet system.

Failure to observe the following may cause damage to the magnet system and components.

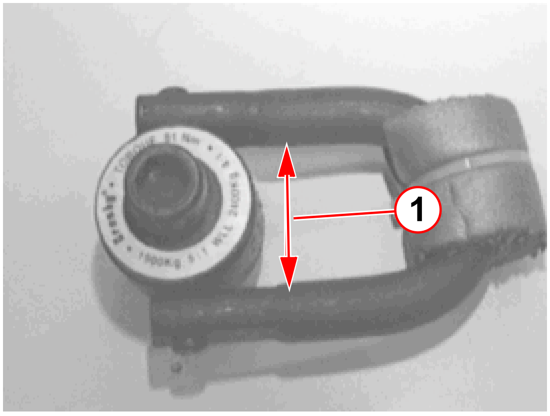
» **Note the Safety Information in chapter 1.**

» **Use 4 suitable rated slings or chains (minimum of 3 m in length) for lifting the full system weight.**

» **Check the magnet components are not fouled by the lifting straps or chains.**

» **Do not stand on the magnet. Damage to components may occur.**

Fig. 19: Lifting eye



(1) 45 mm

- The lifting eyes are part of the magnet shipment.
- The width of the lifting eyes is 45 mm.

2.2.1 Positioning the steel plates

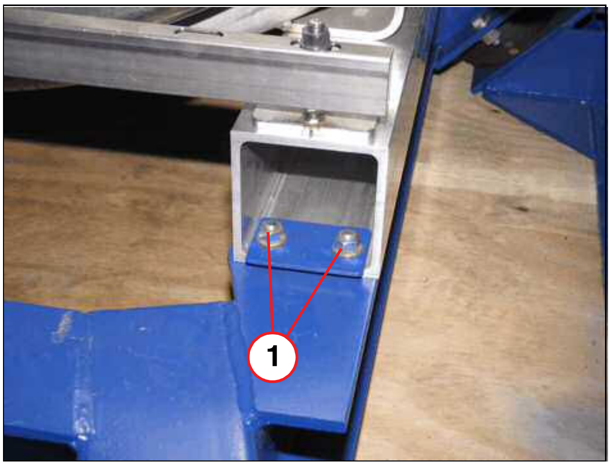
Fig. 20: Positioning the steel plates

- Position the steel plates used for the magnet.

2.2.2 Lifting the magnet from the Kilian pallet

- Remove the cover of the Kilian pallet using the crane.
- Remove the pallet´s exterior walls.
- Remove the mounting used for the magnet on the Kilian pallet.

Fig. 21: Attachment to Kilian pallet



(1) Nuts

- Remove the nuts and plates from the four corners of the box section.

Fig. 22: Four chains at the magnet

- Hook the four shackles into the lifting eyes. The opening of the shackle has to be secured by a safety slide.

To prevent the shackle from getting twisted, it should have enough space in the lifting eye.

Proceed with care so that the cables, connections, and components of the magnet turret are not damaged.

Fig. 23: Lifting the magnet

- Lift the magnet slowly.

Prior to lifting the magnet off the pallet, verify that the shackles and lifting eyes are attached correctly.

Proceed slowly when unloading the magnet from the pallet.

2.2.3 Lifting the magnet from the truck

- Remove the mountings from the magnet used to hold it on the truck.

Fig. 24: Four chains at the magnet

- Hook the four shackles into the lifting eyes. The opening of the shackle has to be secured by a safety slide.

To prevent the shackle from getting twisted, it should have enough space in the lifting eye.

Proceed with care so that the cables, connections, and components of the magnet turret are not damaged.

Fig. 25: Lifting the magnet

- Lift the magnet slowly.

> Prior to lifting the magnet off the truck, verify that the shackles and lifting eyes are attached correctly.

> Proceed slowly when unloading the magnet from the truck.

Fig. 26: Lifting the magnet, truck

- Proceed slowly when unloading the magnet from the truck.

2.2.4 Lowering the magnet

DANGER

Never stand underneath the magnet during this procedure.

» Working under a suspended load is prohibited.

Fig. 27: Setting down the magnet

- Use the crane to suspend the magnet above the steel plates.
- Carefully place the magnet on the steel plates.
- Unhook the shackles from the lifting eyes at the magnet.

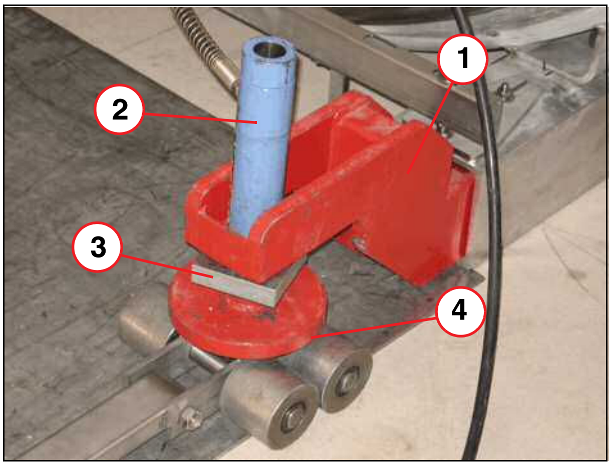
The magnet has to stand uniformly and securely on the steel plates covering the floor.

2.2.5 Installing the magnet transport kit

Fig. 28: Magnet transport device

> The lifting elements must be carefully inserted in the box section; do not damage the cover.

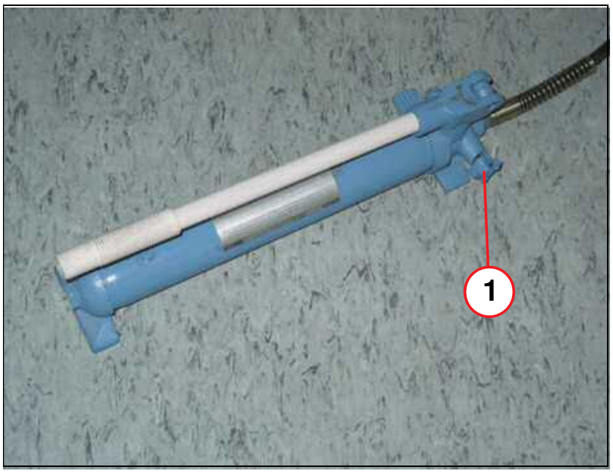
Fig. 29: Magnet transport device



(1) Lifting element
(2) Hydraulic cylinder
(3) Base plate
(4) Roller

- Carefully insert the four lifting elements (1) at the four corners of the box section.
- Insert the four hydraulic cylinders (2) in the lifting elements (1).
- Mount the four base plates (3) to the hydraulic cylinders (2).

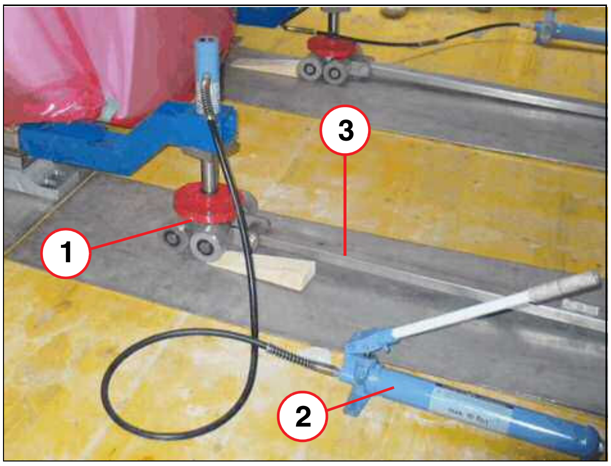
Fig. 30: Hydraulic pump



(1) Setscrews

- Turn (to the right) the set screw (1) at the four hydraulic pumps so that the magnet can be lifted.

Fig. 31: Magnet transport set



(1) Role
(2) Hydraulic lifts
(3) Center pole

- Position the four rollers (1) under the four hydraulic cylinders.
- Attach the four towing bars (3) to the rollers (1).
- Lift the magnet by simultaneously activating the four hydraulic pumps (2).
- Lift the magnet slowly and uniformly at all four corners.

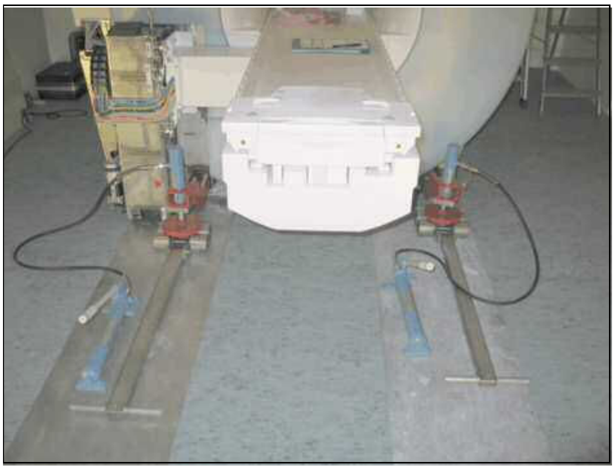
2.3 Moving the magnet

WARNING

Do not push or pull on the system (covers) or exert any pressure.

» Damage to the covers is possible.

Fig. 32: Transport with the transport device



- Using the four transport rollers, the magnet is turned in the direction desired.

During transport, ensure that the lifting elements do not detach from the box section. Carefully move the magnet into the examination room.

2.3.1 Positioning the Magnet

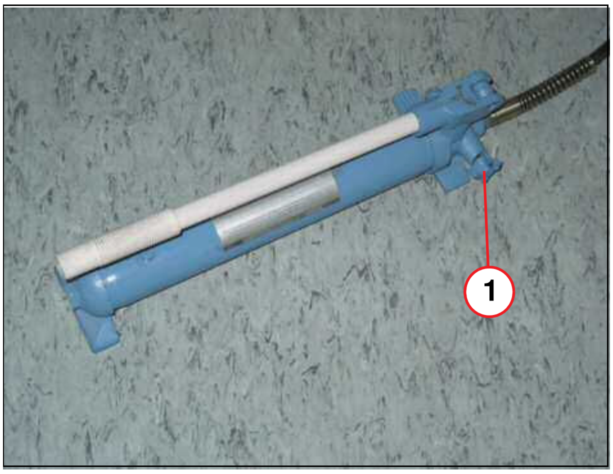
Position the magnet exactly over the markings on the floor. In case of questions, contact the project manager. In particular, maintain the proper distances from the walls.

The distance of the magnet to the wall is measured with the patient table. Measure the distance while the table is all the way in the magnet or out of it.

- Place the Sylomer pads and shim plates under each end of the magnet box sections.
- Aligned with the markings, carefully set the magnet down on the floor.

Lower the magnet uniformly at all four corners.

Fig. 33: Hydraulic pump



(1) Setscrews

- Lower the magnet by slowly opening (turn to the left) the set screws (1) simultaneously at the four hydraulic pumps.

When lowering the magnet, ensure that the covers do not touch the floor.

2.3.2 Adding the Sylomer pads and aluminum shim plates

The magnet is delivered with eight Sylomer pads 150 x 200 x 25 mm for the box sections and two Sylomer pads 100 x 100 x 25 mm for the patient table support disk.

Depending on how much the magnet center height is above floor level, use the corresponding kick plate set.

Sylomer kit contents

Sylomer pads delivered with the system		
Sylomer T (black)	8 pieces 200 x 150 x 25 mm	Box section
Sylomer HD 300 (green)	8 pieces 200 x 150 x 25 mm	Box section
Sylomer (black)	3 pieces 100 x 100 x 25mm	Support disk for patient table and cover frame

New name of item: Sylodamp HD 300 = Sylomer HD 300

- Slide the aluminum shim plates between the Sylomer pads and the floor. The number and the thickness of the plates depend on how much the magnet alignment deviates from the values of the water hose level.

Fig. 34: Sylomer pads and shim plates for leveling the magnet

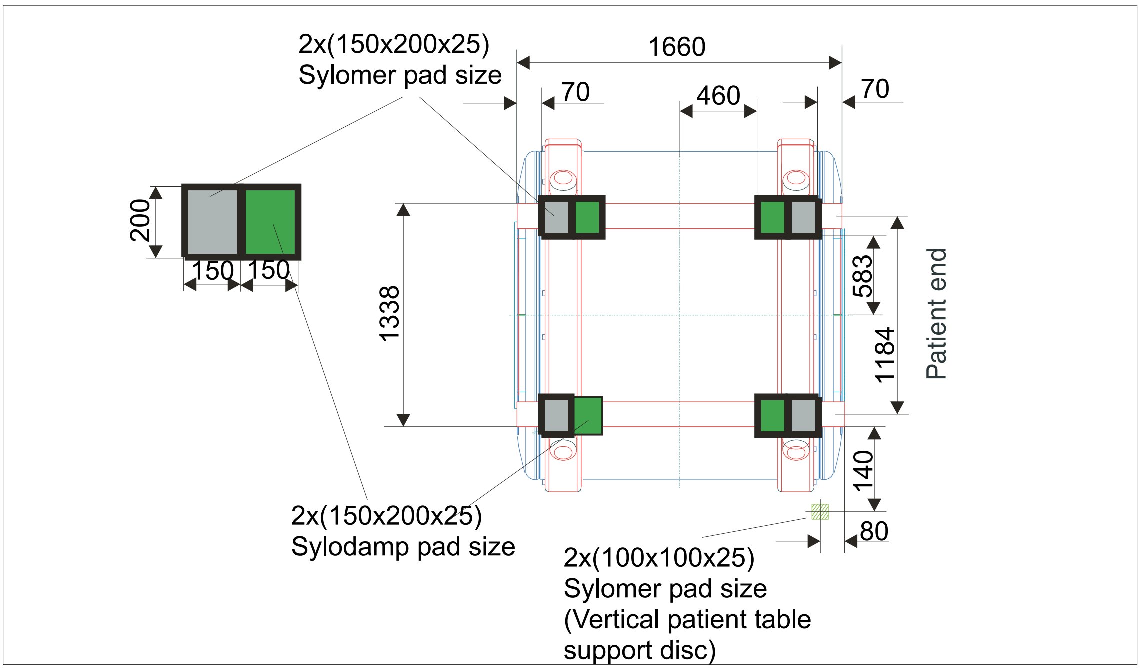


Fig. 35: Sylomer / Sylodamp position without MDF plates

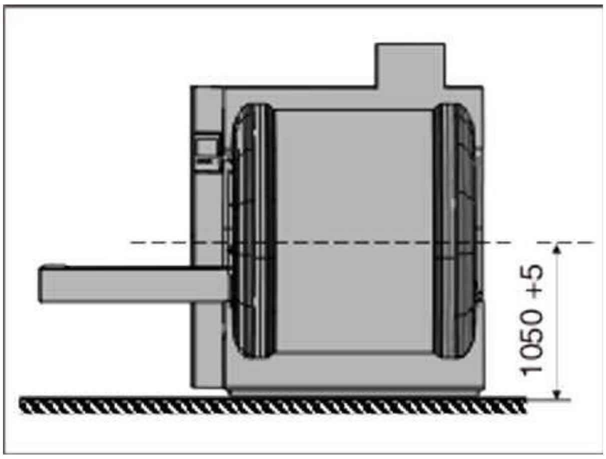
(1) Box section
(2) 5 mm aluminium plate
(3) Sylomer (black)
(4) Sylodamp (green)
(5) Alu shim plates
(6) Floor

2.3.3 Setting the center height of the magnet

The quench pipe must not be installed on the magnet (it may have to be removed) when the magnet is lifted. It may be necessary to loosen the cables, otherwise the system can be damaged.

- Either add or take away the same number of aluminum plates at each corner of the magnet until the center height of the magnet is 1050 + 5 / - 0 mm.

Fig. 36: Center height



> Check once more the horizontal alignment of the magnet, the distance to the walls on either side, as well as the center height. The magnet must rest squarely on the floor.

> If you have to bolt the magnet to the floor due to earthquake requirements adhere to (→ Areas prone to earthquakes / Page 47)!

2.3.4 Leveling the magnet

The objective is to align the magnet in the horizontal and to set the center height of the magnet.

To align the magnet, you need the following:

- Water hose level
- Sylomer pads
- Aluminum shim plates (shipped with the magnet)

Procedure:

- Use the water level hose to level A with B and C with D (in the X direction).
- Raise the magnet with the hydraulic lifts and slide the aluminum shim plates between the floor and the Sylomer pads (→ 5/Fig. 35 Page 42).
- Lower the magnet and verify that A,B and C,D are aligned with one another. If they are not , add more shim plates until the markings match the water level.
- Subsequently, check whether A is level with C (on the left side of the magnet) and B is level with D (on the right side of the magnet). Align the magnet in the Z direction as shown above (markings on the floor).

- Finally, check the alignment of all four points, A,B,C, and D, to one another. The markings must be on the same level according to the water hose level. Acceptable tolerance is 2 mm.

Fig. 37: Leveling the magnet

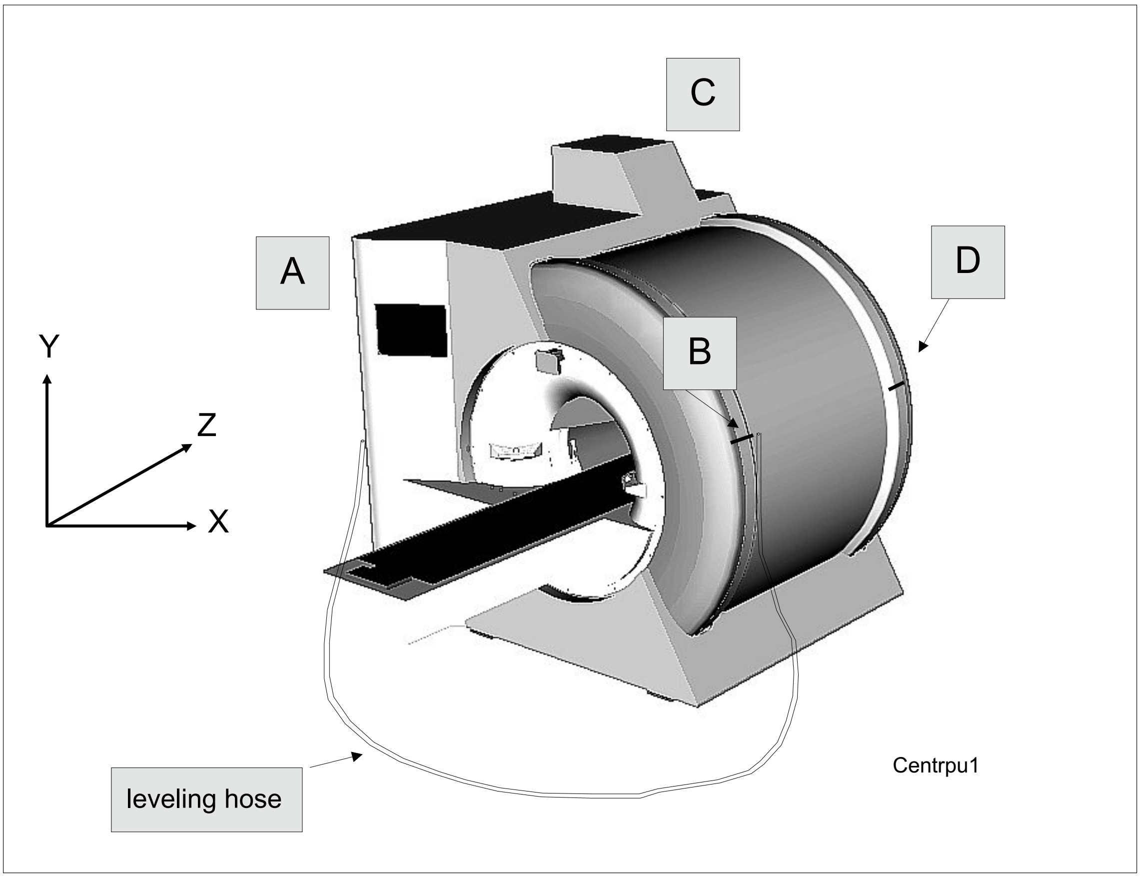


Fig. 38: Markings

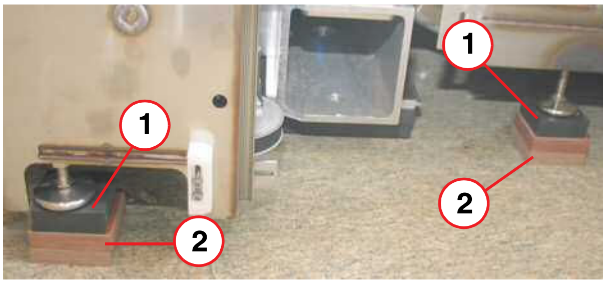
- Use the water hose level with the markings at the magnet.

» When the magnet is aligned correctly, remove the hydraulic lifts. Store the remaining aluminum shim plates in a safe place.

2.3.5 Supporting the patient table

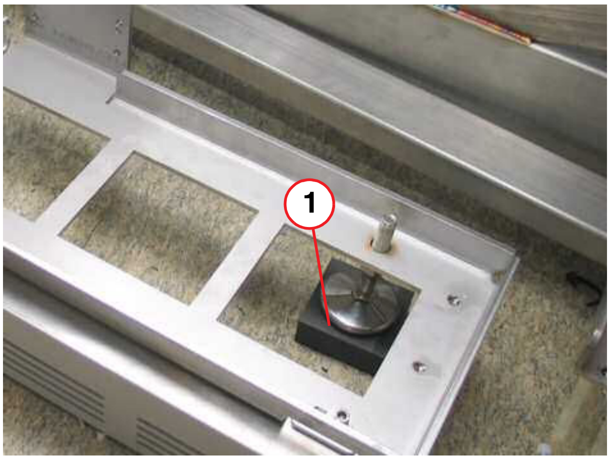
After the magnet is aligned with the patient table, the patient table has to be supported.

Fig. 39: Patient table



(1) Support foot rubber
(2) Support foot wood

Fig. 40: Patient table support arm

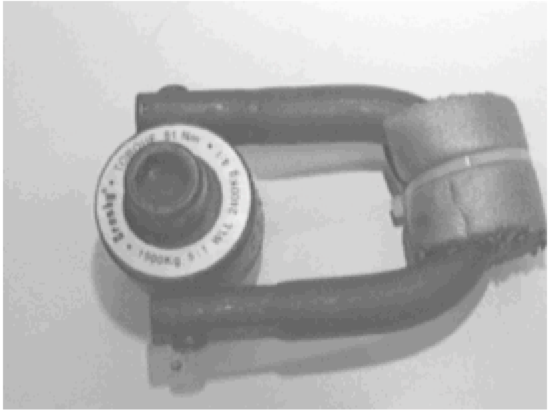


(1) Support feet rubber

- Positioning the Sylomer pads (1) and the blocks (2) included in the delivery volume under the washer (2) of the support disks.
- Turn the washer (2) in clockwise direction to lower it on to the Sylomer pad (1).
- Turn the washer an additional rotation using a flat wrench. The washer should be now be tight against the Sylomer pad.
- Tighten the counternut.

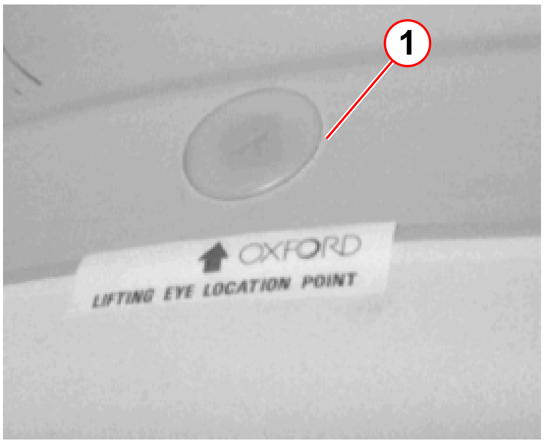
2.4 Removing the lifting eyes

Fig. 41: Lifting eye



- Remove the lifting eyes after the magnet is in its final position.
  » You will need a **14 mm** Allen key.

Fig. 42: Plastic caps



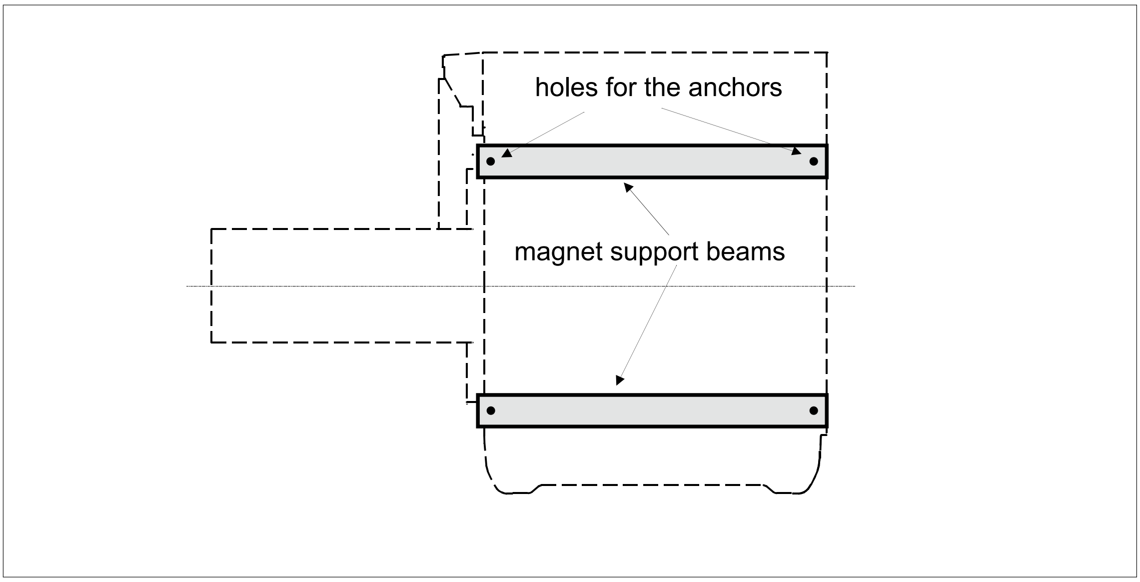
(1) Plastic cap

- Replace the lifting eye with the delivered plastic caps.

2.5 Areas prone to earthquakes

2.5.1 Bolting down the magnet

Fig. 43: Magnet



Procedure:

- Position the magnet at its final position.
- Lift the magnet with the hydraulic lifts off the rollers.
- Mark the holes for the anchors. There is one hole in each of the four corner joints of the magnet support beams.
- Lift the magnet on the rollers again.
- Move the magnet. Drill the holes, and insert the respective anchors (parts like the screws and the anchors have to be obtained locally).
- Position the magnet again.
- Align the magnet vertically and horizontally.
- Bolt the magnet to the floor.

3.1 General

The following steps have to be performed before you start with the installation of the components and the cabling.

Concluding steps	
⇒	Determining the helium level
⇒	Sea freight / air freight kit removal instructions
⇒	Return parts
⇒	Connecting the quench tube
⇒	Galvanic separation of the magnet
⇒	Checking the distance between patient table and the magnet cover
⇒	Positioning the container for dripping water
Component installation in the RF room	

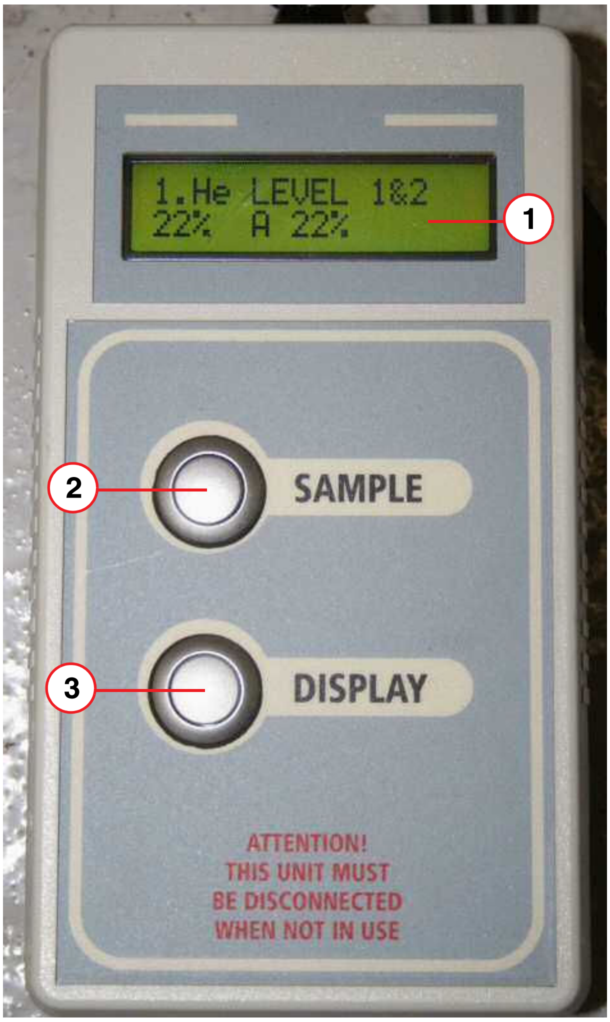
3.2 Determining the Helium Level

On arrival at the installation site, the **helium level** must be checked. Ensure that mains power has been connected to the MSUP.

The helium level can be checked using:-

- Magnet Supervisory (MSUP-Display Unit) (connected to MSUP **X9**)
  - The hand-held unit is a **service tool** (order part number **07758522**).

Fig. 44: MSUP-Display Unit



(1) LCD display
(2) Sample button
(3) Display scroll button

Determining the **helium level**:

- **Connect** the MSUP-Display Unit to the MSUP (**X9**).
- Press both the **Sample** (2) and **Display** (3) buttons (at the same time), for approximatley **5 sec**.
- Press **Display** (3).
- The unit will **automatically** perform a helium level test. The **helium fill level** is shown in the display (1).

Alternatively:

- If mains power is connected to the MSUP **download MSUP data** to the host or laptop computer.
- **NOTE**: With battery power, **only 2 minutes** of power is available for download.

> If site power is NOT available for a period of days, it is advisable to fit the External 36V PSU kit (08396371) to the MSUP.

> Helium level readings below 45% can be inaccurate when the Magnet Display Unit is on BATTERY power. If site power is not available for a number of days then a cross-check with the carbon resistors may be required to confirm the magnet temperatures.

If the Magnet Display Unit shows a helium fill level of < 30%:-

- Refill with helium.

Before commencing a helium fill:

- **Remove** the **Transportation kit**.
- **Re-configure** the magnet **venting** (operational configuration).
- **Connect** the **quench pipe**. Refer to **Quench pipe connection**

> Ensure that the Quench Line has been correctly installed and safe before performing Installation and Startup procedures.

If Magnet Display Unit does not indicate a liquid helium level:-

- If the helium level reading is **zero**, the bottom **carbon resistors** should be checked with the MSUP-Display Unit (screens 4 and 5).
  - » Using a multimeter, check pins **d - e** on 41 way turret electrical connector, or screens 4 and 5 on the hand-held MSUP-Display.
  - » Check that the magnet is at **helium temperature** by comparing the readings with the **Magnet Acceptance Document**.

> Inform Magnet HSC immediately. (→ Magnet HSC, mailto:magnet.wscshsc.healthcare@siemens.com)
>
> A bottom helium fill may be required.

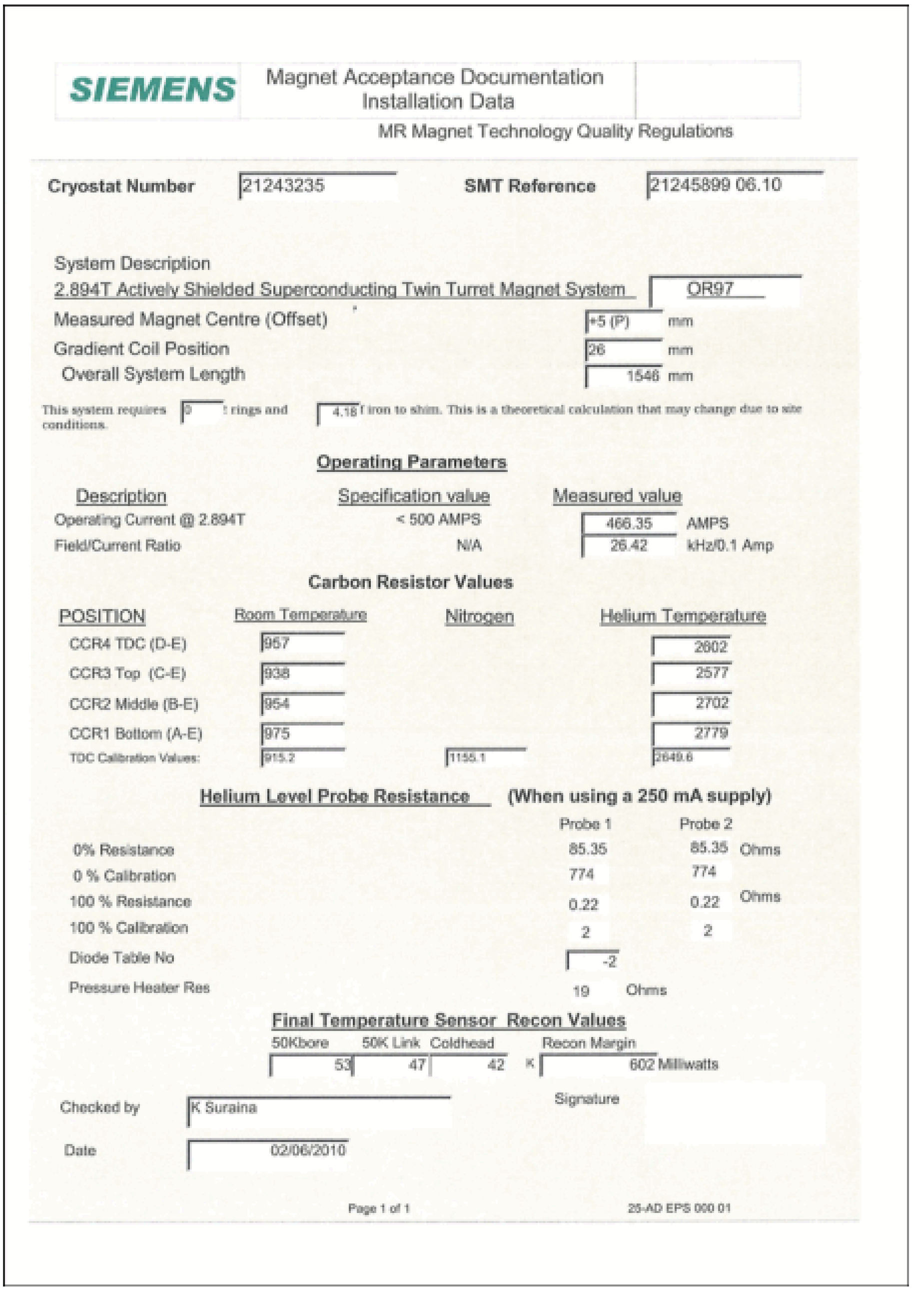
> Use the DISPLAY button on the MSUP-Display Unit to scroll through the parameters.

Checking magnet temperatures

- Use the **MSUP-Display Unit** to measure:-
  - **Helium level** values
  - **Carbon resistor** values

> Ceramic Carbon Resistors
>
> Carbon resistors have been replaced with Ceramic Carbon resistors (CCRs), which have a different resistance range.
>
> Refer to the Magnet Acceptance Document.

Fig. 45: OR97 Magnet Acceptance Document


SIEMENS Magnet Acceptance Documentation
Installation Data
MR Magnet Technology Quality Regulations

Cryostat Number 21243235 **SMT Reference** 21245899 06.10

System Description
2.894T Actively Shielded Superconducting Twin Turret Magnet System OR97
Measured Magnet Centre (Offset) +5 (P) mm
Gradient Coil Position 26 mm
Overall System Length 1546 mm

This system requires 0 rings and 4.18 iron to shim. This is a theoretical calculation that may change due to site conditions.

Operating Parameters

Description	Specification value	Measured value
Operating Current @ 2.894T	< 500 AMPS	466.35 AMPS
Field/Current Ratio	N/A	26.42 kHz/0.1 Amp

Carbon Resistor Values

POSITION	Room Temperature	Nitrogen	Helium Temperature
CCR4 TDC (D-E)	957		2802
CCR3 Top (C-E)	938		2577
CCR2 Middle (B-E)	954		2702
CCR1 Bottom (A-E)	975		2779
TDC Calibration Values:	915.2	1155.1	2649.6

Helium Level Probe Resistance (When using a 250 mA supply)

	Probe 1	Probe 2	
0% Resistance	85.35	85.35	Ohms
0 % Calibration	774	774	
100 % Resistance	0.22	0.22	Ohms
100 % Calibration	2	2	
Diode Table No	-2		
Pressure Heater Res	19	Ohms	

Final Temperature Sensor Recon Values

50Kbore	50K Link	Coldhead		Recon Margin
53	47	42	K	602 Milliwatts

Checked by K Suraina Signature

Date 02/06/2010

Page 1 of 1 25-AD EPS 000 01

Tab. 1 Expected Carbon Resistor Values

41 way plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon One a - e (Bottom)	140 - 180	170 - 200	700 - 1500
Carbon Two b - e	120 - 160	130 - 180	700 - 1500
Carbon Three c - e	120 - 160	130 - 180	700 - 1500

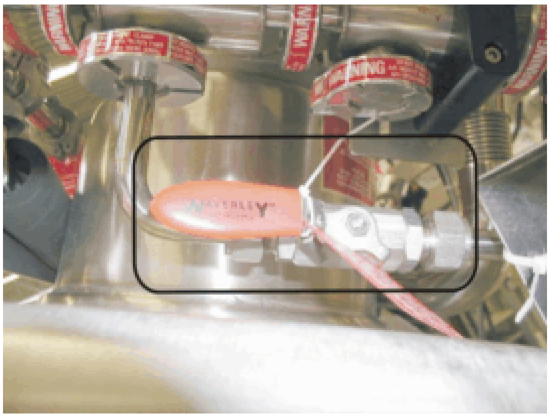
41 way plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon Four d - e (Top)	120 - 160	130 - 180	700 - 1500
All Ceramic Carbon Resistors (CCRs)	850 - 1050	950 - 1350	2300 - 3200

The above table is a guide only. Always compare the readings with those on the Magnet Acceptance Document.

3.3 Sea freight / air freight kit removal instructions

The magnet is shipped with the bypass valve (→ Fig. 46 Page 53) open!

Fig. 46: Valve open



The bypass valve is open during the transport of the magnet!

Fig. 47: Valve closed

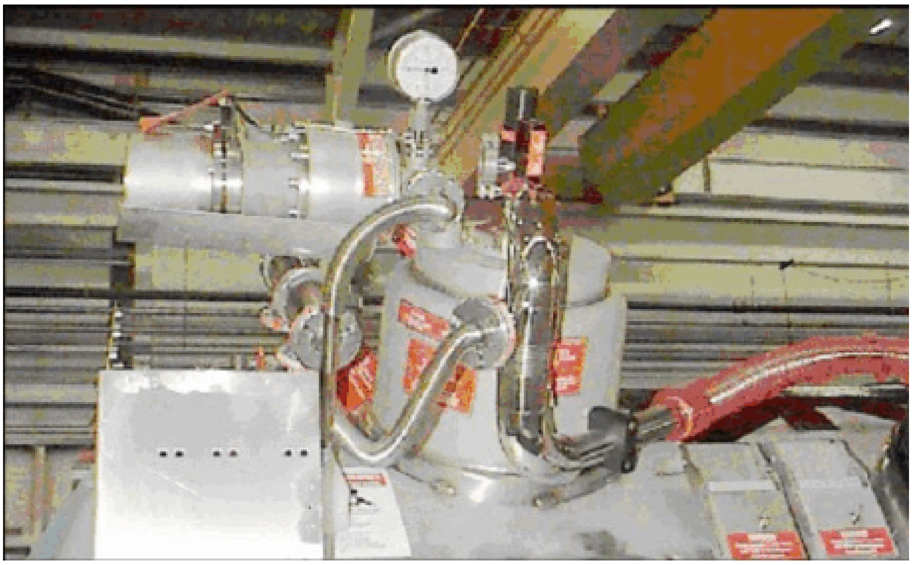
- Depressurize the magnet before removing the air/sea freight kit.
- After depressurization of the magnet close the bypass valve.

Close the bypass valve after depressurization of the magnet (→ Fig. 47 Page 53)!

3.3.1 View of the magnet turret after arrival

No ice formation is visible; o.k.

Fig. 48: Sea ship kit; magnet under normal conditions



Ice formation visible; not o.k.

On arrival, any ice that has formed should be removed from the turret using a heat gun.

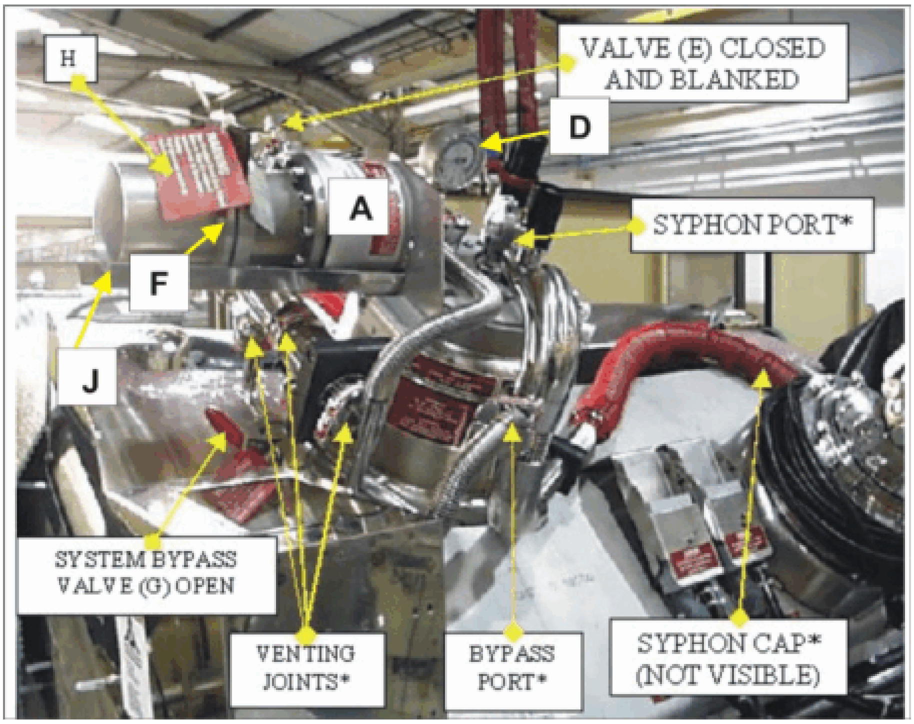
Fig. 49: Sea freight kit

3.3.2 Introduction of the Sea Freight Kit

If the magnet system has been sea freighted with liquid helium contained in the helium vessel, the venting system will be equipped with a relief valve and bursting disk regulating the pressure in the system to 2 bar absolute.

The magnet system can be left safely venting through the relief valve, until depressurization is required. When depressurization is required, follow the procedure described on the following pages ensuring that great care is taken.

Fig. 50: Sea freight kit



-	A-	Quench valve with integral 10 psi bursting disk
-	D-	Pressure gauge
-	E-	Shut off valve
-	F-	2 bar bursting disk assembly
-	G-	Bypass valve
-	H-	Warning label
-	J-	Quench valve drip tray

3.3.3 Procedure for removing the Sea Freight Kit

This procedure must be performed by trained personnel only. Wear protective clothing.

WARNING

Helium gas should not be allowed to vent into the imaging suite or any closed room.

» **Do not touch cold parts. Do not smoke or use open flames.**

The copper seals are for disposable use only. The seal must be replaced each time the clamp is removed.

- Check that the venting is configured as shown in (→ Fig. 50 Page 55). In particular, ensure that the gauge indicates positive pressure in the system (indicator in the orange /green range)
- If the gauge reads zero, contact the local Siemens service engineer. He should check e.g. the bursting disk in the assembly (**F**) (→ Fig. 50 Page 55) or the turret for damage or air leaks.
- If the configuration appears incorrect contact the MR-HOTLINE; Phone: ++49 (0) 9191 18-8080-0. **Do not** continue with the procedure.
- Check that the shut-off valve (**E**) (→ Fig. 52 Page 57);(→ Fig. 50 Page 55) fitted to the venting manifold is in the closed position (the handle is at a right angle to the valve axis).
- Remove the NW25 blank from the shut off valve (E) (→ Fig. 52 Page 57) and connect the on-site vent line.
- Close the bypass valve (**G**) (→ Fig. 47 Page 53) **if not done yet.**
- Confirm that the bypass valve (**G**) **is now closed (see** (→ Fig. 47 Page 53)**!**
- Open the shut-off valve (**E**) (→ Fig. 52 Page 57) in a controlled manner to approximately 15 degrees from the closed position. This will minimize the amount of helium lost during this procedure.

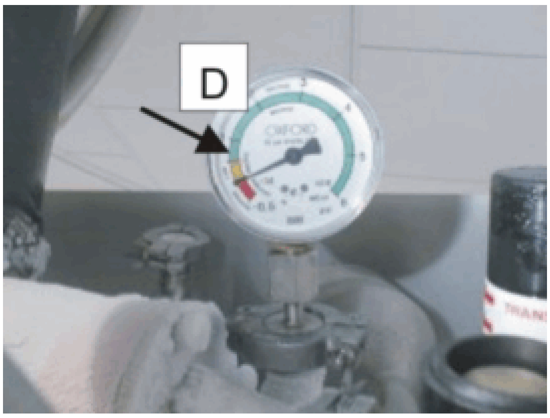
DO NOT OPEN THE VALVE (E) (→ Fig. 52 Page 57) ALL THE WAY. THIS COULD CAUSE DAMAGE TO THE MAGNET.

OPEN THE SHUT-OFF VALVE STEP BY STEP.

This procedure can take from three to approx. eight hours

- Monitor the pressure gauge. Helium gas should now be heard escaping. The vent line will cool down in a few minutes until it becomes very cold.

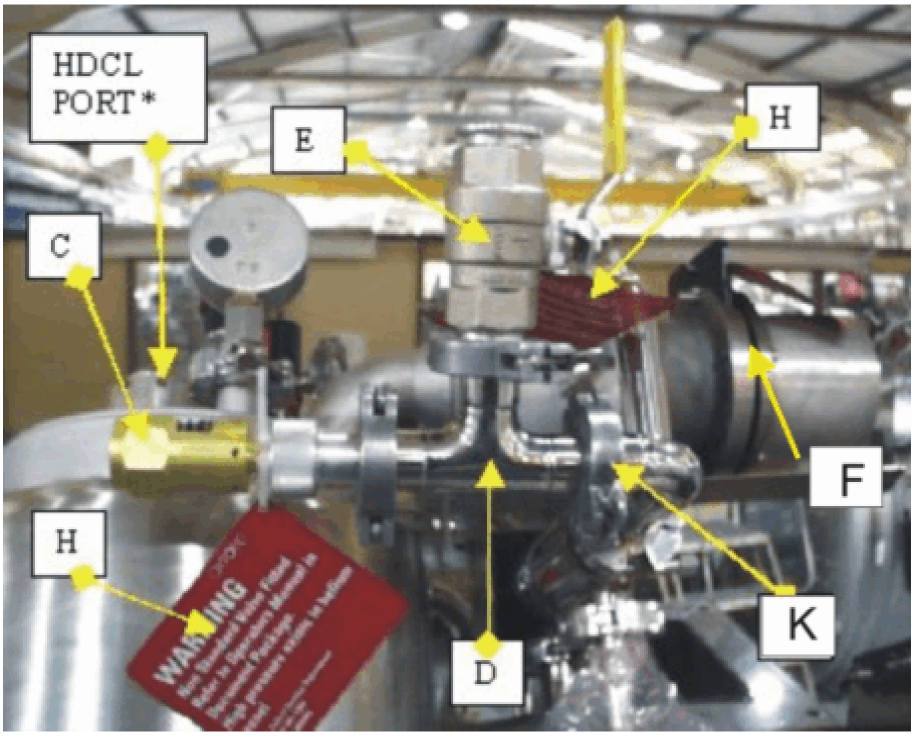
Fig. 51: Pressure gauge



- When the pressure indicated on the pressure gauge (**D**) (→ Fig. 51 Page 56) has fallen to 0.5 psi, open the shut-off valve in a controlled manner further until it is fully open.
- Allow the system to depressurize until the depressurization line has thawed, indicating that the system is now at normal operating pressure.

- De-ice the turret and venting using a heat gun before removing any of the components.
- Remove the quench valve drip tray. (**J**) (→ Fig. 50 Page 55) **This will make it easier to remove the bolts from the bursting disk assembly.**
- Remove the bursting disk assembly (**F**) (→ Fig. 52 Page 57) from the quench valve and connect the site venting to the quench valve. Reinstall the quench valve drip tray.
- Remove the t-piece (**D**) (→ Fig. 52 Page 57) with the shut-off valve (**E**) (→ Fig. 52 Page 57) and the 30 psia relief valve (**C**) (→ Fig. 52 Page 57) from the 90 degree elbow (**K**) (→ Fig. 52 Page 57) and blank the outlet (**K**) (→ Fig. 52 Page 57) with a NW25 blanking flange, PTFE seal, and clamp is supplied in the accessory case.

Fig. 52: Vent valve

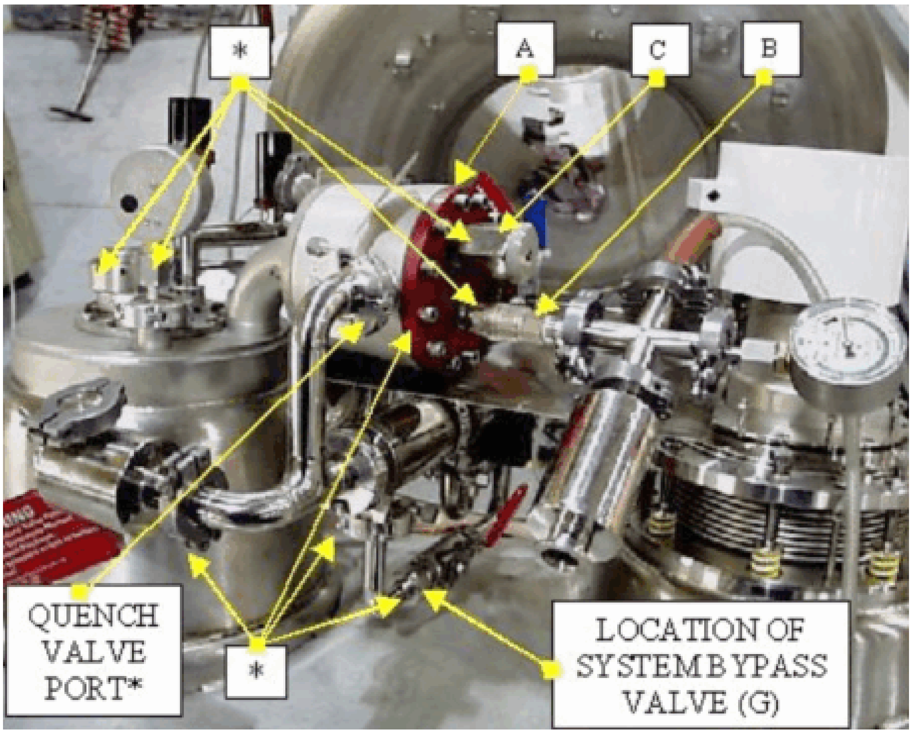


3.3.4 Introduction of the Air Freight Kit

If the magnet has been air-freighted with liquid helium contained in the helium vessel, the venting system will be fitted with relief valves which ensure that the pressure within the helium vessel is kept above atmospheric pressure.

The magnet system can be left safely venting through the relief valves until depressurization is required.

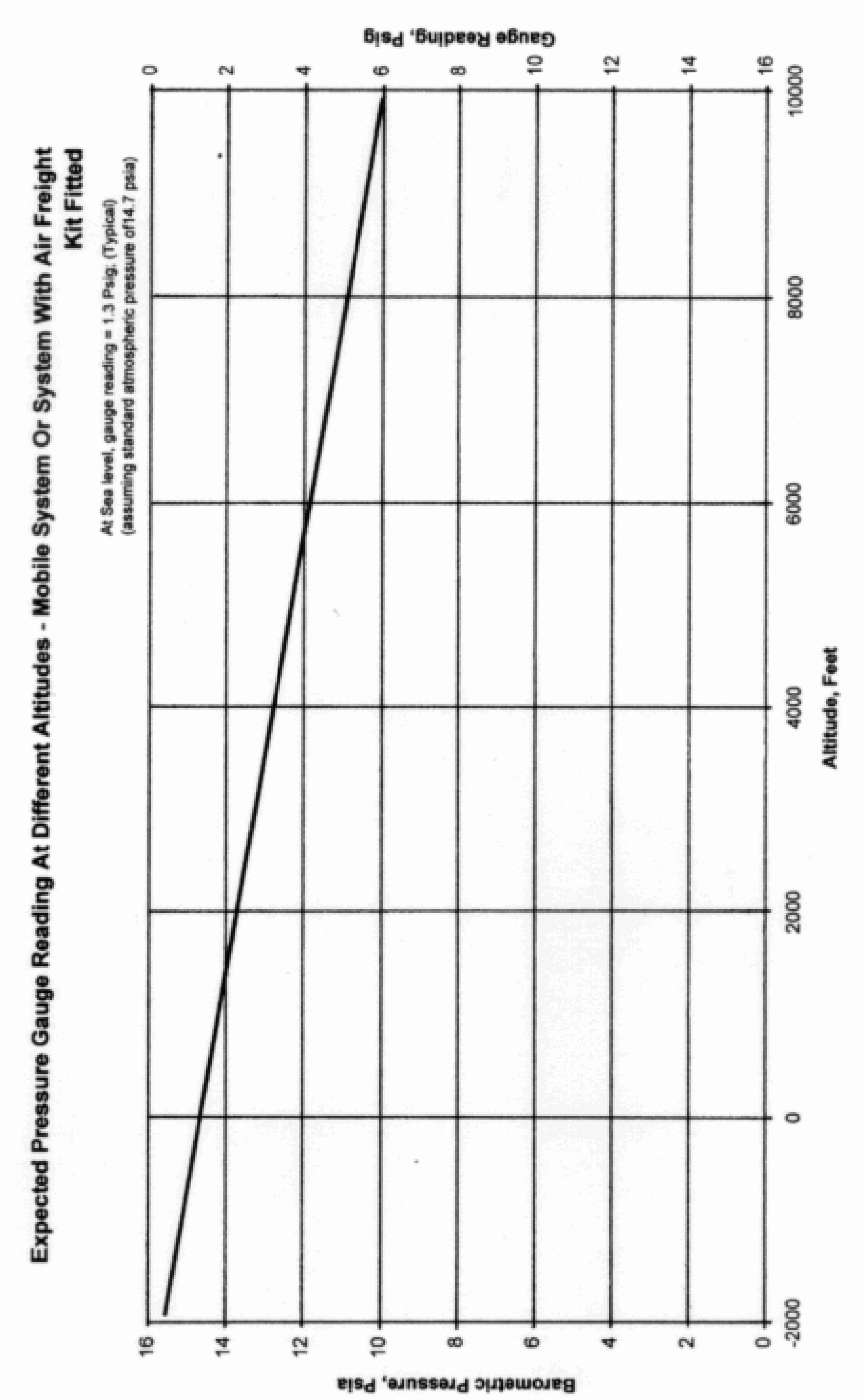
Fig. 53: Air freight kit 1



A	Plate
B	Ball valve
C	13 psi relief valve
G	Bypass

3.3.5 Procedure for removing the air freight kit

Fig. 54: Pressure table



This procedure must be performed by trained personnel only. Wear protective clothing.

WARNING

Helium gas should not be allowed to vent into the imaging suite or any closed room.

» **Do not touch cold parts. Do not smoke or use open flames.**

The copper seals are for disposable use only. The seal must be replaced each time the clamp is removed.

- When the system arrives at its destination check the following points:
  - The ball valve (**B**) (→ Fig. 53 Page 58) (→ Fig. 55 Page 61) is closed (handle at right angle to the valve axis).
  - The bypass valve (**G**) (→ Fig. 53 Page 58) on the system's venting is open (handle parallel to the valve axis)
- Check that the venting is configured as shown in (→ Fig. 53 Page 58). In particular, ensure that the gauge indicates positive pressure with the indicator in the orange/ green range. If the gauge reads zero, contact the local Siemens service engineer. He should check e.g. the bursting disk in the assembly **(D)** (→ Fig. 55 Page 61) or the turret fittings for damage or air leaks.
- If the configuration appears incorrect contact the MR-HOTLINE; Phone: ++49 (0) 9191 18-8080-0. **Do not** continue with the procedure.

  » The table (→ Fig. 54 Page 59) gives an overview of the expected pressure related to the altitude.

The system can be left safely venting through the relief valve, until depressurization is required. However, if the pressure gauge reading is off the top of the scale and/or both the 16 psia and the 13 psi valves are venting, begin the depressurization procedure immediately.

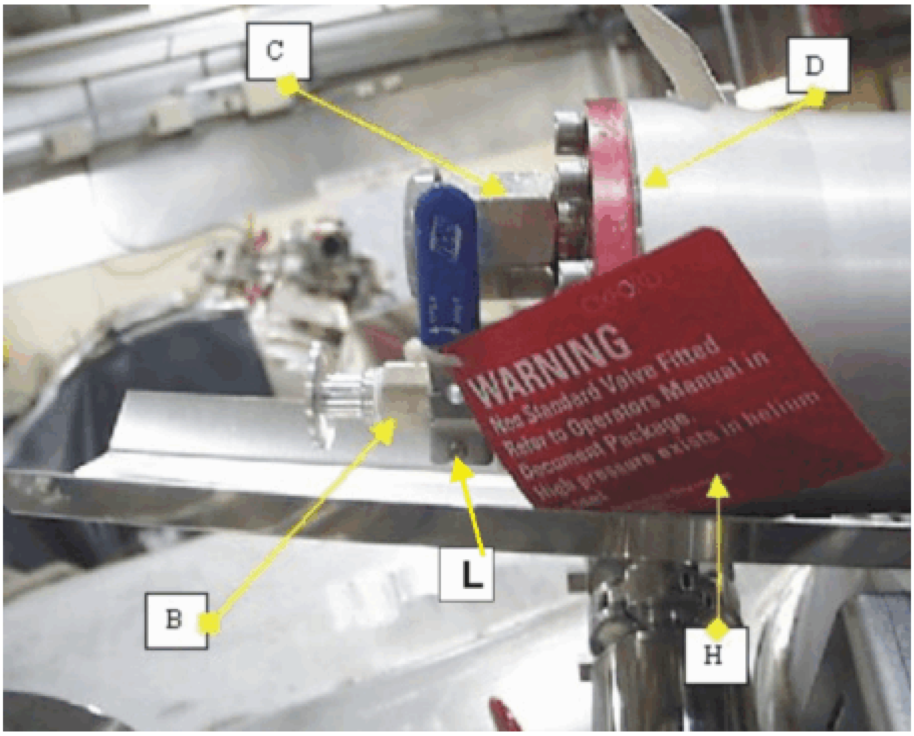
- Close the bypass valve (**G**) (→ Fig. 53 Page 58) (→ Fig. 47 Page 53) **if not done yet!**
- Confirm that the bypass valve (**G**) (→ Fig. 47 Page 53) **is now closed (see)!**
- Fit a vent pipe to the ball valve (**B**) (→ Fig. 55 Page 61) exit, exhausting to the suite vent or through a suitable vent tube running out of the suite.
- On the air freight plate (**A**) (→ Fig. 55 Page 61) release the lock on the ball valve handle (**L**) (→ Fig. 55 Page 61).
- To minimize helium "flash" loss, the pressure release must be controlled. Open the hand lever of the release valve (**B**) (→ Fig. 55 Page 61) slowly to approx. 15 degrees from the closed position (the helium gas should be heard escaping).

DO NOT OPEN THE VALVE (B) (→ Fig. 55 Page 61) ALL THE WAY. THIS COULD CAUSE DAMAGE TO THE MAGNET.

OPEN THE VALVE SLOWLY IN SMALL STEPS.

This procedure can take from one to approx. two hours.

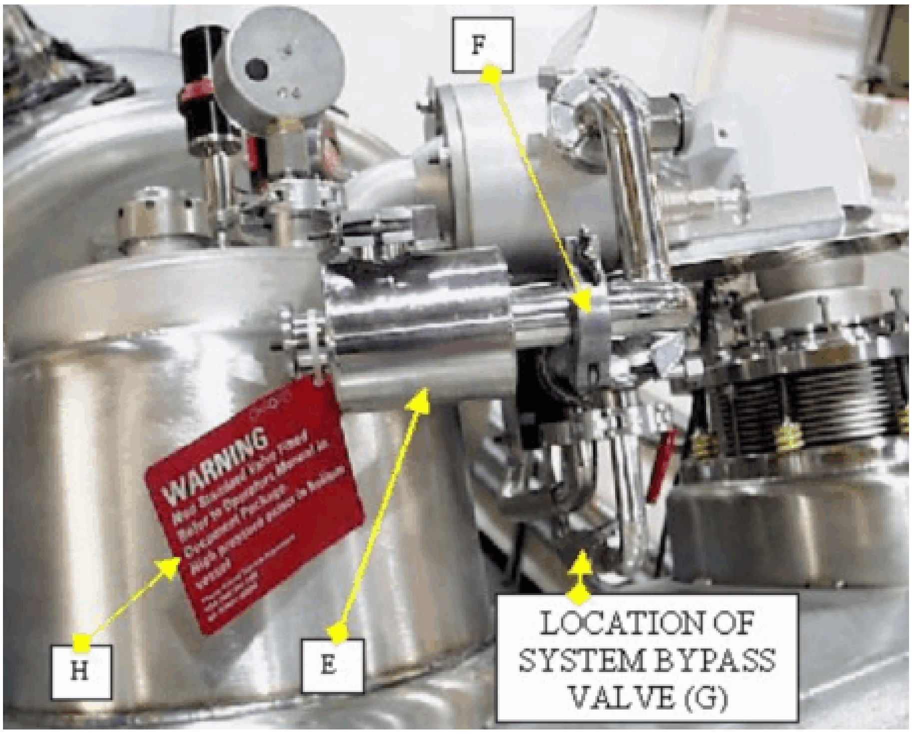
Fig. 55: Air freight kit 2



- When the pressure has fallen back to 5 psi or the pressure reading is back on scale, the valve handle can slowly be opened further.
- Allow the system to depressurize until the depressurization line has thawed, indicating that the system is now at normal operating pressure.
- De-ice the turret and venting using a heat gun before attempting to remove any components.
- The quench valve drip tray (J) (→ Fig. 55 Page 61) can be removed to allow easier access to the air freight plate bolts. The drip tray can be put back after the system has been connected to the suite's vent pipe.

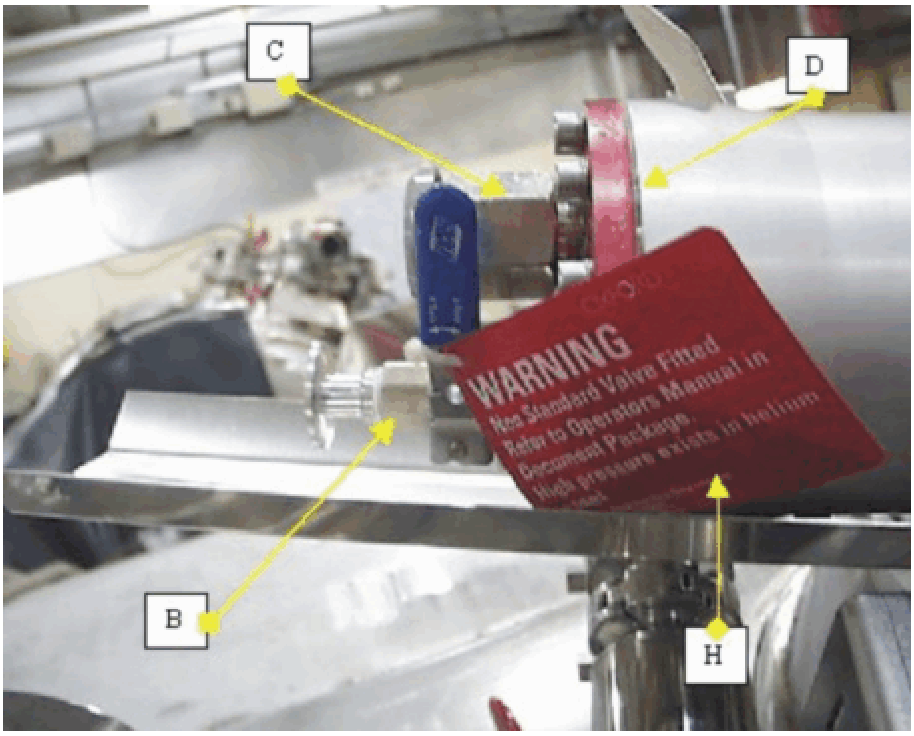
- Remove the non-return valve (**E**) (➔ Fig. 56 Page 62) and blank the connection at the vent port (**F**) (➔ Fig. 56 Page 62) with a NW25 blanking flange.

Fig. 56: Air freight kit 3



- Remove the valve plate (**D**) (➔ Fig. 57 Page 62).

Fig. 57: Air freight kit 4



3.4 Return parts

Pack the removed parts into the delivered return box and send the box to MRL F2. The parts to be returned in the photo are shown enclosed in the case (→ Fig. 59 Page 64).

The address is:

SIEMENS AG

Healthcare Sector

MRL F2

attention Reinhold Soutschek

Allee am Röthelheimpark 2

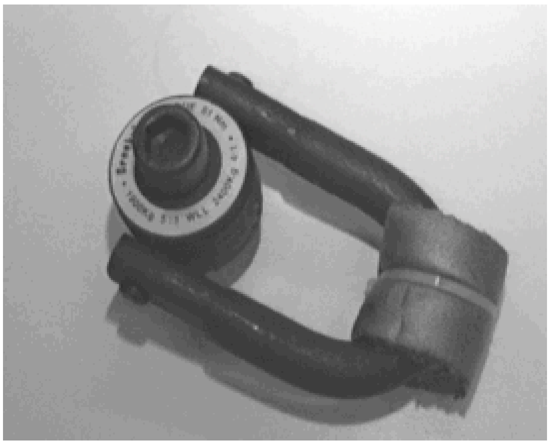
(FH5 / Gate 5)

91052 Erlangen

Fig. 58: Air freight box

Air Freight box to be returned to MRL F2

Fig. 59: Shackle



Return the four shackles as well.

- Store the shackles in the return box.

3.5 Connecting the quench tube

> **i** Prior to refilling liquid helium and depressurization for removing the air/sea freight kit, install the on-site quench tube or an equivalent so that escaping gas can be exhausted to the outside of the building.

Fig. 60: Connecting the quench tube

- **The magnet delivery contains a Teflon seal, a flange, shaft inserts, and screws with nuts and washers for connecting the on-site quench tube.**
- The figure shows the 90˚ elbow mounted to the quench valve; the 90˚ elbow is part of the delivery.

Fig. 61: 90° elbow

- The 90˚ elbow is not mounted to the quench valve. It is supplied separately with the "Interface Kit"

Fig. 62: Standard parts

- The parts shown in (➔ Fig. 62 Page 66) are for installing the quench tube.

Fig. 63: Shaft inserts

- The picture to the left (➔ Fig. 63 Page 66) shows the shaft inserts for the galvanic separation of the magnet to the on-site quench tube.
  » The shaft inserts have to be installed into the screw holes of the adapter flange and the 90° elbow (if used) for the galvanic separation.

Fig. 64: Seal

- Install the red seal (left in the picture (➔ Fig. 64 Page 66) between the 90° elbow and the quench valve.
- The white Teflon seal (right in the picture (➔ Fig. 64 Page 66) has to be installed between the 90° elbow and the quench tube for the galvanic separation.

Do not forget the shaft inserts (→ Fig. 63 Page 66) for the screw holes for the galvanic separation of the magnet to the on-site quench valve.

3.6 Galvanic separation of the magnet

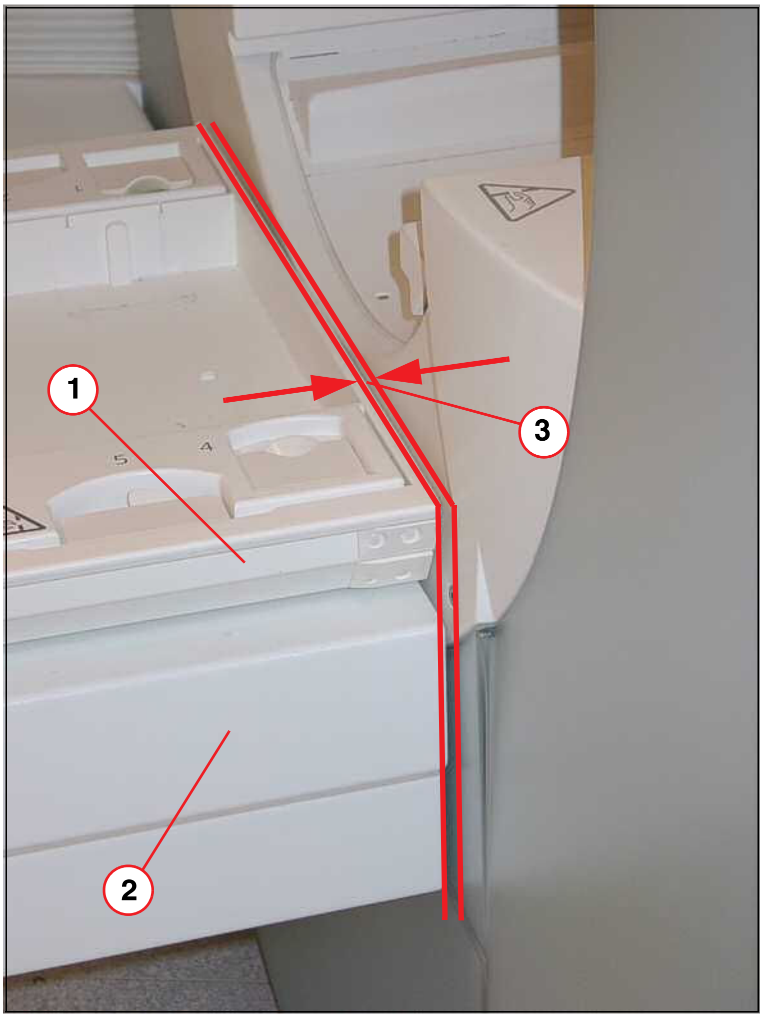
- Check the galvanic isolation of the magnet, the RF room, and the on-site quench tube.
- Use a short-circuit tester (Testaphon) between the central grounding point of the magnet and a metal component of the RF room.
- Enter it in the installation protocol

The magnet installation must be galvanically isolated from the RF room. Check the galvanic isolation between the magnet and the quench tube.

3.7 Checking the distance between patient table and the magnet cover

Check the distance between the support frame, tabletop, and magnet cover. The distance has to be 3-6 mm (➔ 3/Fig. 65 Page 69).

Fig. 65: Clearance between patient table and magnet cover



(1) Tabletop
(2) Support frame
(3) Clearance 3-8 mm between table and magnet cover

Perform this test prior to switching on the patient table. Ensure that the support frame and tabletop are flush and do not collide with the magnet cover.

- Use the hand crank and move the patient table into the highest position, then move it down 80 mm.

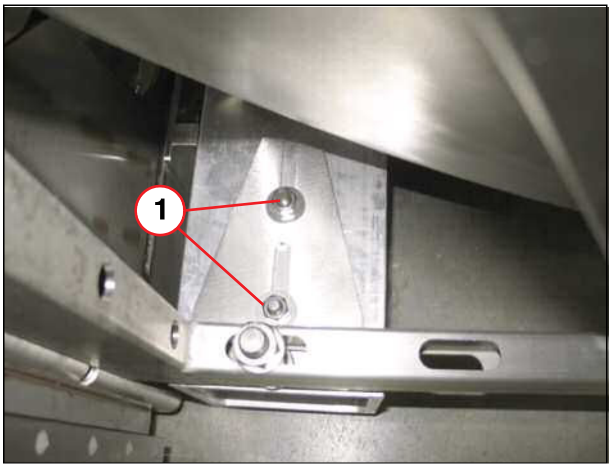
- If the distance at the narrowest location is **less than 3 mm**:
  » Readjust the patient table.

3.7.1 Readjusting the patient table

3.7.1.1 Setting the Z axis

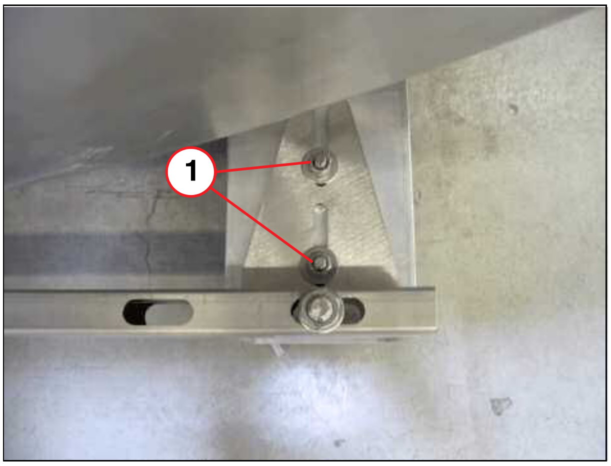
- Move the patient table to the position where the gap between the table and the cover panel is the narrowest.
- Loosen the nuts (1).

Fig. 66: Patient end, left side



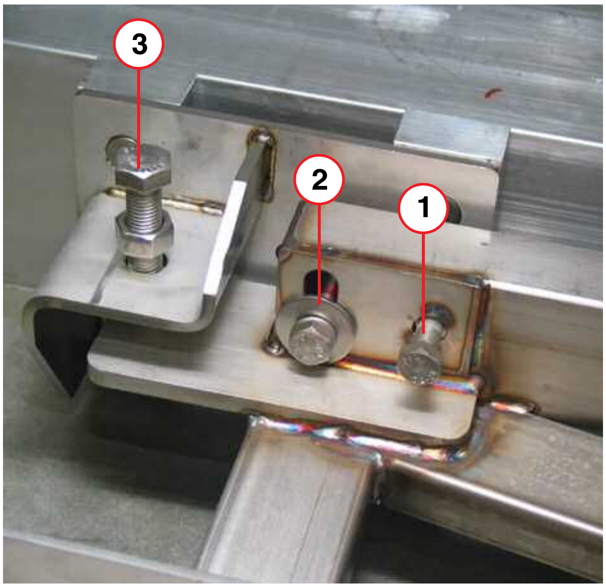
(1) Nuts

Fig. 67: Patient end, right side



(1) Nuts

Fig. 68: Back adjustment

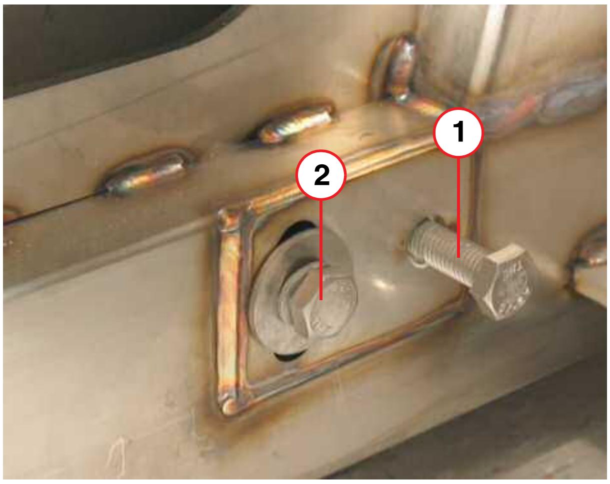


(1) S1
(2) S2 - pull
(3) S3 - push

On the left side of the magnet:

- Loosen screw S2.

Fig. 69: Front adjustment



(1) S4 - push
(2) S5 - pull

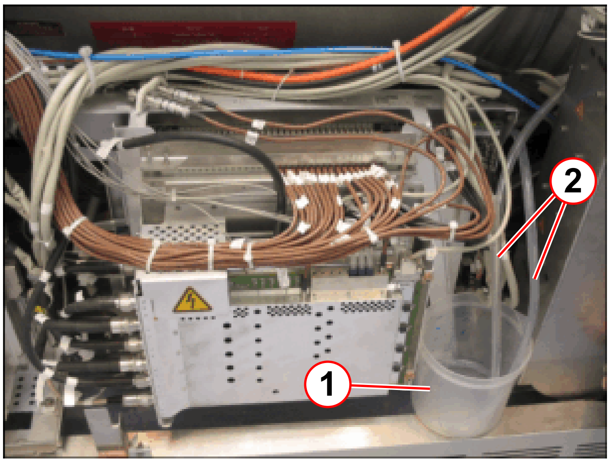
- Loosen screw S4.

- Adjust the distance between the patient table and the magnet cover so that it falls between **3 mm and 8 mm**.

  In the Z axis. the patient table must be moved manually (or with the auxiliary tool).
- Tighten the screws and nuts that were loosened earlier.
- Retighten the patient table support.
- Apply Loctite 221 to the alignment screws and counter nuts.
- Check to make sure the distance is between **3 mm and 8 mm**.

3.8 Positioning the container for dripping water

Fig. 70: Positioning the container for dripping water



(1) Container for dripping water
(2) Hoses for dripping water

Magnet:

- Position the container for dripping water (1).
- Size the hoses of the container (2) so that the dripping water can flow into the container.

4.1 Read first

The table provides an overview of the components to be installed in the RF room.

RF room	
⇒	Remote magnet stop switch
Cabling is covered in chapter 6	

The RF shielding in the examination room may not be damaged e.g. during installation of the magnet or cable ducts.

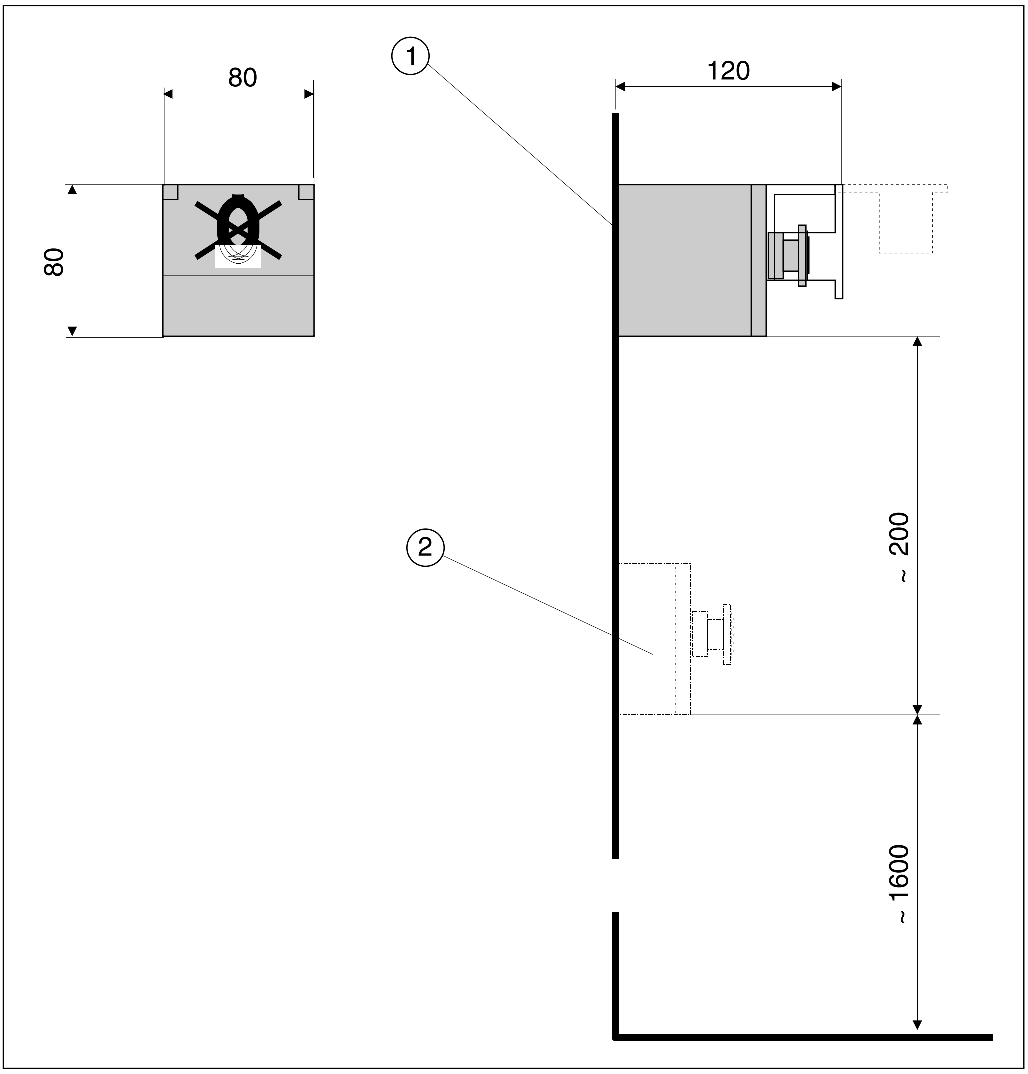
The room itself is grounded only through the system via the cable included in the delivery. Do not create any other ground, either intentionally or unintentionally

Only non-magnetic materials may be used and installed in the RF room.

4.2 Remote Magnet Stop switch

Position the magnet stop switch carefully so that it cannot be pressed by accident.

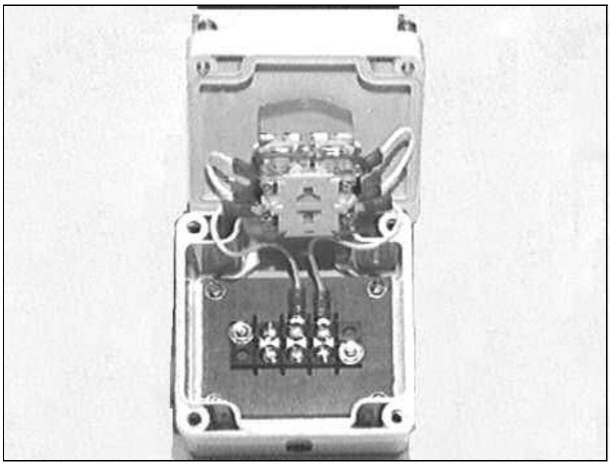
Fig. 71: Remote magnet stop button



(1) Remote magnet stop button
(2) EPO (emergency power off)

- Install the Magnet Stop switch at an easily accessible location in the examination room (fig.)

Fig. 72: EMERGENCY OFF



- Use only non-magnetic screws for attachment.

Do not damage the RF shielding of the magnet room while installing the magnet stop.

The cables (the individual lines) are insulated for connection to the ERDU button.

Remove the insulation only when connecting them to the ERDU button. In case of short-circuit, the magnet will lose helium.

Fig. 73: ERDU buttons

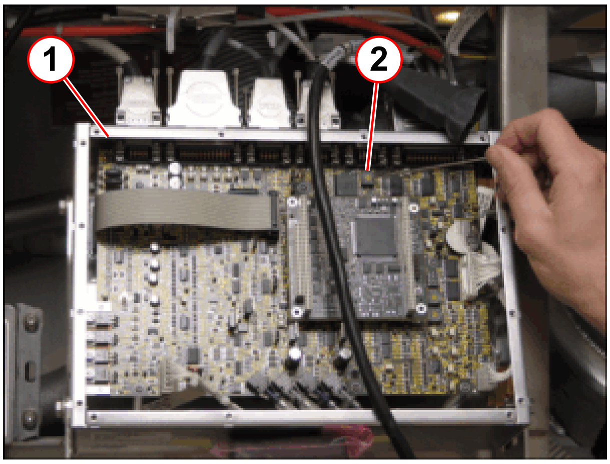
Connecting the ERDU button:

- Disconnect plug X6 at the MSUP.
- Remove the insulation from the lines and connect them to the ERDU button.
- Connect plug X6 at the MSUP again.

4.2.1 Second remote Magnet Stop switch

The MSUP is located on the left of the magnet.

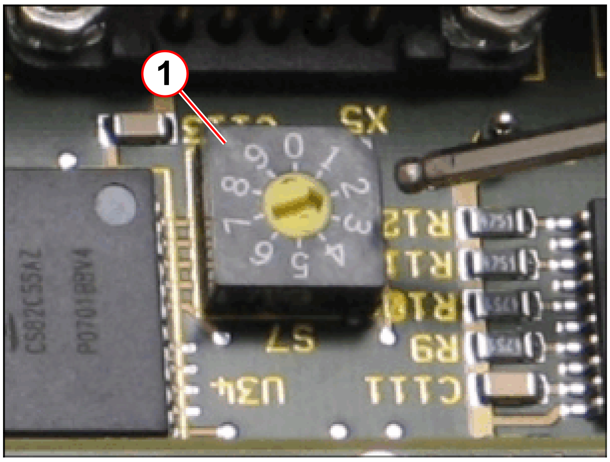
Fig. 74: MSUP (ERDU No SET SWITCH)



(1) MSUP
(2) Switch

- Remove the cover at the MSUP (1).

Fig. 75: MSUP (ERDU No SET SWITCH)



(1) Switch

- Set the switch (1) from position **2** to position **3**.
- Reinstall the cover at the MSUP.

For mobile systems, the second remote Magnet Stop switch has to be installed behind the magnet (service side) near the Electronic-Cabinet-Mobile.

5.1 General

Install the electronics cabinet (ACC, GPA) and the RCA with the helium compressor in the equipment room. Route the cables after you finished the mechanical installation of the cabinets and components.

5.1.1 Components

The table provides you with an overview of the components to be installed in the equipment room.

Equipment room	
⇒	Filter plate
⇒	GPA/ACC cabinet
⇒	RCA with helium compressor
⇒	Earthquake anchoring (optional)
⇒	Removing the covers
⇒	Connecting the water hoses
Refer to Chapter 6 for cabling	

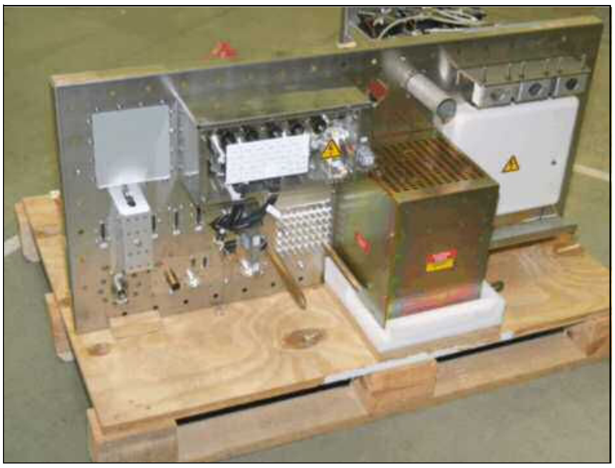
5.2 Unloading and positioning the components

5.2.1 Filter plate

5.2.1.1 Unpacking

The filter plate is delivered mounted on a palette.

Fig. 76: Filter plate



The filter plate is installed from the exterior of the magnet room (→ Fig. 78 Page 80).

5.2.1.2 Installation

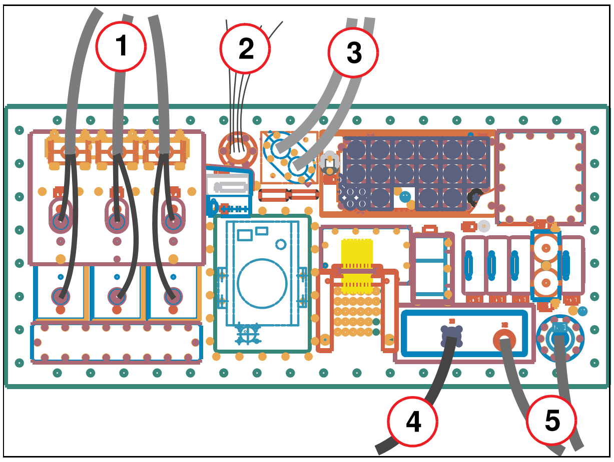
When installing the filter plate, the contact surfaces have to be cleaned with alcohol.

Do not apply Molikote (lubrication) or similar to the screws at the filter plate, since these substances affect the transfer resistance.

In a modular RF room or a conventional one (e.g. copper RF room), the **filter plate** can be installed in the recess provided using **RF springs**.

If the RF filter plate is installed vertically, the gradient filters must face upward.

Fig. 77: Filter plate



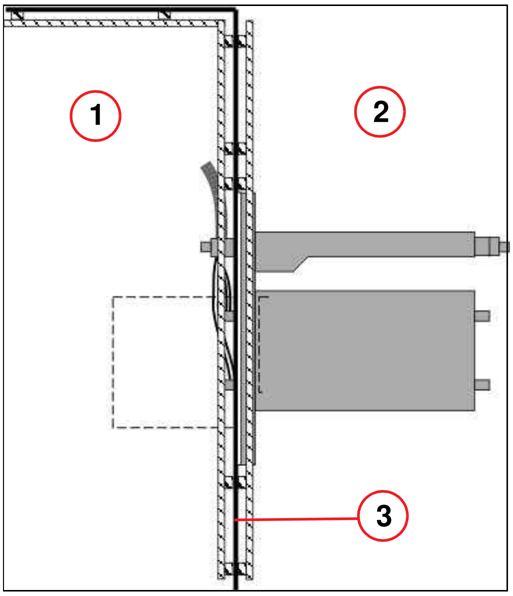
(1) Gradient cable
(2) Fiber optic cable
(3) Helium hoses
(4) RF cable
(5) Cooling water supply and return lines

i

The screws and material needed for installing the filter plate at the RF room are not included in the delivery volume for the system. These have to be procured on-site.

The correct torque values must be used for each screw size when securing the RF filter plate (e.g. M10 = 40 Nm).

Fig. 78: Filter plate installation



(1) Examination room
(2) Electronics room
(3) RF room

5.2.2 GPA/ACC cabinet

Since the GPA/ACC cabinet is on wheels, it allows for easy unloading with the hydraulic tail lift of the truck.

You have the option to use the lugs on the red transport rails to lift the double cabinet off the truck using a crane.

Fig. 79: GPA/ACC cabinet

- Transporting the GPA/ACC cabinet to its final position.

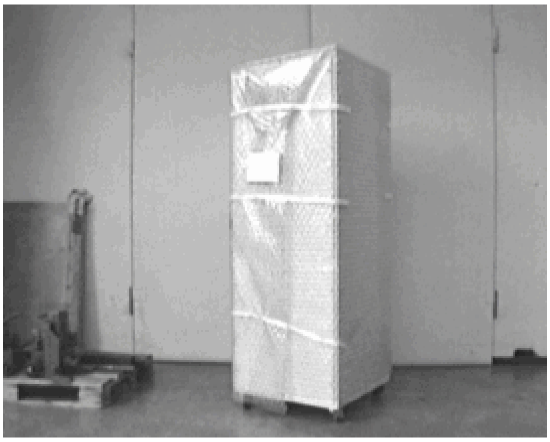
Remove the red transport rails from the GPA/ACC. If you remove the red transport rails, ensure that they remain with the system (attach them with the screws provided). Depending on the instructions of the project manager, the transport rails can remain installed.

5.2.3 RCA cabinet

Since the RCA cabinet is on wheels it is easy to unload the cabinet via the hydraulic ramp, which is part of the truck.

If the truck is not equipped with a hydraulic ramp, use the eye lugs on top of the cabinet to lift the RCA cabinet off the truck with a crane.

Fig. 80: RCA cabinet



- Unload the RCA cabinet from the truck.
- Move the RCA cabinet to its final position

AVOID CONTAMINATION. Do not tip the compressor more than 10 degrees from the horizontal to avoid contamination.

When checking the compressor for shipping damage, do not connect gas lines or the cold head. Components may become contaminated with compressor oil.

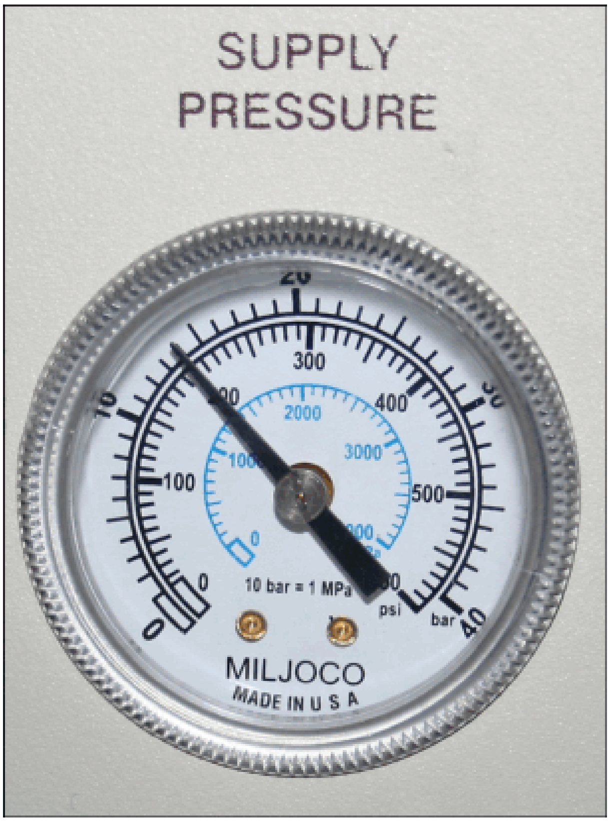
Before installation, the RCA cabinet should be inspected for any **shipping damage** or **mishandling.**

Fig. 81: F70 Compressor Tip-N-Tell indicators

- Inspect the **'Tip-N-Tell'** sensor for signs of activation. The sensor turns blue if the unit has been tipped or mishandled during transport.
- If the sensor has been activated, refer to **Troubleshooting Guide - Refrigerator.**

5.2.3.1 Compressor Pre-Connection Checks

Fig. 82: F70 Static pressure reading



- **Check** the **static pressure** is **positive**.
  - If the gauge indicates **zero pressure** the compressor **cannot** be used.
  - Arrange a replacement compressor.

NOTICE

If the compressor arrives at ZERO BAR - DO NOT CONNECT THE HE PRESSURE LINES TO THE COMPRESSOR OR COLD HEAD.

Connecting the lines to the compressor at zero bar may cause contamination of the whole refrigeration system.

» **Check the compressor pressure before attempting to connect the He pressure lines.**

» **Contact Siemens Magnet HSC for advice.**

Tab. 2 F70 static pressure on arrival

Pressure	Ambient temperature
207 - 212 psi (G)	20^0C (68^0F)
1430 - 1460 kPa	
13.5 - 14.0 bar	

- If the static pressure is **positive**, move on to the processes described under **Connecting the cold head to the He compressor F70.**

5.3 Earthquake anchoring (optional)

5.3.1 Anchoring the MR cabinets to the floor with bolts

In areas prone to earthquakes, the cabinets must be bolted to the floor in accordance with local guidelines.

Fig. 83: Earthquake kit

The earthquake kit includes anchors and the screws.

The bolts require a 24-mm wrench.

Fig. 84: Anchoring with wheels

The cabinets can be anchored to the floor with their wheels still installed.

- Bolts for anchoring the cabinets to the floor.

Fig. 85: Lifting the cabinets

Another option is to lift the cabinets and remove the wheels.

In this case, the cabinets are directly bolted to the floor.

How to proceed:

- Position the electronics cabinet and the SEP cabinet in their final locations.
- Raise the cabinets with the hydraulic lift.
- If necessary, remove the wheels from the cabinets.
- Install the floor anchors.
- Lower the cabinets slowly and carefully.
- Drill the floor holes and anchor the cabinets to the floor.

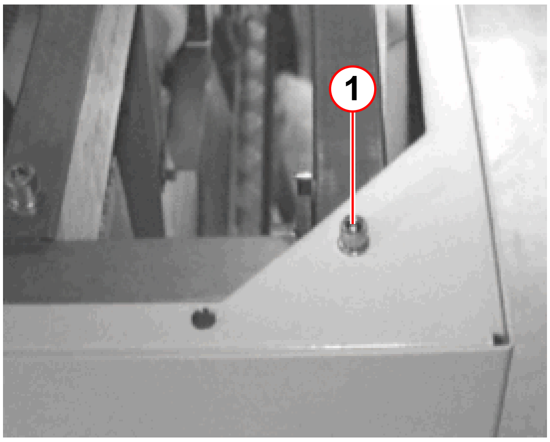
5.4 Removing the covers

You should have completed the following before installing the system cables:

- All system components have been installed at their final position.
- The on-site power connection has been installed.
- The on-site chilled water connection is available.

» **Before you start with the cabling remove the following magnet covers:**

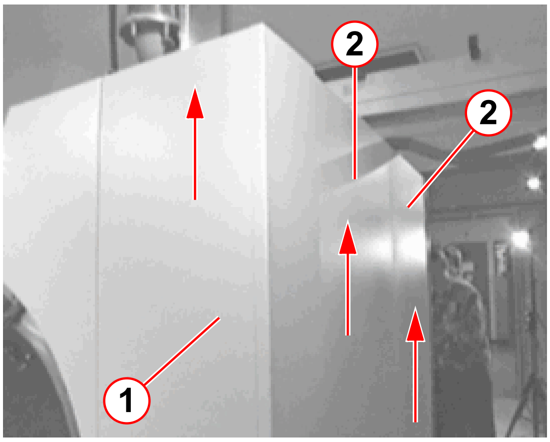
Fig. 86: Back corner cover



(1) 6 mm Allen screw

- Remove the 6 mm Allen screw on top of the back corner cover.

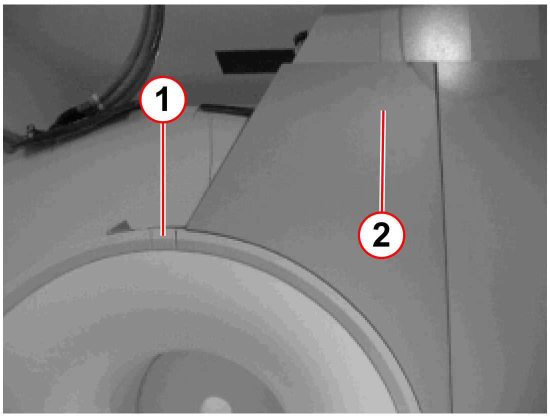
Fig. 87: Magnet cover



(1) Back corner cover
(2) Side panels

- Slide the back corner cover upwards (arrow direction) and remove it.
- Slide the two side panels upwards and remove them.

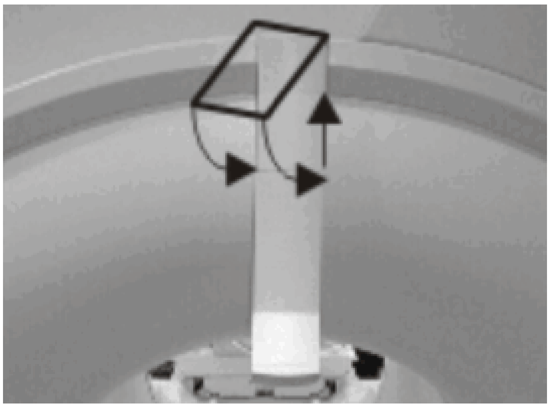
Fig. 88: Top back cover



The gradient cable connecting plate is behind the back top cover **(2).**

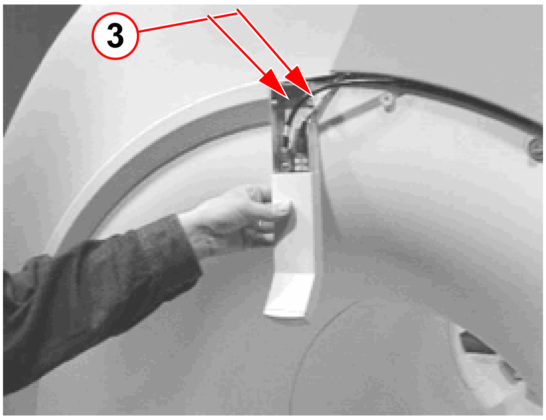
- If there is no patient camera installed remove the small cover **(1)**.

Fig. 89: Patient camera



- If there is a patient camera installed open the cover of the camera by sliding the upper part of the cover upwards and then towards you.

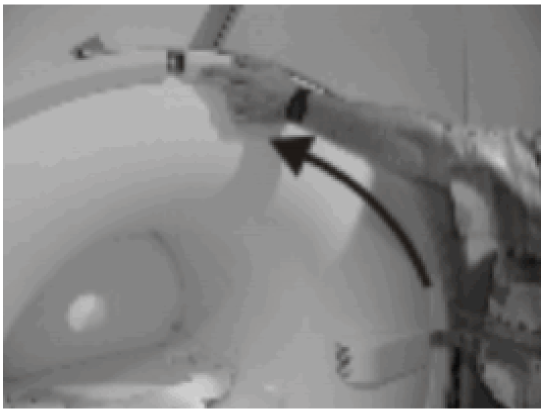
Fig. 90:



- Disconnect the two cables
- Unscrew the two Allen screws **(3)**.
- Store the camera on a safe place.

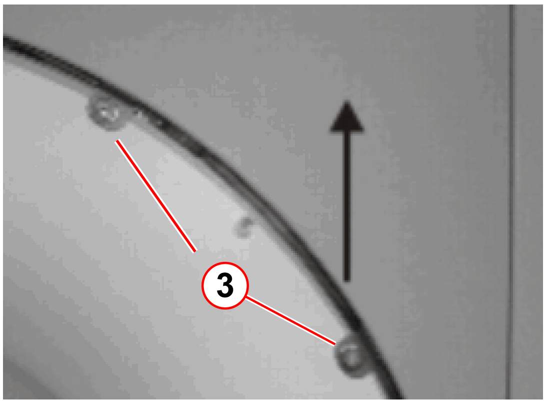
The bending radius for the patient supervision fiber optic cable is:**R_B ≥ 45 mm.**

Fig. 91: Ring segment



- Slide the ring segment in the arrow direction until it is unlocked.
- Remove the segment by pulling it towards you.

Fig. 92: Top cover



- Unscrew the two 6 mm Allen screws **(3)** and slide the cover upwards to remove it.

5.5 Connecting the water lines

The screw connections for the water connections should be tightened with a torque of

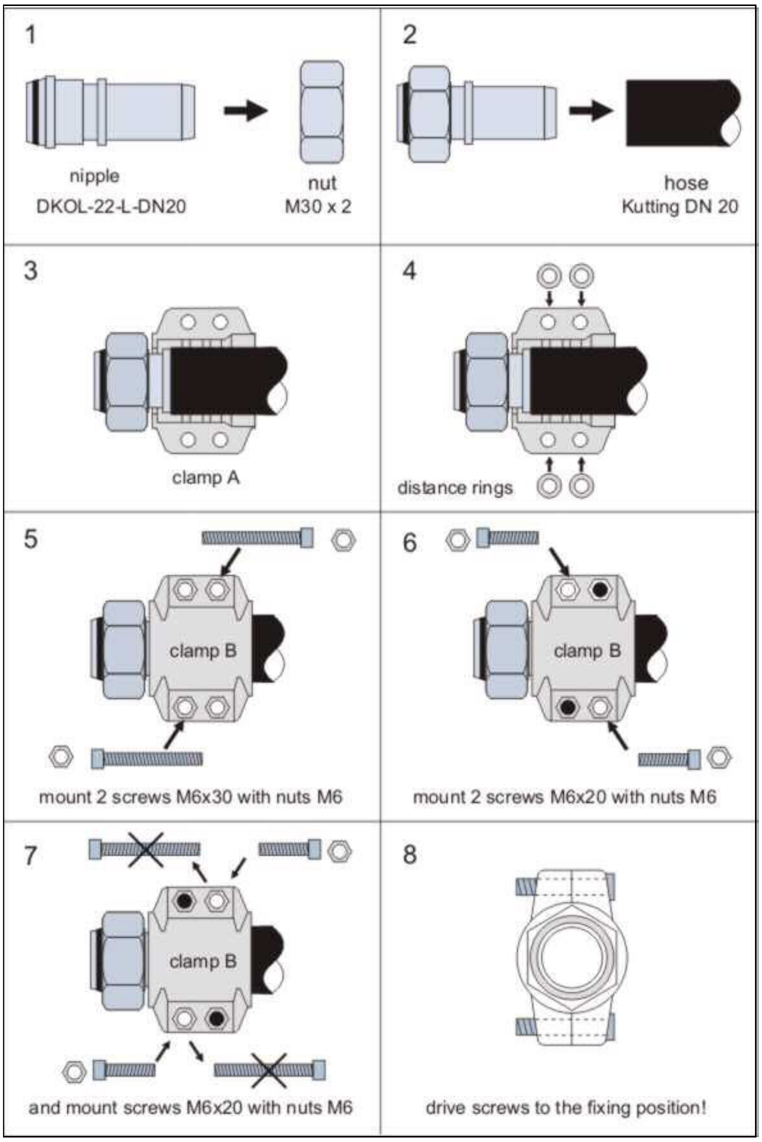
25 Nm (wrench size: 36 and 42 mm) and

50 Nm (wrench size: 60 mm).

Excess lengths of water hoses cannot be routed or stored behind the cabinets due to the heat build up in the cabinets. In addition, the hoses cannot be routed along the cabinet walls, including their side walls.

5.5.1 Assembling the water connections

Fig. 93: Assembling the water couplings



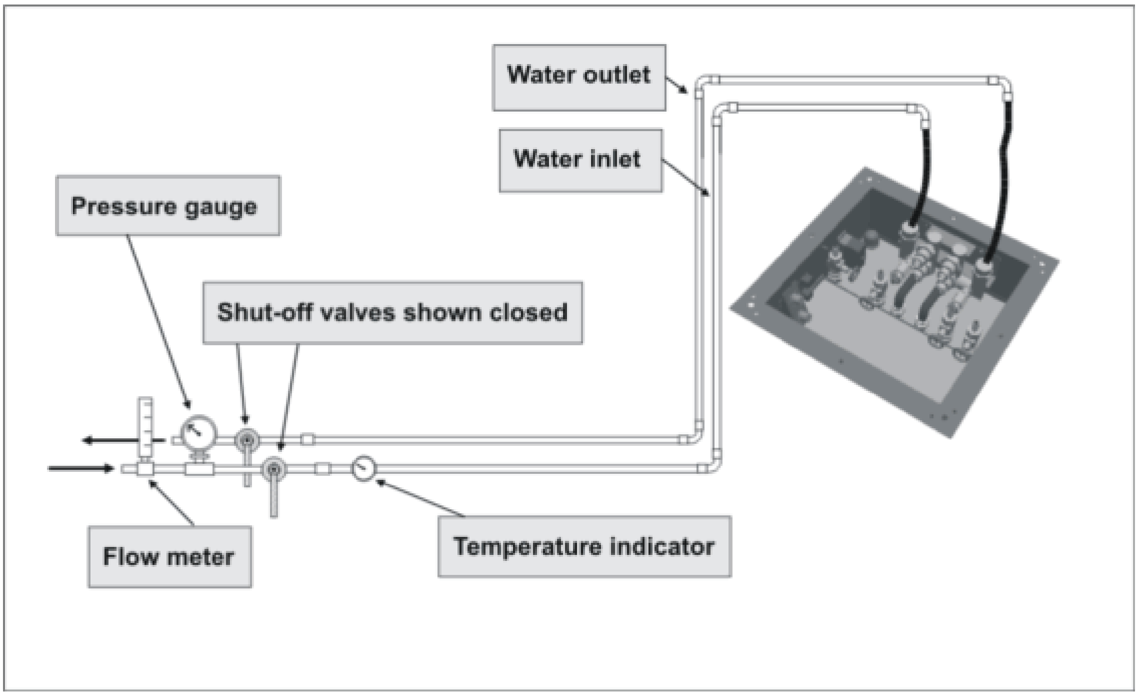
For water connections with 90 degree elbows, the clamps have to be installed the same way.

5.5.2 Installation of the RCA (Refrigeration Cabinet)

The RCA cabinet is an essential component for the operation of the high performance gradient system. Because of the heavy heat dissipation of the gradient coil and gradient amplifier it is necessary to use water cooling. In addition to the water cooler, the cabinet also contains the helium compressor for the cold head of the magnet.

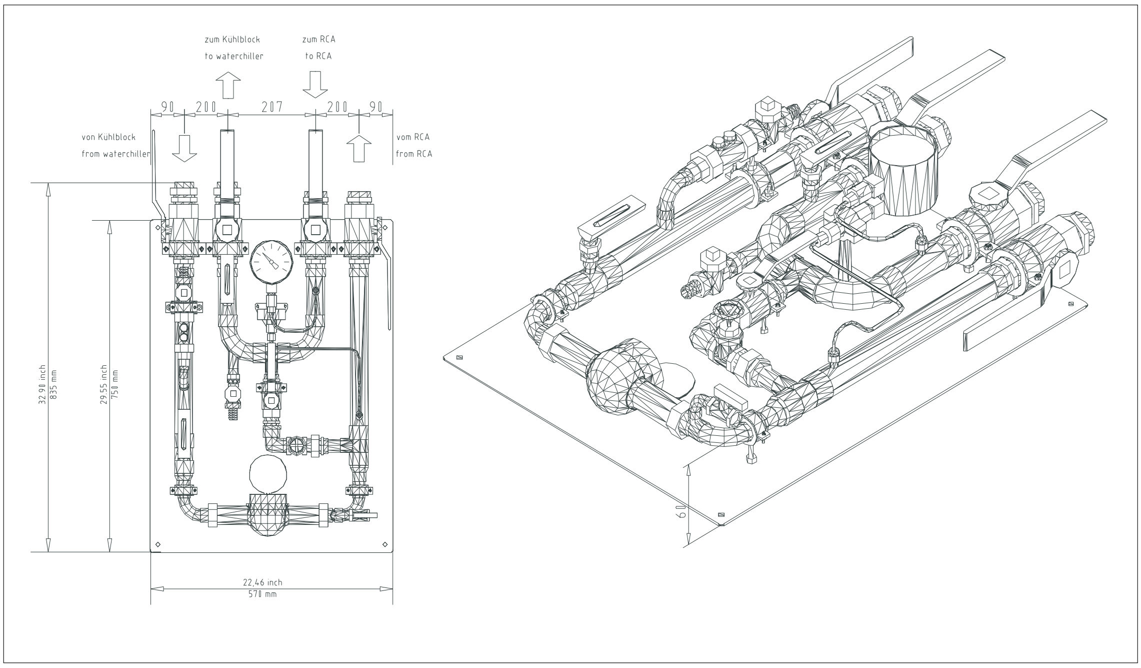
5.5.2.1 Overview of the on-site water supply

Fig. 94: On-site water supply, RCA



5.5.2.2 Water transfer station (optional)

Fig. 95: Water transfer station



5.5.2.3 Connecting the on-site water supply

Preliminary steps

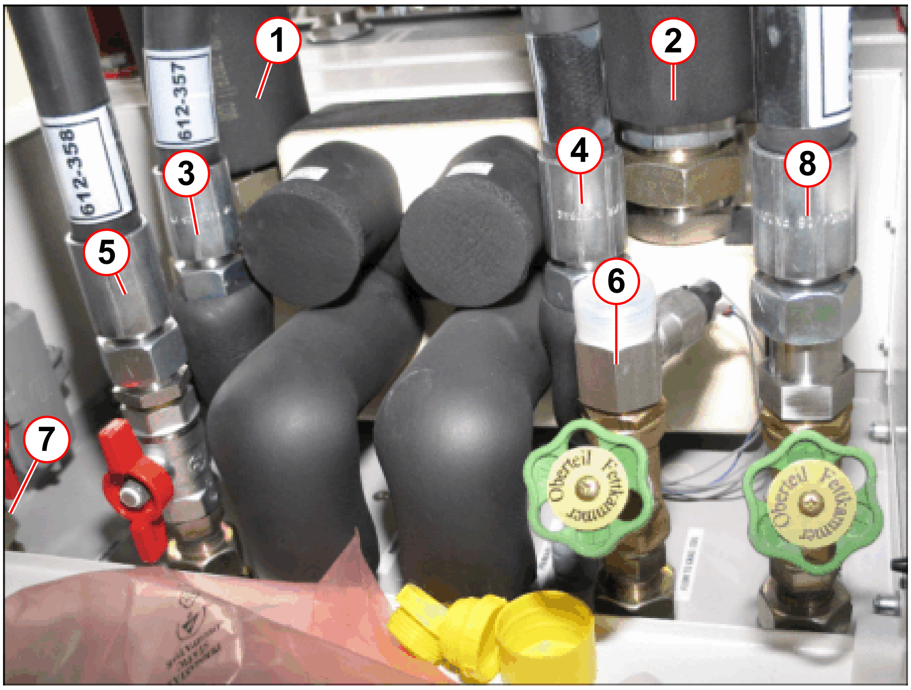
- The hoses for supplying the RCA with primary chilled water should have been routed up to the final location of the RCA cabinet
- Ensure that the on-site chilled water inlet hose includes a flow rate meter.
- One shut-off valve each should be provided for the chilled water inlet and outlet of the on-site water providing panel.
- One temperature indicator should be provided for the on-site chilled water inlet.

i While connecting the hoses on top of the cabinet, do not mix up the water inlet with the water outlet connections.

5.5.2.4 Overview

(→ Fig. 96 Page 92) provides you with an overview of the water connections for the RCA cabinet.

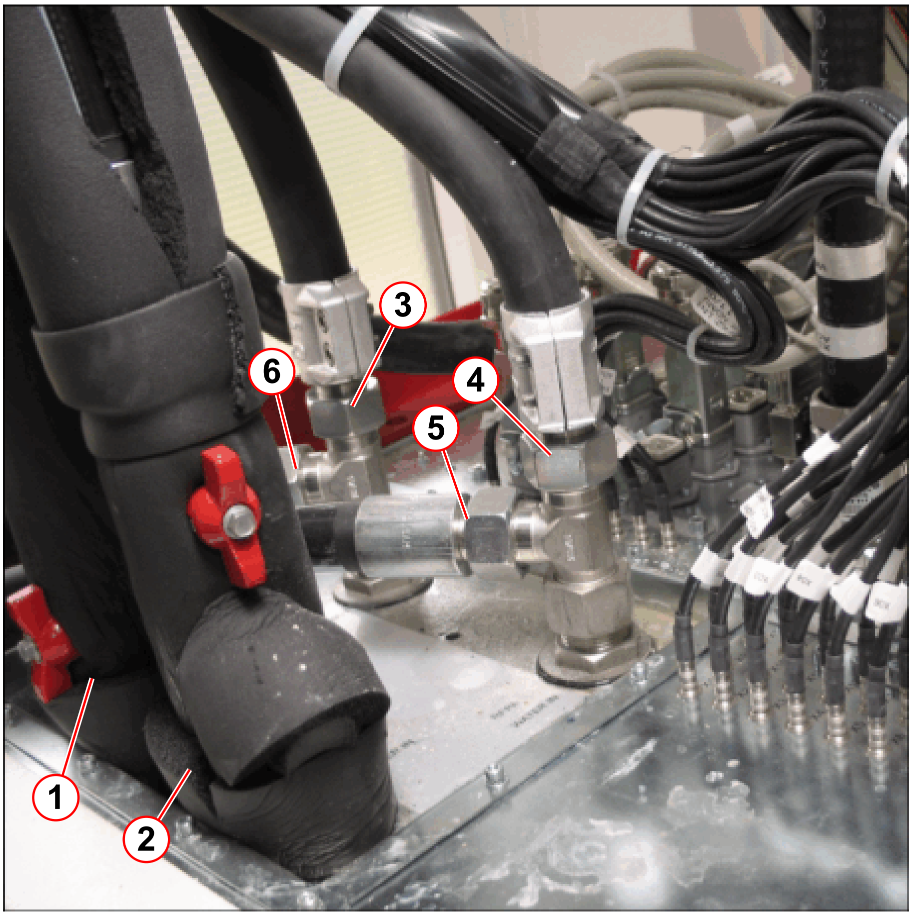
Fig. 96: Water connection at the RCA



(1) PRIMARY WATER IN
(2) PRIMARY WATER OUT
(3) CCS FLOW
(4) CCS FROM
(5) FROM GPA/CCA
(6) FLOW GPA/CCA
(7) FROM GRAD. COIL
(8) FLOW GRAD. COIL

5.5.3 Installing the water hoses to the CCS / ACC / GPA

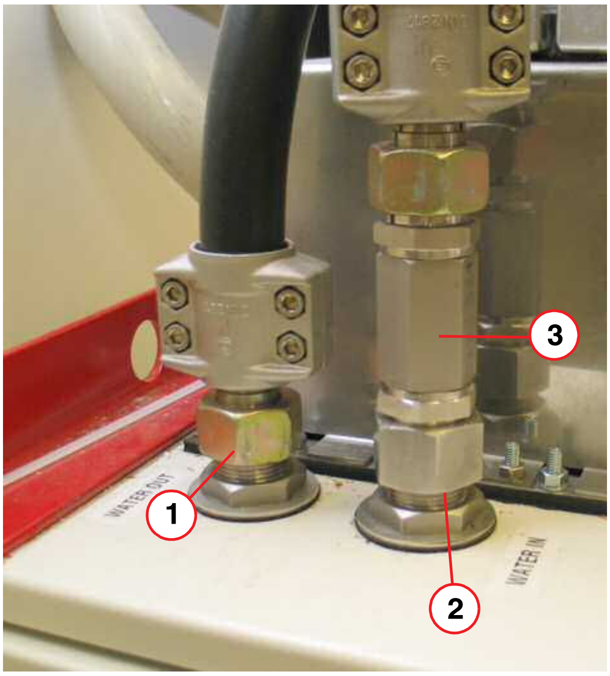
Fig. 97: Water connection at the ACC



(1) CCS WATER OUT, primary water
(2) CCS WATER IN, primary water
(3) RFPA WATER IN, secondary water
(4) RFPA WATER OUT, secondary water
(5) Flow to GPA, secondary water
(6) Flow from GPA, secondary water

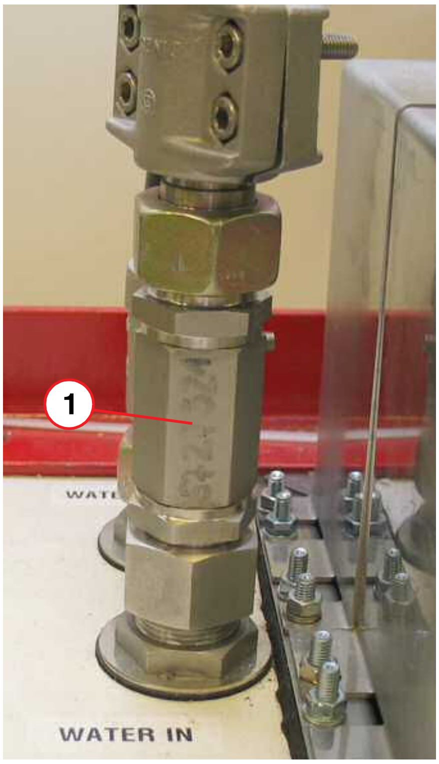
- Connect the primary water hoses from the RCA to the CCS (➔ 1,2/Fig. 97 Page 93):
  - » CCS WATER IN
  - » CCS WATER OUT
- Connect the secondary water hoses from the RCA to the ACC (➔ 3,4/Fig. 97 Page 93):
  - » Water hose “Flow to GPA/CCA” to “RFPA WATER IN”
  - » Water hose “Flow from GPA/CCA” to “RFPA WATER OUT”
- Connect the secondary water hoses from the ACC to the GPA (➔ 5,6/Fig. 97 Page 93):
  - » Mount the flow reducer to the GPA “Water in”. (➔ 1/Fig. 99 Page 94)
  - » Water hose “RFPA WATER IN” to “GPA WATER IN”
  - » WATER HOSE “RFPA WATER OUT” to “GPA WATER OUT”

Fig. 98: Gradient cabinet



(1) Water out
(2) Water in
(3) Flow reducer

Fig. 99: Gradient cabinet



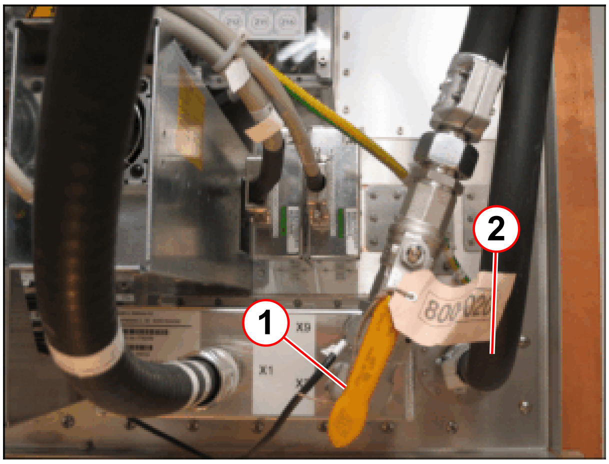
(1) Flow reducer

- **Insulate the primary water hoses.**

> Ensure that the connection to “WATER IN” and “WATER OUT” is correct.
> Incorrect water flow to the RFPA can damage the RFPA.

5.5.4 Connecting the waters hoses to the filter plate

Fig. 100: Water connection at the filter plate



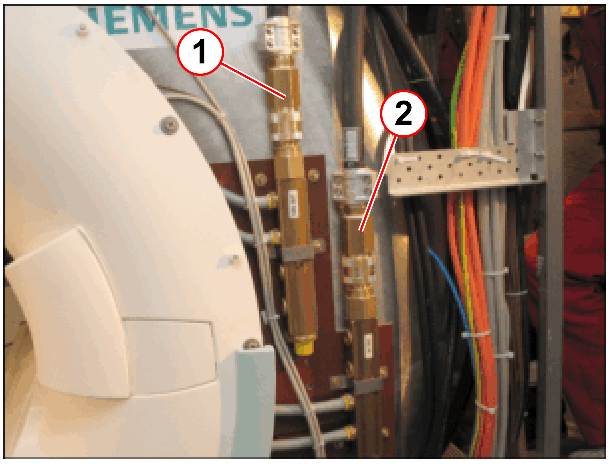
(1) X2
(2) Return OUT

Equipment room:

- Connect the water hose to the TAS_3TM

5.5.5 Connecting the water hoses to the magnet

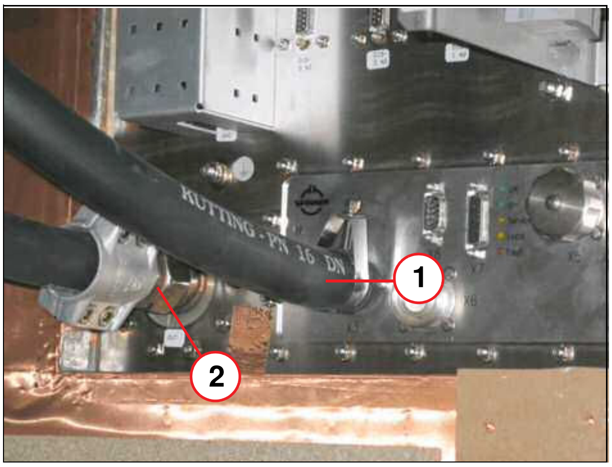
Fig. 101: Water connection at the magnet



(1) COIL OUT
(2) COIL IN

- Route both water hoses from the magnet water couplings to the filter-plate water couplings
  - Magnet COIL OUT(1) to filter plate Return Out
  - Magnet COIL IN (2) to filter plate X3

Fig. 102: Exam room filter-plate water connections

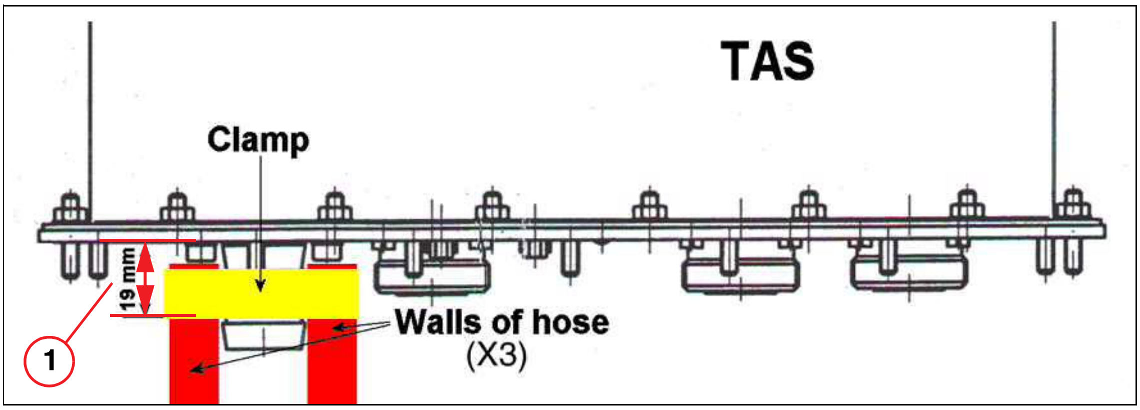


(1) X3 (supply)
(2) Return out

- Trim the water hoses to the required length.

  The water hoses must be cut off straight.
- Attach the clamp to the water hose (see figure: Assembling the water couplings)
- Connect the X3 (supply) filter plate to the COIL IN coupling on the magnet.
- Connect the Return Out filter plate with the COIL Out coupling on the magnet.
- Attach the hose clamp at X3 with screws (➔ Fig. 103 Page 96).

Fig. 103: Water hose connection at the RF filter plate TAS X3



(1) Dimension ≤19 mm

6.1 Cabling table

Cabling of the components	
⇒	On-site line voltage connection
⇒	Protective conductors
⇒	Ground test
⇒	Line voltage cable
⇒	Connecting the cold head and the helium compressor:
⇒	Gradient cable ▪ Connecting the GPA with the 1.5 m cable set **or** ▪ Connecting the GPA with a cable set larger than 1.5 m
⇒	RF cables
⇒	Fiber-optic cables
⇒	Installing the alarm box
⇒	Host PC
⇒	Console components
⇒	Footswitch (option)
⇒	Alarm contact
⇒	Door contact
⇒	System cabling » Use the "top sheets" in the binder "Technical Documents Volume 2 of 2", Chapter"Diagrams"
Starting magnet cooling	

6.2 On-site voltage connection

Please ensure that the line voltage at the on-site power distributor is switched off (interrupt the fuses and lock the switches) prior to starting cable installation.

Apply the 5 safety rules:

- Disconnect
- Ensure against switch-on
- Make sure that the power is switched off.
- Ground and short-circuit
- Cover or rope off adjacent parts where voltage is present.

The on-site line voltage connection at the ACC cabinet (mains box) may be performed only by a trained technician.

WARNING

Risk of injury due to electric shock.

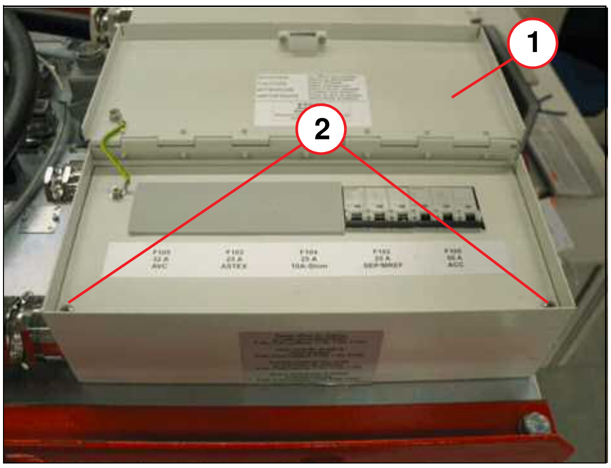
» **Observing the general safety regulations!**

» **Ensure that line voltage is not present at the ACC, GPA, the compressor or the SEP (optional).**

» **Switch off the line voltage supply on-site and secure it against accidental switching on (sign or lock).**

The line voltage connection is located on top of the ACC cabinet..

Fig. 104: ACC power supply box

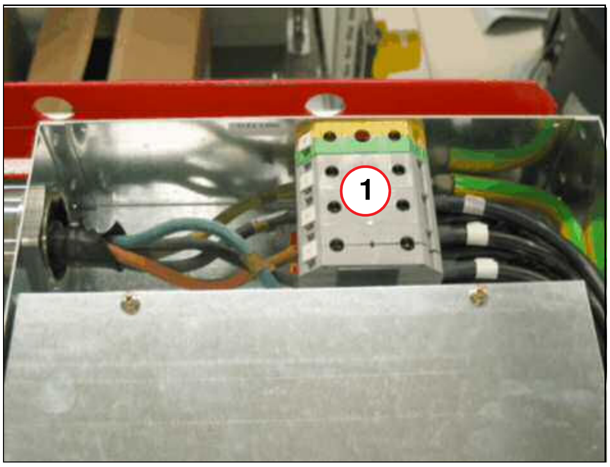


(1) Power supply box
(2) Screws

Open the line voltage connection box:

- Remove the two screws (2).
- Remove the power supply cover (1).

Fig. 105: ACC line voltage connection



(1) Terminal block

- The picture on the left shows the terminal block for connecting the line voltage cable (L1, L2, L3, ground) from the on-site power distribution.

- Torque: 17.5 ± 2.5 Nm

> The shielding of the line voltage cable from the on-site power distributor to the line voltage connection box has to be connected to the strain relief of the line voltage connection box.

6.3 Cabling

When laying or working close to the cables and hoses, ensure that these are not damaged (e.g., when mounting the ceiling and wall panels).

Most of the cabling to the magnet was connected at the factory, i.e., the cabling was routed from the magnet and connected as labeled (e.g., to filter plate, ACC cabinet, etc.).

The cables cannot touch the quench pipe. Additionally, the cables cannot be located in areas where the quench pipe shows drops of water condensation.

The cables, routed from the magnet into the ceiling, must not touch the ceiling.

This is to prevent the transmission of vibration.

The cables must be kept more than 2 cm away from the ceiling.

Fig. 106: Cable routing

Routing cables on the magnet tower

- Route the cables and wires so that the tower cover can be attached (e.g., see picture at left).

6.3.1 Using the wiring diagrams

The wiring diagrams for the individual components are located in the "Diagram" manual.

Route the cable and system connections according to the cable descriptions and the circuit diagrams (make sure that the ends of the cables are marked).

6.3.2 Bend radius of the cables

Bend radii	
Gradient cable	RB >= 135 mm
Transmitter cable	RB >= 120 mm for single bend RB >= 360 mm for multiple bends
Glass-fiber cable	RB >= 150 mm
Glass-fiber cable for patient observation	RB >= 45 mm

6.4 Installing/routing cables to prevent flashover

6.4.1 Cable routing

To ensure error-free operation of the imaging system, the cables have to be routed separately.

- Route the gradient cables separately from the other cables at a distance of at least **300 mm**.
- Route the optional shim cable separately from the gradient cables (the electrical shim is an option).
- Attach the cables securely so they cannot shift.
- Route an additional cable harness, if necessary.

> ℹ You are permitted to cross the cables at one or two places near the filter plate, on the cabinets, or on the magnet. The cables must be positioned as far away from one another as local conditions permit. In all cases, the distance between the cables in the cable conduit must be 300 mm. If possible, use existing materials.

> ℹ Keep the individual cables separate, i.e., tie the gradient cables, RF cables, the signal and power cables separately (minimum distance 300 m).
>
> DO NOT TIE ALL CABLES TOGETHER.

> ℹ In case of remaining cabe lengths: Do not coil up the remaining cable lengths. Instead, let excess cable lengths meander in the ceiling.

Fig. 107: Cable Separation

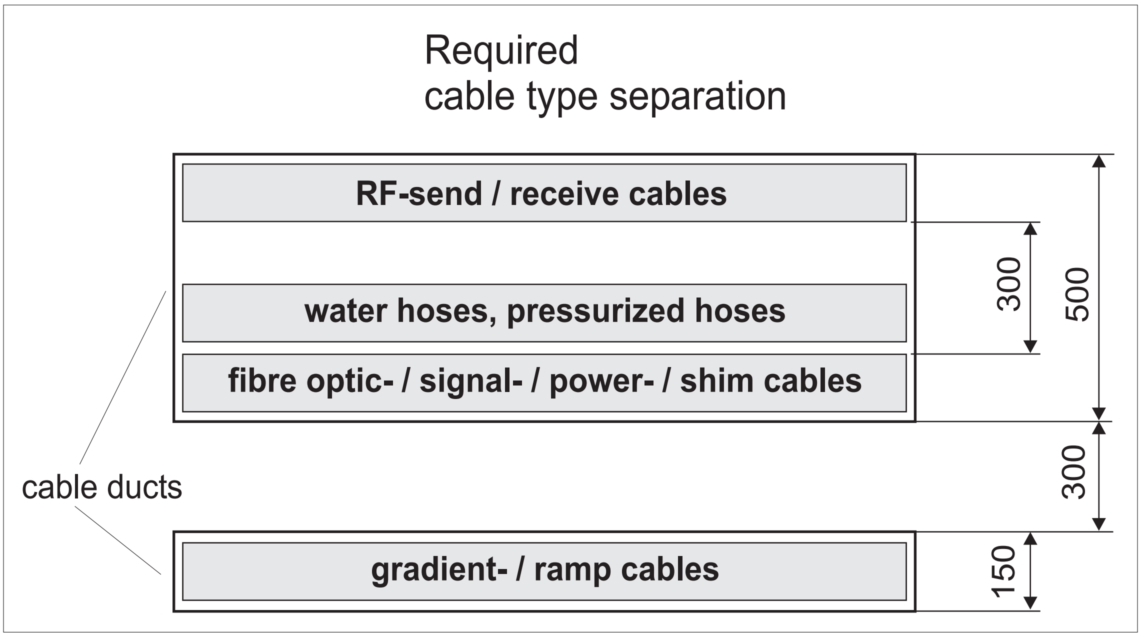


Fig. 108: Duct

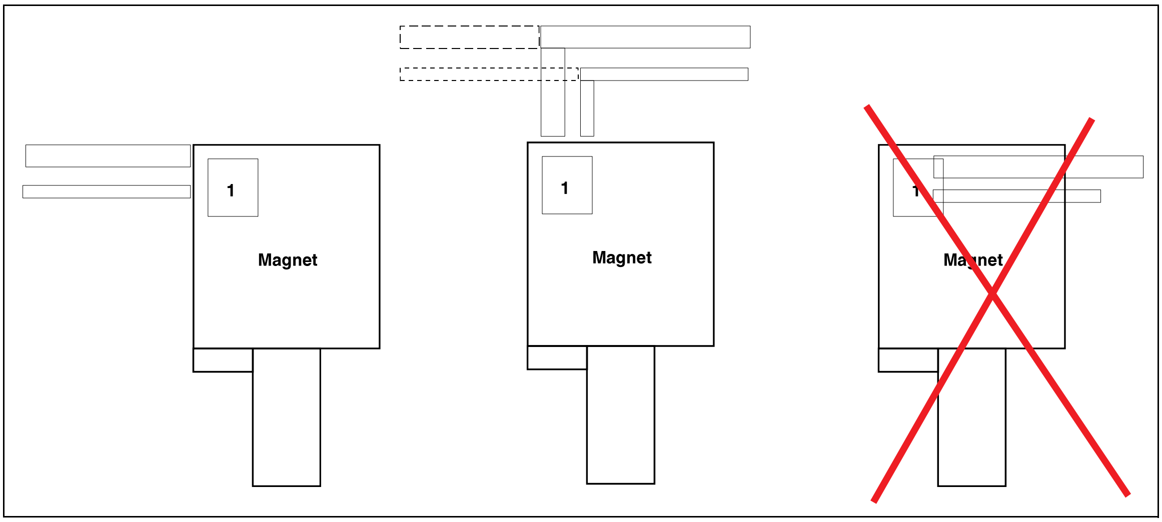
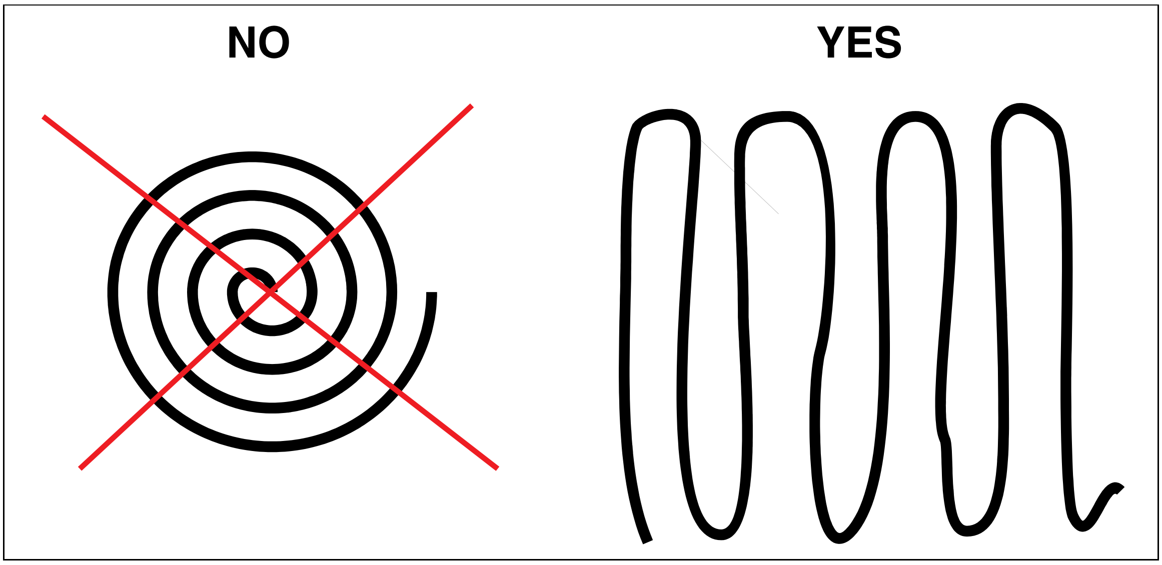


Fig. 109: No loops



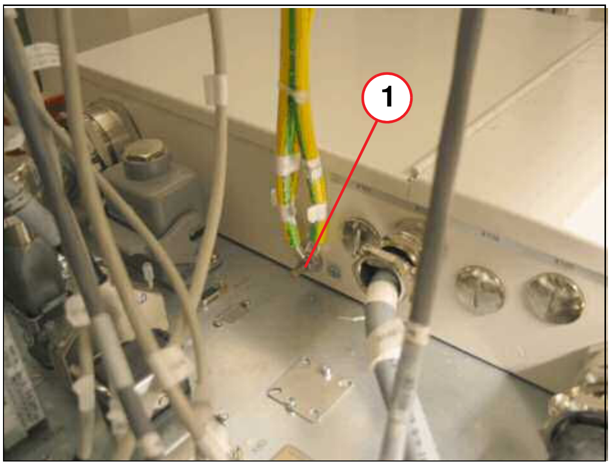
6.5 Ground

Prior to beginning magnet cooling, start-up, and ramping of the magnet, a secure protective conductor connection must be established and proven.

The country-specific regulations and laws must be observed.

6.5.1 Protective conductor connection

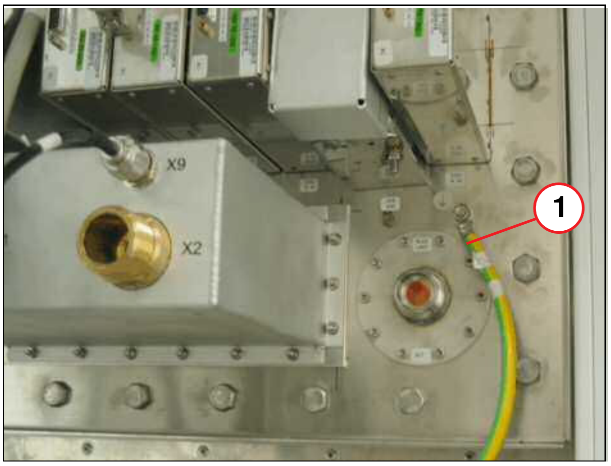
Fig. 110: ACC protective conductor terminal



(1) ACC cabinet protective conductor terminal

- Connect all 16 mm2 protective conductors to the protective conductor terminal of the ACC cabinet

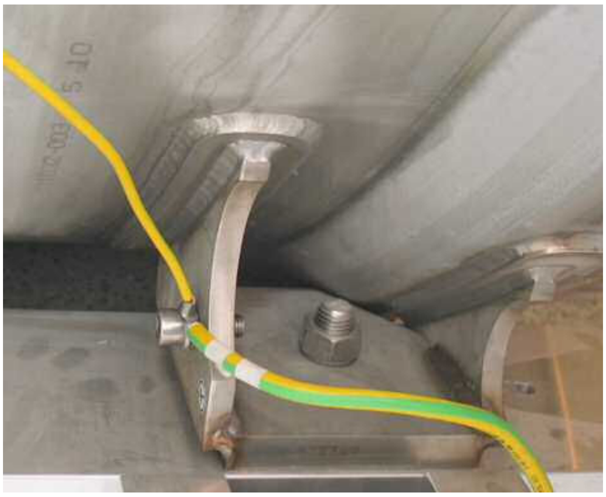
Fig. 111: Electronics room filter-plate protective conductor terminal



(1) Filter-plate protective conductor

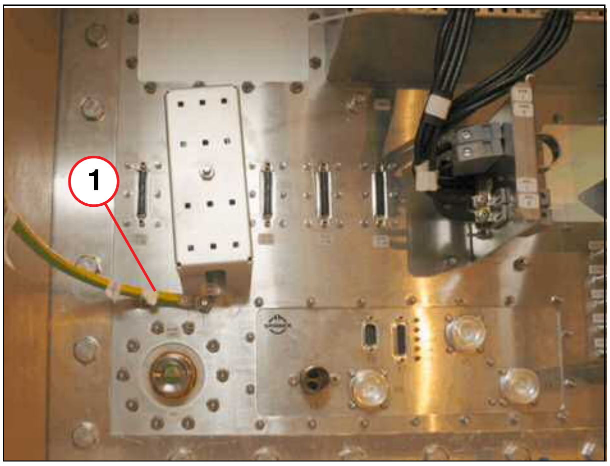
- Route the protective conductor (1) from the filter plate (electronics room) to the protective conductor terminal at the ACC cabinet.

Fig. 112: Magnet ground



- Route the protective conductor (1) from the magnet to the filter plate.

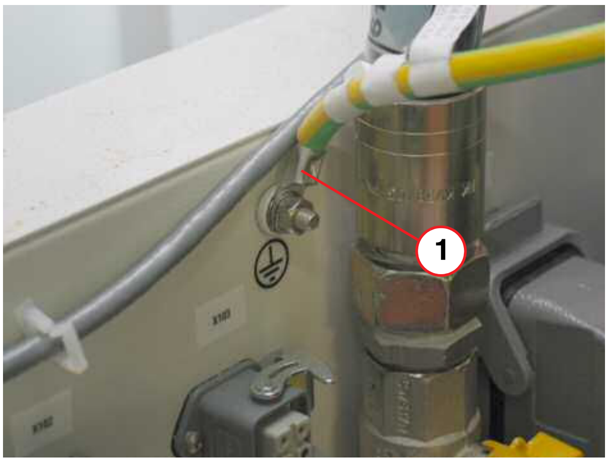
Fig. 113: Exam room filter-plate protective conductor terminal



(1) Filter-plate protective conductor

- Connect the ground (1) to the ground terminal of the filter plate (exam room) at the magnet.

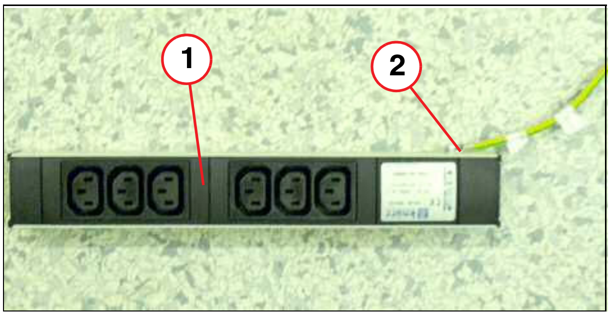
Fig. 114: RCA



(1) Protective conductor

- Connect the protective conductor W0821 from the RCA to the ACC.

Fig. 115: Protective conductor at line voltage distribution



(1) Line voltage distribution
(2) Protective conductor

Equipment room:

- Connect the protective conductor to the ACC protective conductor terminal and route it to the power distribution (1).

Operator room:

- Connect the protective conductor (2) to the power distribution (1).

6.6 Ground test

6.6.1 General

> Read the operating instructions for the meter before performing the protective conductor test!
>
> The country-specific regulations and laws must be observed.

> All ground cabling must be installed before you can perform this test. The system cabling (e.g., gradient and RF cables, etc.) may NOT be installed or connected.

6.6.2 Procedure

Perform the ground test.

> In addition, the protective conductor test has to be performed from the on-site line power distribution to the line voltage connection box of the ACC cabinet. This confirms that there is a protective conductor connection at the ACC cabinet connection point.

Use the protective conductor meter to measure the resistance in ohms between the protective conductor terminal on top of the ACC cabinet and an uninsulated piece of metal on the:

- ACC / GPA cabinet (cabinet frame)
- RCA / compressor (cabinet frame)
- Magnet (magnet housing)
- Patient table (screw)
- Filter plate (grounding bolt)
- Power distribution for the console component (screw connected to ground)
- On-site ground (power distribution)

Enter the measurements in the installation protocol.

> If the resistance exceeds 0.2 Ohm at any given location, check the protective conductor in question for adequate galvanic connection.

6.7 Line voltage cable

⚠ WARNING

Risk of injury due to electric shock.

» Observing the general safety regulations!

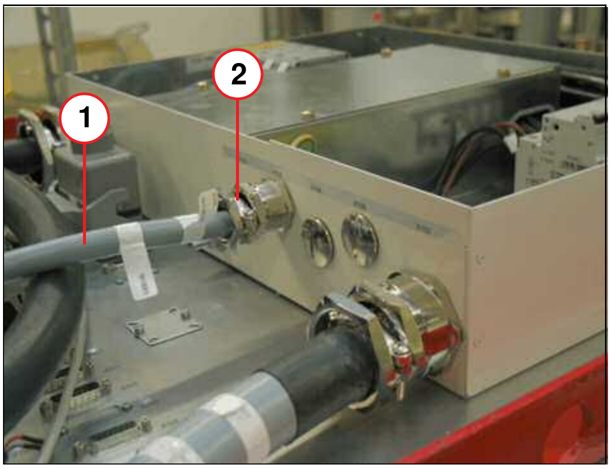
» Ensure that line voltage is not present at the ACC, GPA, the compressor or the SEP (optional).

» Switch off the line voltage supply on-site and secure it against accidental switching on (sign or lock).

6.7.1 Line voltage connection RCA

- Connect the power supply plug to RCA and route to line voltage connection box ACC X102.

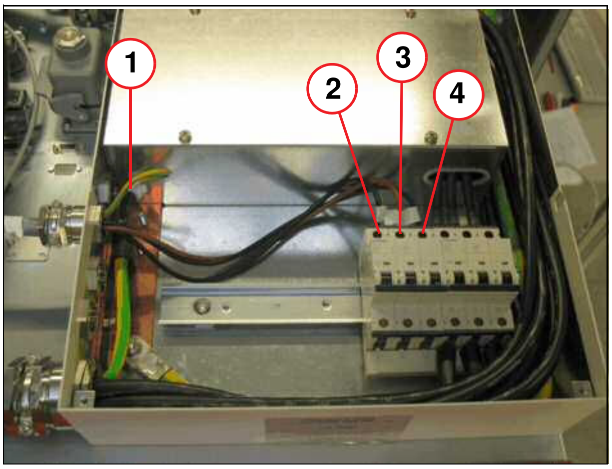
Fig. 116: Strain relief



(1) Power cable
(2) Strain relief

- Route the line voltage cables (2) into the line voltage connection box.
- Attach the strain relief.

Fig. 117: Power cable



(1) Protective conductor
(2) L1
(3) L2
(4) L3

- Connect the ground wire (1).
- Connect voltage cable X102 to fuse F102 (3).
- Check the torque $3.0^{+0.5}$ Nm.

6.8 Connecting the cold head to the He compressor F70

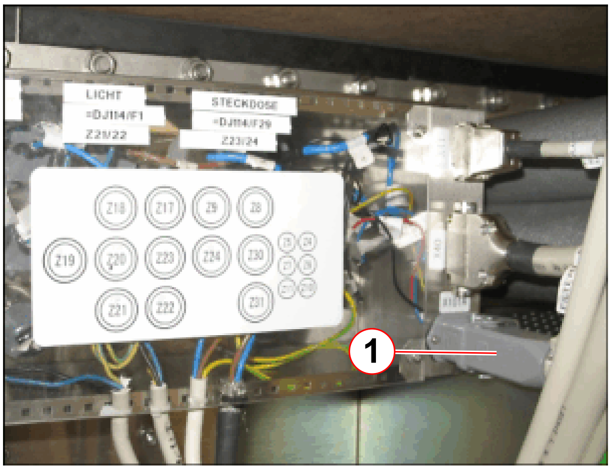
6.8.1 General

The **He pressure lines** are routed from the **helium compressor** to the **magnet cold head** via the **filter plate** and **RF feed-through**.

Any excess length of line must be routed in the **equipment room, NOT** in the magnet room. If a separate closed cable duct has been provided for the He pressure lines, they should be routed in this duct. The minimum bend radius of the pressure line is > 300 mm.

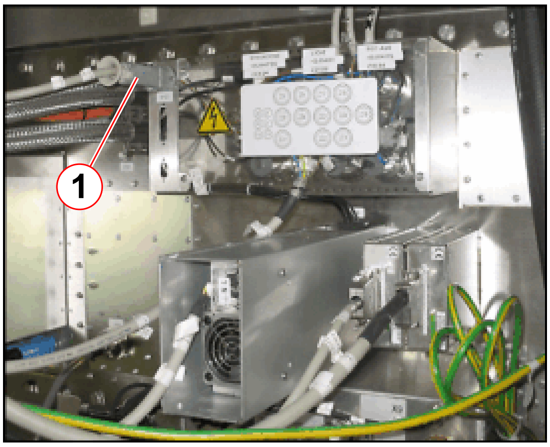
The 90° elbow is permanently attached to one end of the He pressure lines. Fit this end to the helium compressor.

Fig. 118: Magnet room - Cold head power connection



(1) Cold head power cable connector X101A

Fig. 119: Equipment room - Cold head power connection



(1) Cold head power cable connector X101

The **power connection** between the **cold head** and the helium **compressor** is made at the **X101** connector on the **filter plate**.

On installation and/or replacement of the compressor, a test run must be performed (to clean up any oil migration) before the He pressure lines and cold head power cable are connected.

Do NOT connect the:

- HE PRESSURE LINE

- COLD HEAD POWER CABLE

to the compressor until after the test.

Refer to Installation - Starting the helium compressor F70.

6.8.2 Overview

The magnet **cold head** must be connected to the **Helium compressor**. The connections are established in two steps:

1. He pressure lines - **from** the compressor **through** the filter plate and RF feed-through **to** the cold head.
2. Power cables -
   - **Magnet room** - from the cold head to the filter plate **connector X101A** (→ 1/Fig. 118 Page 110).
   - **Equipment room** - from the filter plate **connector X101** to the compressor (→ 1/Fig. 119 Page 110)

Do not confuse the HP and LP pressure lines when connecting them.

Red (SUPPLY) = High pressure (HP)

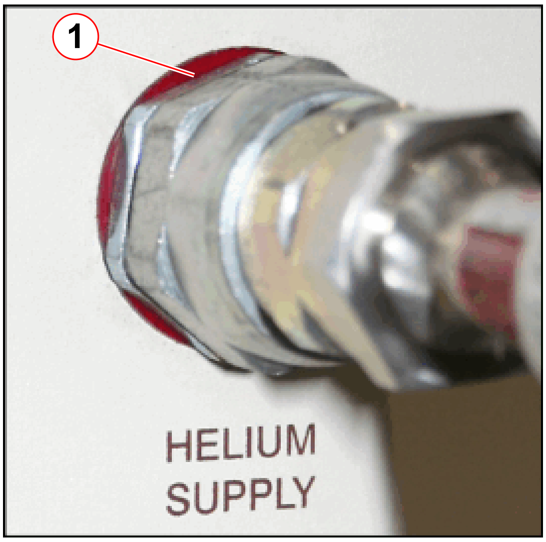
Green (RETURN) = Low pressure (LP)

The helium pressure lines are color coded as shown in the table below.

Tab. 3 Gas line color connections

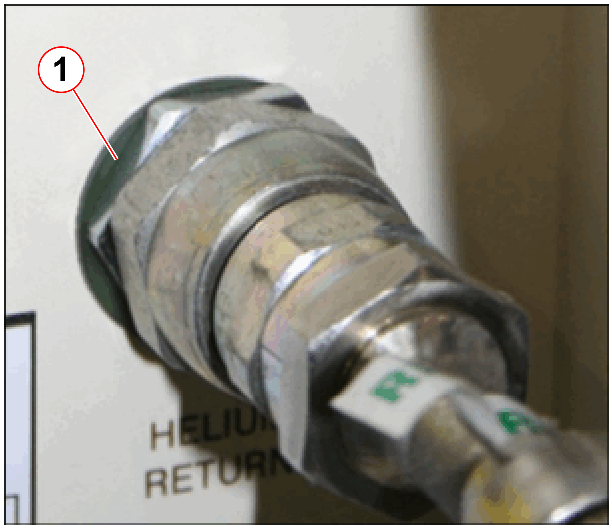
System	Gas line color	F70 He connection
4K (HP)	Yellow	Red
4K (LP)	Green	Green
10K (HP)	Red	Red
10K (LP)	Green	Green

Fig. 120: F70 Helium supply connections - color coding



(1) Helium supply = RED

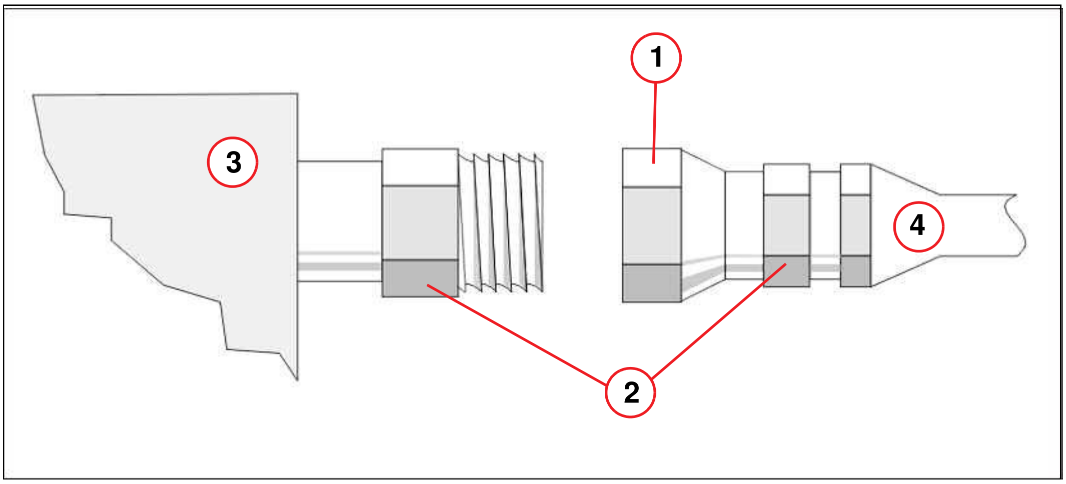
Fig. 121: F70 Helium supply connections - color coding



(1) Helium return = GREEN

6.8.2.1 Helium pressure line connections

Fig. 122: Helium line connection

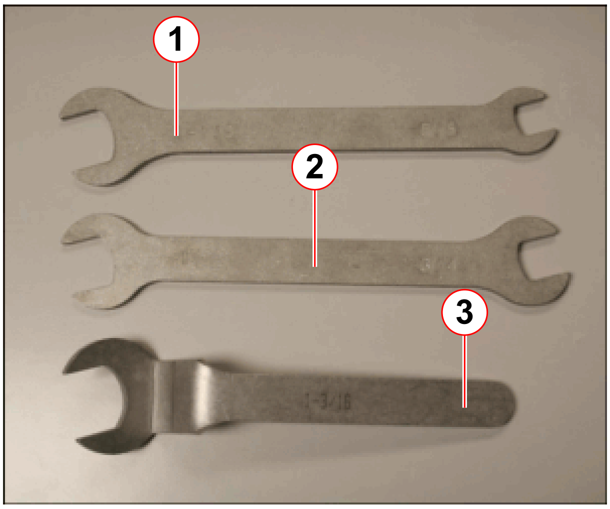


(1) Turn
(2) Hold
(3) Cold head/helium compressor
(4) Helium line

> ℹ Please clean surface of quick couplings.
> Clean connector with clean paper towel before connecting the lines.

> ℹ Ensure a good seal of connectors.
> Before connecting lines ensure the o ring is present in the connector.

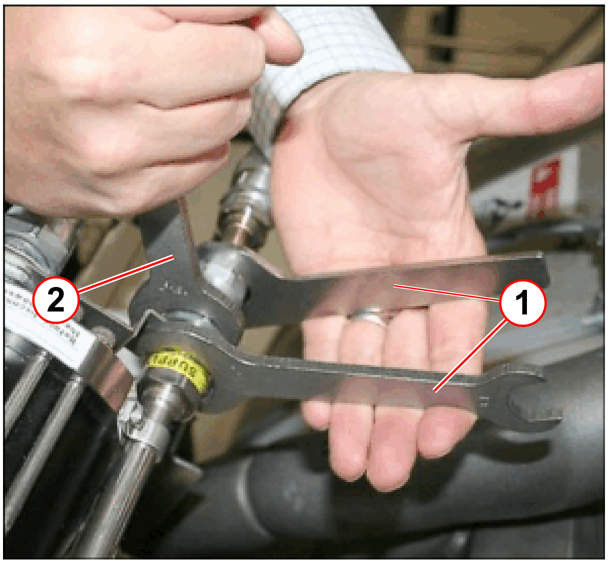
Fig. 123: F70 Spanner kit (part no. 10125650)



(1) 1 1/8" open-end and 5/8" open-end spanner/wrench
(2) 1" open-end and 3/4" open-end spanner/wrench
(3) 1 3/16" open-end crowfoot spanner/wrench

- Always use **three** open-end wrenches when connecting and disconnecting the **He pressure lines** to the **cold head.**
  - » **Tighten** down all couplings as far as possible and then turn them back by 1/4 turn to relieve strain.
  - » If disconnected **incorrectly**, the coupling at the cold head may become **loose** and cause a **leak**.
  - » As a result, the cold head will need to be **replaced**.

Fig. 124: Connecting the cold head pressure lines



(1) Hold
(2) Turn

Cold head helium pressure lines

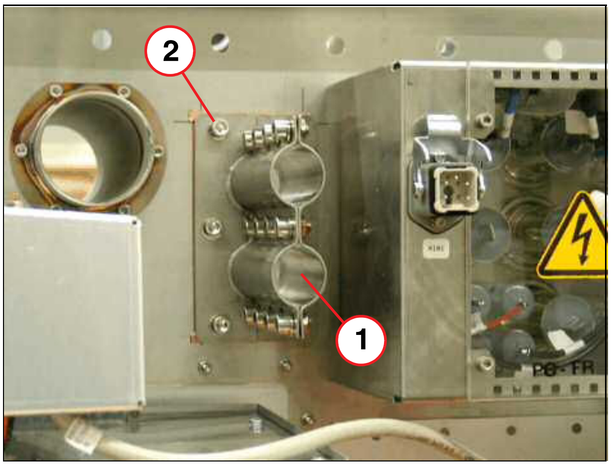
- **Tighten** down all couplings as far as possible and then turn them back by 1/4 turn to relieve strain.

6.8.3 Routing the He pressure lines

6.8.3.1 Removing the RF feed-through on the filter plate

The **RF feed-through** for the high- and low-pressure He lines is installed at the **filter plate.**

Fig. 125: Filter plate, electronics room



(1) RF feed-through, helium lines
(2) Screw, lock washer

Equipment room:

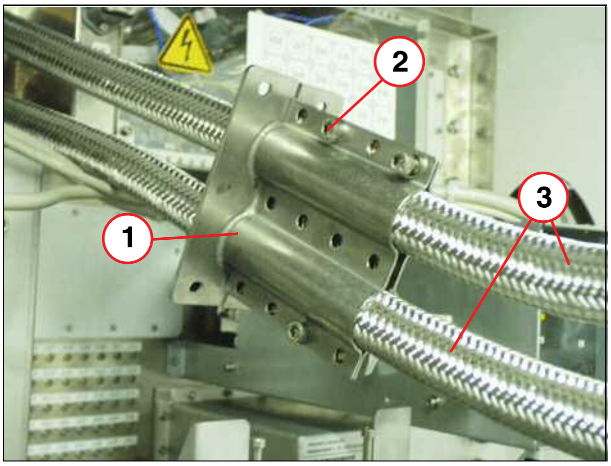
- **Remove** the **RF feed-through** (1) by removing the 6 screws (2).

- Starting at the magnet **cold head**, route the **He pressure lines** carefully through the **filter plate** to:
  - the He compressor in the **RCA cabinet**.

> DO NOT connect the He pressure lines to the RCA until test running is complete.

6.8.4 Reinstalling the RF feed-through at the filter plate

Fig. 126: RF feed-through, helium lines

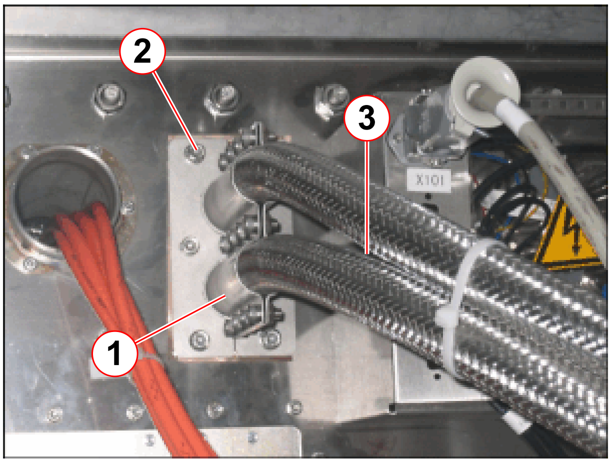


(1) RF feed-through
(2) Screw, lock washer, washer
(3) Helium lines

Equipment room:

- **Remove** the 12 clamp screws (2) from the **RF feed-through**.
- **Fit** the RF feed-through around the **He pressure lines** and refit the screws.
- **Tighten** the screws **loosely** to allow the He pressure lines to pass through easily.

Fig. 127: Filter plate - RF feed-through



(1) RF feed-through
(2) Screw, lock washer, washer
(3) Helium lines

Equipment room:

- **Refit** the **RF feed-through** (1) to the **filter plate** using the 6 screws (2).
- Fully **tighten** the screws on the **RF feed-through** (1).

Fig. 128: Insulating the helium pressure lines

Magnet room:

- **Insulate** the **He pressure lines** using the supplied insulation material from the **magnet cold head** to the **filter plate**.
- Use an insulating adhesive tape to cover the edges of the insulation material.

Provide additional insulation for the He pressure lines in the monitoring room and equipment room if excessive noise is audible.

CAUTION

Danger of contamination!

The compressor must be operated for 10 minutes to clear up any oil contamination incurred during transportation.

» **Do not connect the helium pressure lines to the compressor until AFTER test running of the compressor.**

6.8.5 Connecting the cold head power cable

6.8.5.1 Cold head

Fig. 129: F70 He gas connection

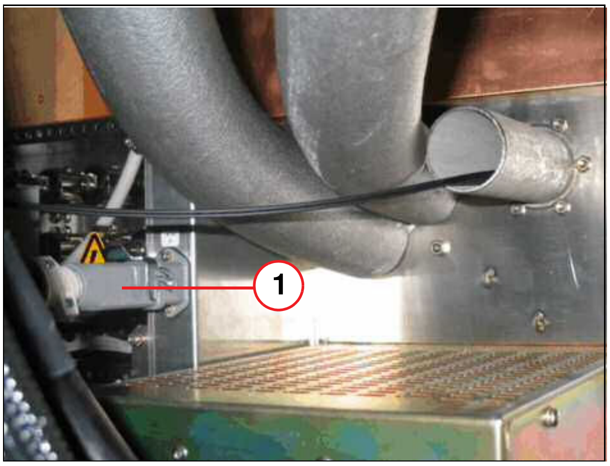
(1) High pressure line (SUPPLY/RED)
(2) Low pressure line (RETURN/GREEN)
(3) Power connector

Magnet room:

- Route the **cold head power cable** from the **cold head** (1) to **filter plate.**

6.8.5.2 Filter plate

Fig. 130: Filter plate cold head power cable

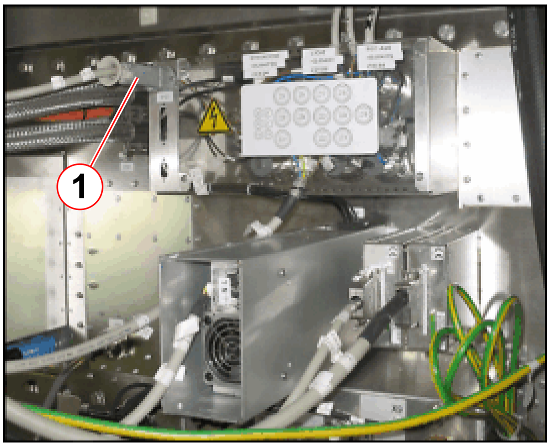


(1) Power cable

Magnet room:

- **Connect** the **cold head power cable** to **connector X101A** (1) at the **filter plate**.

Fig. 131: Equipment room - Cold head power connection



(1) Cold head power cable connector X101

Equipment room:

- **Connect** the **cold head power cable** to **connector X101** (1) at the **filter plate.**

  DO NOT CONNECT THE POWER CABLE TO THE COMPRESSOR UNTIL AFTER THE 10 MINUTE TEST IS COMPLETE.

  Refer to Starting the helium compressor F70 - **Test running the compressor before startup.**

DO NOT connect the power cable to the RCA / compressor!

6.9 Connecting the gradient cables

6.9.1 General Remarks

Prevent damage to the insulation by ensuring that the cables cannot come into contact with any bolts. Use cable ties to secure any excess lengths. Secure all gradient cables using a torque wrench.

Gradient cable	Torque
GPA	40 ± 4 Nm
Gradient connection plate at the magnet	40 ± 4 Nm
RF filter plate	22 ± 1 Nm

WARNING

Physical injury or property damage.

The gradient cables have to be accurately connected.

» **The torque and polarity of all gradient cable connections have to be checked by a second person.**

» **The gradient cable connections fastened with the correct torque have to be marked using a waterproof pen**

» **Record this in the protocol.**

NOTICE

Damage to the cable sheathing.

» **Do not secure the gradient cables to the mounting plates (magnet, RF-filter plate, and GPA) using cable ties.**

WARNING

System damage

Make sure not to mix up the polarity (+/-) or the three axes (x, y, z) of the gradient system when connecting the cables. Note the labeling.

» Z+ = yellow

» Y+ = blue

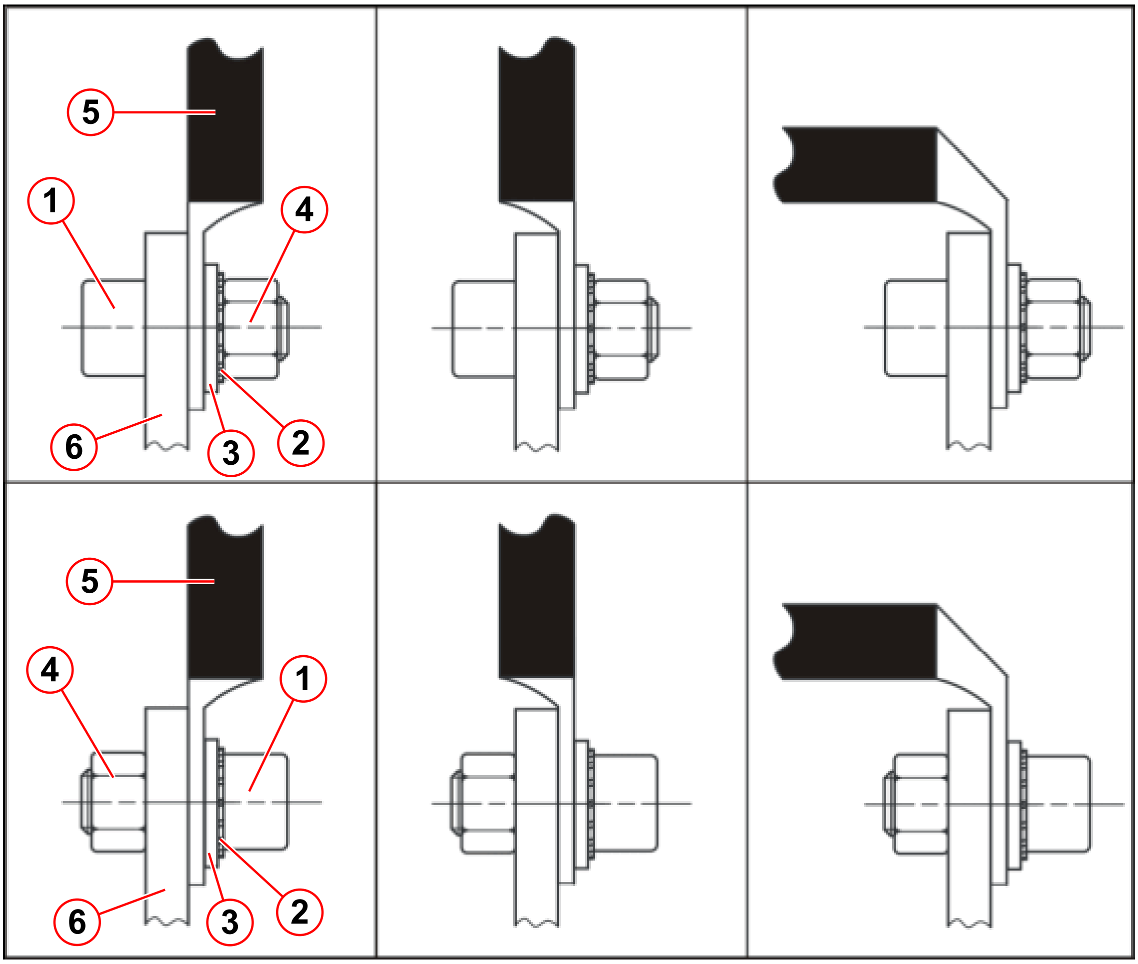
» X+ = red

The position of the cable lug can vary (rotation by 180°):

The flat side of the cable lug can be directly in contact with the copper bar or the step side of the cable lug can be directly in contact with the copper bar, refer to (➔ Fig. 132 Page 120).

The lock washer and the washer must be on the side of the gradient cable lug, refer to (➔ Fig. 132 Page 120).

Fig. 132: Examples of gradient cable connection to the GPA



(1) Screw
(2) Lock washer
(3) washer
(4) Nut
(5) Gradient cable
(6) GPA connection

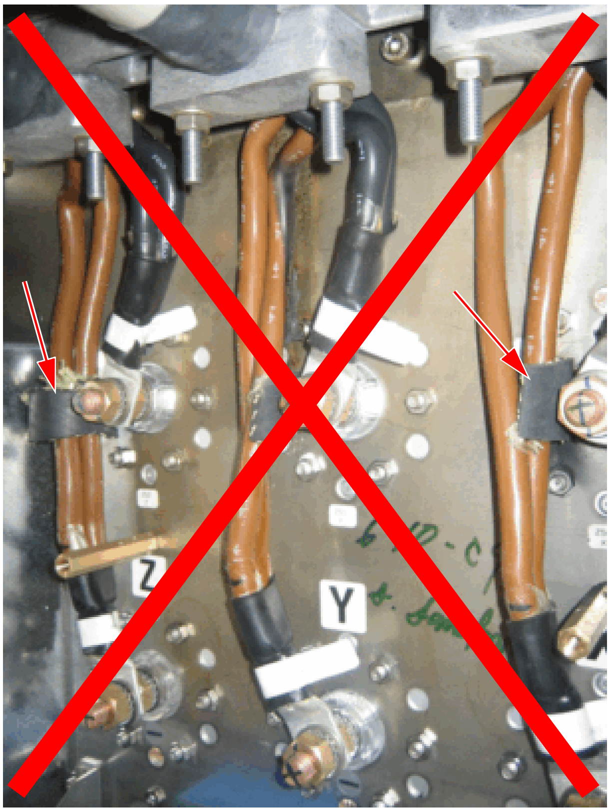
NOTICE

Risk of short circuit!

The insulation of the brown cable (-) must not touch the connecting bolt or cable lug of the black cable (+).

- » **Ensure sufficient air space between the (-) and (+) cable.**
- » **If necessary, rotate the entire gradient cable in the cable ground clamp to achieve sufficient space.**
- » **Never use additional material for extra insulation.**

Fig. 133: Never use additional materials for insulation



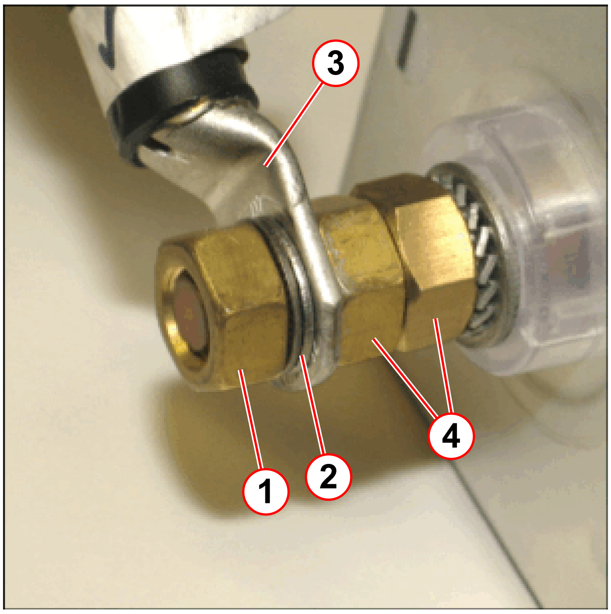
Take care not to turn the bolts when you tighten the gradient cables at the gradient filter.

If the bolts turn, the gradient filters may be damaged.

The cable lugs can point toward the back or front; do not rotate them >180 degrees.

Connection of the gradient cable to the filter plate:

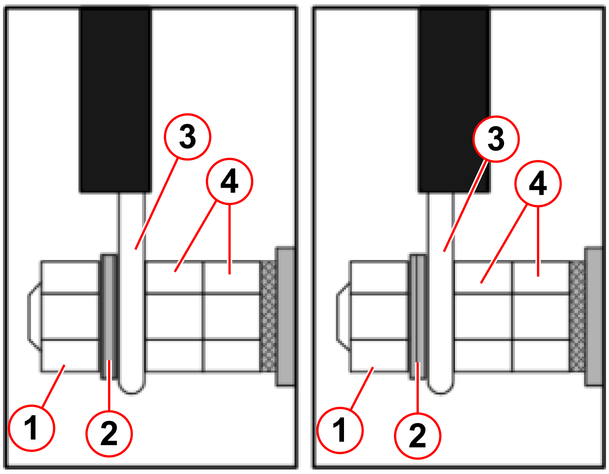
Fig. 134: New connection at the filter plate



(1) Nut
(2) Nordlock washer
(3) Cable lug
(4) Nuts

In the newer systems, the gradient cable connection to the filter plate has changed; refer to the pictures.

Fig. 135: New connection at the filter plate



(1) Nut
(2) Nordlock washer
(3) Gradient cable lug
(4) Nuts

> **i** The gap between the cable clamp segments has to be the same on the left and right side, refer to (→ Tab. 4 Page 123).
>
> However, the absolute gap size can vary and depends on the cable type used (different manufacturer).

Tab. 4 Gradient cable clamps

Fig. 136: Correct cable clamps

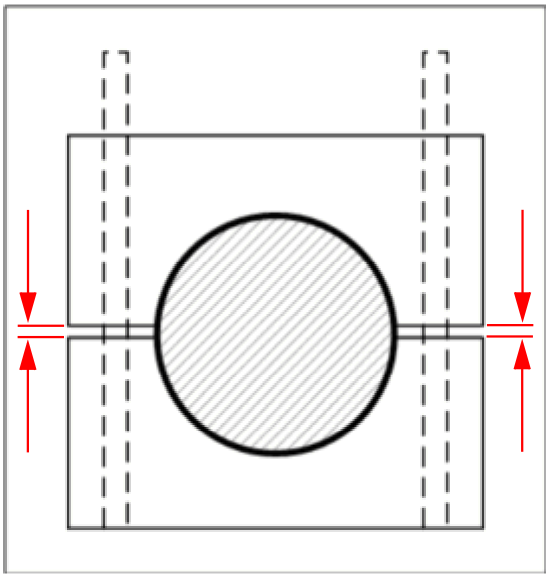


Fig. 137: Wrong cable clamps

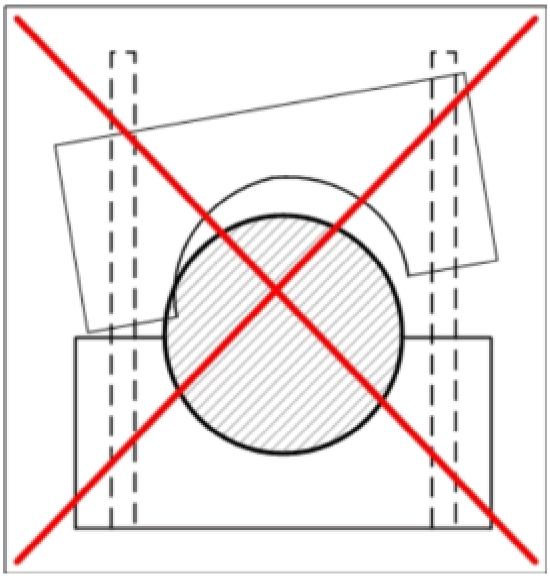
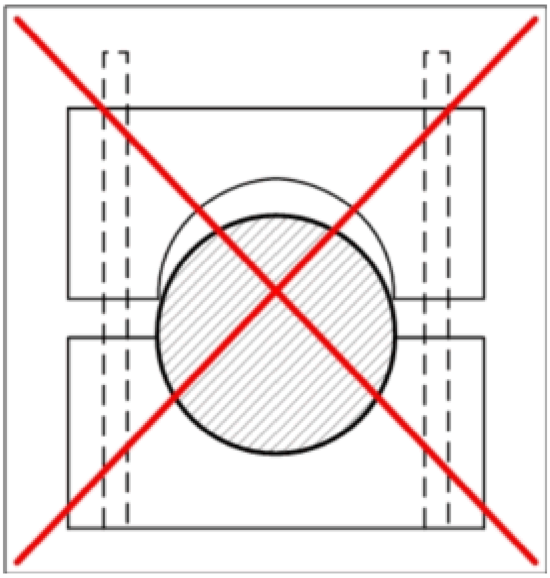


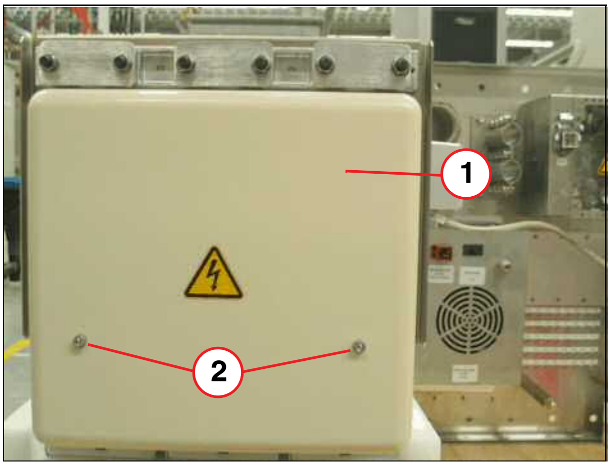
Fig. 138: Wrong cable clamps



6.9.2 Gradient Cables outside the RF Room

Connect the gradient cables on the RF filter plate (electronics room) to the GPA.

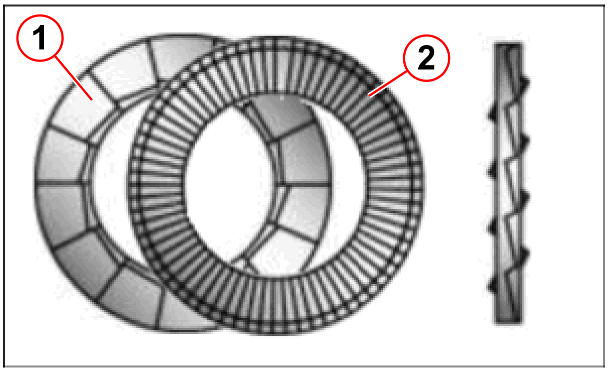
Fig. 139: Protective cover, gradient filter (technical room)



(1) Protective cover
(2) Screws

- Remove the protective cover (1) of the gradient filters on the RF filter plate.

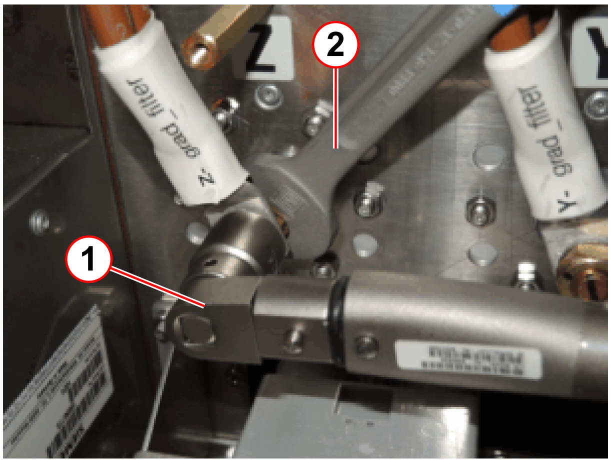
Fig. 140: NORD-LOCK washer



(1) Cam side
(2) Ridged side

You must insert a Nord-Lock washer (comprising two parts) between the eye bolt and the nut. Align the washer parts correctly (see the figure on the left).

Fig. 141: RF-filter plate connection

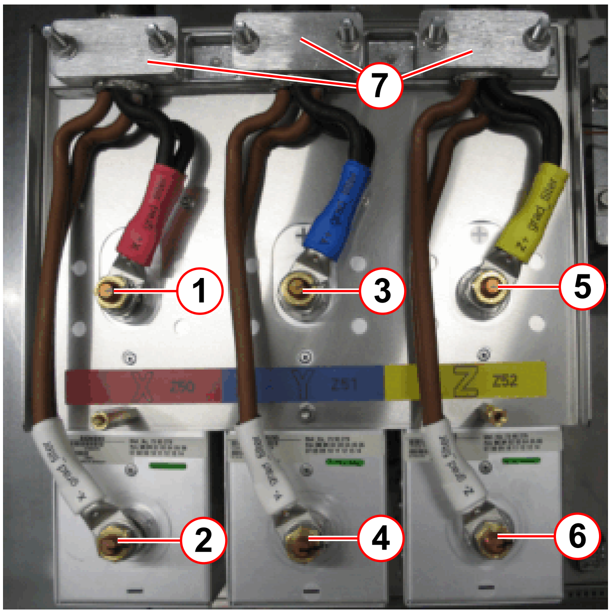


(1) Torque wrench
(2) Open-end wrench

Use a torque wrench and a open-end wrench to secure the gradient cables.

» To avoid that the bolt turns at the filter plate.

Fig. 142: Gradient cable connection at RF-filter plate (equipment room)



(1) X+
(2) X-
(3) Y+
(4) Y-
(5) Z+
(6) Z-
(7) Cable clamps

- Connect the gradient cables (Z, Y, and X) to the gradient filters (1-6).

  Use a torque of 22 ± 1 Nm (1-6)

- Secure the gradient cables using cable clamps (7).
- Check that the torque is

  22 ± 1 Nm (1-6).

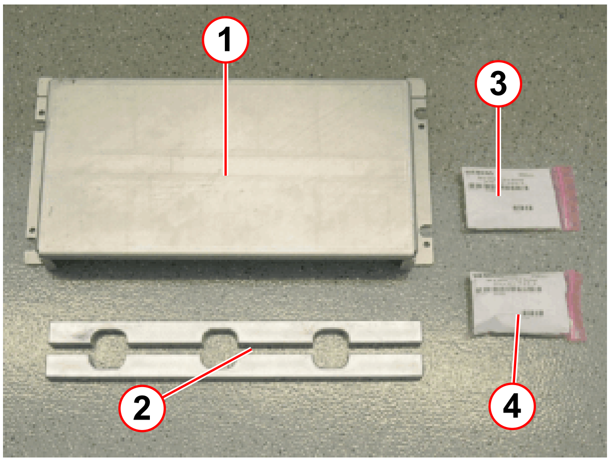
  » The torque and polarity must be verified and confirmed by two different people (four-eyes principle).
- Mark the connections using a waterproof pen.
- Replace the protective cover.

The cable clamps (→ 7/Fig. 142 Page 125) must clamp the shield only.

Ensure that the insulation is not clamped.

6.9.2.1 Connecting the GPA (Shorter Cable Kit: 2.5 m)

Fig. 143: GPA accessories

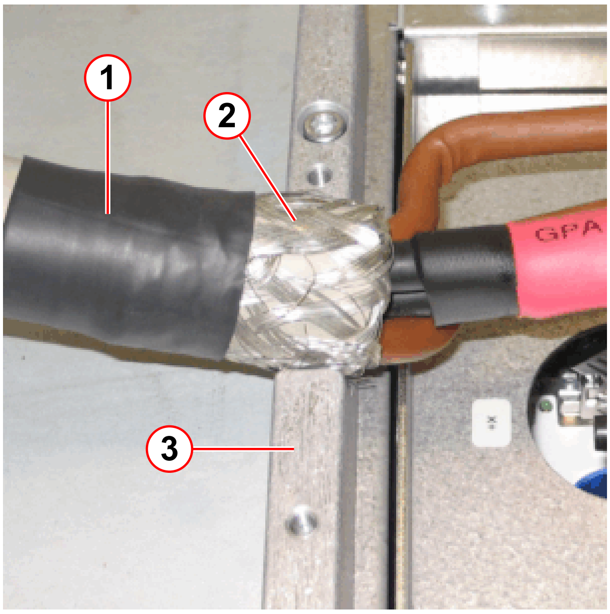


(1) Cover
(2) Rails
(3) Gradient cable accessories
(4) Accessories: cover and rails

The figure to the left shows the accessories for connecting the gradient cables to the GPA and installing the cover and guide rails.

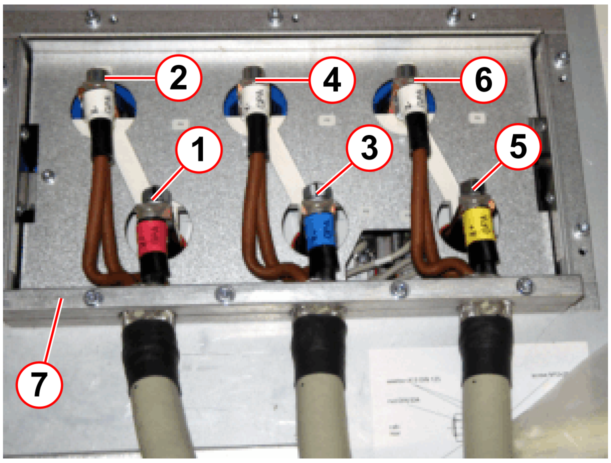
Ensure that the gradient cables are connected as shown in the figure (→ Fig. 144 Page 126). The shield must be clamped between the rails.

Fig. 144: GPA gradient cable (short cable)



(1) Gradient cable
(2) Cable shield
(3) Clamp rail

Fig. 145: GPA gradient cable connection



(1) X+
(2) X-
(3) Y+
(4) Y-
(5) Z+
(6) Z-
(7) Rail

- Connect the gradient cables (Z, Y, and X) to the GPA using Allen key screws, washers, serrated lock washers, and nuts.

  Use a torque of 40 ± 4 Nm (1-6).

- Install the guide rails (7) using the screws provided.
- Check that the torque is

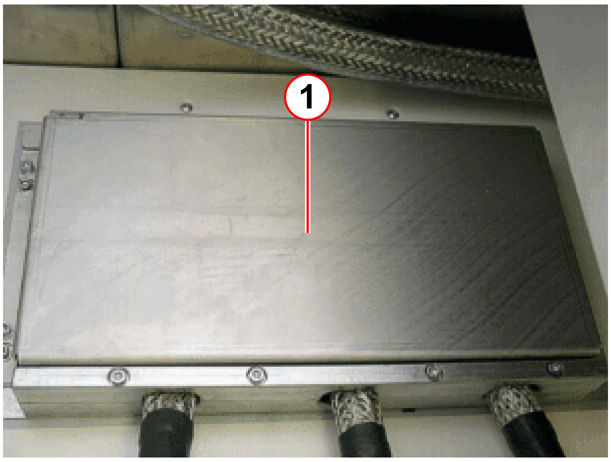
  40 ± 4 Nm (1-6).

  » The torque and polarity must be verified and confirmed by two different people (four-eyes principle).

- Mark the connections using a waterproof pen.

> **i** The cable clamps (→ 7/Fig. 145 Page 127) must clamp the shield only.
>
> Ensure that the insulation is not clamped.

Fig. 146: GPA gradient cover



(1) Gradient cover

- Attach the GPA cover (1).

6.9.2.2 Connecting the GPA (Longer Cable Kit: >2.5 m)

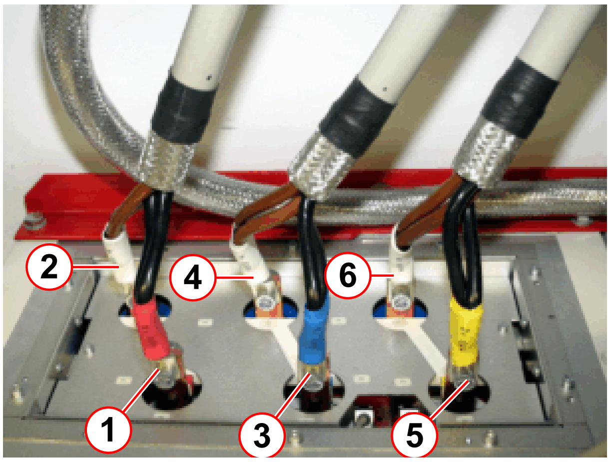
Fig. 147: GPA Accessories

(1) Gradient wire accessories
(2) Accessories: cover and clamps

Accessories for connecting the gradient cables to the GPA and installing the cover and clamps.

> **i** The cable lugs can point toward the back or front; do not rotate them through 180 degrees.

Fig. 148: GPA gradient cable connection



(1) X+
(2) X-
(3) Y+
(4) Y-
(5) Z+
(6) Z-

- Connect the gradient cables (Z, Y, and X) to the GPA using Allen key screws, washers, serrated lock washers, and nuts.

  Use a torque of 40 ± 4 Nm (1-6).

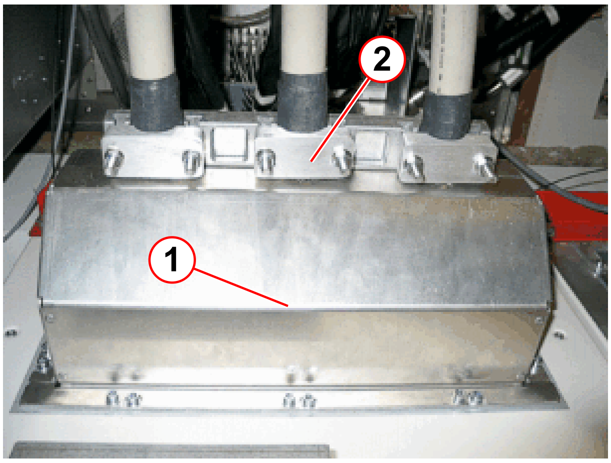
  

- Attach the rear GPA covers and align the gradient cables so they can be secured using the cable clamps.
- Remove the rear GPA cover again.
- Check that the torque is

  40 ± 4 Nm (1-6).

  » The torque and polarity must be verified and confirmed by two different people (four-eyes principle).
- Mark the connections using a waterproof pen.

Fig. 149: GPA gradient cover



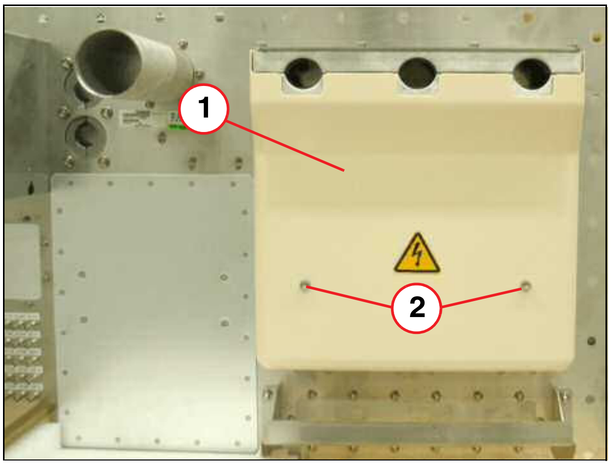
(1) Gradient cover
(2) Clamps

- Attach the GPA cover.
- Install the cable clamps using the screws, washers, serrated lock washers, and nuts provided and then secure the gradient cables using the cable clamps.

> The cable clamps (→ 2/Fig. 149 Page 129) must clamp the shield only.
>
> Ensure that the insulation is not clamped.

6.9.3 Connecting the gradient cables in the RF room

Fig. 150: Protective cover, gradient filter (examination room)

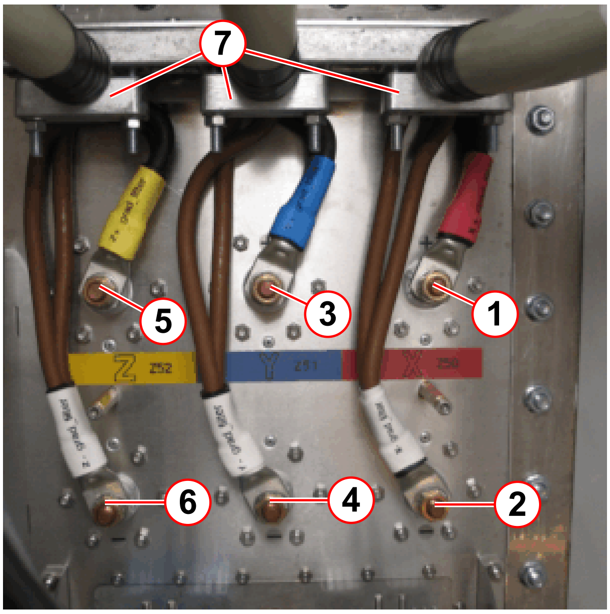


(1) Protective cover
(2) Screws

- Remove the protective cover (1) from the RF filter plate.

> The cable lugs can point toward the back or front; do not rotate them through 180 degrees.

Fig. 151: Gradient connection at RF-filter plate (examination room)



(1) X+
(2) X-
(3) Y+
(4) Y-
(5) Z+
(6) Z-
(7) Clamp

- Connect the gradient cables (Z, Y, and X) to the gradient filters (1-6).

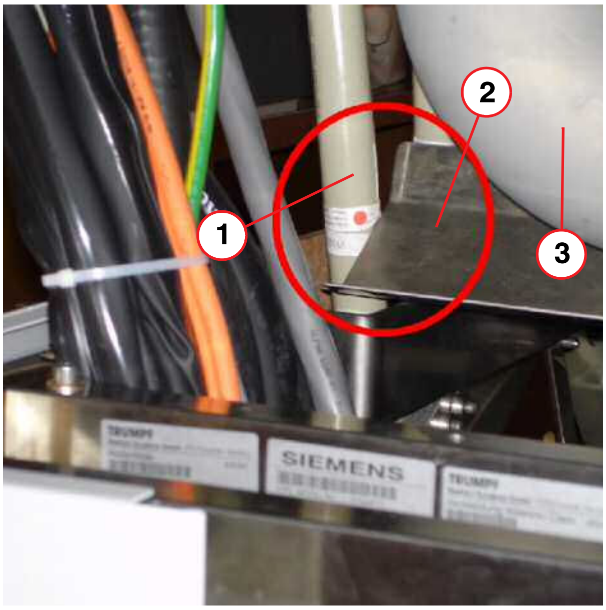
  Use a torque of 22 ± 1 Nm (1-6).

- Secure the gradient cables using cable clamps (7).
- Check that the torque is

  22 ± 1 Nm (1-6).

  » The torque and polarity must be verified and confirmed by two different people (four-eyes principle).
- Mark the connections using a waterproof pen.
- Reattach the protective cover.

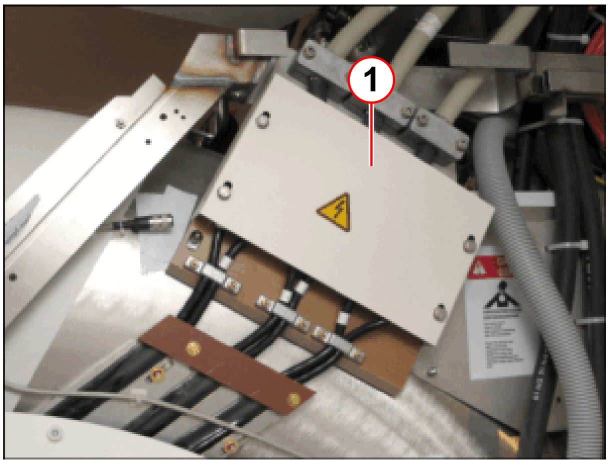
Fig. 152: Cable routing



(1) Gradient cable
(2) Trip tray
(3) Quench pipe

- Gradient cable should not touch the trip tray.

Fig. 153: Gradient cable connection plate at the magnet

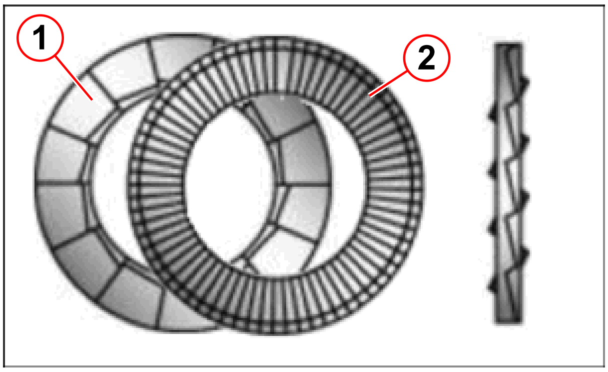


(1) Protective cover

- Remove the protective cover (1) from the magnet.

When routing the gradient cables from the magnet and connecting them to the connector panel, ensure that they are not up against the base frame of the covers and that they do not get caught on the base frame.

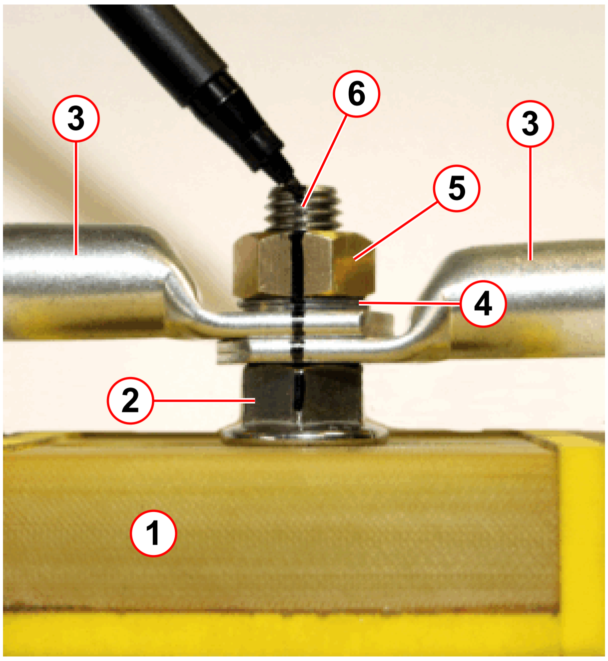
Fig. 154: NORD-LOCK washer



(1) Cam side
(2) Ridged side

You must insert a Nord-Lock washer (comprising two parts) between the eye bolt and the nut. Align the washer parts correctly (see the figure on the left).

Fig. 155: Gradient connection plate at the magnet (example Aera)

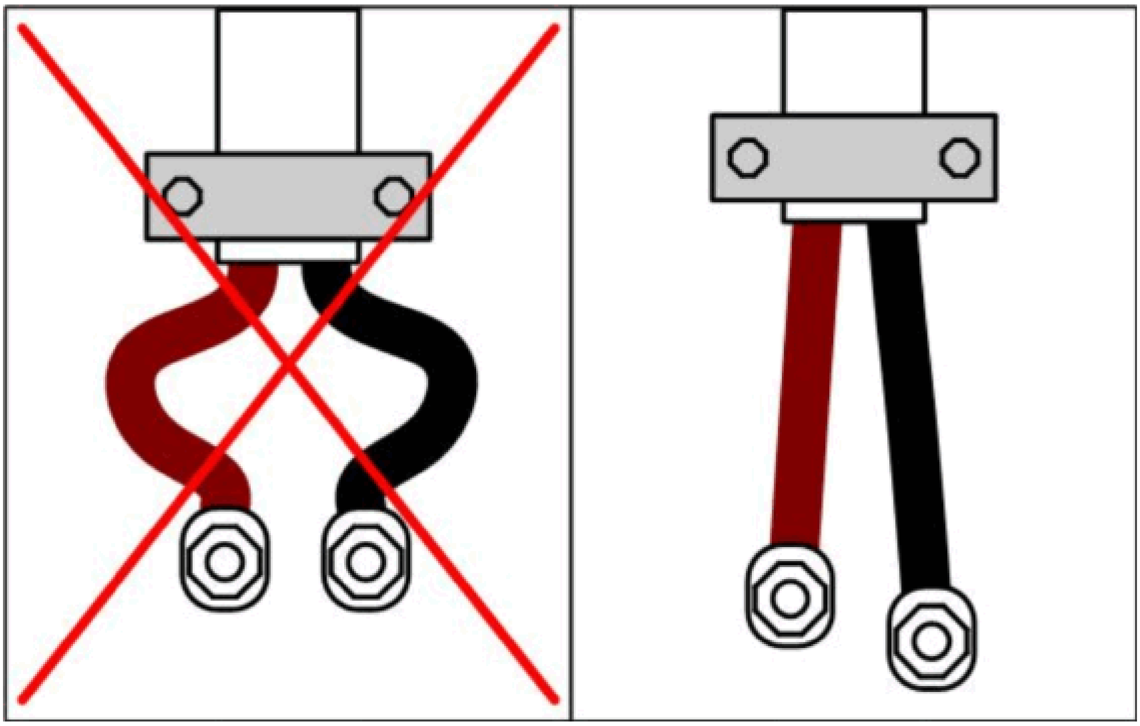


(1) Connection plate
(2) Range nut
(3) Gradient cable lugs
(4) Nord-Lock washer
(5) Nut
(6) Mark with waterproof pen

- Attach the **lower nut (2)** with a torque of 21 ± 2 Nm at the X,Y, and Z connection using the open-end wrench.

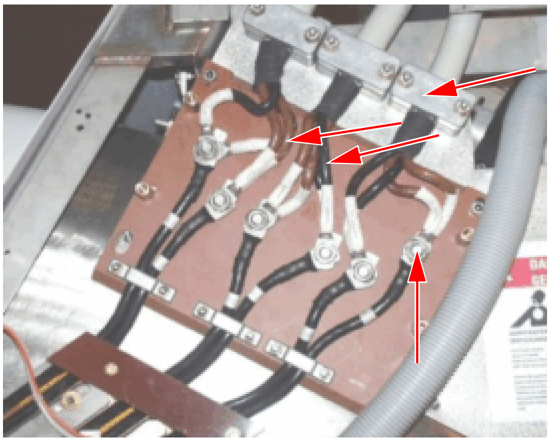
(If necessary, move the screw in the elongated hole to achieve a strain-free mounting of the coax-cable).

Fig. 156: Gradient cable routing at the magnet connection plate



Route the gradient cables as straight as possible, they should not be bent. Refer to (→ Fig. 156 Page 132).

Fig. 157: Gradient connection plate

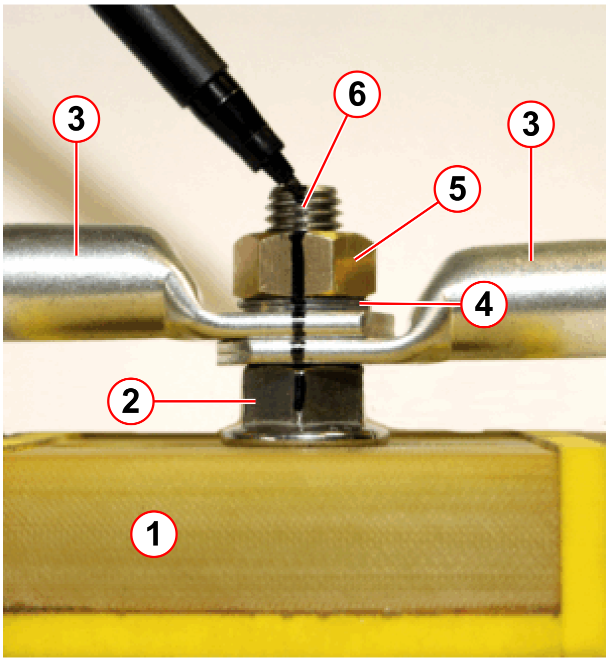


- Connect the gradient cables to the gradient mounting plate.

  Use a torque of 35 ± 2 Nm.
- Clamp the gradient cables using the cable clamps. Position the clamps so that the gradient cables are correctly clamped.
- Check to make sure the torque is 35 ± 2 Nm

  » The torque and polarity must be verified and confirmed by two different people (principle of dual control).

Fig. 158: Gradient connection plate at the magnet (example Aera)



(1) Connection plate
(2) Range nut
(3) Gradient cable lugs
(4) Nord-Lock washer
(5) Nut
(6) Mark with waterproof pen

- Tighten the nut again and have a second person check the torque and polarity.

  Check that the torque is 40 ± 4 Nm

  » Record this in the protocol.
- Mark the connections (5) using a waterproof pen (6).
- Replace the protective cover.

Do not allow the cables or cable lugs to touch the mounting materials.

Check the cable insulation for defects to prevent short circuits. Make absolutely sure that the cables are not in contact with any conductive materials.

6.10 RF connection

Designation	Torque:	Wrench size:
7/16[1]	25 Nm.	27 and 32 mm
HN[2]	6 Nm.	18 mm.

1. Designation for the transmit cable connection
2. Designation for transmit and/or receive cable connections

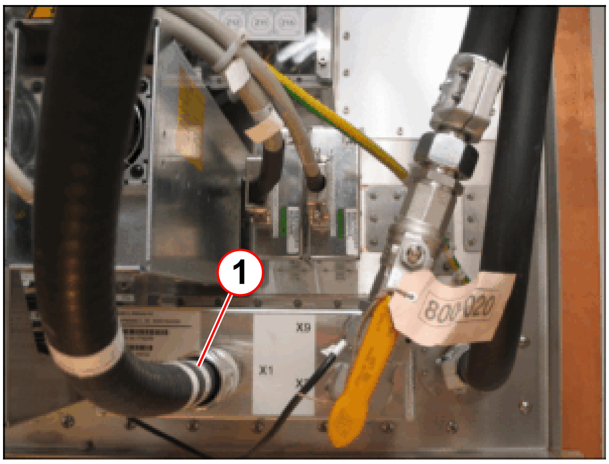
Tighten all other RF connections by hand.

6.10.1 RF cable connection

The RF cables are sensitive to mechanical forces such as bending, twisting and squeezing.

When connecting the RF cables, make sure that the connector threads fit their counterparts exactly and are not damaged.

Fig. 159: RF connection at the filter plate

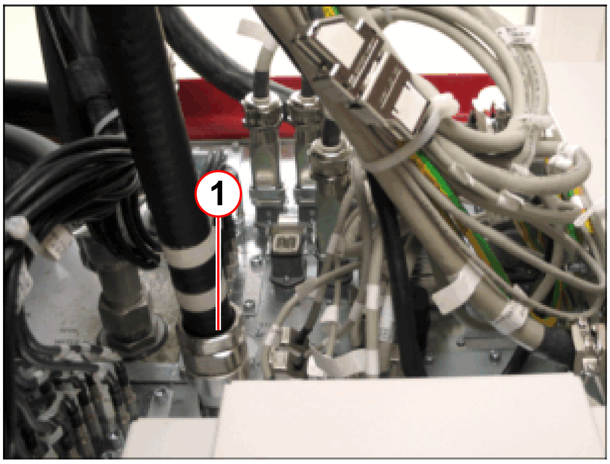


(1) X1

RF cables on the filter plate in the equipment room:

- Connect the RF cable (W0190) to filter plate X1 TAS (1).

Fig. 160: RF connection at the ACC

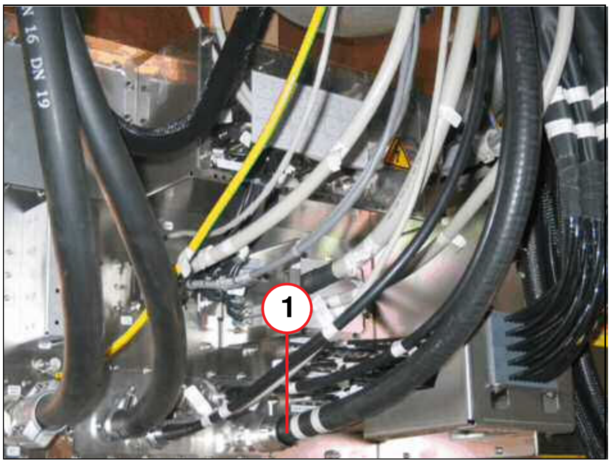


(1) X38

RF cables on the filter plate in the equipment room:

- Connect the RF cable to the top of the ACC X38 (1).

Fig. 161: RF filter plate UR

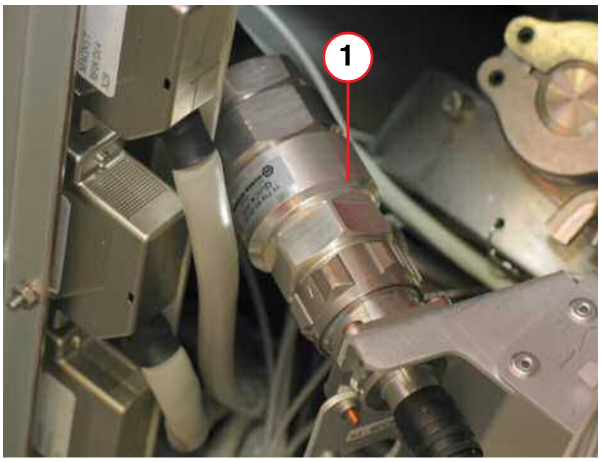


(1) X4

RF cable on the filter plate in the examination room:

- Connect the RF cable to filter plate X4 (1).

Fig. 162: Cable routing

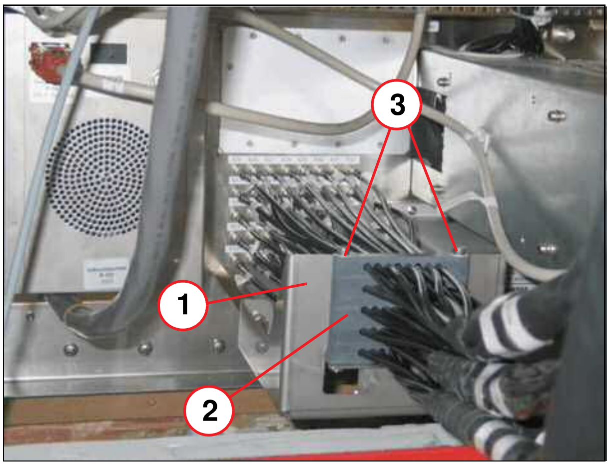


(1) RF transmit cable W1190

RF cable to the magnet

- Connect the RF cable:
  - » W1190 to X1, BCCS
  - » W1200 to X5, TALES

Fig. 163: Electronics room filter-plate RF cable



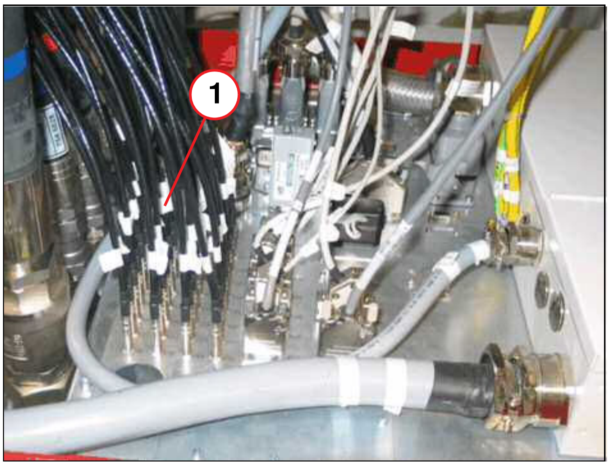
(1) Bracket
(2) Guide rails
(3) Screws

RF cables on the filter plate in the examination room.

- Install the bracket (1) on the filter plate.
- Connect the RF cable (W0185, W0181, W0182, W0183, W0184) to the filter plate

  (the number of RF cables depends on the number of channels, optional).
- Feed the guide rails (2) into the guide.
- Attach the guide rails with two M4 screws (3).

Fig. 164: RF cable ACC



(1) RF cables

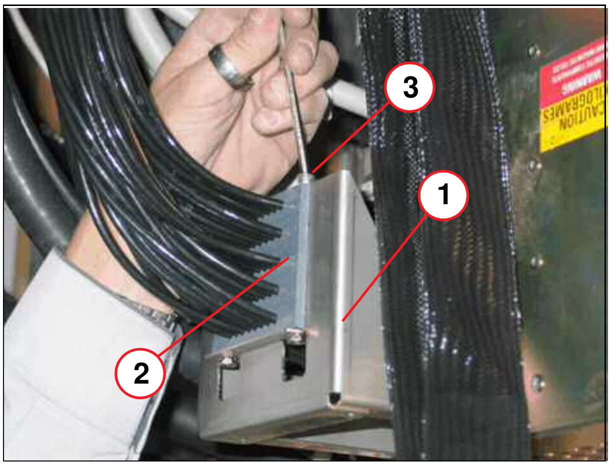
RF cables on top of the ACC cabinet

- Connect the RF cables (W0185, W0181, W0182, W0183, W0184) to the top of the ACC cabinet (1).

RF cables to the magnet

- Route the RF cables (1) already connected to the magnet to the filter plate.

Fig. 165: *Exam room filter-plate RF cable*



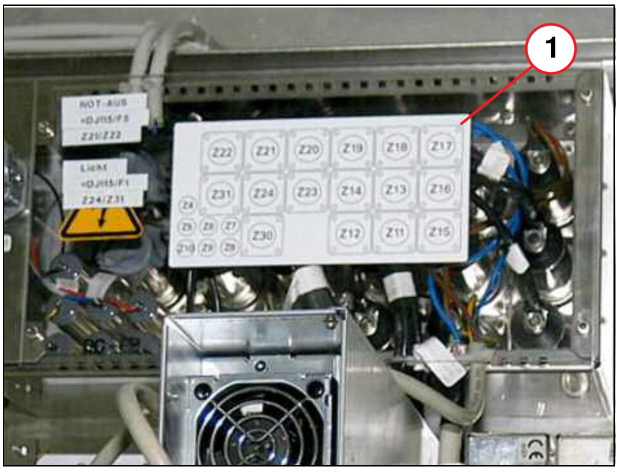
(1) Bracket
(2) Guide rails
(3) Screws

RF cables on the filter plate in the examination room

- Install the bracket (1) on the filter plate.
- Connect the RF cables that come from the magnet to the filter plate.
- Feed the guide rails (2) into the guide.
- Attach the guide rails with two M4 screws (3).

6.10.2 RF filter connection

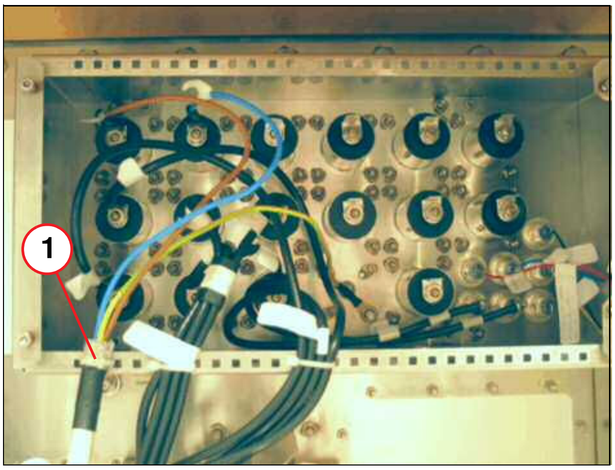
Fig. 166: *RF filter connection*



(1) RF filter

An overview of the RF filters is shown on the cover (1).

Fig. 167: RF filter cables



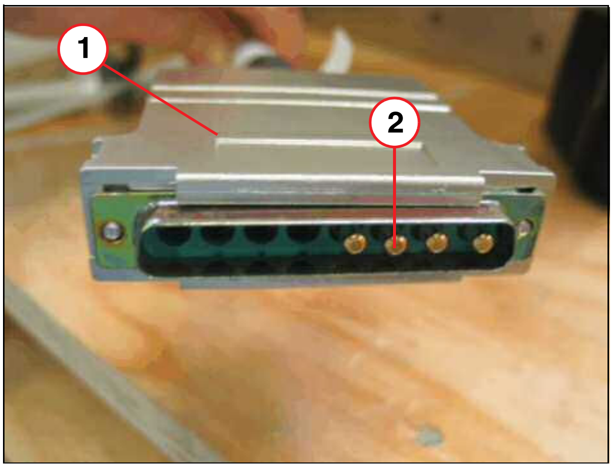
(1) Shielding

- Secure the shielding with a cable tie as illustrated on the left (1).

6.11 Filter plate

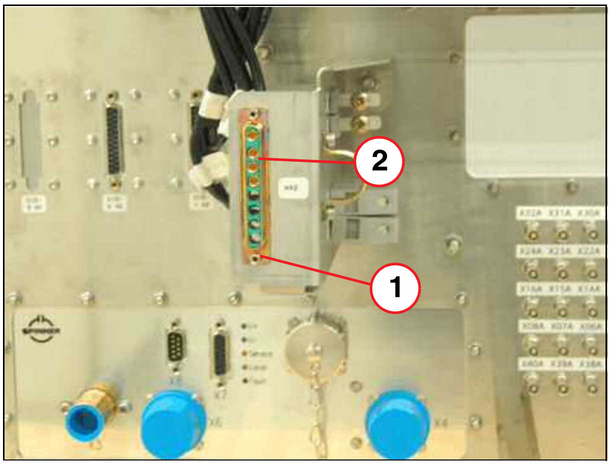
When connecting plug X42 at the filter plate (magnet room), ensure that the plug is not connected in a rotation of 180˚. Note the contacts (above), refer to
(➔ Fig. 168 Page 140) and (➔ Fig. 169 Page 140).

Fig. 168: Filter plate, connector X42



(1) Connector X42
(2) Contacts

Fig. 169: Connection RF filter plate X42



(1) Connection X42
(2) Contacts

6.12 Fiber optic cable

Pay attention to the bending radii.

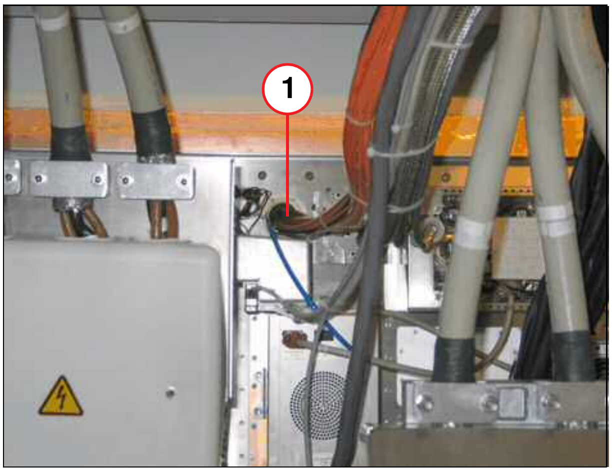
There must be no tractive forces present when routing fiber optic cables.

The excess lengths can be routed in the magnet room.

6.12.1 Fiberoptic cable

Do not attach fiber optic cables too tightly using cable ties. If the cable ties are too taut, they can damage the fiber optic cables.

Fig. 170: Filter-plate feed-through

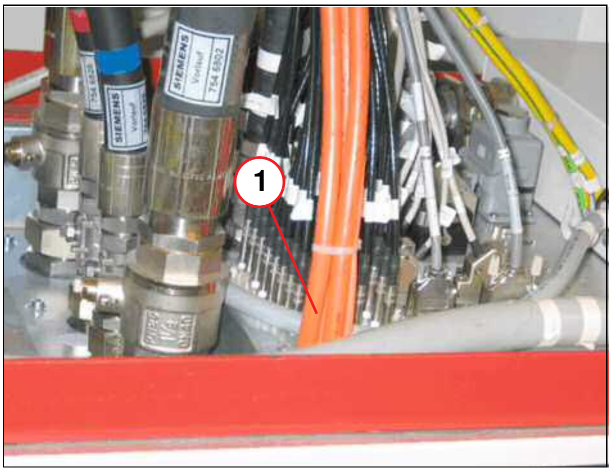


(1) Opening

Feed the fiber optic cables through the opening (1) in the filter plate.

Feed the hose (blue) for the vacuum pump (optional) through the opening (1) in the filter plate.

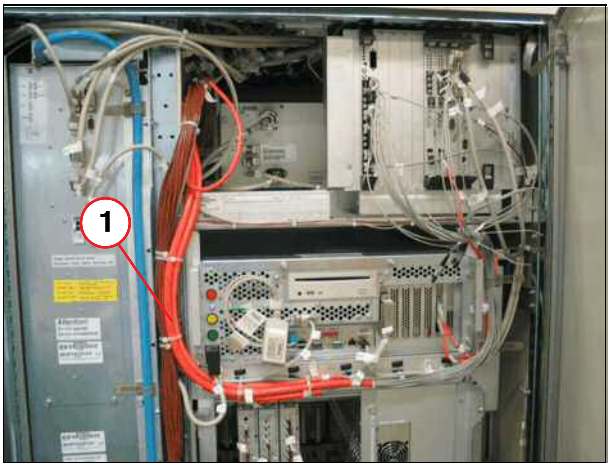
Fig. 171: ACC feed-through



(1) Opening

The opening for the fiber optic cable (1) is on top of the ACC cabinet.

Fig. 172: Fiber optic cable ACC



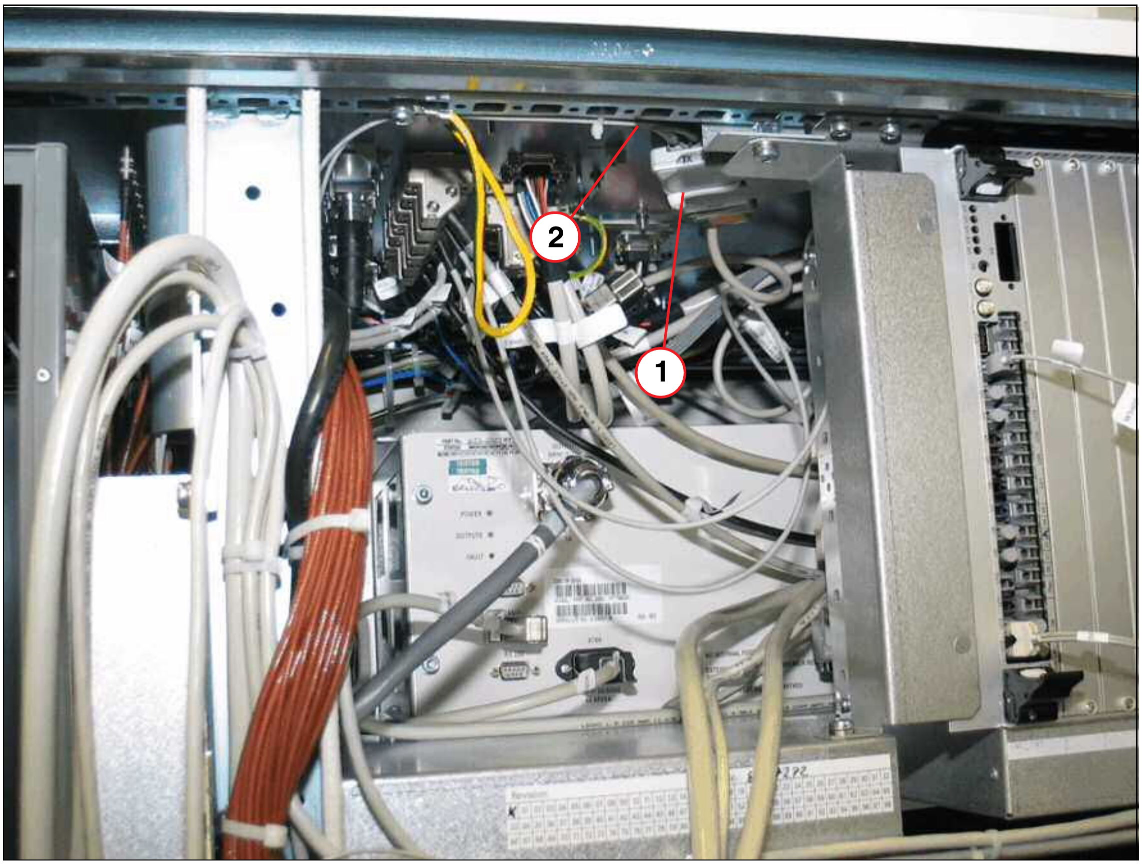
(1) Fiber optic cable

- Route the fiber optic cables in the ACC cabinet as pictured on the left.

Closing the door must not damage the fiber optic cables.

- The FOC CONVERTER is installed in the ACC. The fiber optic cable (W2008 TX/RX) is routed through the opening in the ACC top cover, connected to the FOC CONVERTER and secured with cable ties.

Fig. 173: ACC inside view, top part, FOC CONVERTER



(1) FOC CONVERTER
(2) FOC W2008 TX/RX

6.13 Installing the alarm box

The alarm box, when installed, provides an indication of the MRI system operating condition, along with some basic level of control, i.e **"System ON/OFF"**.

Clear access and visibility are required when positioning the alarm box as the emergency **STOP** button is also an integral function of the unit.

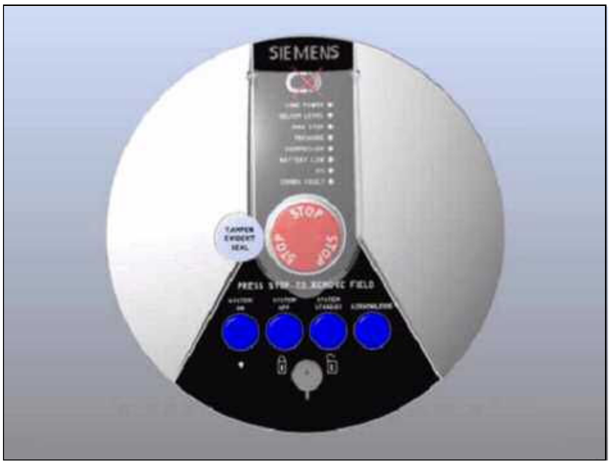
CAUTION

Failure of the magnet stop button

Failure to follow the installation procedure could lead to the malfunction of the magnet safety device.

» **The alarm box must be installed in an easily accessible area outside the 5mT zone.**

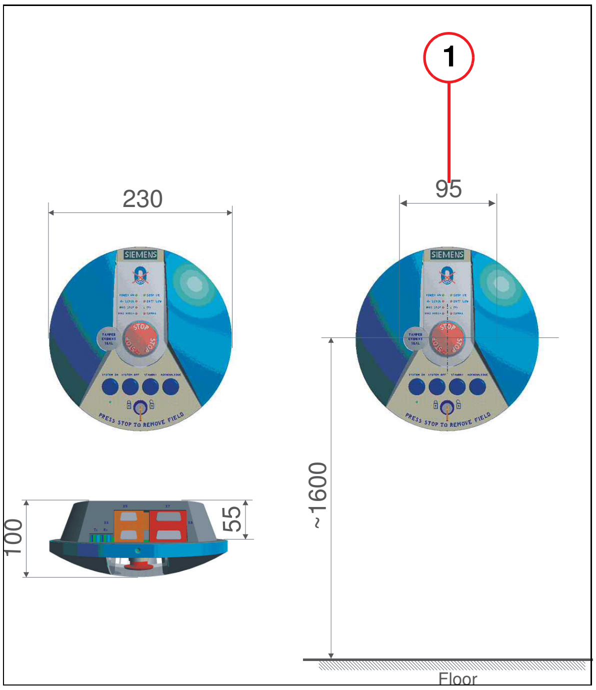
Fig. 174: Alarm box



The alarm box, as supplied, comes fitted with its control cable internally connected to **X7**. It should be disassembled prior to installation to gain access to the:

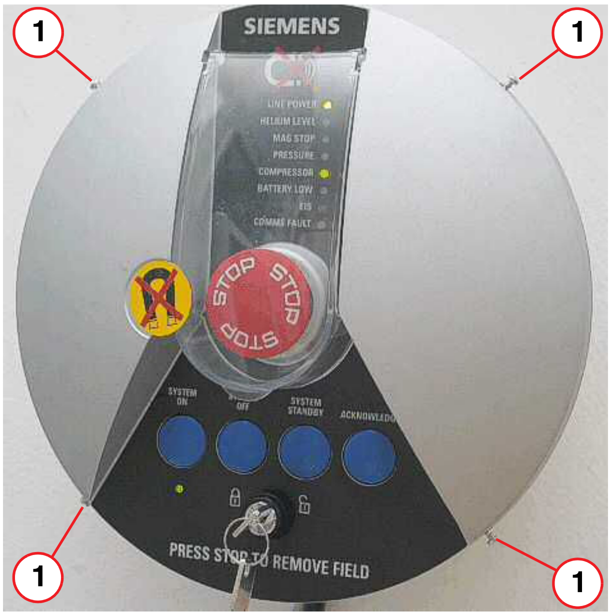
- Fixing plate (→ Fig. 178 Page 146)
- RS232 communications cable (→ 3/Fig. 178 Page 146)

Fig. 175: Dimensions of alarm box



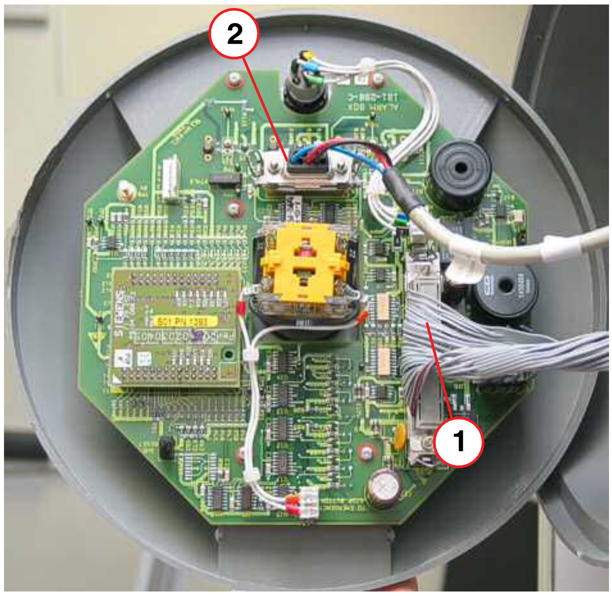
(1) Distance of screws

Fig. 176: Alarm box



(1) Securing screws

Fig. 177: Alarm box connectors



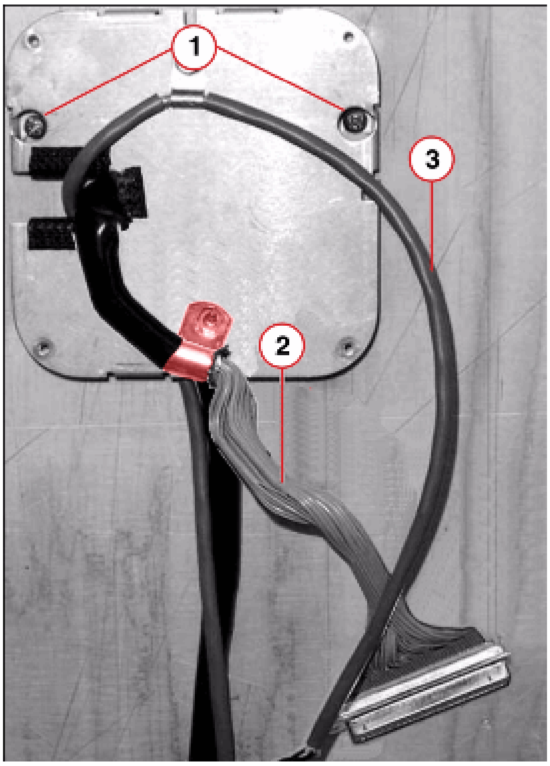
(1) Connector X7
(2) Connector X5

Preparation

- Remove the 4 screws around the outer rim of the alarm box. (➔ 1/Fig. 176 Page 145)
- Separate the two halves of the alarm box.
- Remove the alarm box front from the base.
- Disconnect the control cable from **X7**. (➔ 1/Fig. 177 Page 145)

- Remove the fixing plate from the rear casing. (→ Fig. 178 Page 146)

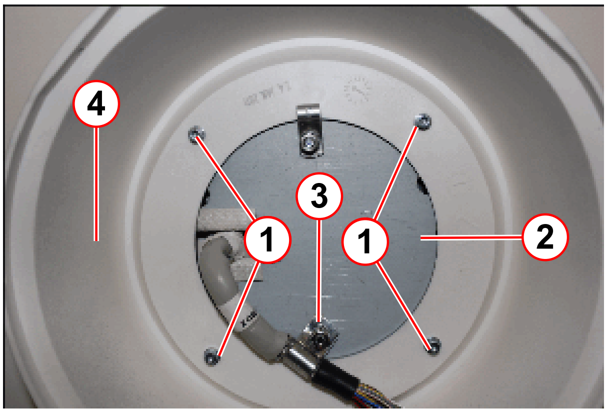
Fig. 178: Alarm box fixing plate



(1) Securing screws
(2) CAN cable (X7)
(3) RS232 cable (X5)

- Fix the two fixing plate mounting screws to the wall.
  - The screws must be positioned 95 mm apart, horizontally.
  - The screw heads must protrude from the wall to allow installation of the fixing plate (min 2 mm, max 9.5 mm).
- Place the fixing plate over the two mounting screws and tighten.
- Install the control cable between the alarm box **(X7)** and the ACC.
- Install the RS232 communications cable between the alarm box **(X5)** and the host PC.

Fig. 179: Back of Alarm Box (inside)



(1) Screws (4 off)
(2) Fixing plate
(3) 'P' clip facing outwards
(4) Plastic base

- Check the cable direction to ensure correct orientation required for the fitment of the alarm box rear casing.
- Fit the rear casing of the alarm box to the fixing plate with 4 screws.

- Reconnect cables to **(X5)** and **(X7)** on the alarm box. (→ Fig. 177 Page 145)
- Place the alarm box front onto the base and secure with the 4 fixing screws. (→ 1/Fig. 176 Page 145)

- Connect the cables from the alarm box to the ACC and the host PC.

6.14 Syngo Acq WP (Host PC)

syngo Acq WP (syngo Acquisition Workplace)

syngo MR WP (syngo MR Workplace)

Fig. 180: Host PC

Host PC (example only; actual hardware may differ)

- The host PC should be set up directly next to the host console.

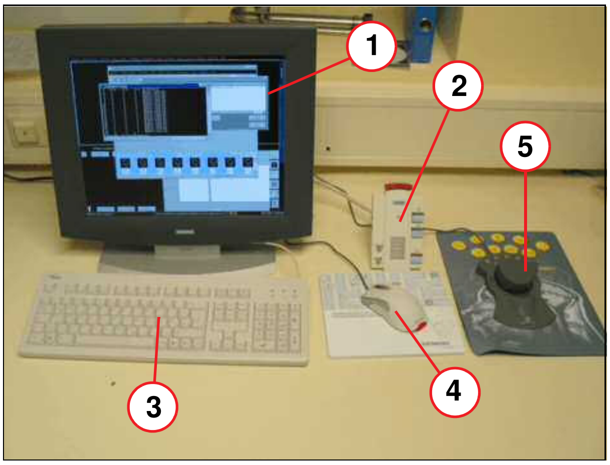
Fig. 181: Host housing

If a system cabinet (option) is included, the host PC must be installed in the cabinet.

6.15 Console Components

Route and attach the cables for the components so as to prevent damage to the cables or them being pulled out during system operation.

Fig. 182: Console components



(1) Color screen
(2) Intercom
(3) Tastatur
(4) Computermouse
(5) Space mouse (optional)

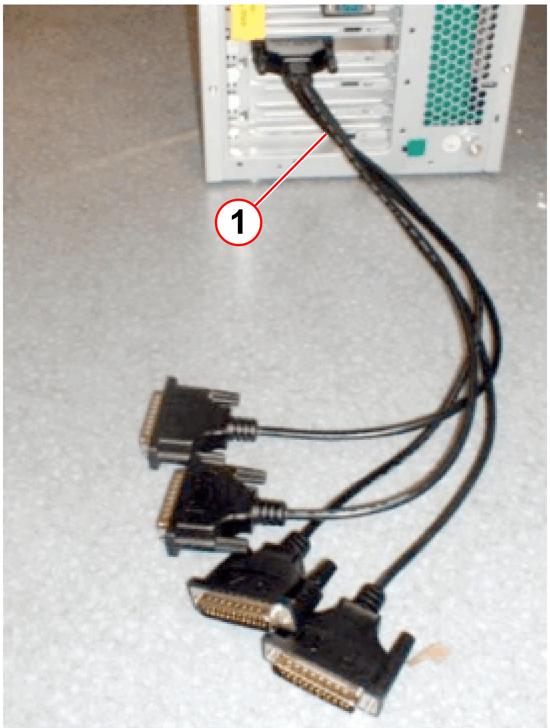
- Place the components on the console table.

The ICU (→ 3/Fig. 187 Page 152) must be installed horizontally for the required air intake.

New 1 to 4x adapter/connector cable	Assigned device	Assigned COM port	Previous port number on the old Ser_Distribution box
P4	Alarm box	COM9	port 7
P3	ACC (MPS)	COM8	port 6
P2	Space Mouse [1]	COM7	port 5
P1	Foot switch [2]	COM6	port 4

1. Optional
2. Optional

Fig. 183: Host PC with 1 to 4x-adapter/connector cable



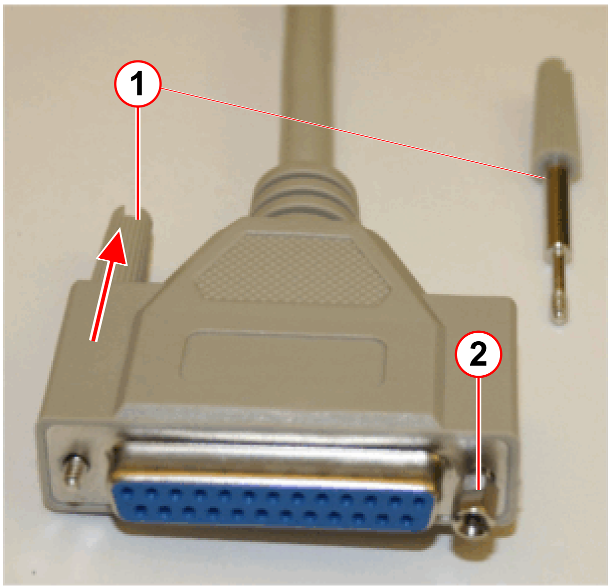
(1) New 1 to 4x adapter/connector cable

Fig. 184: Hex-headed extension bolts

Optional:

The hex-headed extension bolts are delivered with the system (accessory box) and used if the female connector is equipped with screws.

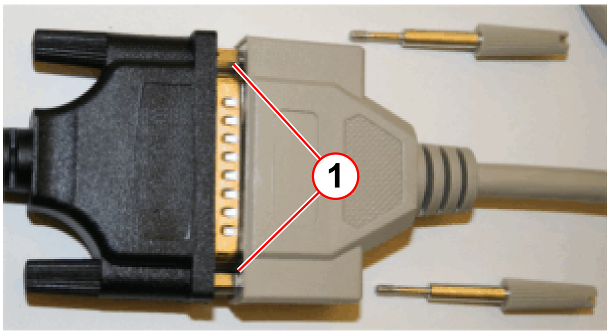
Fig. 185: Female connector of the cable



(1) Attachment screw for connector
(2) Hex-headed extension bolt

- If the female connector is equipped with screws, remove the attachment screws of the female connector (pull out and turn counter-clockwise).
- Fix the attached hex-headed extension bolts to the front side of the female connector.

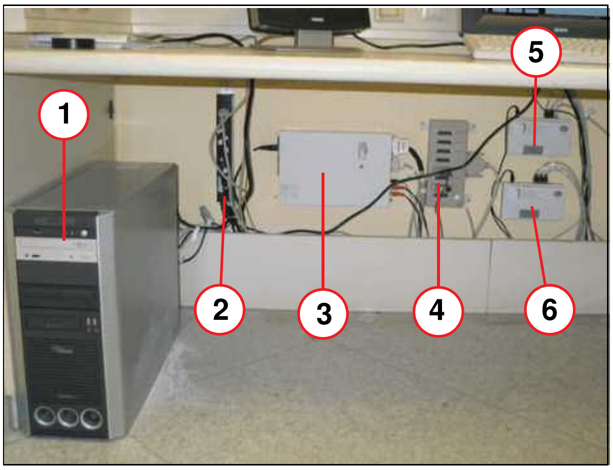
Fig. 186: Connected cable



(1) Hex-headed extension bolt

- The picture shows when the cable is connected to the 4x-adapter/ connector cable.

Fig. 187: Console components 1



(1) Host PC
(2) Power distributor
(3) ICU
(4) Ser_Distribution
(5) MRC switch
(6) MRSC switch (optional)

- Installing and cabling the components.

6.15.1 MSUP/MPS cabling

The W2005 cable (617 309 25T) is used for ramping the magnet.

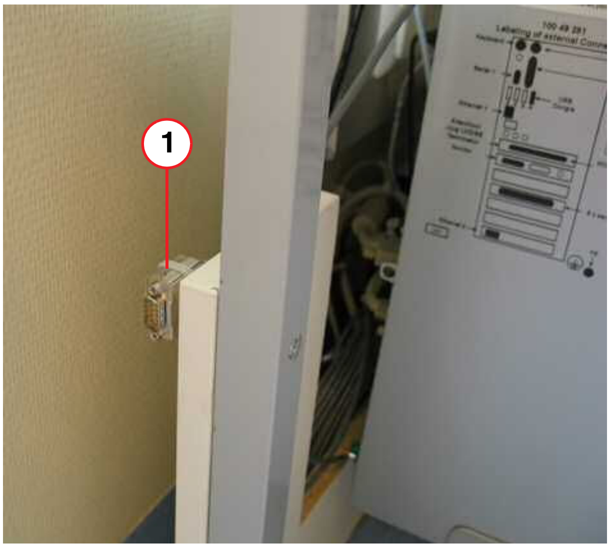
- Connect the cable to Ser_Distribution port 6

Fig. 188: Ser_Distribution Box

- Route it to the control room close to the examination room door (connection for the magnet power supply).

  The connection must be visible to the service engineer.

Fig. 189: MPS cable



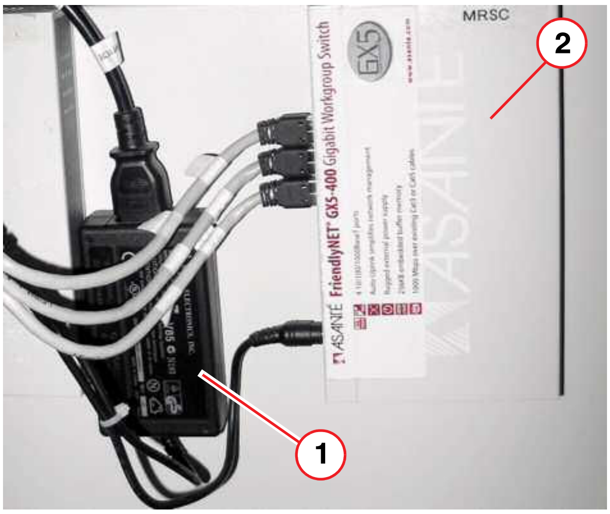
(1) MPS cable

6.15.2 Ethernet switch

> Do not place the power supply in a cable duct or unventilated housing as this could result in overheating.

Attach the power supply to the back wall of the console table with cable ties.

Fig. 190: Ethernet switch, power supply (example)



(1) Power supply
(2) Power supply

The figure to the left shows an example of how to install the power supply.

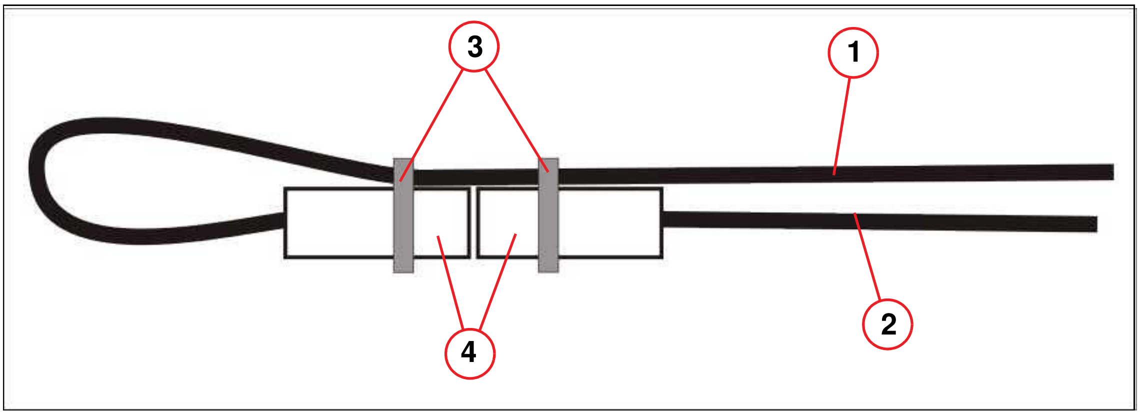
There are various types of Ethernet switches and corresponding power supplies.

To ensure proper function, the power supply should be replaced with the Ethernet switch.

6.15.3 Mouse and keyboard extension cables

Secure the extension cables for the mouse and the keyboard (→ Fig. 191 Page 154) to prevent contact interruption.

Fig. 191: Mouse and keyboard extension cables



(1) Extension cable
(2) Mouse/keyboard cable
(3) Cable ties
(4) Plugs

6.15.4 Syngo MR WP (Optional)

- Connect the syngo MR WP to the designated socket in the wall.

The voltage supply for the satellite console is not provided by the MR system (dimensioning of transformer).

- Use a multiway connector together with the switch.
- Install the MRSC switch (→ 6/Fig. 187 Page 152).
- Connecting the monitor, mouse, and keyboard to the syngo MR WP.
- Connecting the Ethernet cable to the switch.
- If a console table is included in the delivery, route the cables into the cable ducts (table frame).

6.16 MR Workplace Table (Optional)

If the system is equipped with the optional MR Workplace Table, the components must be installed as described in the document (→ MR Workplace Table / MR00-000.814.33) under “Options” in the “Technical Documents Vol. 2” folder.

Fig. 192: MR Workplace Table

(1) MR Workplace Table
(2) MR Workplace Container, 50 cm (optional)

6.17 Gradient Supervision

6.17.1 General Information

6.17.1.1 Functional Description of the VLM VESDA Device

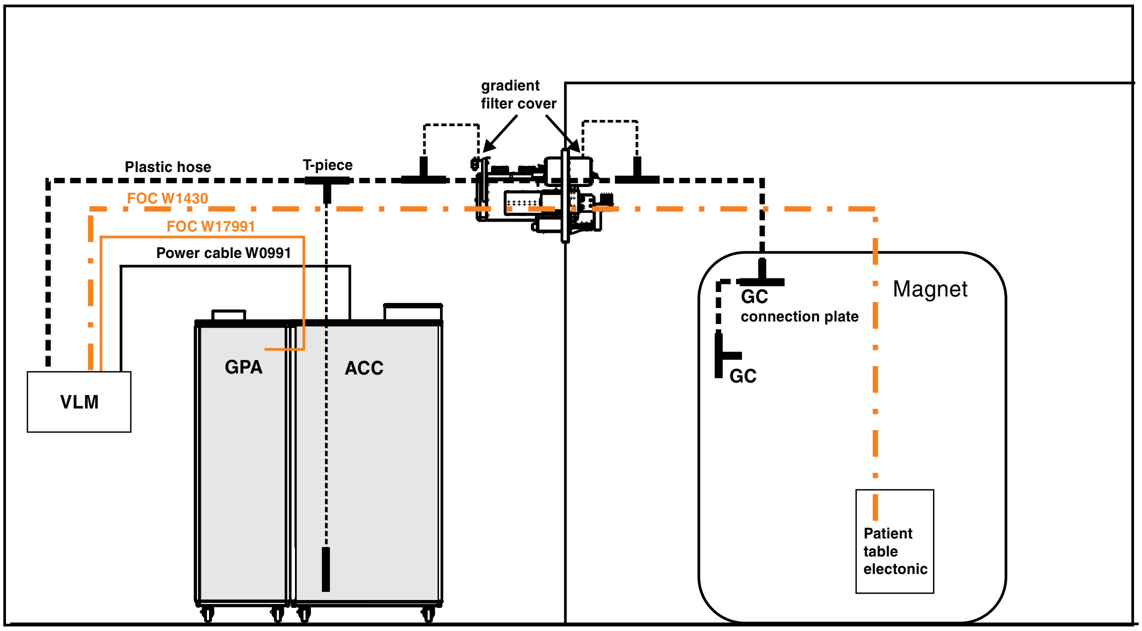
The unit adapted to the MR system with this UI is basically a very sensitive laser-driven smoke detector which is used to switch off the gradient power whenever exposed pilot gases are detected at a very early stage of over-heated gradient cables.

The unit has a little fan which permanently sucks air through a hose system from the different sample points (ACC cabinet, gradient filters, gradient connections at the magnet).

The aspirated air-flow is fed through a filter into the detector chamber.

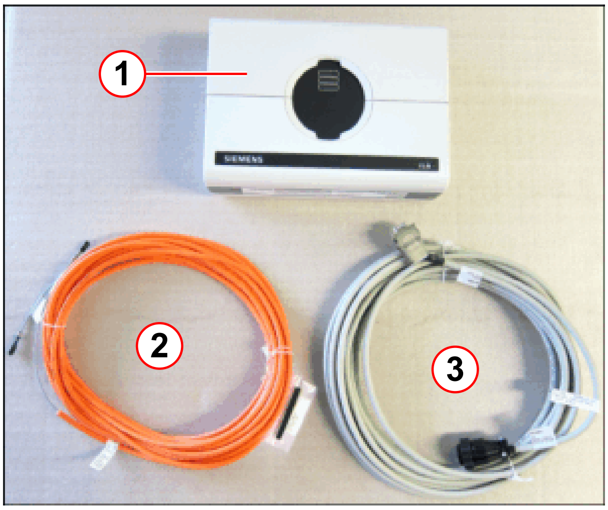
> The VLM device is not supposed to work as or to replace a fire warning system, but rather as an additional safety precaution to power-off the MR-system in case of over-heated gradient cables.

Fig. 193: Gradient Connection Supervision (Tim systems, Biograph mMR)



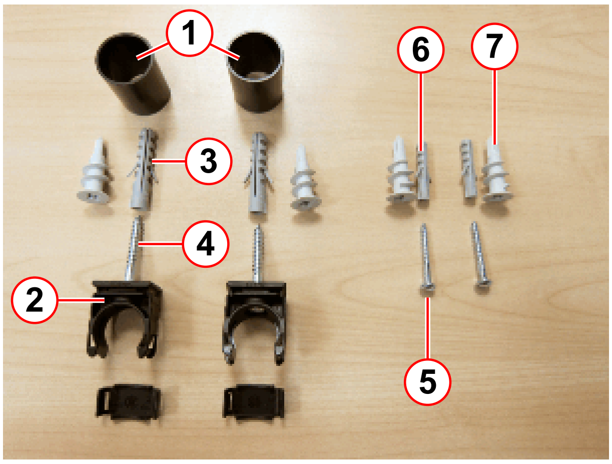
6.17.1.2 Information on Materials

Fig. 194: VLM device, FOC and power cable (Aera and Skyra)



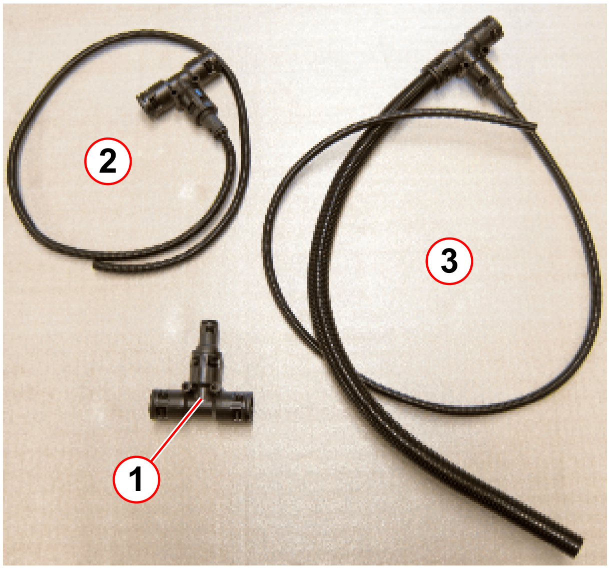
(1) VLM device
(2) FOC W17991
(3) Power cable W10991

Fig. 195: Mounting material for VLM device



(1) Bushing (for hose connection)
(2) Hose clip
(3) Dowel 8 mm (for hose clips)
(4) Wood screw 5x40 (for hose clips)
(5) Wood screw 4x30 (for mounting the VLM device))
(6) Dowel 6 mm (for mounting the VLM device)
(7) Drywall dowel (replacement for Pos.3 and 6 on dry-walls)

Fig. 196: Filter plate hose kit



(1) T-piece (10683922)
(2) T-piece, preconfigured (10684497)
(3) T-piece, preconfigured (10684507)

Fig. 197: 30m Flexa hose plus hose cutter

Fig. 198: Magnet exhaust Symphony, A Tim System

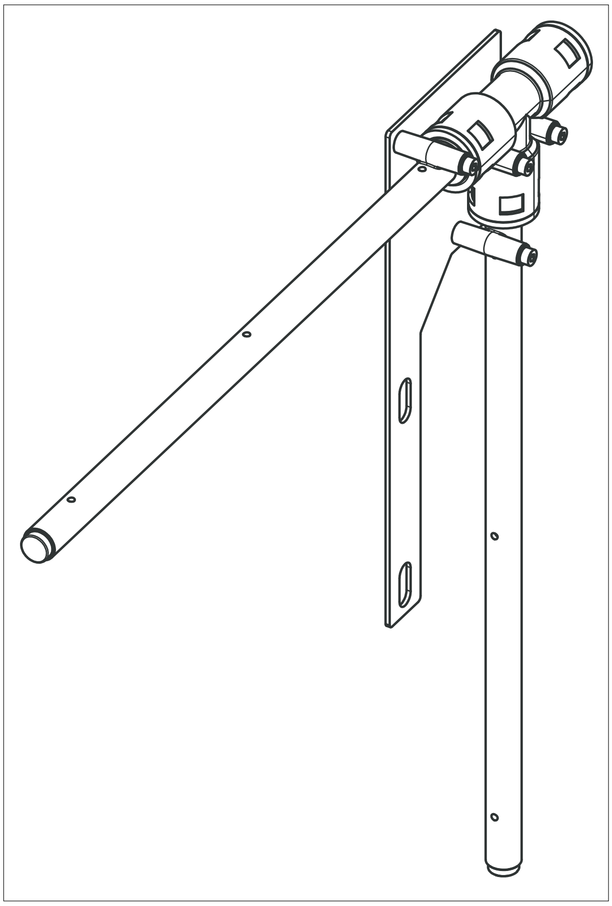


Fig. 199: Cabinet mounting bracket

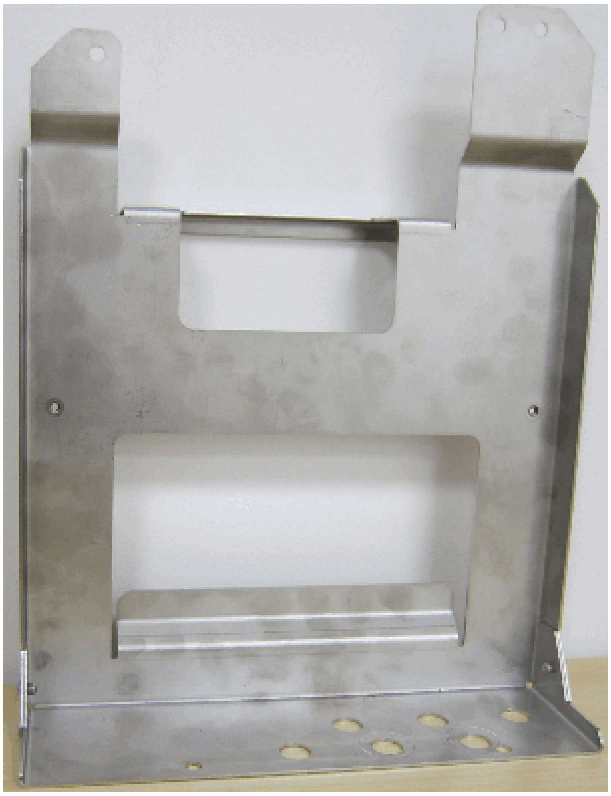


Fig. 200: Additional Wave Duct (10915434)

For Symphony, A Tim Systems, the additional Wave Duct (shown left) usually has to be fitted to the RF-filter plate (instead of the blind plate), in order to feed the Flexa hoses of the Gradient Supervision through the RF-filter plate.

6.17.2 Installation

6.17.2.1 Installing the SIEMENS VLM Device

> The cabinet mounting bracket is the preferable solution for attaching the VLM device. This universal bracket allows a mounting on top or on the side panel of the cabinet. This bracket is not part of the Aera/Skyra UI. In case there is insufficient space on the cabinet the device can be mounted to the wall alternatively.

Tab. 5 Preparing the mounting bracket

Fig. 201: Attach the VESDA mounting plate to the mounting bracket (2x M4 Allen screws)

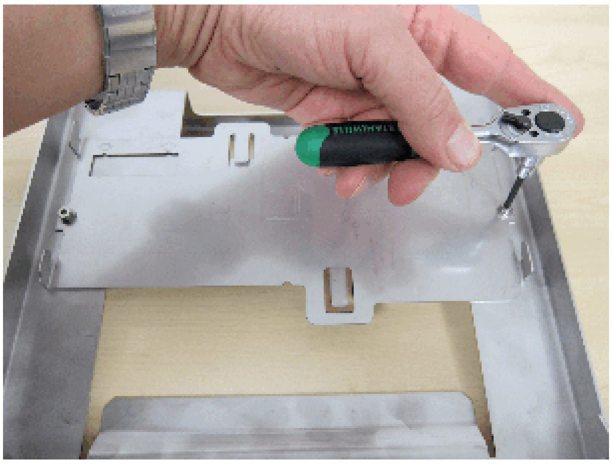


Fig. 202: Attach the hose clamp to the bracket (M6 Allen screw plus nut)

Fig. 203: Cable tie fixing clamps

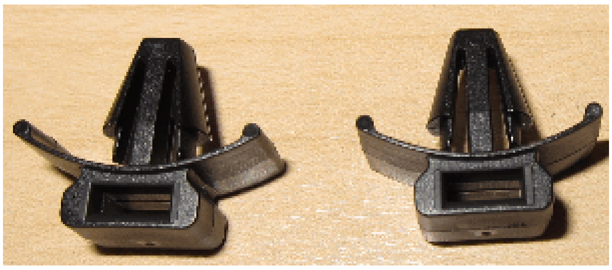
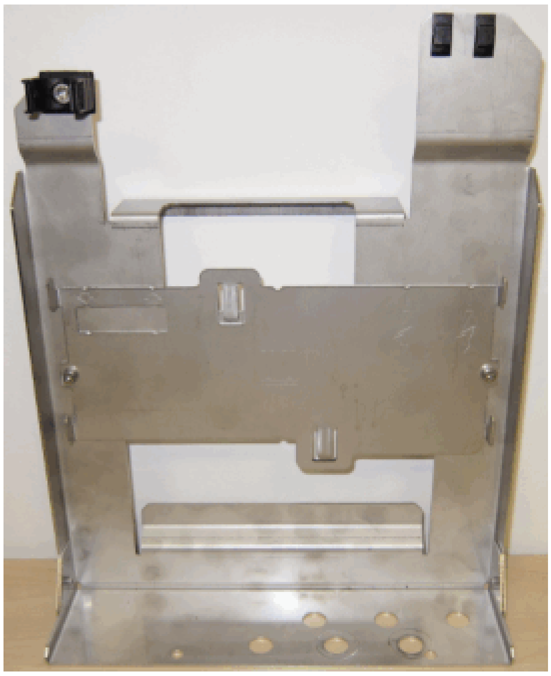


Fig. 204: Attach the cable tie fixing clamps to the right side of the mounting bracket



- Either mount the bracket to the side panel (remove at least the upper 4 screws of the side panel and hang the bracket onto the side panel).
- Or attach it to the cabinet roof on the front side between GPA and ACC/EPC (use the existing SW19 screws holding the red transport bracket).

Tab. 6 Final mounting of the bracket on the cabinet

Fig. 205: Side panel mounting of the bracket

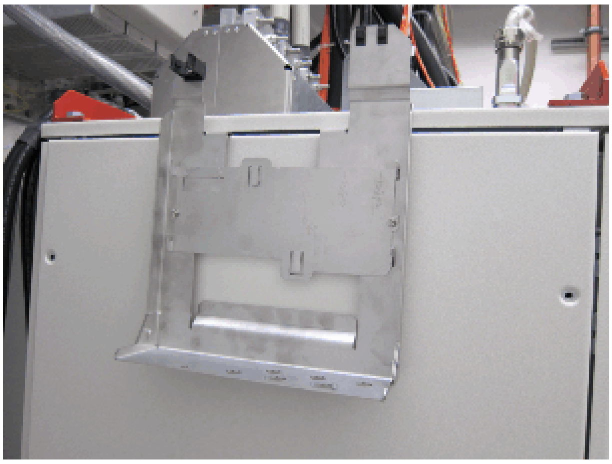
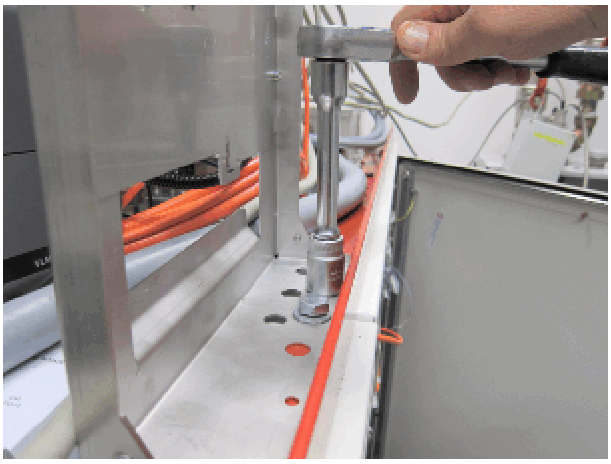
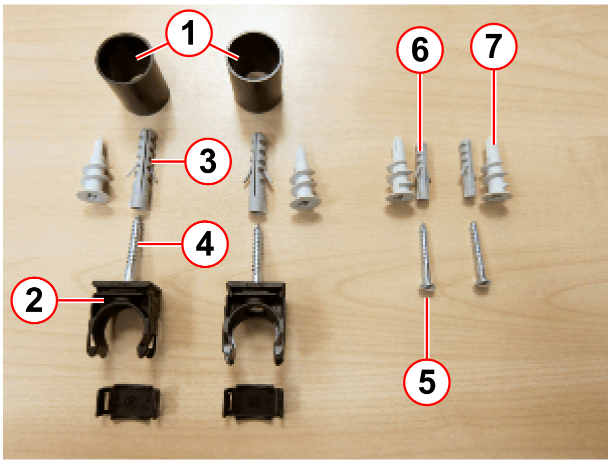


Fig. 206: Cabinet roof mounting of the bracket



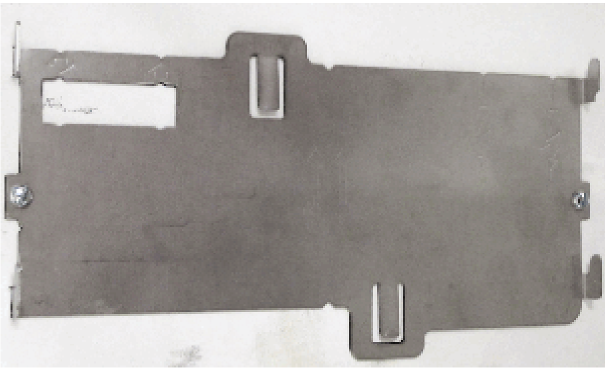
- Finally hang in the VLM-device onto the mounting bracket

Fig. 207: *Mounting material for VLM device*



(1) Bushing (for hose connection)
(2) Hose clip
(3) Dowel 8 mm (for hose clips)
(4) Wood screw 5x40 (for hose clips)
(5) Wood screw 4x30 (for mounting the VLM device))
(6) Dowel 6 mm (for mounting the VLM device)
(7) Drywall dowel (replacement for Pos.3 and 6 on dry-walls)

Fig. 208: *VESDA mounting plate*



Optional wall mounting of the VLM device

- Find an appropriate place for mounting the VLM device **in the equipment room.**
  » The device should be mounted at eye level.
- Use a level and mark the position of the holes for the mounting plate on the wall with a pencil.
- **Standard dowels** (used for concrete walls):

  Drill 2x 6 mm holes, insert the standard dowels.

  or

  Special drywall dowels (used for drywall):

  Drill the special drywall dowels directly into the wall.
- Attach the mounting plate to the wall.
- Hang the VLM device onto the mounting plate.

Fig. 209: *Mounting the VESDA device to the wall*

- Hang the VLM device onto the mounting plate.

Fig. 210: Hose clip installation

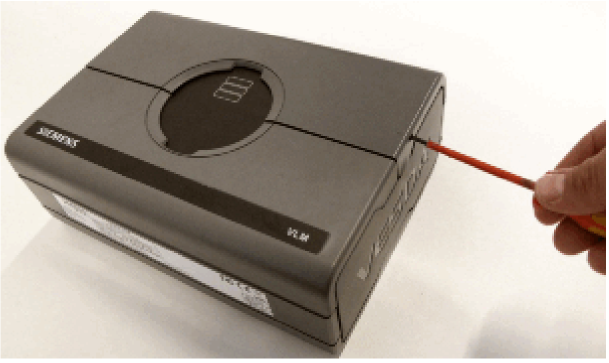
(1) Hose clip

Install the hose clip:

- Mark the position of the hose clip (1) approx. 200-250 mm above the VLM device hose inlet using a level.
- Drill holes, insert the dowels, and attach the hose clip to the wall.

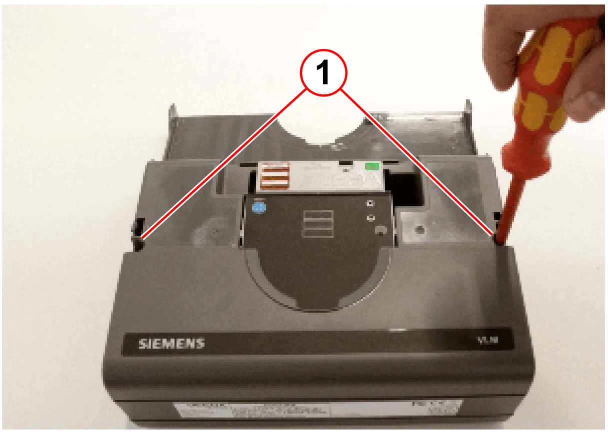
VLM Jumper Settings

Fig. 211: Opening the upper cover



- Open the upper cover of the device.

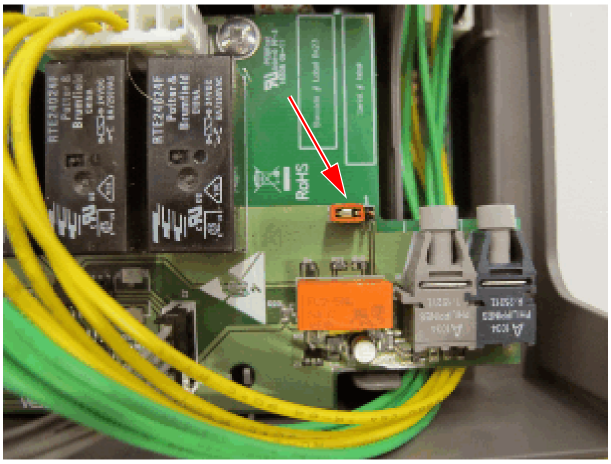
Fig. 212: Opening the lower housing



(1) 2x Phillips screws

- Remove the 2 Phillips screws (1) and swing out the housing.

Fig. 213: Jumper on 2-3 default setting - correct for Tim systems and Biograph mMR

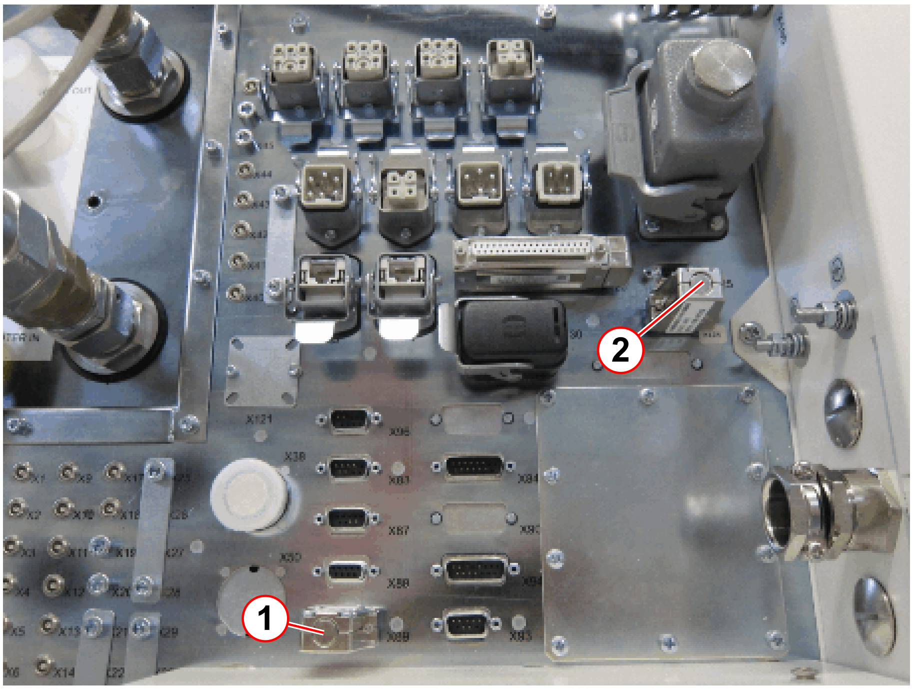


- There is one jumper left from the fiber optic connectors. The default jumper setting is **2-3**, which is appropriate for all **Tim systems**.

Electrical and FOC Connections to the VLM Device

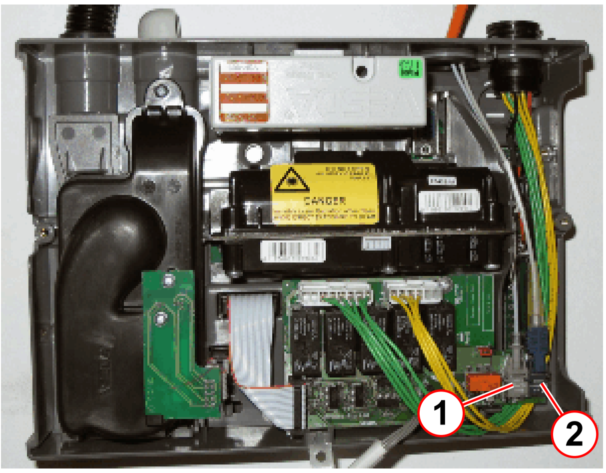
- If there is a blind plug attached to X89 and X124, replace it with cable W0991.
  » This means, that the cabinet is internally wired up for the VLM device ex factory.
  » The blind plugs can be used for trouble shooting and should not be scrapped.
- Attach cable W0991 to X89 and X124 on the ACC roof.
- Connect the other end of the cable to the VLM device, X1.

Fig. 214: ACC094 cabinet roof with blind plugs X89 and X124 fitted



(1) X89 blind plug
(2) X124 blind plug

Fig. 215: FOC connection to VESDA device



(1) U1 FOC connection W17991
(2) U2 FOC connection W1430

- Connect the FOC W1430 coming from the magnet to the VLM device at U2 (2).
  » When the system is powered on, light must be visible at W1430 and should result in light being emitted from transmitter U1.
- Relabel it with

  "Grad Cable SV POF RX".
- Connect the FOC W17991 coming from D17/U7 to the VLM device at U1 (1).

6.17.2.2 GPA cabling

Adhere to ESD regulations for the following work steps and use an anti-static wristband.

Fig. 216: Laying the the FOC into the ACC cabinet

Connecting FOC W17991

- Feed FOC W17991 through the wave duct on the ACC094 cabinet roof.
  » Label "U1/D70" must be inside the cabinet.
- Feed W17991 through the side panel of the ACC094 cabinet into the GPA cabinet.

- Remove all front connectors from the GSSU.
- Unscrew the 2 upper bolts holding the GSSU front cover in place (it looks different on some GPA cabinets, and an upper rail may be fitted instead of the front cover).

- Swivel down the GSSU front cover (or remove the upper rail)

Fig. 217: Opening the front cover of the GSSU

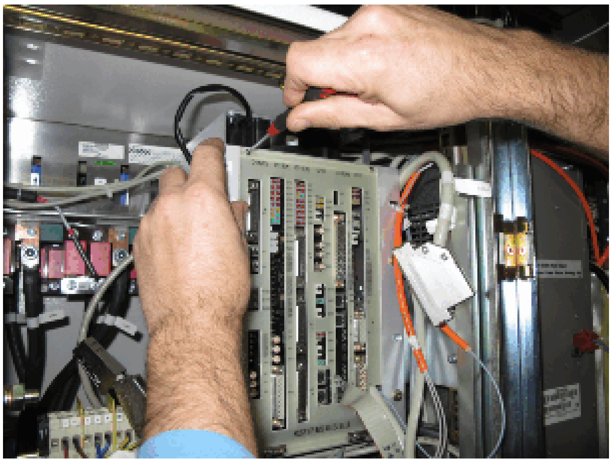
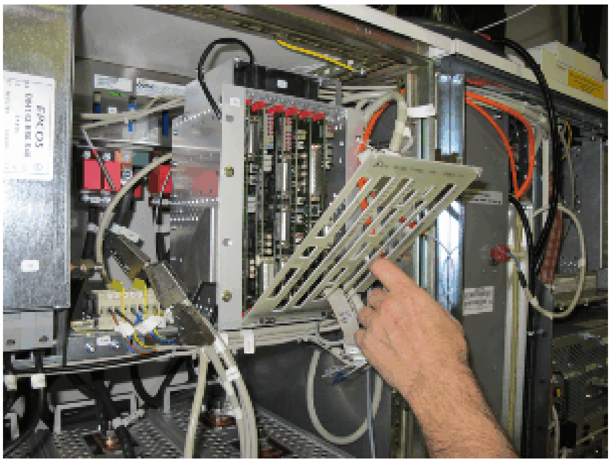


Fig. 218: Swiveling down the fornt cover of the GSSU



- Pull out D17 from GSSU (in Fig.41 connector X3 is used to lighten the removal of D17).
- Unplug FOC U1/D70 from optical receiver U7 (➔ Fig. 220 Page 166).
- Connect W17991 to U7/D17
  » The other end of the FOC will be connected to the VLM device later on.
- Remove the original FOC W1430 from the cabinet.
  » This FOC will be connected to the VLM device later on.

Fig. 219: Removal of D17

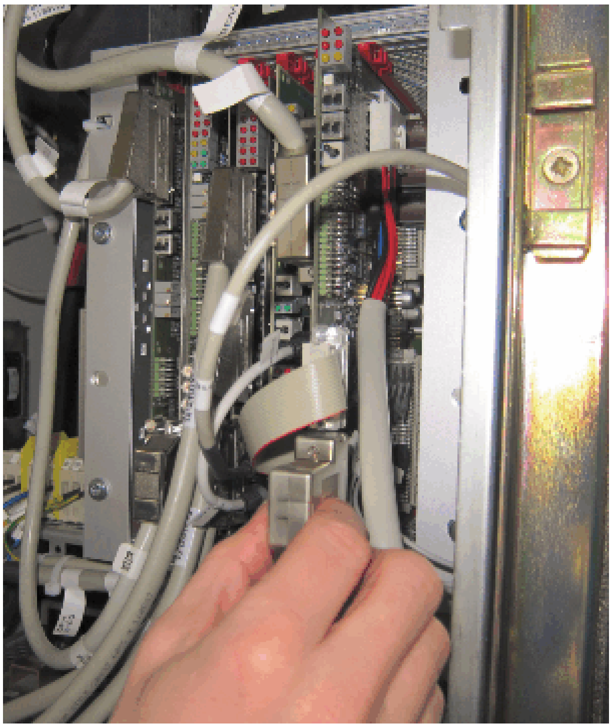
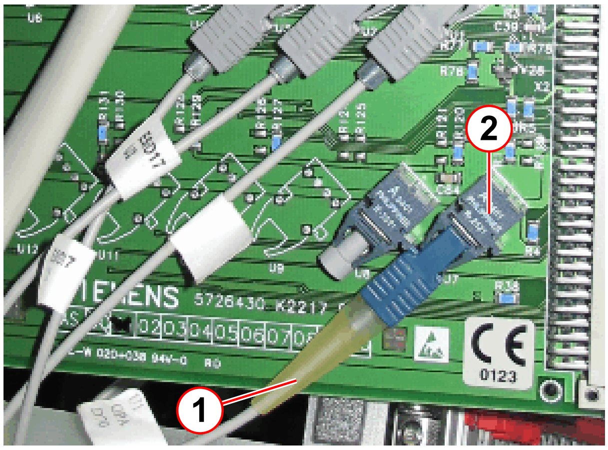


Fig. 220: Replacement of the FOC on D17



(1) Original FOC (labeled with U1 GPA D70)
(2) FO receiver U7

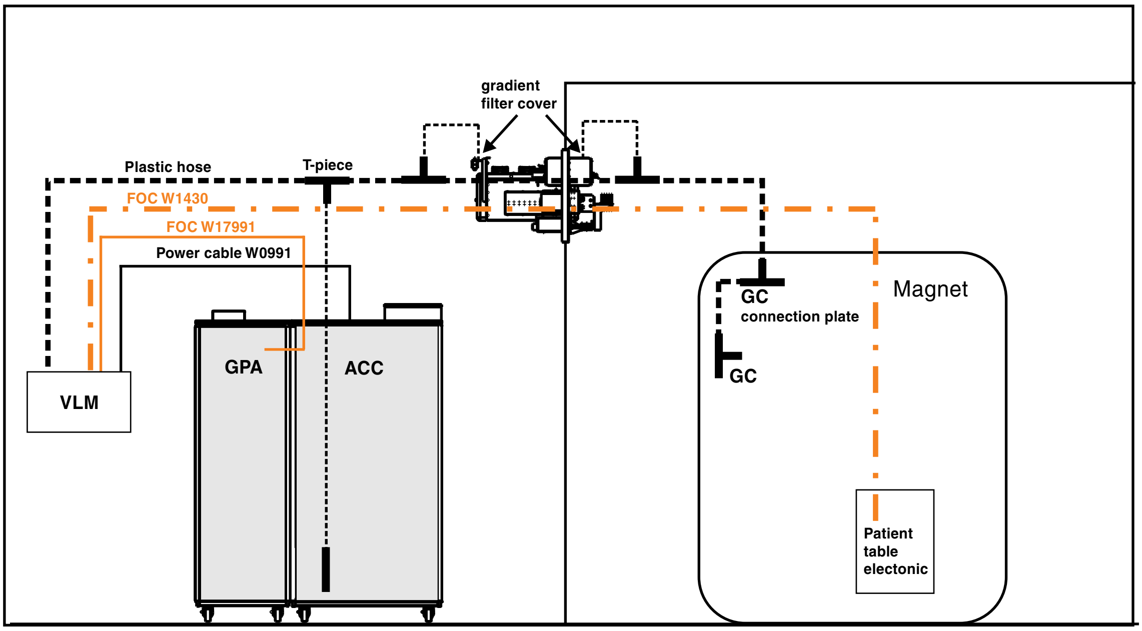
- Re-assemble GSSU in reverse order.

6.17.2.3 Installing the Hoses

Overview for Hoses

The following figure gives you an overview of the hose system to be connected. There are sample points in the EPC cabinet, on both gradient filter covers, and at the magnet gradient connections.

Fig. 221: Gradient Connection Supervision (Tim systems, Biograph mMR)



General Information

The maximum hose length is 30 m.

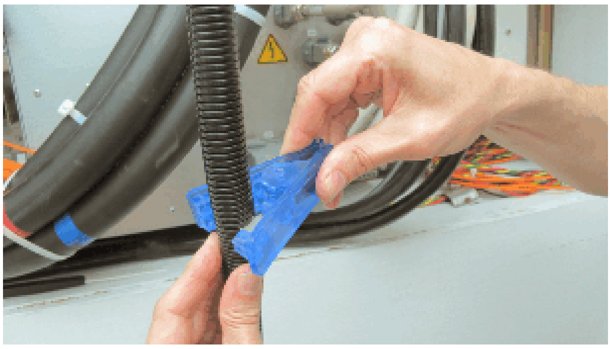
The minimum hose lengths are (this means do not install shorter hose sections than listed below):

- VLM to T-piece on ACC/EPC > 1 m
- T-piece on ACC/EPC to T-piece at filter panel (equipment room) > 1 m
- T-piece at filter panel (examination room) to T-piece magnet > 5 m
- All other lengths are preconfigured and therefore fixed
- Observe the min/max lengths but do not make the hose longer than necessary, because additional length results in a longer reaction time.

Basically a hose system from Flexa is used for all hose connections. There are 10 mm hoses used (inside the ACC/EPC cabinet and for the gradient filter covers) and 22 mm hoses (for all other connections).

Please observe and strictly adhere to the following general hints regarding the proper handling of the Flexa connections being used for the hosing.

Fig. 222: Using the hose cutter



Proper cutting of the Flexa hose

- Prepare a piece of Flexa hose of the required length and use the blue hose cutter which is provided with the Update Kit (→ Fig. 222 Page 168)).
  - » Take care that the blade of the cutter is cutting in the groove of the hose (→ Fig. 223 Page 168) and not on top (→ Fig. 224 Page 168), because the hose can only be plugged into the T-pieces when it is properly cut.

Fig. 223: Proper cut

Fig. 224: Improper cut

Fig. 225: Proper Flexa hose connection

Proper Flexa hose connection

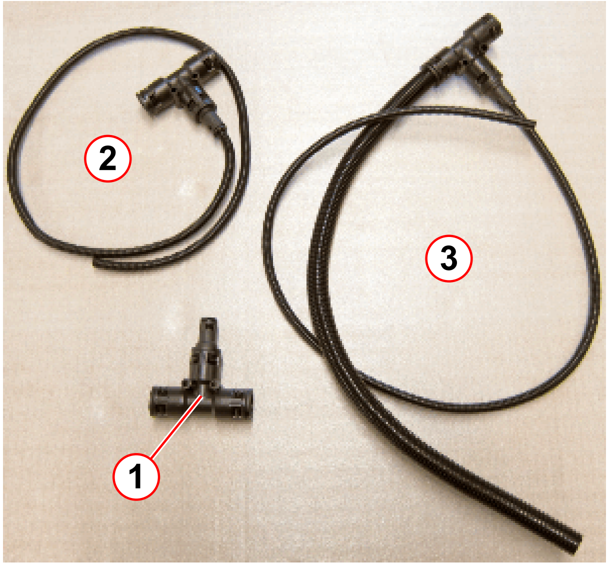
- Push-fit the Flexa hose hard into the T-piece, while twisting it slightly (➔ Fig. 225 Page 169).
  - » It should move into the coupling for about 4 ring-section elements of the hose.
  - » In order to verify the proper connection, try to pull out the hose again. This should not be possible.
  - » Twisting the hose in the coupling should not be easy.

Proper removal of the Flexa hose from the T-piece

- In case you have to remove the Flexa hose from the T-Piece, just push the upper ring segment of the T-piece down while pulling out the hose.

RF-Filter Plate Installation

Fig. 226: Filter plate hose kit (example)



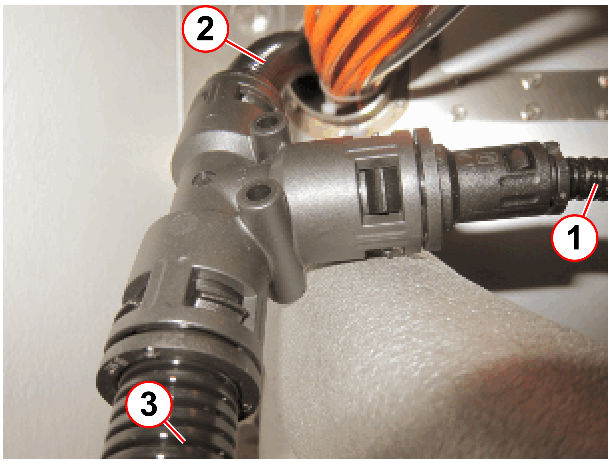
(1) T-piece
(2) T-piece, preconfigured
(3) T-piece, preconfigured

Fig. 227: Feeding the short hose connecting piece through a wave duct

Hose connections at the filter panel

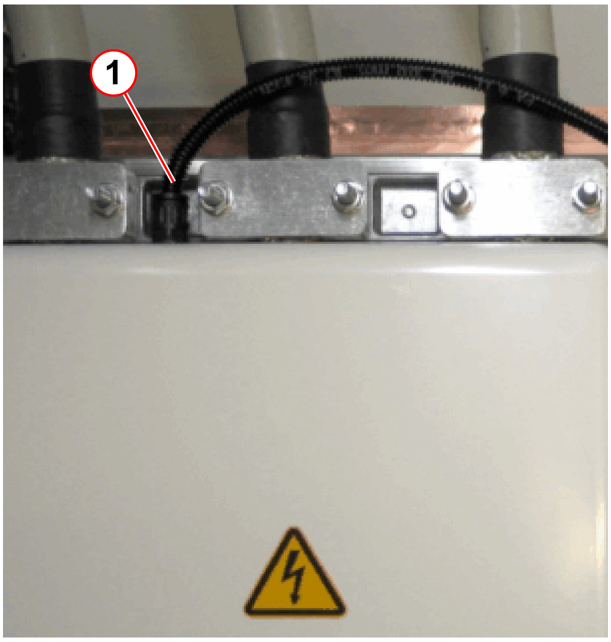
- Take the preconfigured T-piece (3) provided with the UI-kit (the thin hose is actually shorter than shown on the left) and feed the thicker (22 mm) hose through the wave duct of the filter panel (➔ Fig. 227 Page 170) from the examination room side to the equipment room.

Fig. 228: T-piece on both sides of the filter panel



(1) Connection to new gradient filter cover
(2) Hose junction
(3) Connection to magnet (magnet room) and EPC (equipment room)

Fig. 229: Gradient filter connection

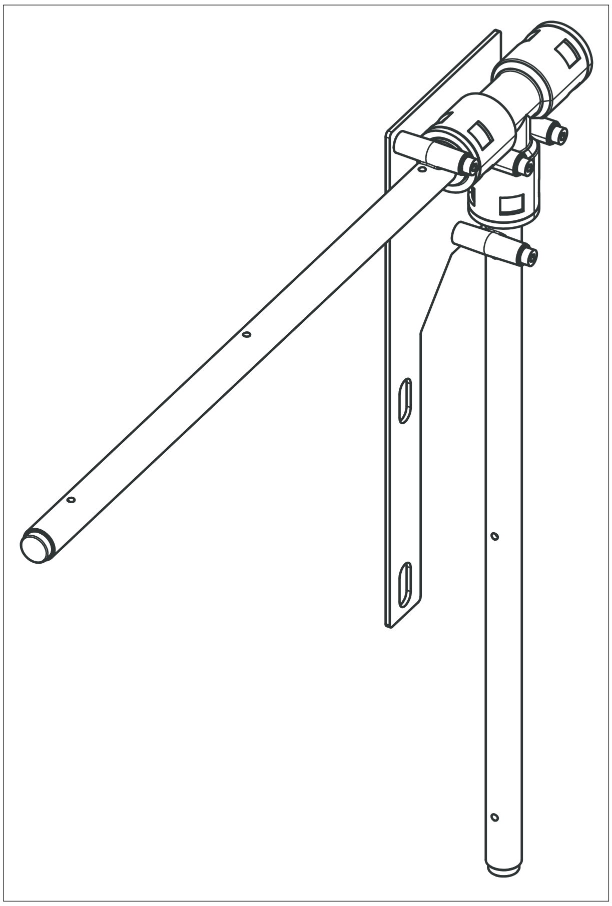


(1) Hose (10mm)

- Press-fit the other preconfigured T-piece (➔ 2/Fig. 226 Page 170) to the end of the hose just fed through the wave duct.
- Push-fit the short 10 mm hose pieces (1) into the couplings at the gradient filter covers.
- Cut a piece of hose to feed from the T-piece at the filter panel to the T-piece on top of the ACC/EPC cabinet **(minimum length = 1 m)**.
- Lay the hose along the on-site cable ducts.
- Press-fit this hose into both T-pieces (ACC/EPC and filter panel).

Magnet Hose Installation

Fig. 230: Magnet exhaust Symphony, A Tim System



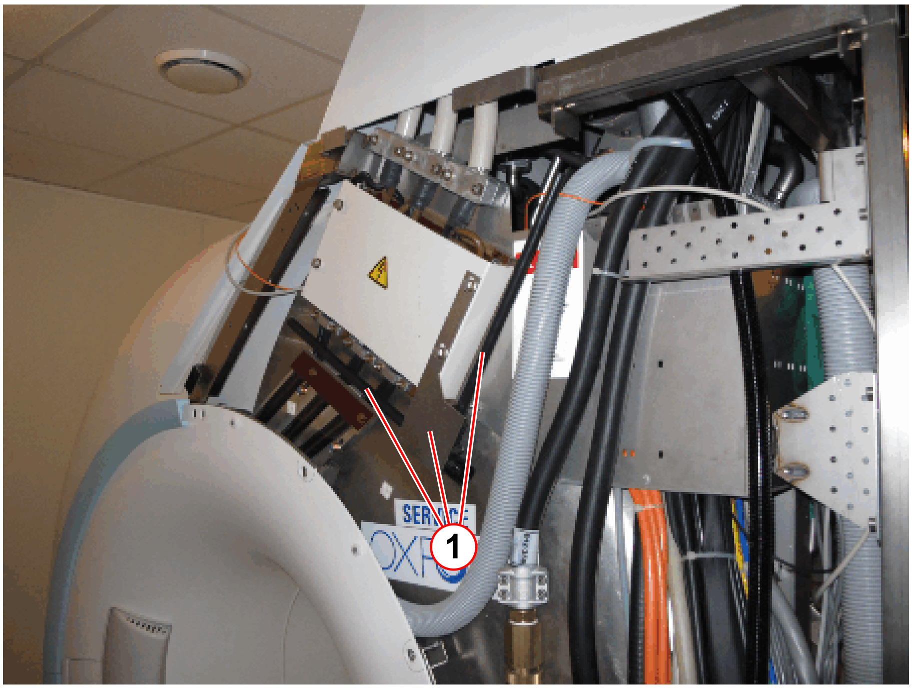
The Symphony, a Tim System magnet exhaust consists of a vertical and a horizontal sample tube combined with a T-piece.

Each of the tubes has 3 sample holes.

- Remove the rear upper right magnet cover.
- Remove the service turret cover.
- Route a piece of hose (using the 30 m roll provided with the UI-kit) from the filter panel T-piece through the ceiling to the magnet **(minimum length = 5 m)**.

- Install the magnet exhaust on top of the gradient cable cover, using the original bolts.

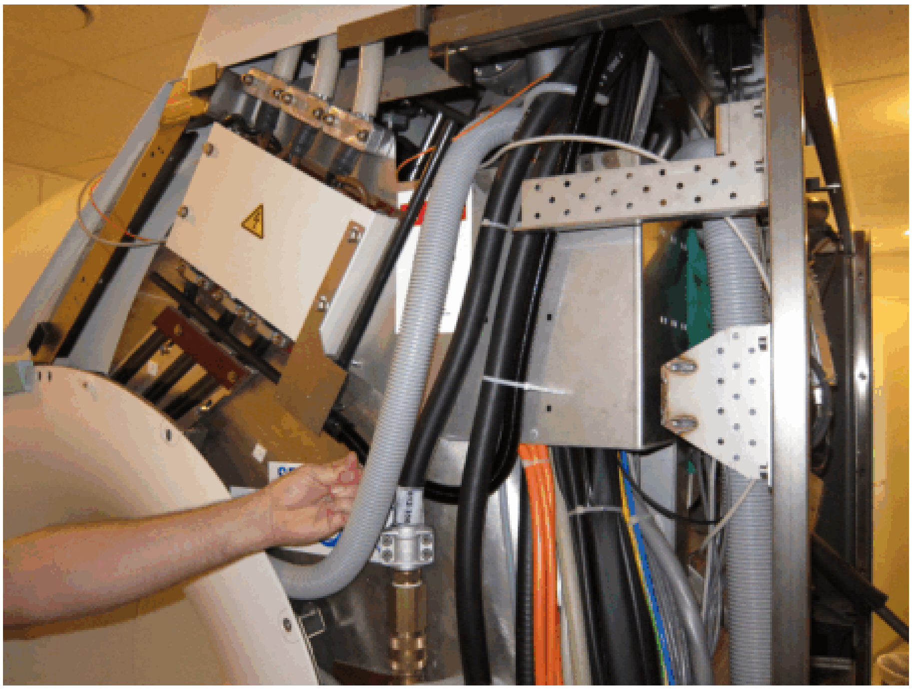
Fig. 231: Rear magnet covers removed and magnet exhaust installed



(1) Magnet exhaust

- Connect the hose coming from the filter panel to the open joint of the T-piece from the magnet exhaust.

Fig. 232: Connecting the hose to the magnet exhaust



VLM Device Hose Connection

Fig. 233: Remove the blind cap from the VLM

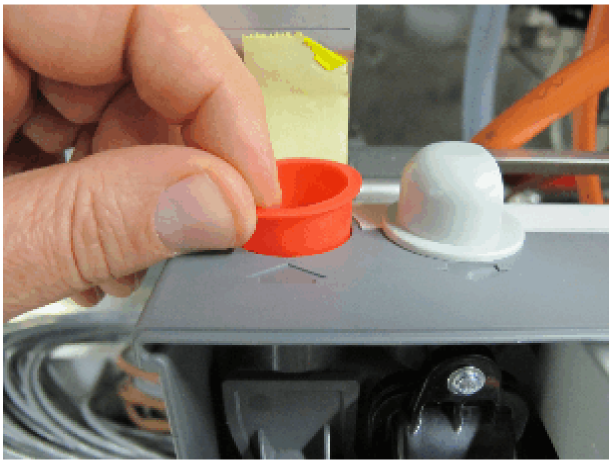
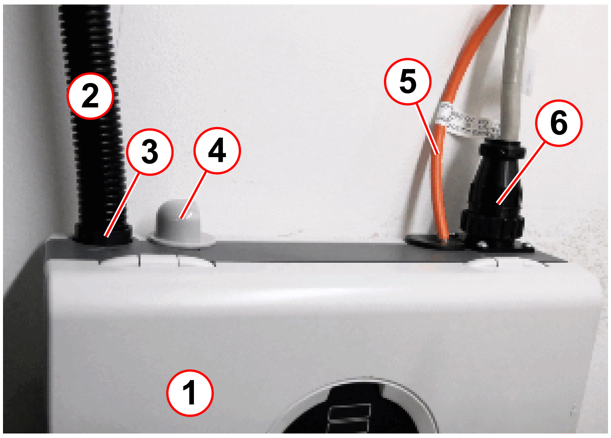


Fig. 234: Connections to the VLM device



(1) VLM device
(2) Hose
(3) Bushing (adaption to hose diameter)
(4) Fan exhaust
(5) FOC
(6) Power cable X1

Hose connection VLM - ACC/EPC

- Remove the red blind cap from the inlet side of the VLM device.
- Prepare a piece of plastic hose **(minimum length = 1 m)** at a length suitable to feed it from the VLM device to the T-piece on top of the ACC/EPC cabinet (cut it to the proper length, using the blue hose cutter provided with the UI-kit).

  » **Take care that the blade of the cutter is cutting in the groove of the hose and not on top, because the hose can only be plugged into the T-pieces when it is properly cut.**

Fig. 235: VLM hose connection

(1) Black plastic bushing
(2) Hose 22 mm

- Push the black plastic bushing (1) onto the hose.
- Press the black plastic bushing (1) into the suction hole of the VLM device.
  » The bushing is cone-shaped and must be fitted with the thinner end first.

Fig. 236: T-piece connection, VESDA to ACC/EPC

- Lay the hose from the VLM-device via the cable ducts to the ACC/EPC roof and push-fit it to the T-piece (➔ Fig. 236 Page 175).

Final Steps

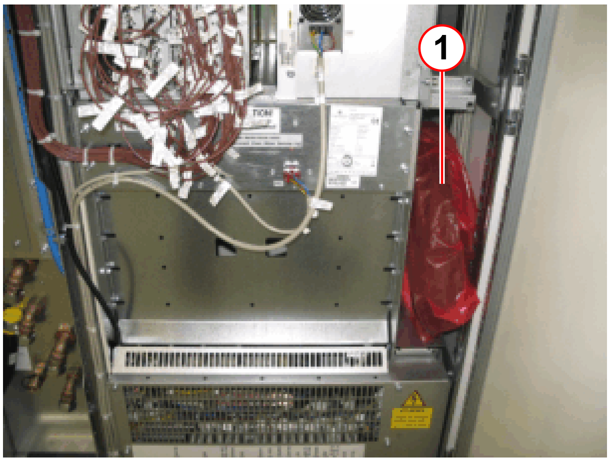
> ℹ The remaining length of the Flexa hose and the second bushing are necessary for the start-up procedure.

Fig. 237: Remaining Flexa hose with second bushing

(1) Second bushing
(2) Remaining Flexa hose

- Attach the second bushing (1) to the Flexa hose (2).

Fig. 238: Position of the plastic bag inside the ACC



(1) Remaining Flexa hose with second bushing

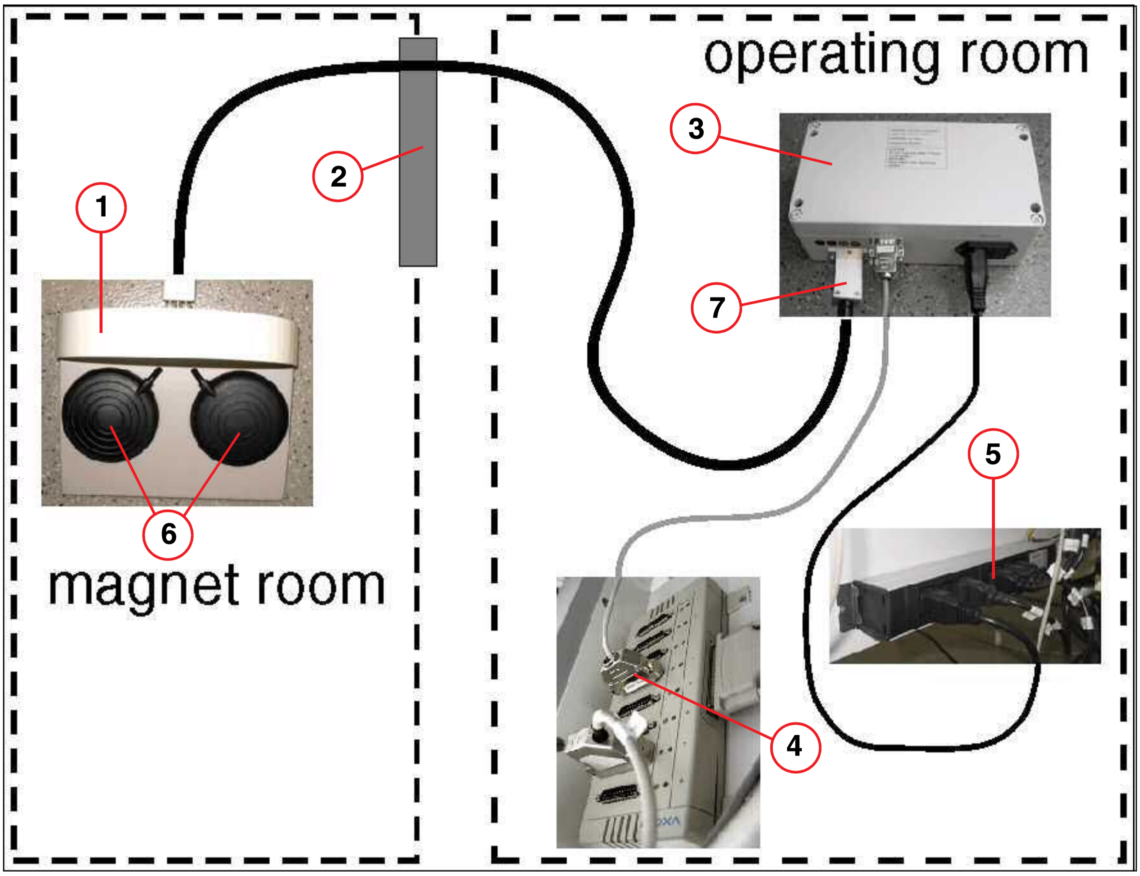
- Position the remaining length of the Flexa hose and the second bushing in a plastic bag inside the ACC cabinet.

6.18 Footswitch (option)

Determine the start/stop (left/rigt) functions of the footswitch together with the customer.

- Route the line (air hose) from the footswitch via the feed-through of the filter plate to the control room and connect it.
- Attach the start/stop labels to the footswitch.

Fig. 239: Cabling for foot switch



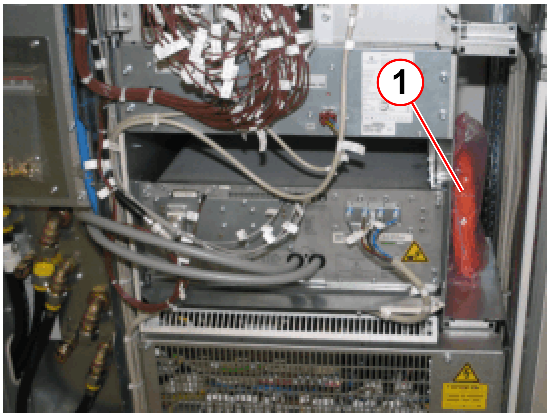
(1) Footswitch
(2) RF filter plate (FOC feedthrough)
(3) Converter
(4) Ser_Distribution (connector: 4)
(5) Line voltage distribution
(6) Start and stop button
(7) Air hoses connection (start/stop button specification, left or right)

6.19 Remote Alarm Contact

The EPC cabinet is equipped with one potential-free alarm contact X119.

- The contact (X119) is used for remote alarm of the following malfunctions (pin 1; 2; 3):
  - Liquid helium level warning and alarm
  - Shield temperature
  - Magnet pressure (high/low)
  - Helium compressor stops working
- The contact (X119) is used for remote alarm of the following malfunctions (pin 4; 5):
  - Quench alarm

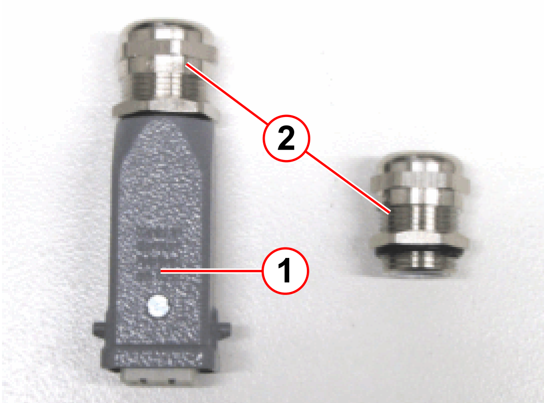
Fig. 240: ACC inside



(1) Remote alarm contact plug and FOC W3470

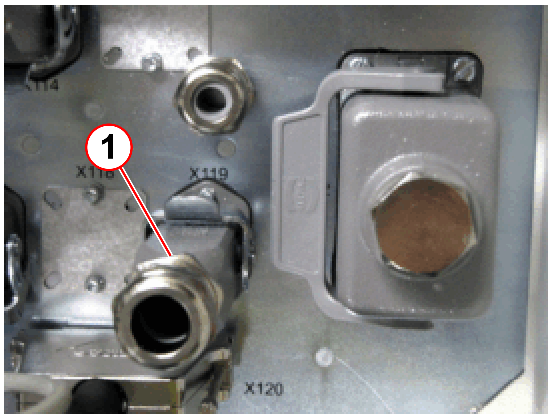
- Connect the remote alarm plug at EPC X119.

Fig. 241: Remote alarm plug Han Q6



(1) Han Q6
(2) Strain reliefs (different diameters)

Fig. 242: EPC X119 (top)

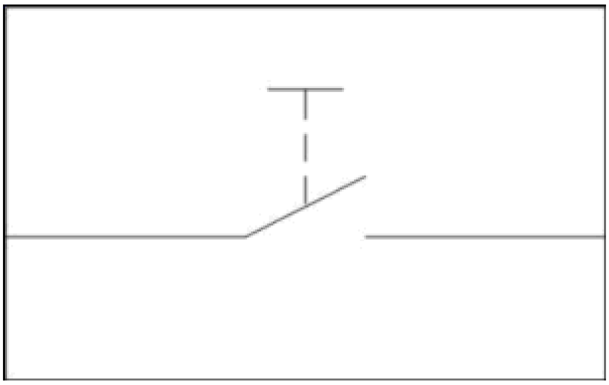


(1) Remote alarm plug X119

6.20 Door Contact

- Check the contacts of the door switch and connect it as a normally open contact.
  - » When the door is opened, the switch is opened.
  - » When the door is closed, the switch is closed.
- ACC plug contact X87.

Fig. 243: Door switch: normally open contact

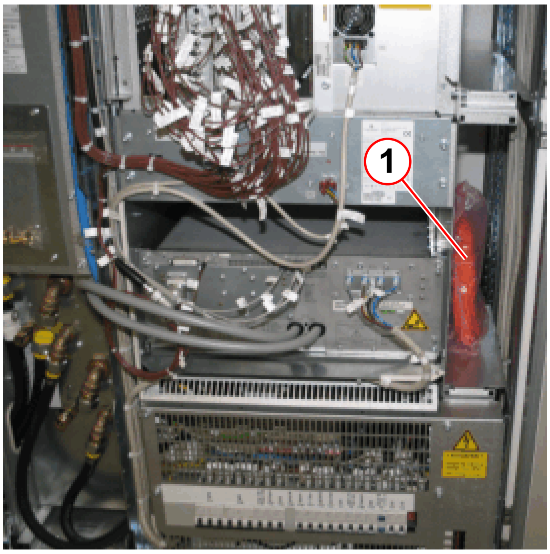


6.21 fMRI Trigger Converter

The fiber-optic cable (part no.: 81 11 887) for the option: “fMRI Trigger Converter Box” is always supplied in the ACC.

- The cable must always be laid (from the ACC to the console components in the operating room), even if the option is not included in delivery.

Fig. 244: ACC inside



(1) W3470 (part no.: 81 11 887) and remote alarm plug

Fig. 245: FOC W3470 (part no.: 81 11 887) and remote alarm plug

6.22 Wiring diagrams

» Use the "top sheets" in the binder "Technical Documents Volume 2 of 2", Chapter"Diagrams"

7.1 Line power adaptation

- Switch off the system in accordance with country-specific requirements
  Prior to switch-on, several line voltage adaptations have to be performed.
- Ask the project manager for the line voltage on-site or measure it.

The line voltage on-site is:	VAC
The line frequency on-site is:	Hz

WARNING

Injury from electric shock.

» **Adhere to the general safety regulations.**

» **Ensure that line voltage is not present at the ACC, GPA, as well as the compressor and the SEP (optional).**

» **Switch off the on-site line voltage supply and secure it against accidental switch-on (sign or lock).**

7.1.1 Checking the ERDU buttons

- Check the settings of the ERDU buttons. They **must not** have been activated.

Fig. 246: Alarm Box

(1) ERDU button
(2) Tamper-proof seal

Fig. 247: ERDU buttons

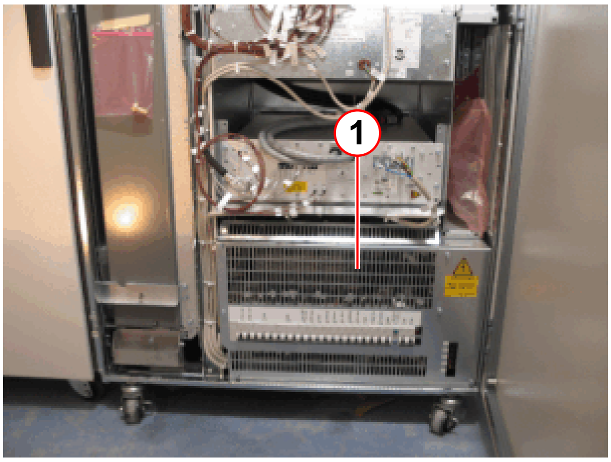
7.1.2 Line voltage adjustment of components

The voltages must be adapted to the voltages on site.

Adapt the line voltage at the following components:	
⇒	ACC
⇒	GPA
⇒	Helium compressor

7.1.2.1 ACC line voltage adaption

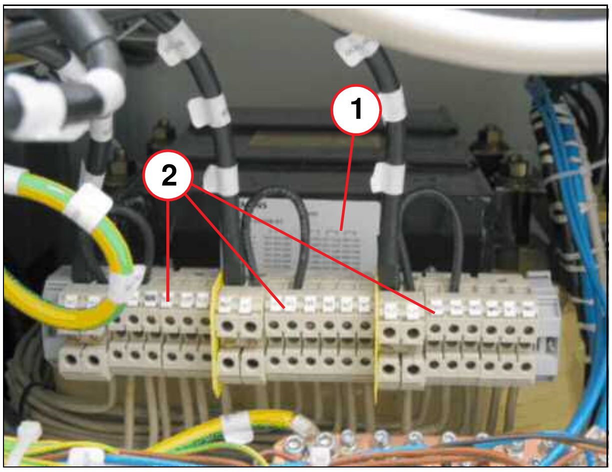
Fig. 248: ACC



(1) Protective cover

- Remove the protective cover (1).

Fig. 249: ACC voltage

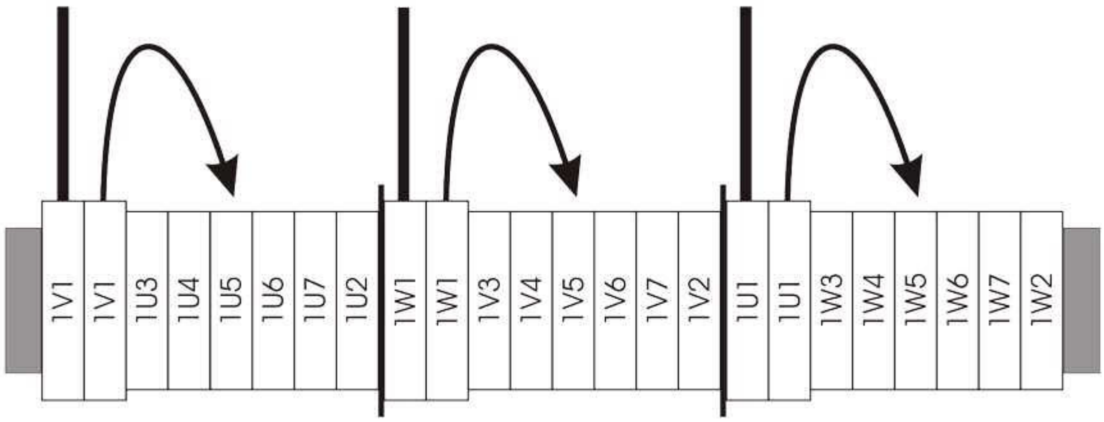


(1) Listing
(2) Voltages

- Adjust the voltage according to the sign on the transformer (all 3 phases).
- Re-attach the protective cover.

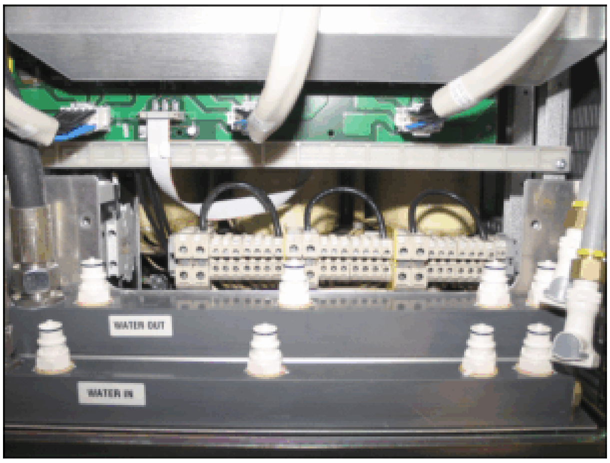
Fig. 250: Adapting the transformer in the ACC to the line voltage

480 V	1U1-1V1-1W1	1U1 - 1W2	1V1 - 1U2	1W1 - 1V2
460 V	1U1-1V1-1W1	1U1 - 1W7	1V1 - 1U7	1W1 - 1V7
440 V	1U1-1V1-1W1	1U1 - 1W6	1V1 - 1U6	1W1 - 1V6
420 V	1U1-1V1-1W1	1U1 - 1W5	1V1 - 1U5	1W1 - 1V5
400 V	1U1-1V1-1W1	1U1 - 1W4	1V1 - 1U4	1W1 - 1V4
380 V	1U1-1V1-1W1	1U1 - 1W3	1V1 - 1U3	1W1 - 1V3



7.1.2.2 GPA line voltage adaptation

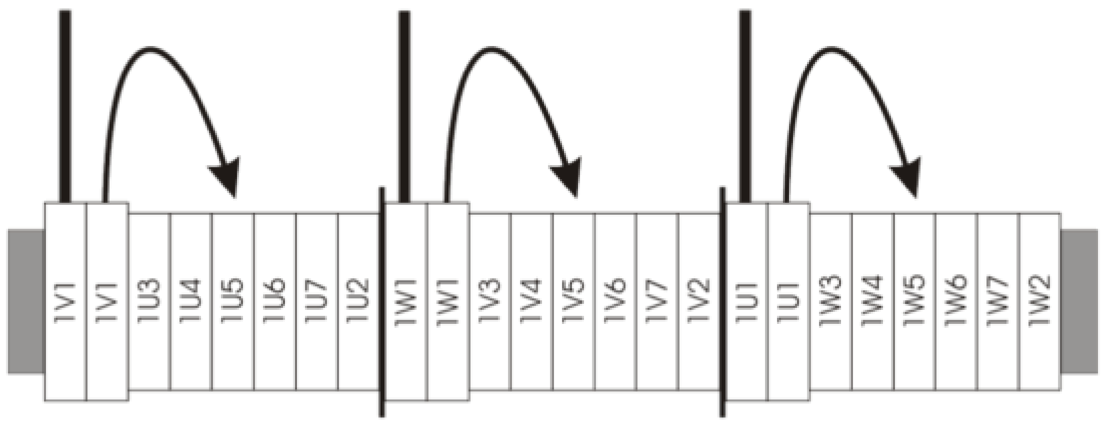
Fig. 251: Transformer GPA



- Disconnect the water hoses.
- Remove the protective cover.
- Adapt the voltage in accordance with the graphics, refer to (➔ Fig. 252 Page 185).
- Reattach the protective cover.
- Connect the water hoses.

Fig. 252: Adapting the transformer in the GPA to the line voltage

480 V	1U1-1V1-1W1	1U1 - 1W2	1V1 - 1U2	1W1 - 1V2
460 V	1U1-1V1-1W1	1U1 - 1W7	1V1 - 1U7	1W1 - 1V7
440 V	1U1-1V1-1W1	1U1 - 1W6	1V1 - 1U6	1W1 - 1V6
420 V	1U1-1V1-1W1	1U1 - 1W5	1V1 - 1U5	1W1 - 1V5
400 V	1U1-1V1-1W1	1U1 - 1W4	1V1 - 1U4	1W1 - 1V4
380 V	1U1-1V1-1W1	1U1 - 1W3	1V1 - 1U3	1W1 - 1V3



7.1.2.3 Line Voltage Adaptation, He Compressor F70

The unit is configured in the factory to 400 V/3~50 Hz

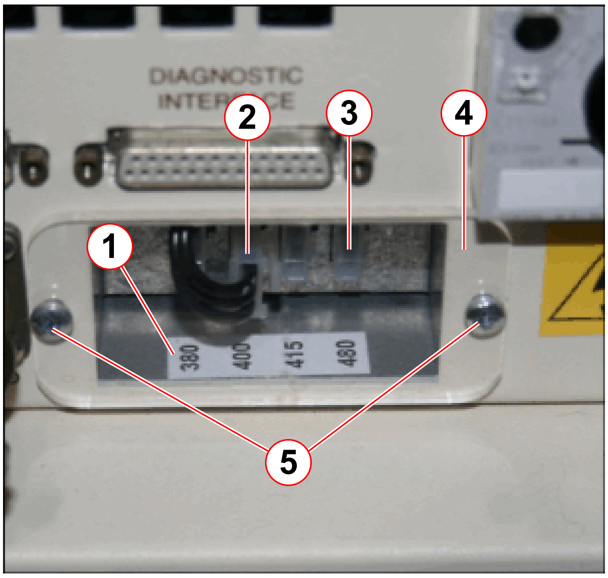
The **F70** has 4 voltage taps to match the incoming power supply.

The following voltage/frequency combinations are supported:

- 480 VAC @ 60 Hz
- 380, 400, & 415 VAC @ 50 Hz

Ensure the compressor is insulated from any power supply.

Fig. 253: F70 Compressor voltage selection access panel



(1) Voltage label
(2) Voltage selection connectors
(3) Voltage tap
(4) Clear plastic cover
(5) Screws (2 off)

- **Remove** the 2 screws securing the **clear plastic cover.**
- **Plug** the **voltage connector** (2) into the required **voltage tap** (all 3 phases).
- **Refit** the **clear plastic cover** using the 2 screws.

7.1.2.4 Stepping Transformer (optional)

The Stepping Transformer is absolutely necessary for all nominal voltage/frequency combinations other than **400V/50Hz** and **480V/60Hz**, refer to (➔ Stepping Transformer / MR00-000.814.44).

7.1.3 Checking the line voltage

Ask the respective personnel to switch on the on-site voltage supply.

WARNING

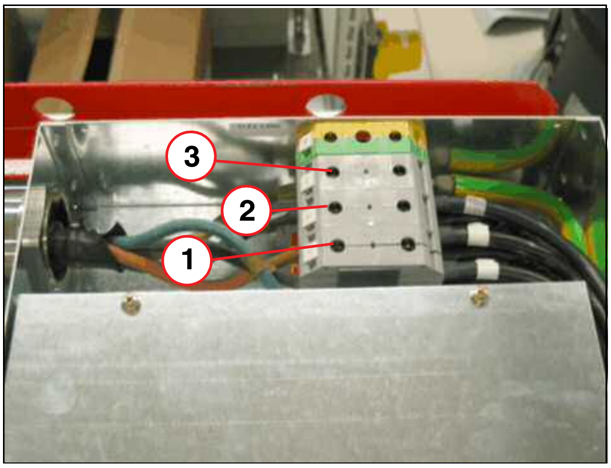
Injury from electric shock.

» **Adhere to the general safety regulations.**

» **Adhere to country-specific requirements when checking the line voltage and frequency.**

ACC:

Fig. 254: ACC power supply line



(1) L1
(2) L2
(3) L3

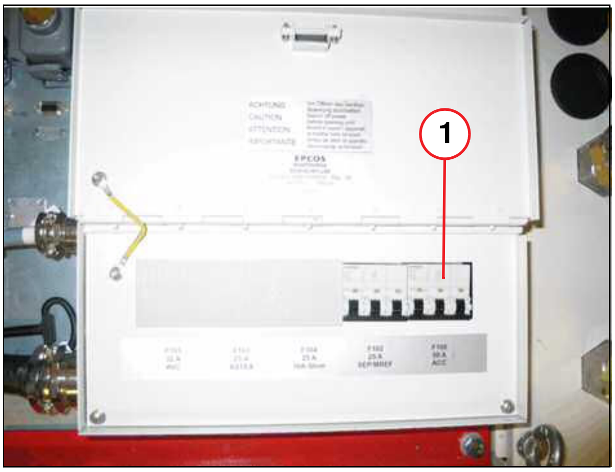
- Measure the line voltage at the terminals (**country-specific 360 - 480V**):
  - **L1-L2**
  - **L2-L3**
  - **L1-L3**

  Enter it in the installation protocol.

- Check the rotating field of the input voltage at terminals L1 to L3 and enter into installation protocol.

ACC X130:

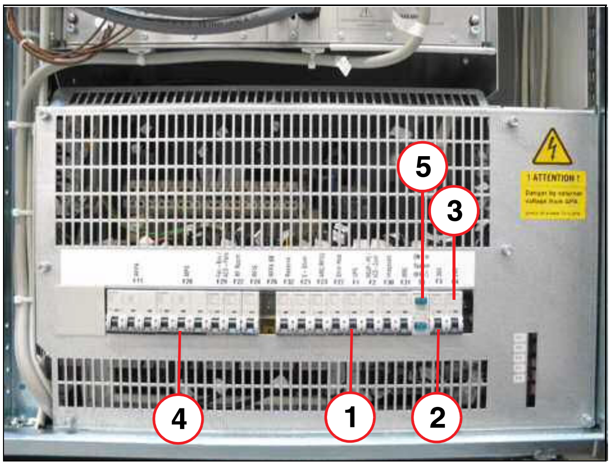
Fig. 255: ACC fuse



(1) F 100

- Switch on fuse F100 (1) at the top of the ACC mains box.

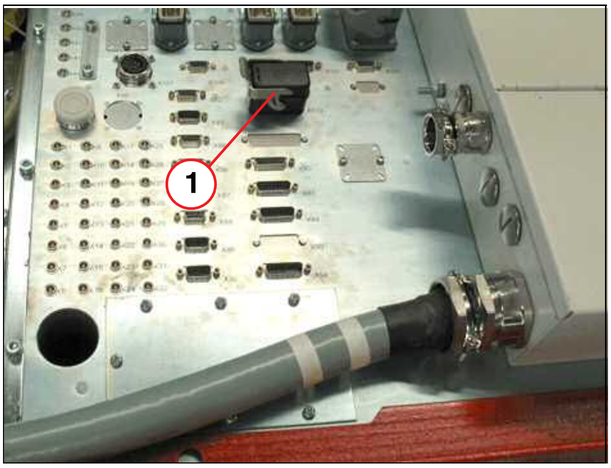
Fig. 256: ACC



(1) F1 UPS
(2) F3 36 V
(3) F4 24 V
(4) F28 MPS
(5) S1 ON

- Switch on the following circuit breakers:
  - **F1**
  - **F2**
  - **F3**
  - **F4**
  - **F28**
  - Press switch **S1**

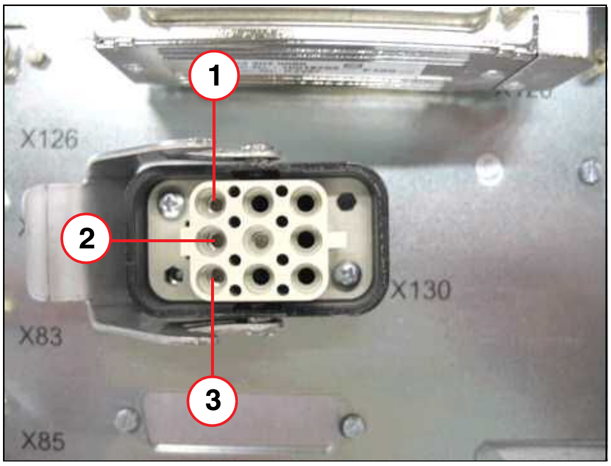
Fig. 257: ACC (view from the top)



(1) X130

Measure the voltages at X130 (1).

Fig. 258: ACC X130



(1) Pin 1, L1
(2) Pin 2, L2
(3) Pin 3, L3

- Measure the line voltage at X130 between:
  - **L1 - L2** (400 V ± 5 %)
  - **L2 - L3** (400 V ± 5 %)
  - **L1 - L3** (400 V ± 5 %)

  Enter it in the installation protocol.

- **Switch off all circuit breakers!**

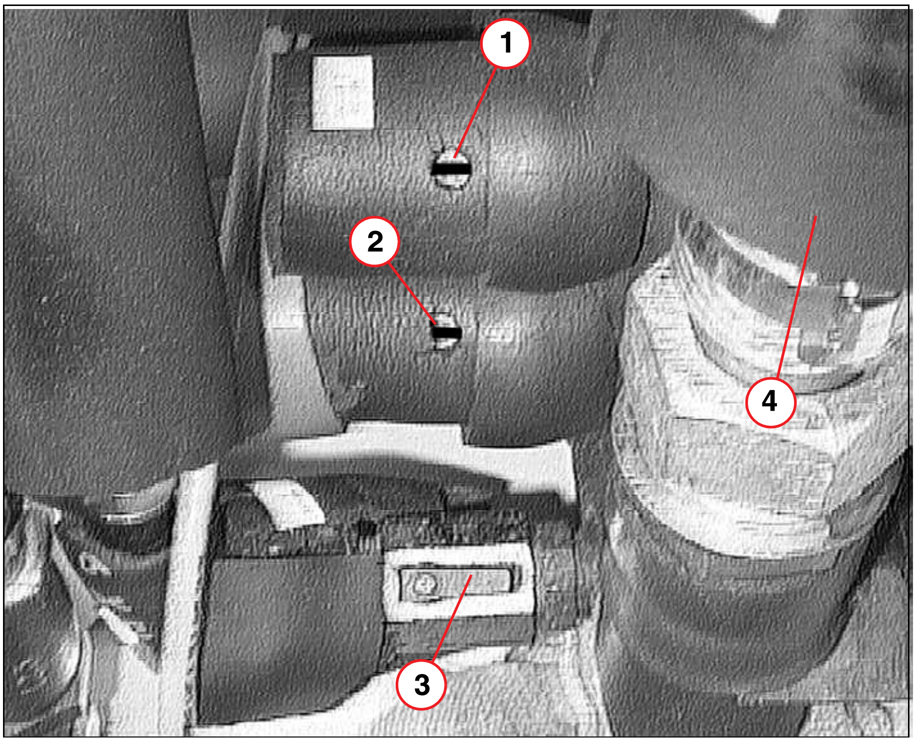
> If the values provided by the project manager do not correspond to the values measured, discuss this discrepancy with the project manager.

7.2 Starting the refrigerator

- Open the main system water valve so that the refrigerator (coolpack) begins cooling.
- Make sure that the flow direction of the primary water circuit (primary water) corresponds to the arrow shown in (➔ Fig. 259 Page 190).
- Open the ball valves on both sides with a 5 mm Allen wrench or screwdriver to cool "Cooler 1". The slot must point in the direction of flow.

> "Cooler 2" (➔ 1/Fig. 259 Page 190) must close (vertically)!

Fig. 259: RCA top view



(1) Cooler 2 opened
(2) Cooler 1 opened
(3) CCS valve
(4) Primary waterflow

- Check if the primary water circuit has to be refilled.

7.3 Connecting the He pressure lines:

Always use three open-end wrenches when disconnecting and connecting the He pressure lines to the cold head.

Fig. 260: F70 He gas connection

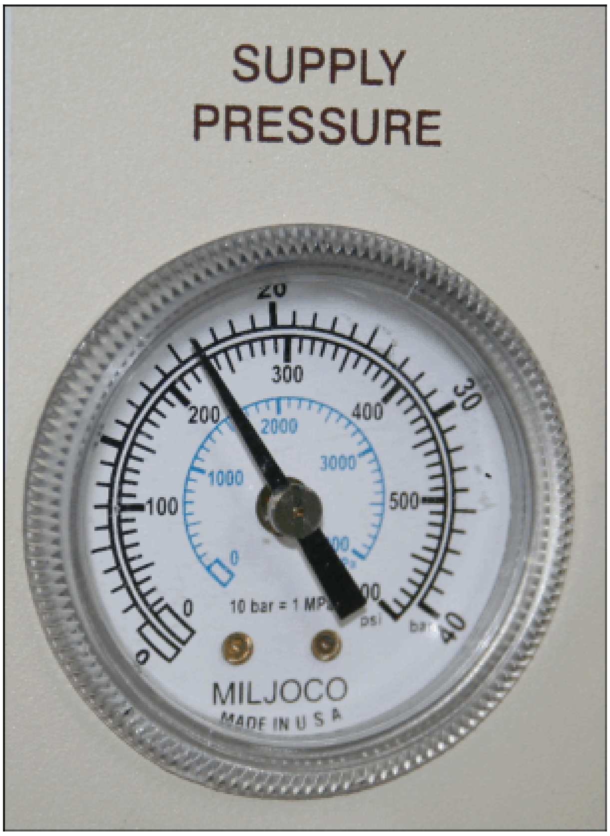
(1) High pressure line (SUPPLY/RED)
(2) Low pressure line (RETURN/GREEN)
(3) Power connector

OR70 magnet with F70 compressor and APD cold head

- Connect the **low pressure** line (RETURN = Pos.2) at the cold head.
- Connect the **high pressure** (Pos.1) line at the cold head.

7.4 Starting the helium compressor F70

Fig. 261: F70 Static pressure reading



- **Check** the **helium pressure** at the compressor. (➔ Tab. 8 Page 198)
- **Enter** the reading in the **Installation Protocol.**

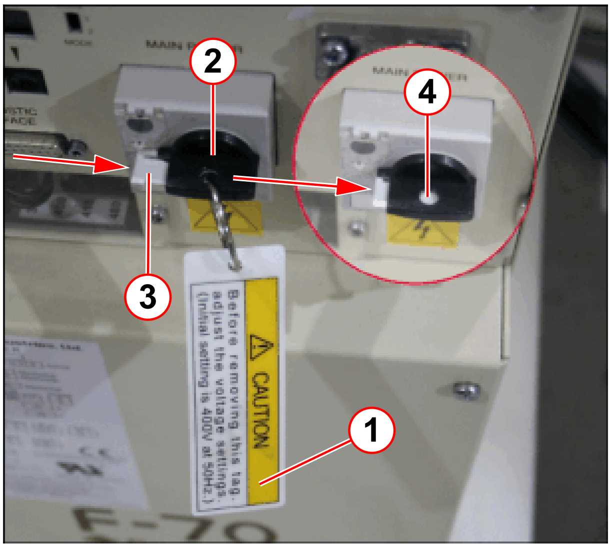
7.4.1 Test running the compressor before start-up

The compressor must be operated for **10 minutes** to clean up any oil migration incurred during transportation, **BEFORE** the **helium lines** and **cold head** are connected. For this test, only the compressor **water** and **power** connections are required.

In the **RCA cabinet**, the **water** and **power** are already connected to the compressor.

Check that the transformer voltage tap is correctly selected.

Fig. 262: F70 Main power switch - locking device



(1) Voltage caution label
(2) Main power switch
(3) White locking device
(4) Device fully engaged (view through hole)

- **Remove** the **voltage caution label** (1) (fitted through the main circuit breaker).
- **Engage** the **white switch locking device** fully **before** attempting to **turn on.**
- **Push in** the white locking device fully.
  » The **device** should be **visible** through the **blocking hole** (pos. 4).
- **Connect** the **grounding cable** to the **ground terminal**.

Ensure that the white switch locking device is FULLY engaged. THE SWITCH WILL BREAK IF FORCE IS USED.

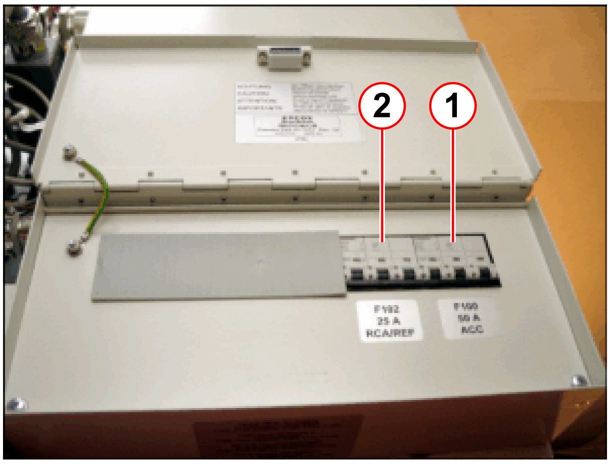
The compressor will become unusable and will have to be replaced.

F70 in RCA cabinet (water and power are already connected to the compressor):-

- Ensure that the **RCA cabinet** is connected to the **water** supply and return, and that the water is turned **on**.
- Ensure the compressor **water shut-off valves** are **open**, and water is flowing through the compressor.
- **Remove** the **voltage caution label** (fitted through the main circuit breaker).
  - **Engage** the **white switch locking device** fully **before** attempting to **turn on.** (➔ Fig. 262 Page 193)

7.4.1.1 Switch on the compressor via the ACC cabinet

Fig. 263: ACC power supply box

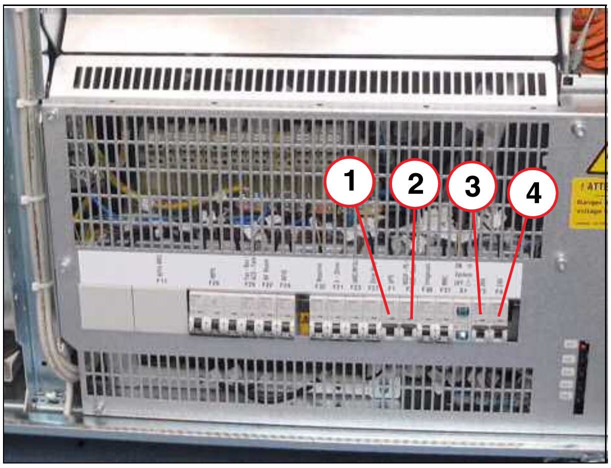


(1) F100 ACC
(2) F102 RCA/REF

ACC cabinet

- Switch **ON** circuit breaker **F100**
- Switch **ON** circuit breaker **F102**

Fig. 264: ACC fuse (inside)



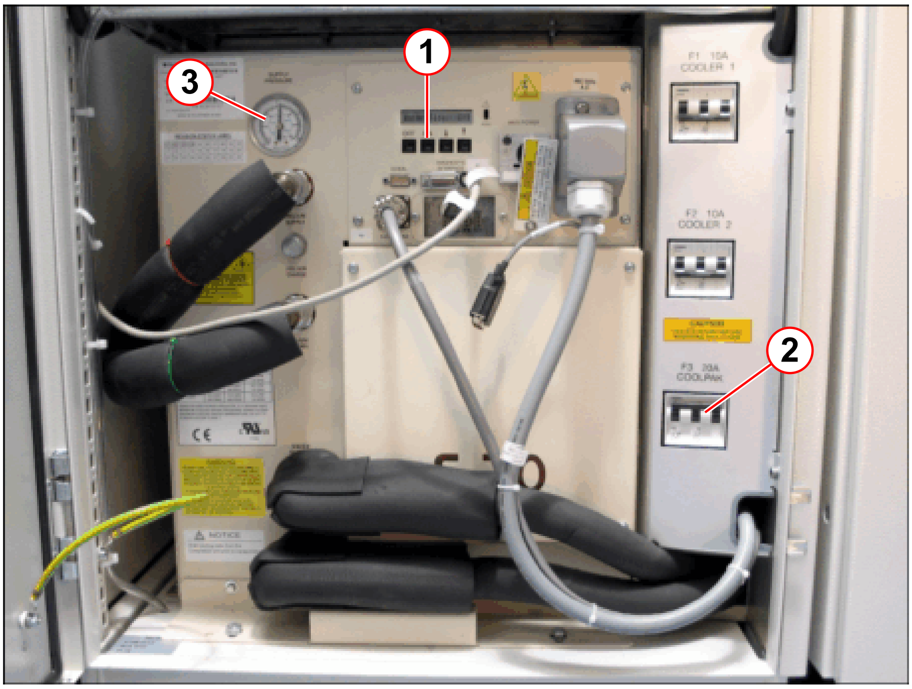
(1) Fuse F1
(2) Fuse F2
(3) Fuse F3
(4) Fuse F4

ACC cabinet

- Switch ON circuit breakers **F1, F2, F4** and **F4.**

- Switch **on** the **circuit breaker F3** (2) in the RCA.

Fig. 265: Helium compressor F70 in the RCA



(1) Switch ON button
(2) Circuit breaker F3
(3) Pressure manometer

7.4.1.2 Setting the configuration selector switch

> Configured in the factory for configuration 1.

Before the compressor's **main power switch** is switched on, the **configuration mode selector switch** must be set to either configuration **1** or configuration **2**.

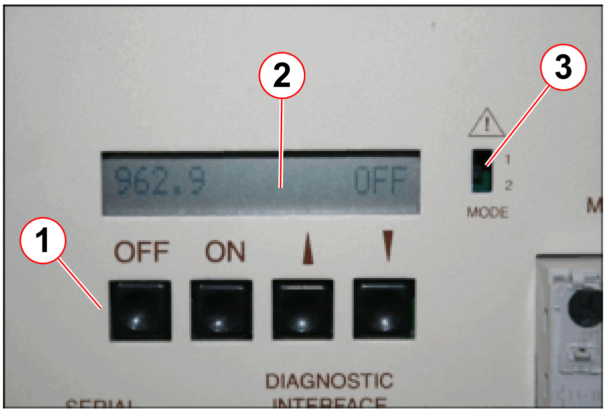
The **ON** and **OFF** buttons are **disabled** in configuration mode **2**, as the compressor is controlled via the pressure within the magnet.

Tab. 7 Configuration mode switch

Configuration	Switch position	Magnet system type	ON/OFF
Mode 1	UP	10KGM shield cooling	Enabled
Mode 2	DOWN	4KGM re-condensing	Disabled
Mode 2	DOWN	4KPT pulse tube	Disabled

> For this test, the configuration mode selector switch must be set to 1 (ON/OFF buttons enabled).

Fig. 266: F70 System status display

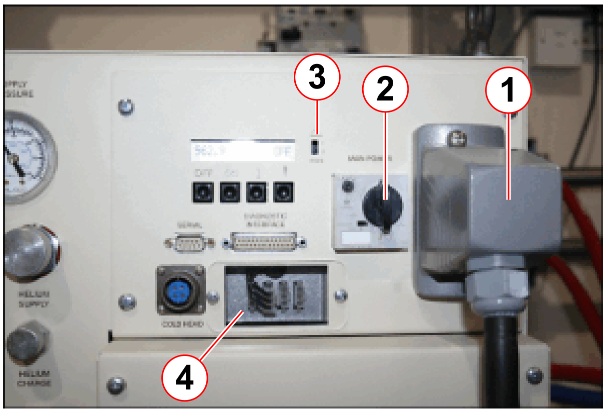


(1) Control buttons
(2) LCD display
(3) Configuration selector switch

- Use a flat-blade screwdriver to move the switch (3) up or down.
  - Switch **up** = **configuration 1**
  - Switch **down** = **configuration 2**

Do not change the switch position after the main power has been switched on.

Fig. 267: F70 Compressor - connections for test running before connection



(1) Main power switch
(2) Power supply
(3) Configuration mode switch
(4) Voltage selector

- **Ensure that the main power switch is OFF before adjusting the selector switch.**
- Check the configuration mode selector switch is set to **1**.

  Use a flat-blade screwdriver to move the switch up and down.

  » The **ON/OFF** buttons will only work in configuration mode **1**.

  » The compressor must **NOT** be run in standby mode in configuration **2**.

  In configuration **2**, both **ON** and **OFF** buttons are **disabled**. Operation is controlled remotely via the **DB25 diagnostic interface.**

- **Turn ON** the **main power** switch.
- Press the **ON** button to start the compressor. The display will read **RUN**.
- Touch the water supply (water in) and return (water out) lines.

  » The **return** (water out) line should be **warmer** than the **supply** (water in).
- **Stop** the compressor and reverse the water lines if the **return** line is **cooler**.
- **Run** the compressor for **10 minutes**.
- **Turn OFF** the **main power** switch.

- **The test run is complete.**

7.4.2 Connect the Supply (HP) and Return (LP) He pressure lines to the RCA

When connecting the He pressure lines, DO NOT OVER-TIGHTEN the couplings. Tighten down all couplings as far as possible, and then back them off by 1/4 turn to relieve strain.

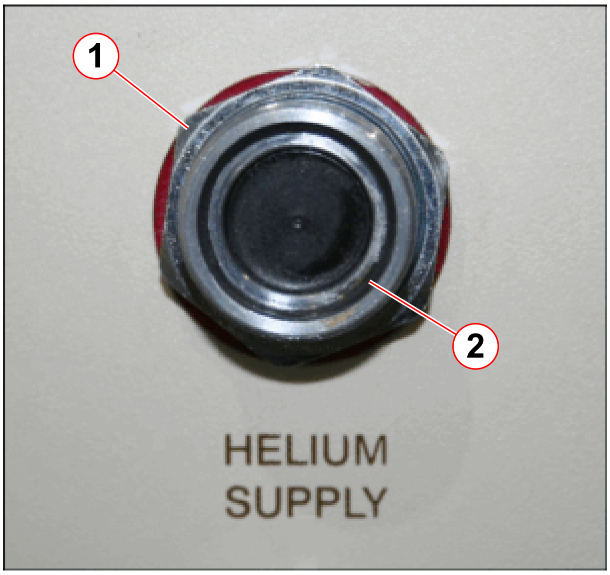
Fig. 268: F70 Helium gas coupling - protective cap

(1) Protective cap

Connection on the top of the RCA:

- **Remove** the protective transit covers from the **Return** & **Supply** connections.

Fig. 269: F70 High-pressure Aeroquip coupling



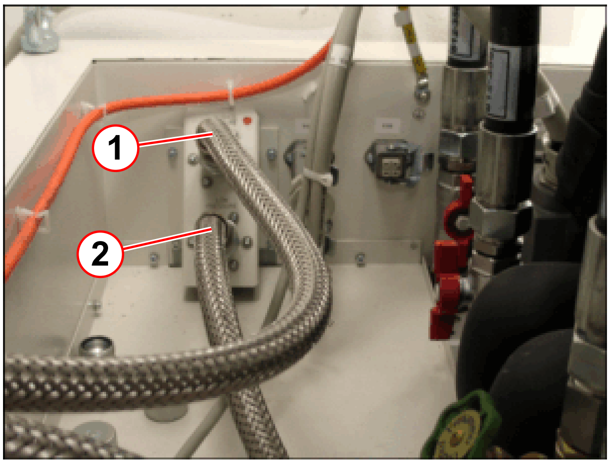
(1) Locknut
(2) Flat rubber gasket

Connection on the top of the RCA:

- **Check** the **rubber gasket** for dirt and dust and ensure it is fitted correctly.

  NOTE: An incorrect fit may cause a helium gas leak.

Fig. 270: RCA He line connections



(1) High pressure line (SUPPLY)
(2) Low pressure line (RETURN)

- Connect the **high pressure** line (1) at the **RCA** cabinet.
- Connect the **low pressure** line (2) at the **RCA** cabinet
- **Connect** the **cold head power cable**.

- **Check** the helium **pressure** at the compressor.

Equalization (static) pressure at 20^0 C (68^0 F) for 12 to 20 meter-long gas lines:

Tab. 8 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
10K - 50Hz	14.0 - 14.5 bar (14.3 nominal) 203 - 210 psig	18.3 - 23.2 bar 265 - 337 psig
10K - 60Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psig	18.0 - 23.0 bar 261 - 334 psig

If the pressure shown on the helium pressure manometer does not match the pressure shown in the table above, refill the helium compressor with helium (grade 99.999%). Refer to Troubleshooting Guide - Gas Charging for more information.

After the compressor has been running for 24 hours, check the static and dynamic pressures again.

If necessary, adjust the pressure (refer to Troubleshooting Guide - Adjusting the equalization pressure).

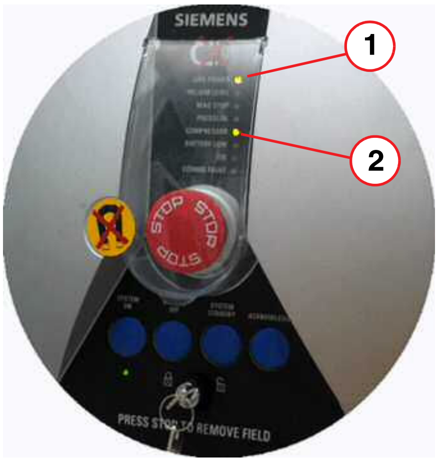
Perform a final check prior to system handover.

- Press **ON** button to start.

Audibly check if the cold head is running.

7.4.3 Status display at the alarm box

Fig. 271: Alarm box



(1) LED: LINE POWER
(2) LED: COMPRESSOR

After starting the compressor, the **alarm box** will indicate:

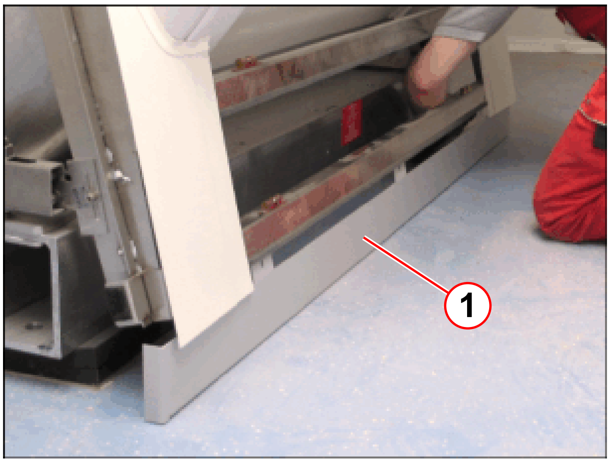
- **LINE POWER** (1):
  - » LED lights up green:
- **COMPRESSOR** (2):
  - » LED lights up green:

Check the alarm box status and LEDs. If any red LEDs are illuminated, troubleshoot.

8.1 Installation of the cover

8.1.1 Attach the socket covers

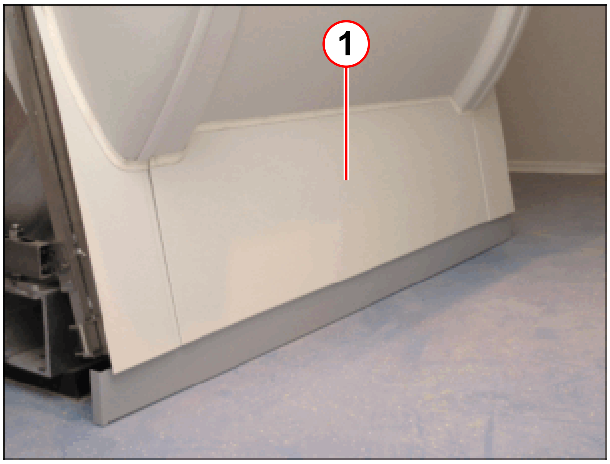
Fig. 272: Right socket cover



(1) Right socket cover

- Attach the right socket cover (1).

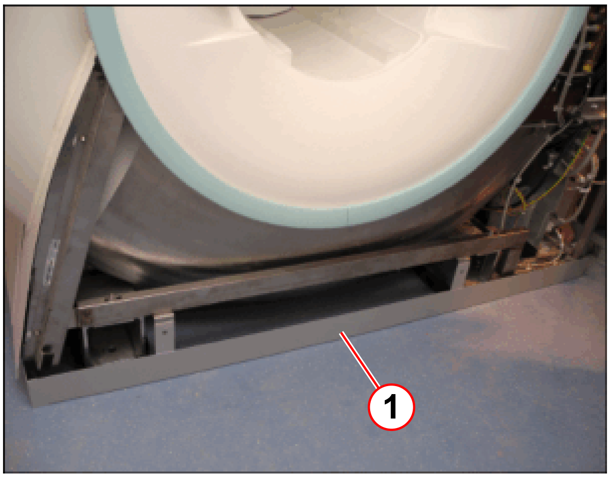
Fig. 273: Lower right cover



(1) Lower right cover

- Reattach the lower right cover (1).

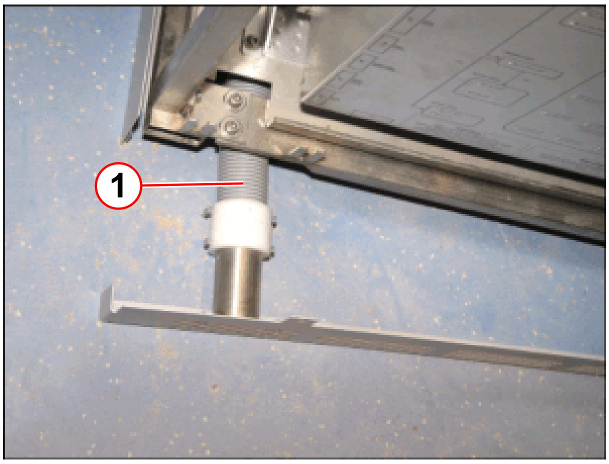
Fig. 274: Back socket cover



(1) Back socket cover

- Attach the back socket cover (1).

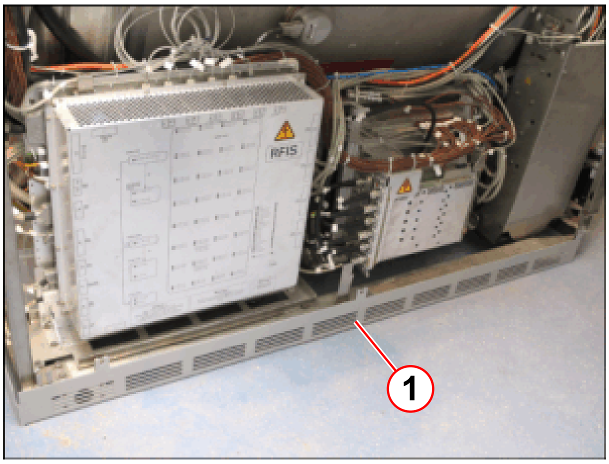
Fig. 275: Left socket cover with air tube connection



(1) Air tube

- Connect the air tube to the left socket cover (1).

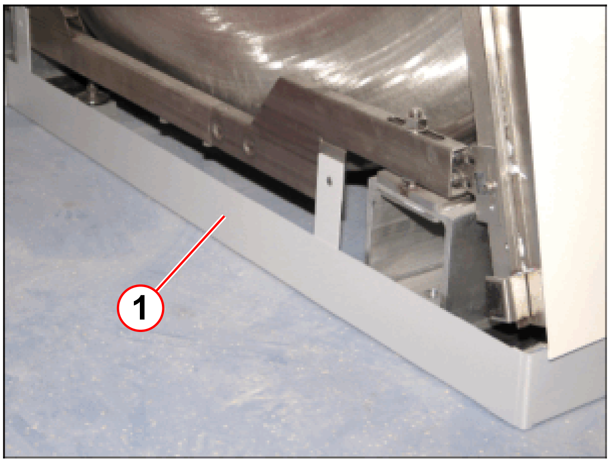
Fig. 276: Left socket cover



(1) Left socket cover

- Attach the left socket cover (1).

Fig. 277: Front socket cover

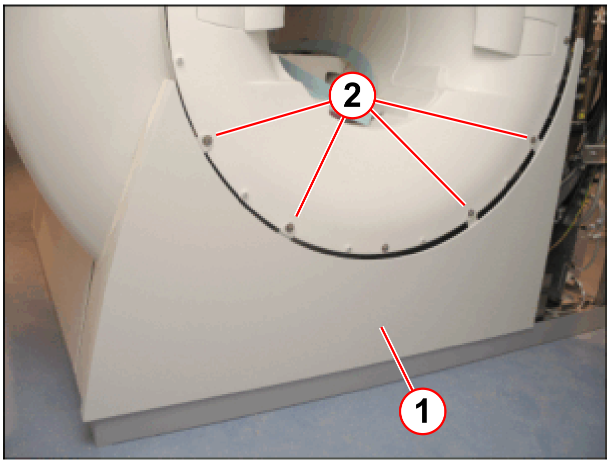


(1) Front socket cover

- Attach the front socket cover (1).

8.1.2 Attach the back covers

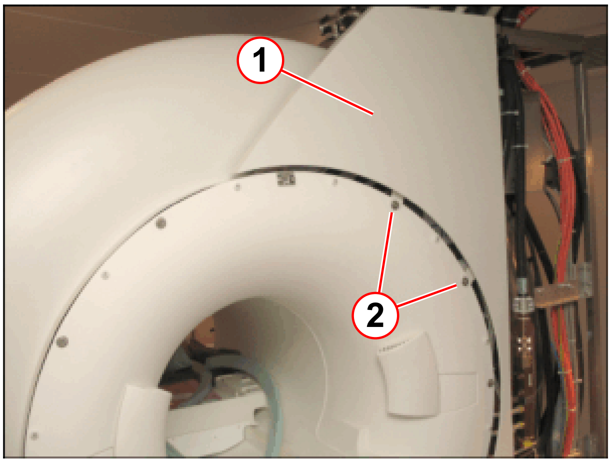
Fig. 278: Rear lower cover



(1) Rear lower cover
(2) Screws

- Attach the back lower cover (1) with the markings at the funnel.

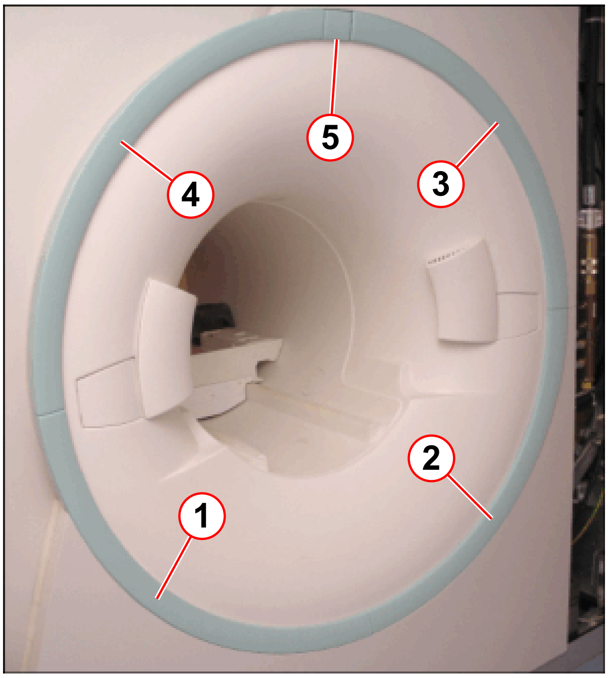
Fig. 279: Top rear cover



(1) Top rear cover
(2) Screws

- Attach the back top cover (1) with the markings at the funnel.

Fig. 280: Back ring segments



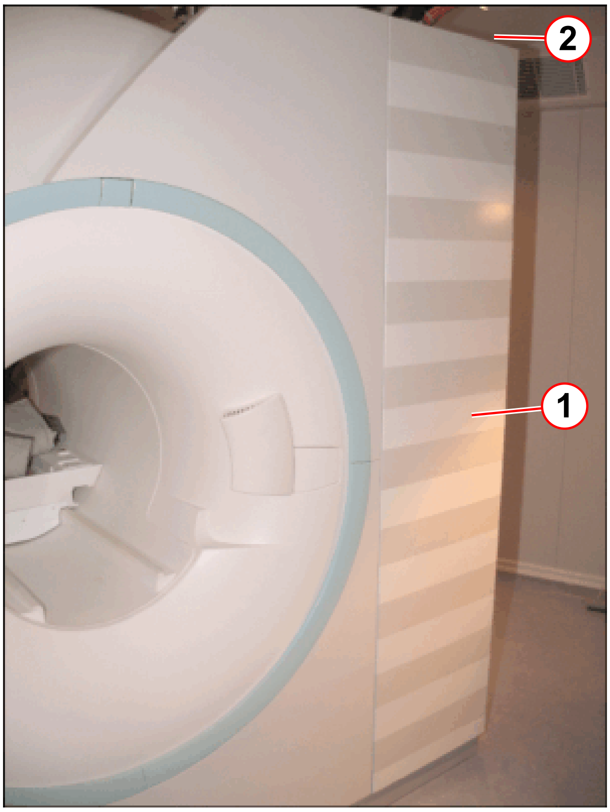
(1) Small cover
(2) Ring segment
(3) Ring segment
(4) Ring segment
(5) Ring segment

- Attach the ring segments in numerical sequence.

Patient camera (optional):

- Connect and attach the patient camera at position 5.

Fig. 281: Left rear cover

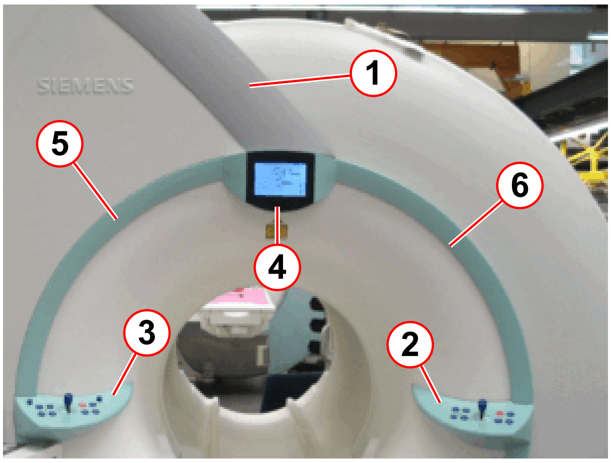


(1) Left rear cover
(2) Screw (at the top)

- Attach the left back cover (1) and secure it with the screw (2) at the top.

8.1.3 Front covers

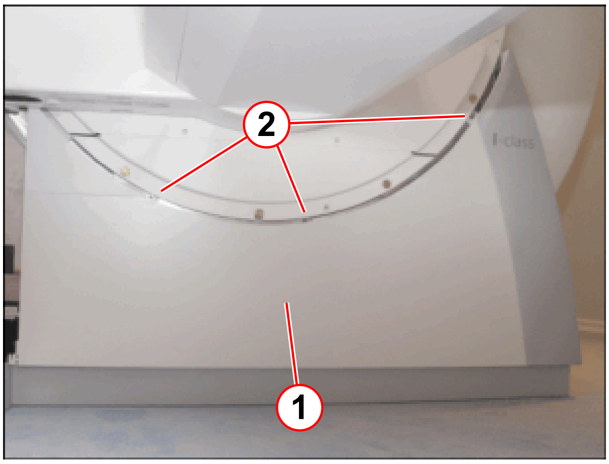
Fig. 282: Patient end



(1) Cover
(2) Right control panel
(3) Left control panel
(4) Display
(5) Left front ring segment
(6) Right front ring segment

- Remove cover (1), it is connected.
- Remove the display (4) and the control panels (2, 3).
- Remove the ring segments (5, 6).

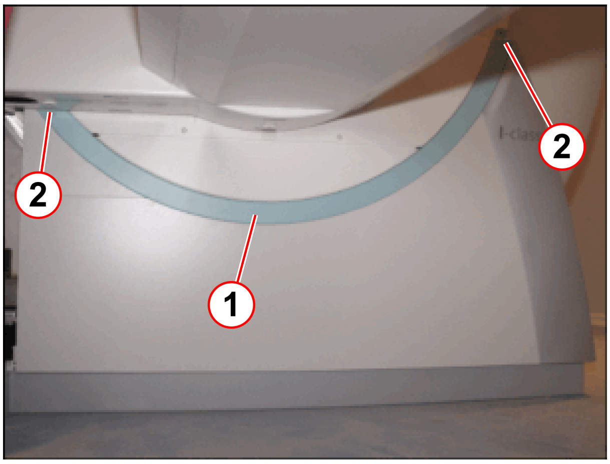
Fig. 283: Front lower cover



(1) Front lower cover
(2) Screws

- Attach the front lower cover (1).

Fig. 284: Front lower ring segment



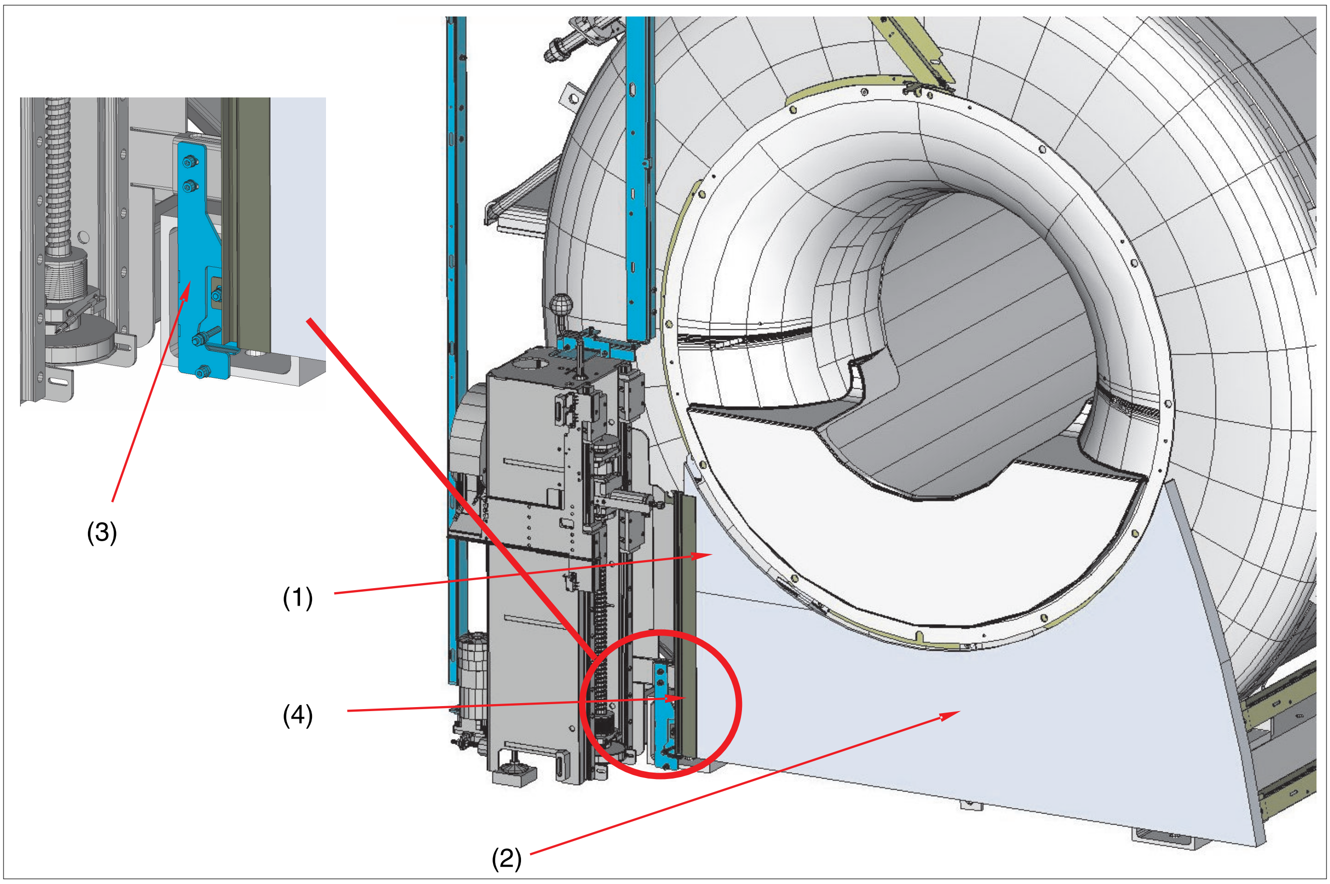
(1) Front lower ring segment
(2) Screws

- Attach the front lower ring segment (1).

8.1.4 Lifting column covers

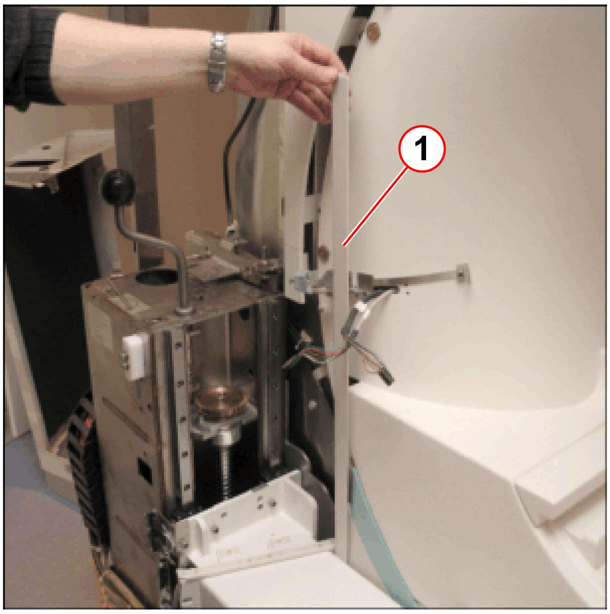
- Install the bracket, part no. 1313105 (→ 3/Fig. 285 Page 207).

Fig. 285: Magnet front cover



(1) Front cover
(2) Front cover
(3) Holder
(4) Bellows guide

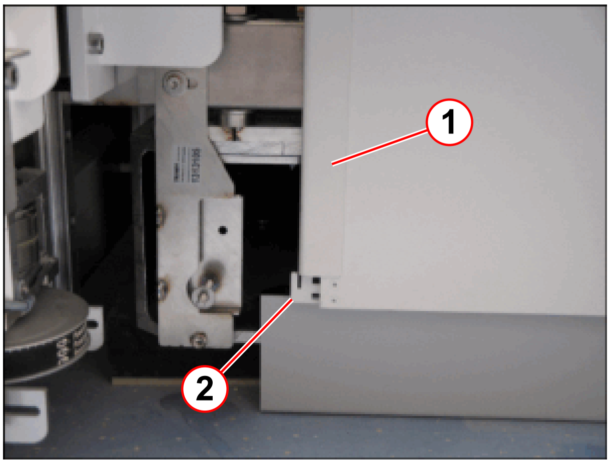
Fig. 286: Bellows guide



(1) Bellows guide (1313106)

- Move the patient table down manually with the hand crank.
- Slide in the bellows guide (1).

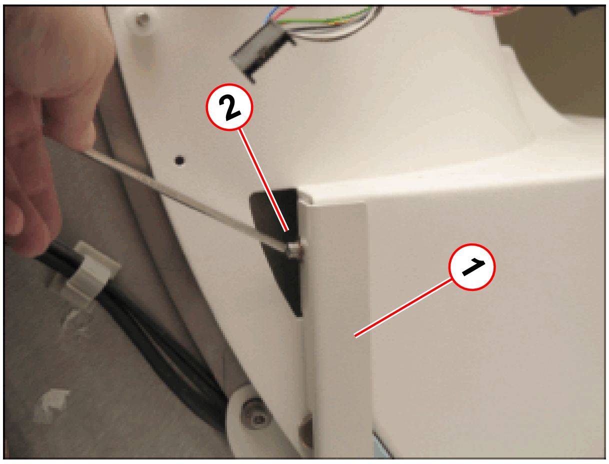
Fig. 287: Bellows guide (bottom)



(1) Bellows guide
(2) Slide in

- Slide in the bellows guide (1) at the bottom (2).

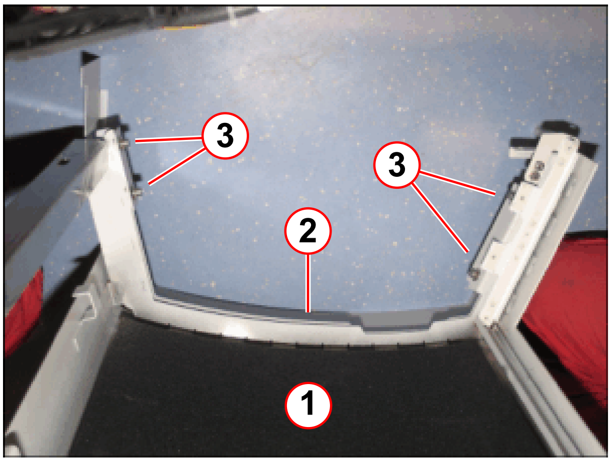
Fig. 288: Bellows guide (top)



(1) Bellows guide
(2) Screw

- Secure the bellows guide (1) with the screw (2) at the top.

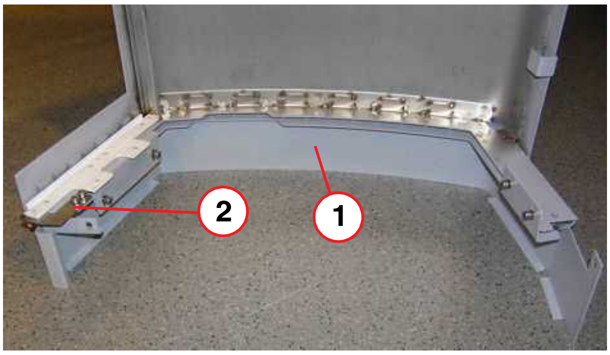
Fig. 289: Lifting column cover (bottom)



(1) Lifting column
(2) Socket lifting column cover
(3) Screws

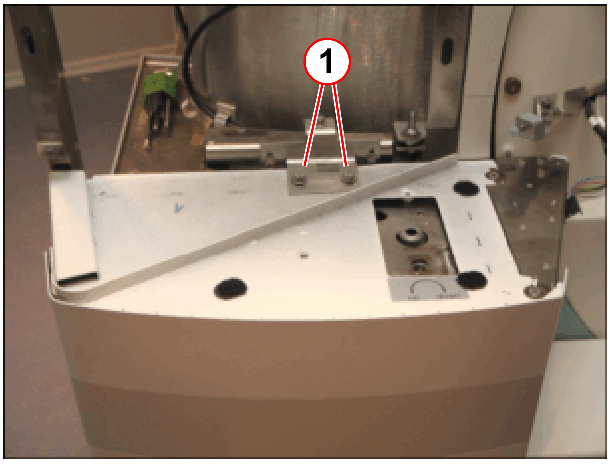
- Attach the socket lifting column cover (2) at the lifting column (1).

Fig. 290: Lifting column cover



(1) Socket cover
(2) Screws

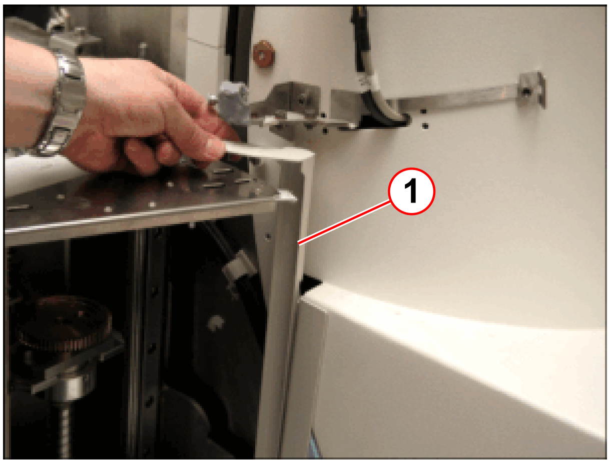
Fig. 291: Lifting column cover (top)



(1) Screws

- Attach the lifting column.
- Secure the lifting column with the screws (1).

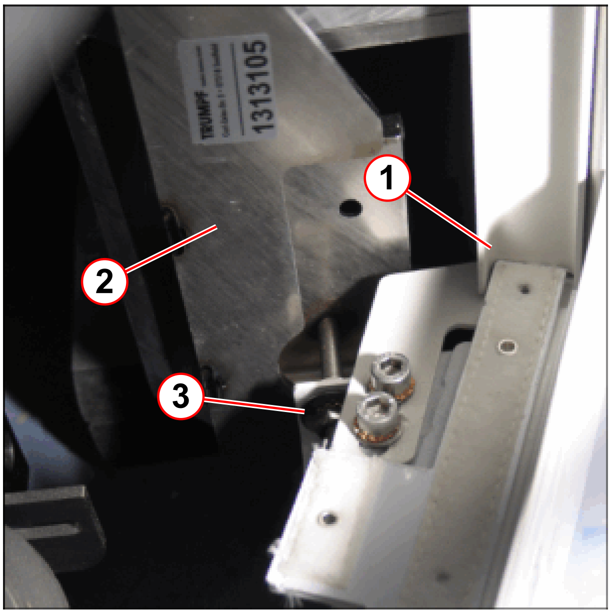
Fig. 292: Bellows guide (at the lifting column)



(1) Bellows guide (1313107)

- Slide in the bellows guide (1).

Fig. 293: Lifting column cover (bottom right side)



(1) Slide in the bellows guide (1313107)
(2) Nut, lock washer, and washer
(3) Bracket (1313105)

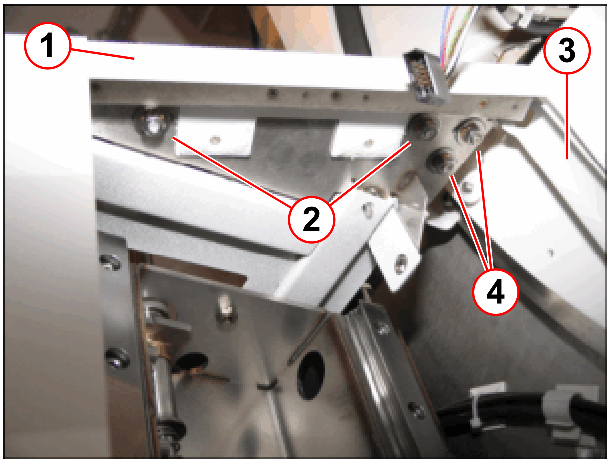
- Slide in the bellows guide (1) at the bottom.
- Secure the lifting column cover with the nut (3) at the bracket.

Fig. 294: Bracket

(1) Bracket (1325116)

- Attach the bracket (1).

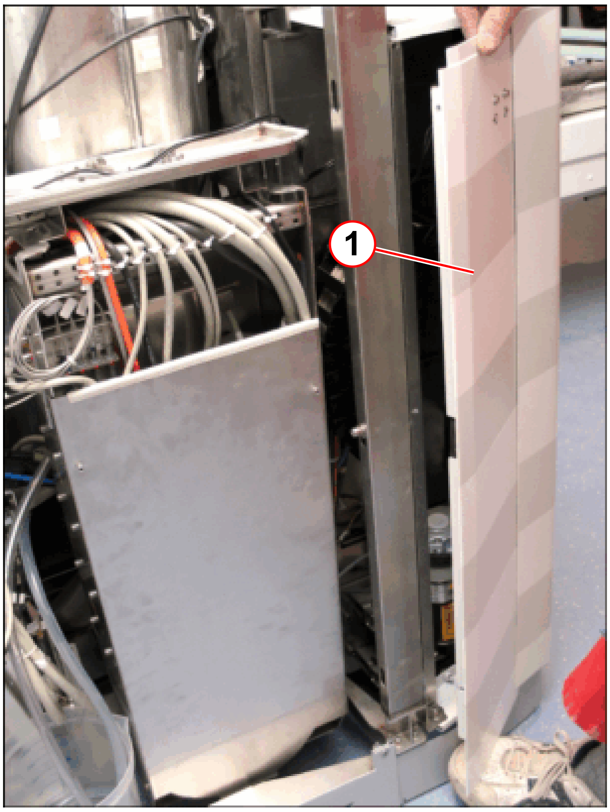
Fig. 295: Bracket attachment (view from the bottom)



(1) Bracket (1325116)
(2) Screws
(3) Bellows guide (1313107)
(4) Screws

- Secure the bellows guide (3) and the bracket (1) with the nuts (2,4).

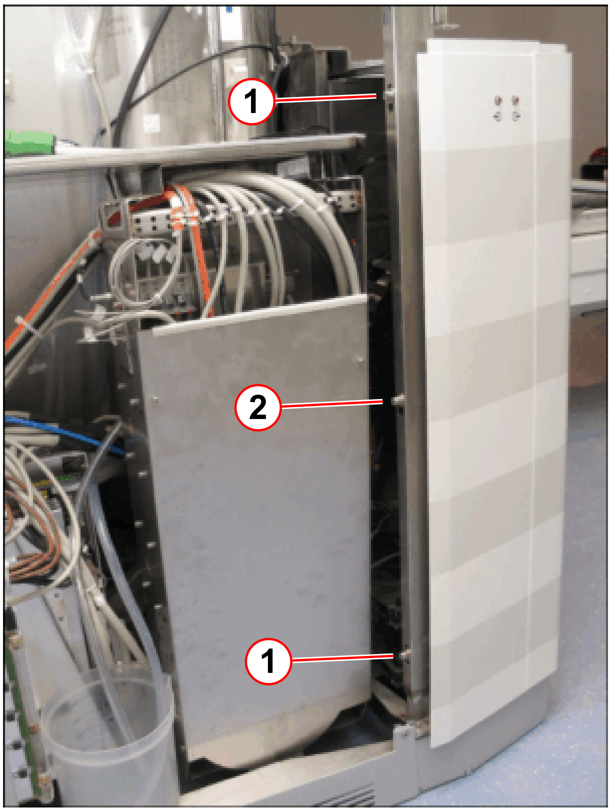
Fig. 296: Lower lifting column cover



(1) Lower lifting column cover

- Slide in the lower lifting column cover (1) at the front.

Fig. 297: Lower lifting column cover (attached)



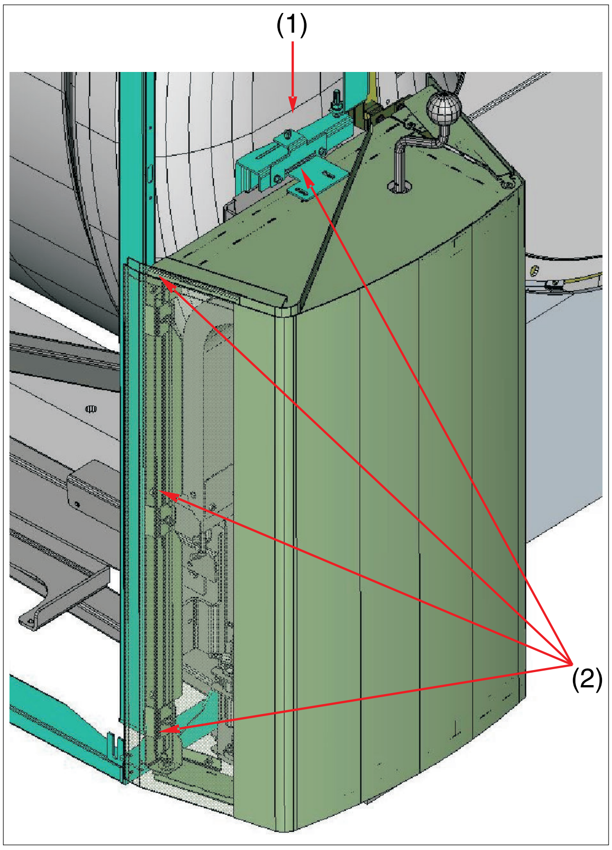
(1) Screws
(2) Screw (for the lifting column attachment)

- Secure the lower lifting column cover with the screws.

External trigger (optional):

- Connect the plug X7 / X8 at the PDAU.

Fig. 298: Lifting column cover

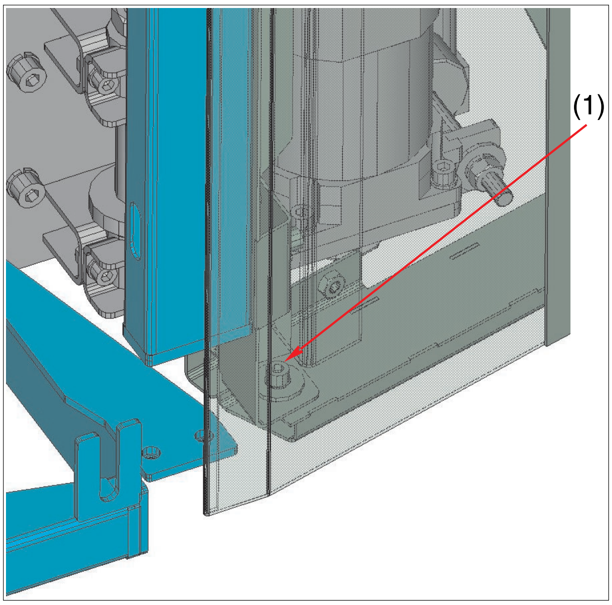


(1) Height adjustment
(2) Screw locations

- Align the lifting column cover vertically to the support arm (the same distance at the upper and lower position).

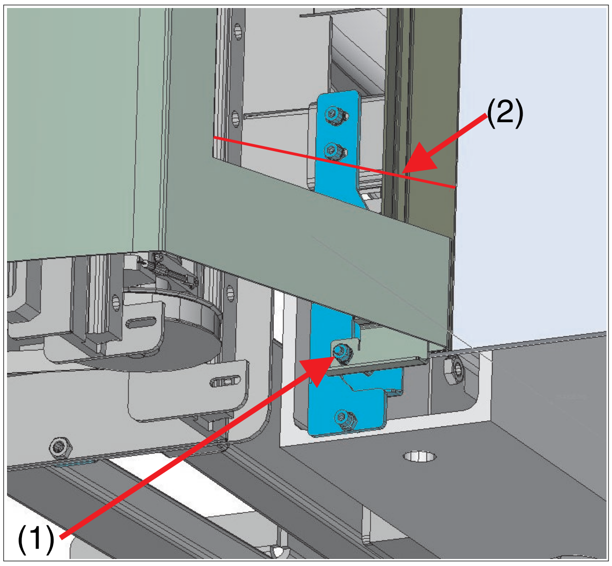
  Distance between cover and lifting column: min. 212 mm, insert bellows to test.

Fig. 299: Lifting column cover



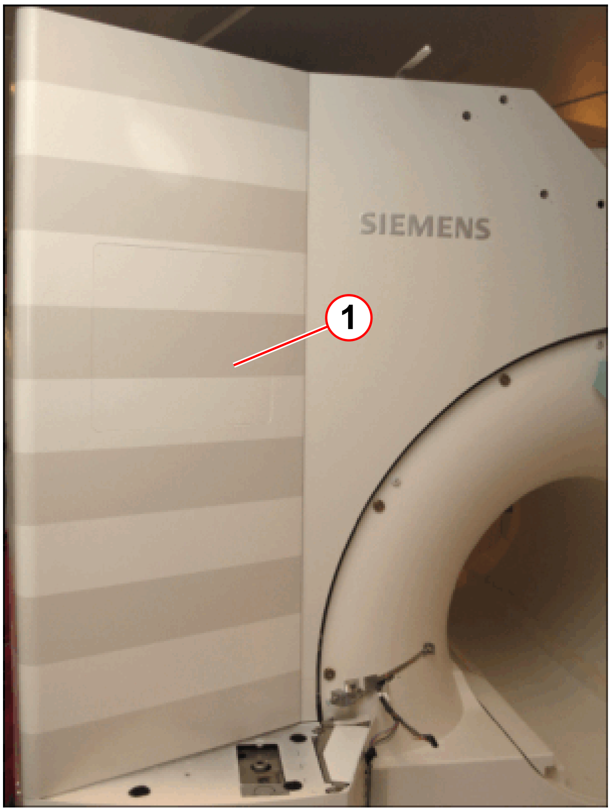
(1) Adjustment range (upper and lower)

Fig. 300: Lifting column cover



(1) Screw locations
(2) Vertical distance to bellows guide min. 212 mm

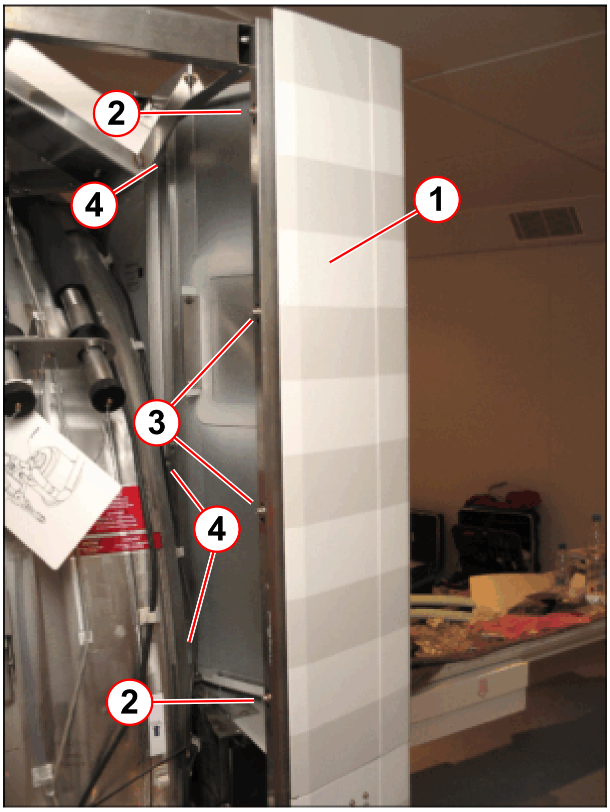
Fig. 301: Top lifting column cover



(1) Lifting column cover

- Attach the top lifting column cover (1).

Fig. 302: Top cover

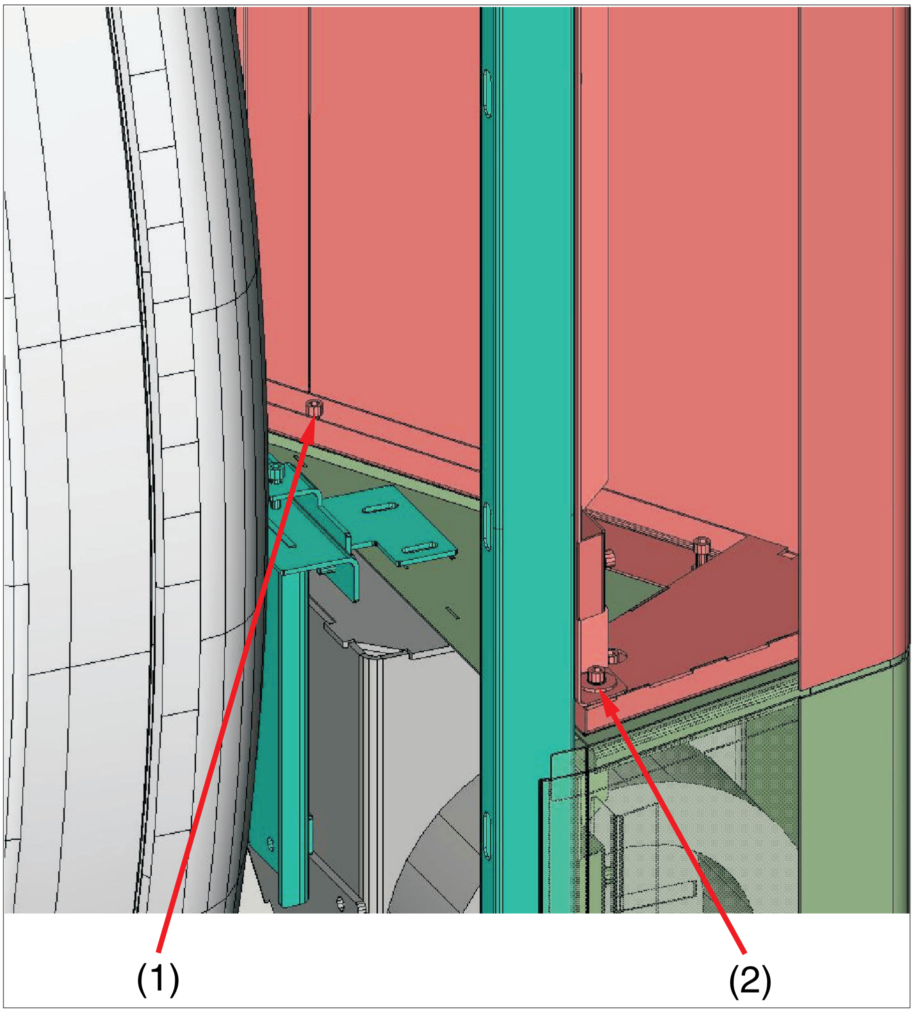


(1) Top cover
(2) Screws
(3) Screws (for the top lifting column cover attachment)
(4) Hexagon screw

- Secure the top cover with the screws (4) at the base frame.
- Slide in the top cover (1) at the front and secure it with the screws (2, 3) at the base frame.
- Adjust the height.

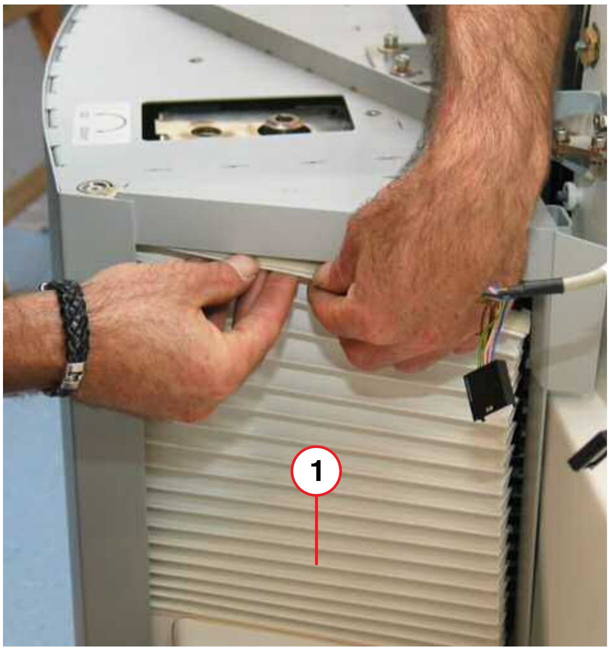
- Align with the lower cover

Fig. 303: Magnet front cover



(1) Height adjustment 3x
(2) Adjustment range (upper and lower)

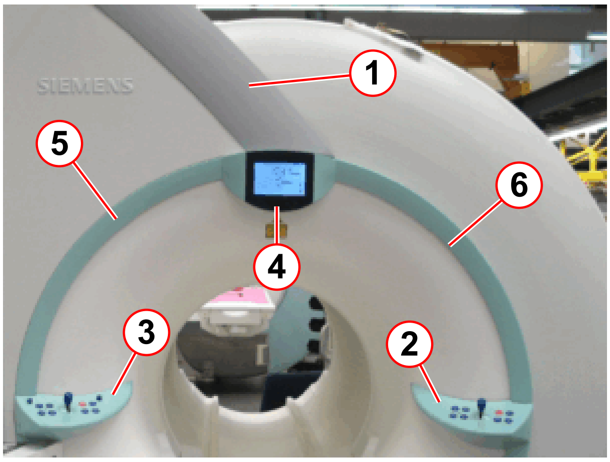
Fig. 304: Column cover



(1) Bellows

- Attach the bellows at the top and bottom.

Fig. 305: Patient end



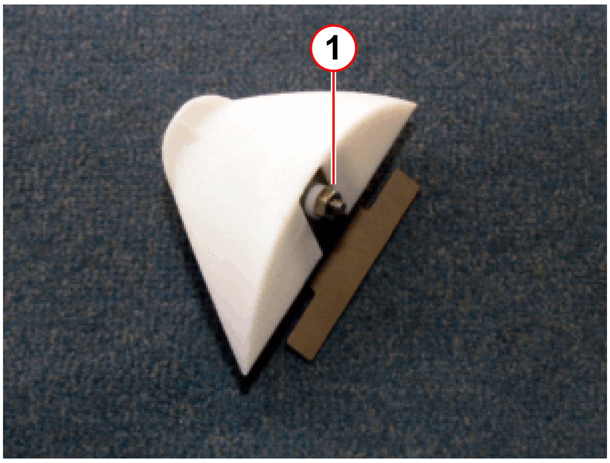
(1) Cover
(2) Right control panel
(3) Left control panel
(4) Display
(5) Left front ring segment
(6) Right front ring segment

- Attach the ring segments (5, 6).
- Connect and attach the display (4) and the control panels (2, 3).
- Attach the cover (1).

8.1.5 Headphone Holder

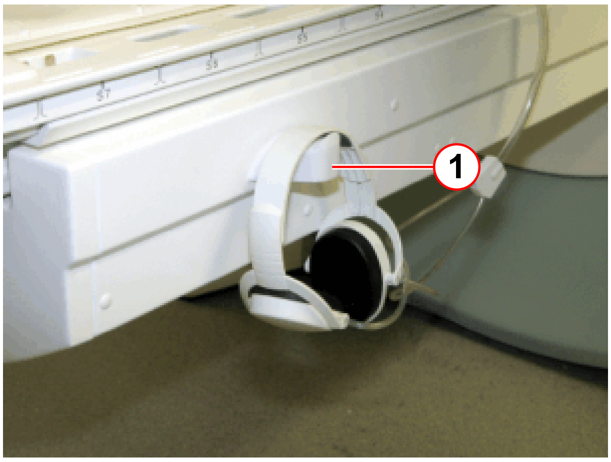
> Prior to installing the headphone holder, determine the side together with the customer.
>
> The headphone holder can be attached on the left or right side of the support frame of the patient table.
>
> If the system is equipped with a “Removable table”, the headphone holder should be attached to the side where the guide bar of the trolley is not attached.
>
> If the system is equipped with the option “Breast Biopsy Device”, the headphone holder must be removed to use the “Breast Biopsy Device”.

Fig. 306: Headphone holder



(1) Nut (only for the transport)

Fig. 307: Headphone holder (example right side of the patient table)



(1) Headphone holder

Patient table support frame (example right side):

- Remove the plastic cap from the support frame.
  - » Only use the upper hole on the feed end position.

    The hole on the left side is the same (mirror-inverted).

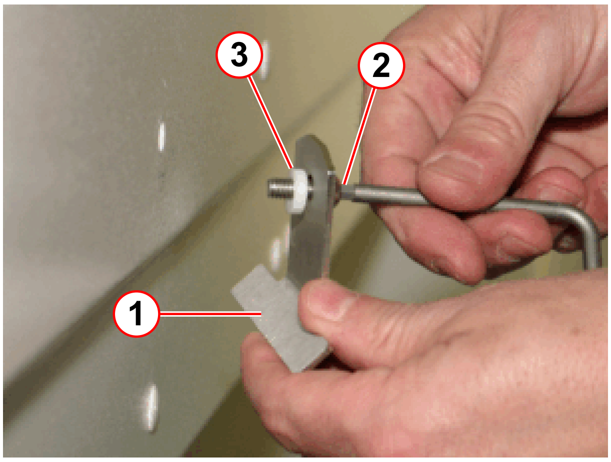
Fig. 308: Position of the headphone holder (example right side of the support frame)

(1) Plastic cap

- Remove the nut from the headphone bracket (the nut is only for transport to secure the screw and the spacer shim).

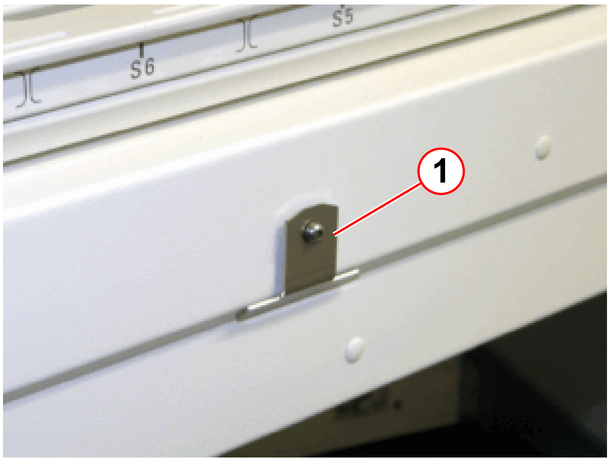
- Attach the headphone bracket with the spacer shim at the support frame.

Fig. 309: Headphone bracket attachment



(1) Headphone bracket
(2) Screw and washer
(3) Spacer shim

Fig. 310: Attached headphone bracket



(1) Headphone bracket

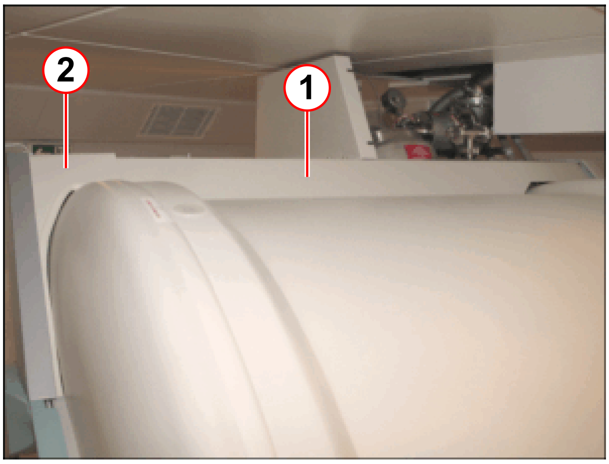
Fig. 311: Attached headphone holder

- Attach the headphone holder to the headphone bracket.
- Attach the headphone to the headphone holder.

8.1.6 Top covers

- Attach the right support bracket (1) at the base frame.
- Attach the front top right cover (2) at the base frame.

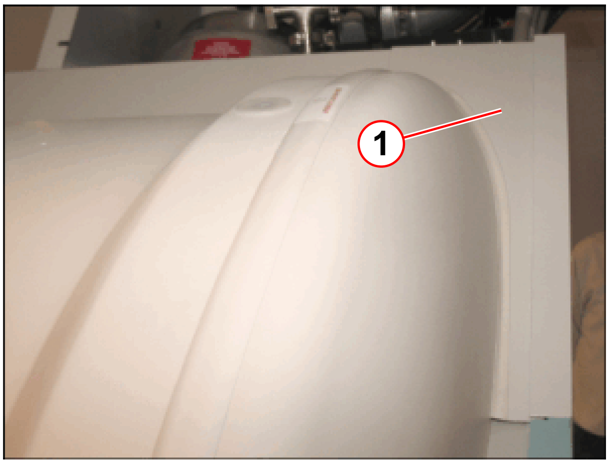
Fig. 312: Top covers



(1) Right support bracket
(2) Front top right cover

- Attach the back top right cover (1) at the base frame.

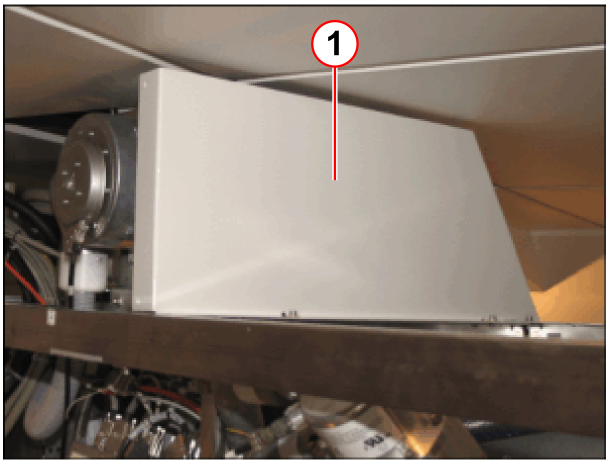
Fig. 313: Top covers (back)



(1) top back right cover

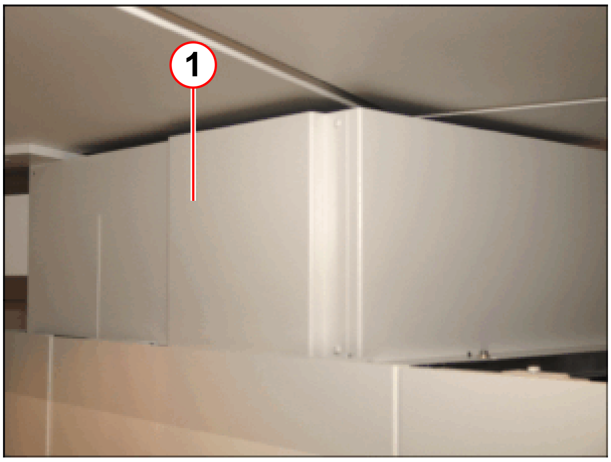
- Attach the turret covers.

Fig. 314: Turret cover 1



(1) Tower front cover

Fig. 315: Turret cover 2



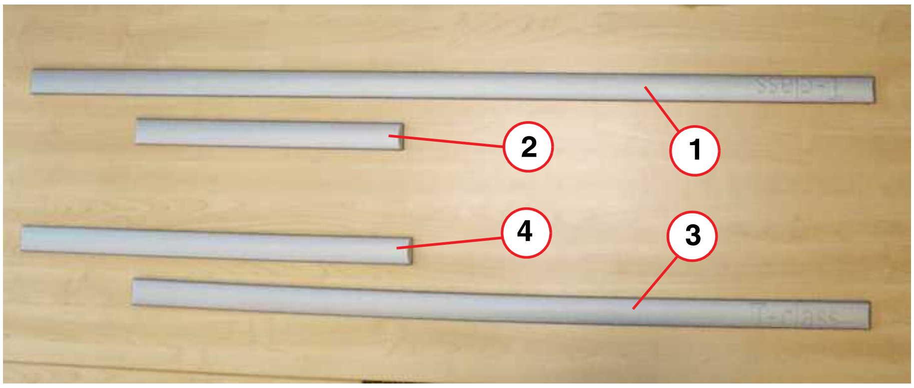
(1) Left turret cover

8.1.7 T-class accent strips and removable table (Optional)

If the system is equipped with the T-class option and removable table, the system is delivered with four T-class accent strips.

The accent strips must be left with the plant for start-up (service technician) (e.g., on the patient table).

Fig. 316: Accent strips for the removable table



(1) Accent strip for the right side of the patient table with docking at the left (length: 1275 mm)
(2) Accent strip for the left side of the patient table with docking at the left (length: 415 mm)
(3) Accent strip for the left side of the patient table with docking at the right (length: 1075 mm)
(4) Accent strip for the right side of the patient table with docking at the right (length: 613 mm)

8.1.8 Cleaning up

Clean system parts with which the patient or users comes in contact in accordance with "Operator´s Instructions, Chapter K1".

9.1 General

9.1.1 Purpose

To facilitate installing the Magnetom Symphony, A Tim system in e.g. small rooms, or moving it around tight corners, it may be necessary to remove the patient table that is attached to the magnet component for delivery.

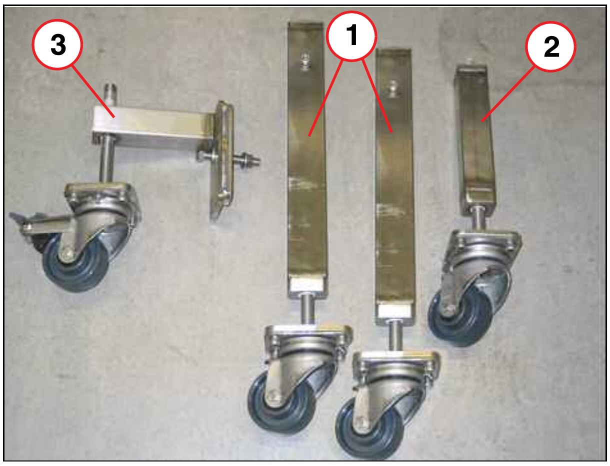
9.1.2 Scope

The following section provides instructions for removing the patient table from the magnet of a Siemens Magnetom Symphony, A Tim System. This procedure should be implemented only if all other possibilities have failed. After participating in a 1-day course, the technician is qualified to remove the patient table.

9.1.3 Tools and auxiliary tools required

- Three transport rollers (➔ Fig. 317 Page 223)
- Transport safety lock (➔ Fig. 318 Page 223)
- Hand crank
- Tool kit
- Patient table lifting element for hoisting the patient table (optional) (➔ Fig. 330 Page 231)

Fig. 317: Table transport device



(1) Transport roller 1
(2) Transport roller 2
(3) Transport roller 3

Fig. 318: Transport lock, patient table

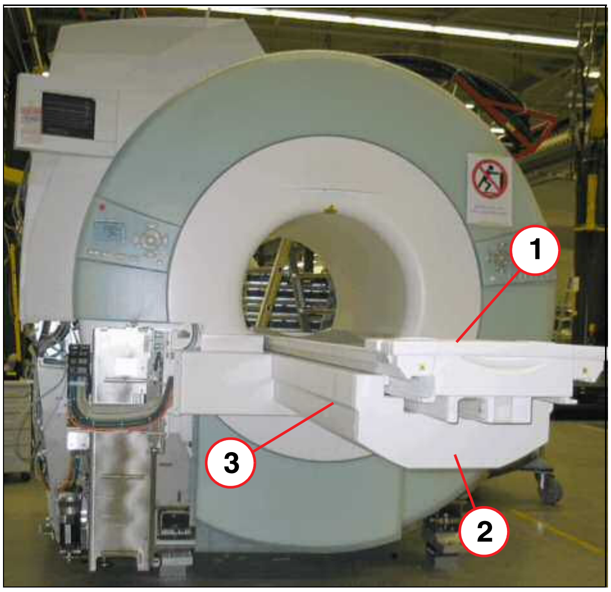
9.2 Removal of the patient table

> **i** Ensure that there is enough room to turn the table around after it has been disconnected from the magnet.

> **i** The magnet has to be positioned on solid, level ground so that the table can be removed.

- Remove the transport ropes and rubber buffers at the patient table.

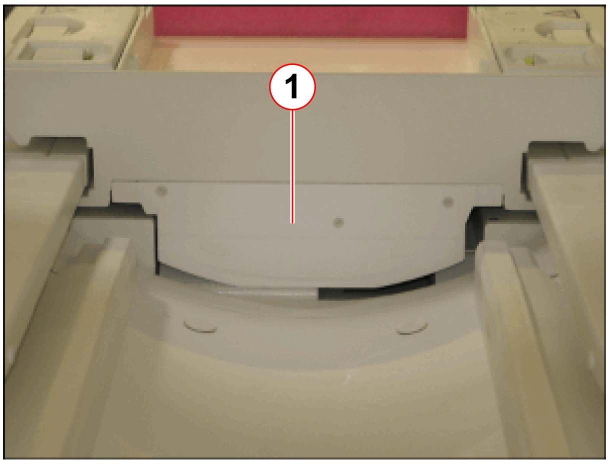
Fig. 319: Magnet, view from patient end



(1) Tabletop
(2) Cover
(3) Emergency release

- Activate the emergency release (3) and slowly move out the patient tabletop (1) by hand. Move it up to the end stop.
- Remove the service access cover.
- Remove the cover (2) at the end of the table (two screws underneath the table).

Fig. 320: Cover of toothed belt mount (head end)



(1) Cover of toothed belt mount

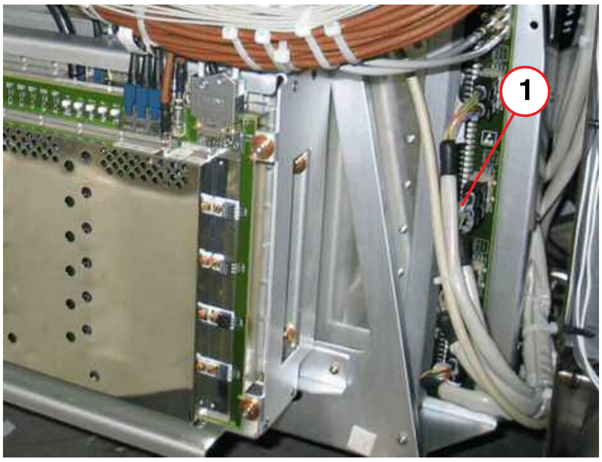
- Remove the cover (1) at the head end on the table top

  to prevent damage during separate transport.

9.2.1 Disconnecting the cabling

> Proceed with care when disconnecting the plugs at the RCCS (➔ 1/Fig. 321 Page 225).
>
> The mechanical locking mechanism of the plugs may break if handled incorrectly.

Fig. 321: Magnet RCCS (example



(1) RCCS connector

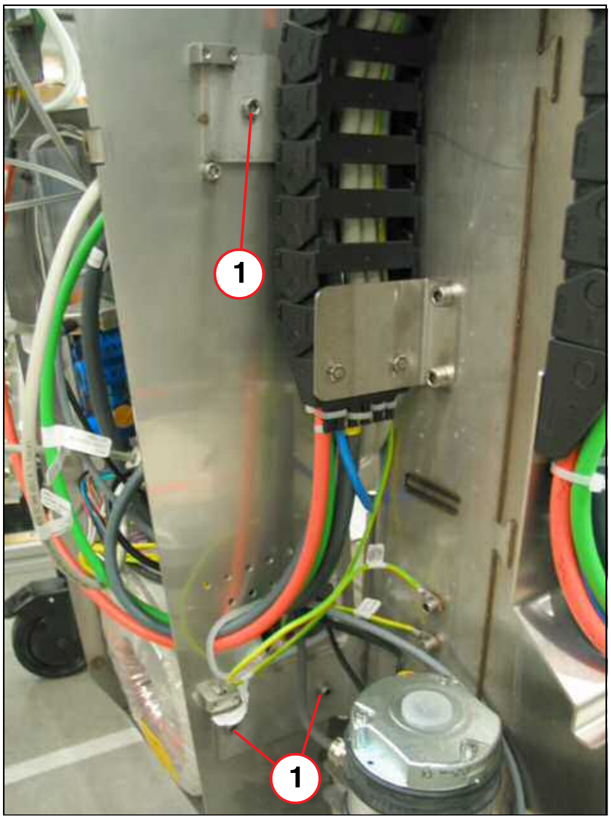
- Remove the cover of the table electronics.
- Disconnect and retract the cables coming from the table (if necessary, remove all cables connected in the electronic housing of the table to ensure that they are not damaged).

When disconnecting the cables, note the routing. When they are reinstalled, the cables have to be routed and connected the same way. Ensure that the cables are not damaged while they are disconnected and pulled out.

9.2.2 Separating the table support arm from the magnet

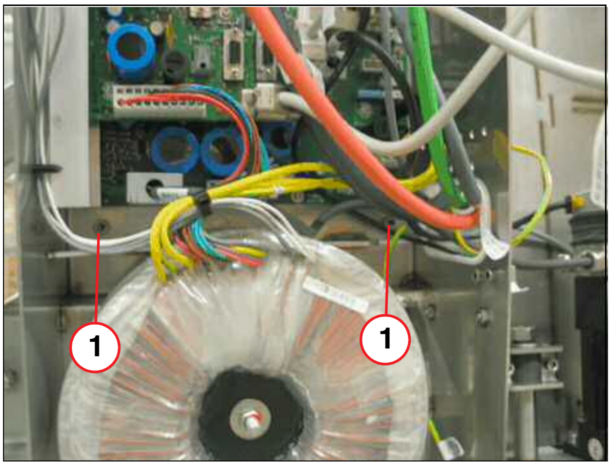
- Remove the screws at the table electronics (➔ 1/Fig. 322 Page 226) and (➔ 1/Fig. 323 Page 226).

Fig. 322: Housing, table electronics



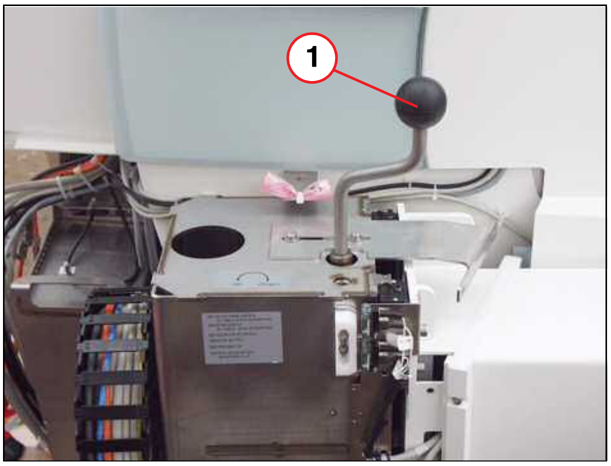
(1) Screws

Fig. 323: Patient table electronics, inside



(1) Screw

Fig. 324: Inserting the hand crank

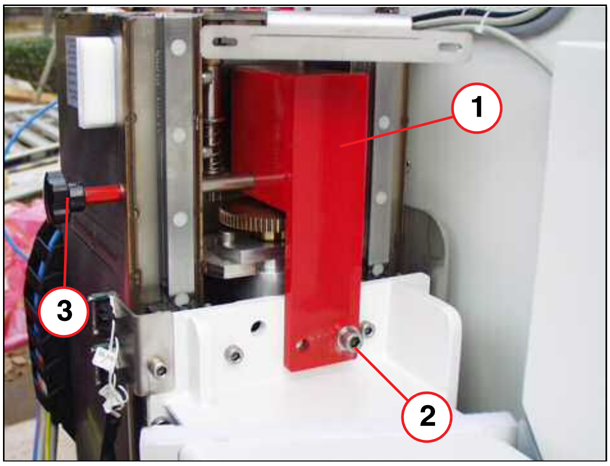


(1) Hand crank

- Insert the hand crank into the table lifting column (1).

  Lower the table using the hand crank (approx. 25 cm).

Fig. 325: Installation, transport lock

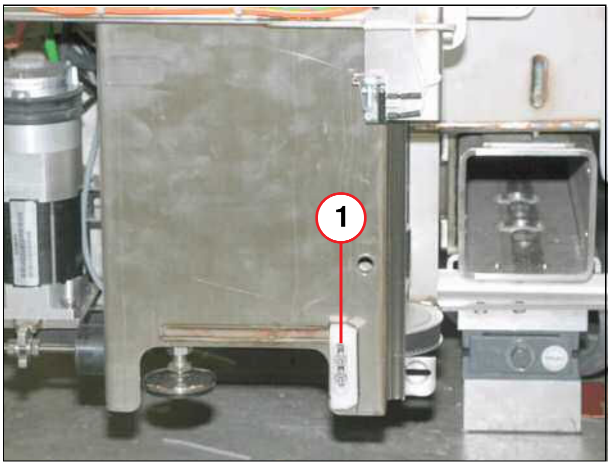


(1) Transport lock
(2) Screw, washer
(3) Safety rod

- Install the transport safety lock (1).
- Raise the table with the hand crank, until screw (2) can be inserted. Insert the screw.

- Mark the position of the switch cam (micro switch end stop) and remove it.

Fig. 326: Limit stop

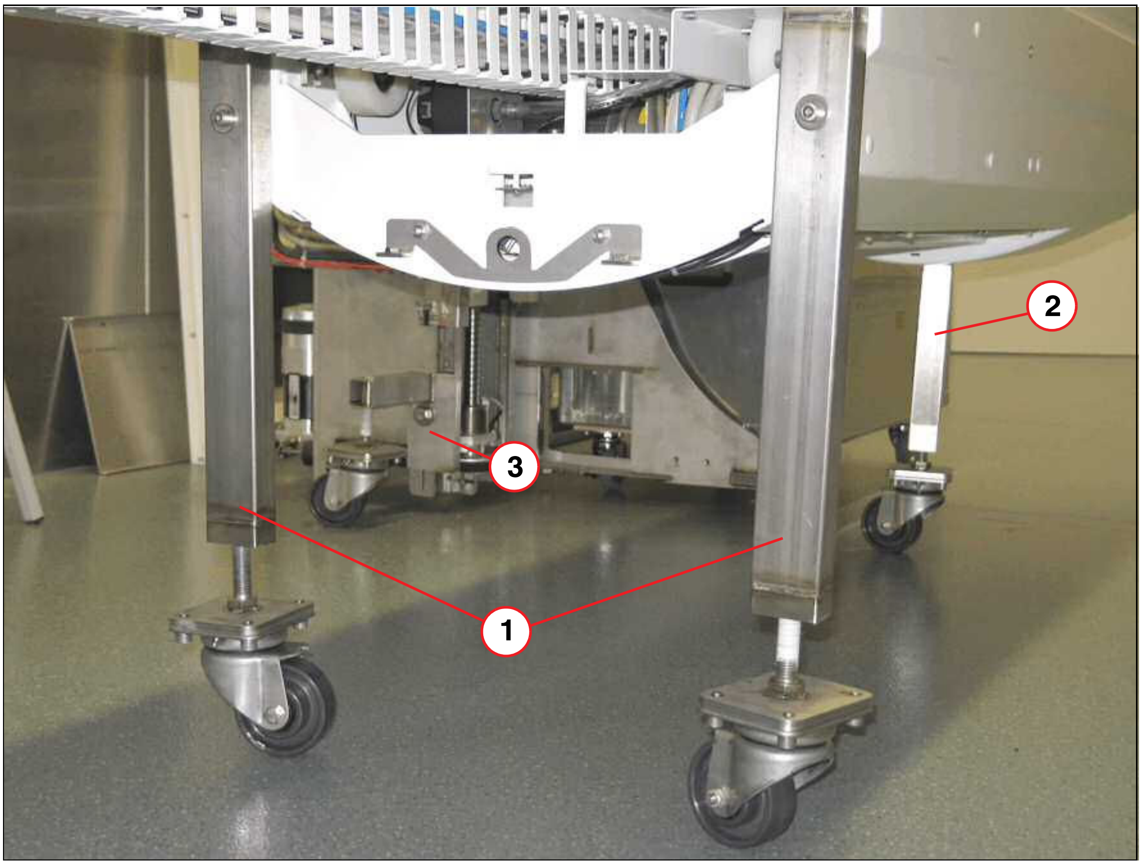


(1) Switch cam

- Install the transport rollers

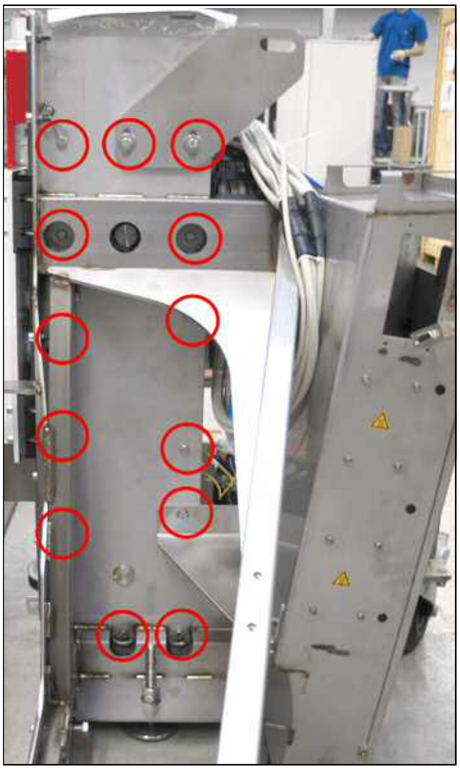
- Turn the rollers of the table's transport rollers down to the floor. They have to be positioned securely and evenly on the floor.

Fig. 327: Table transport device installation



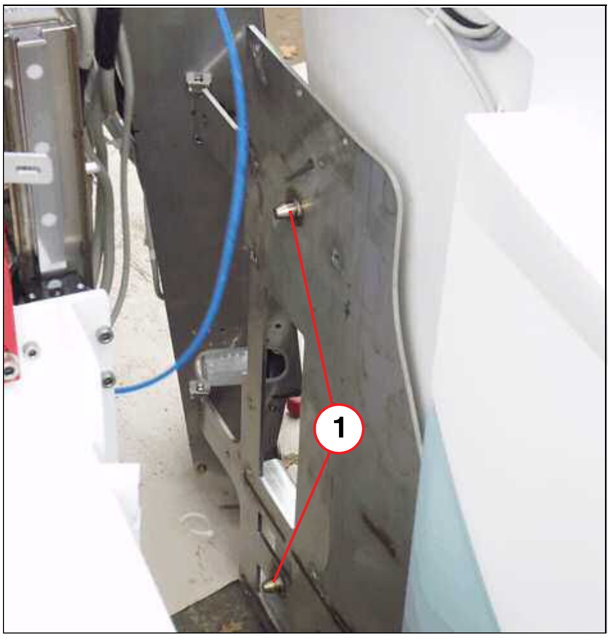
(1) Transport roller 1
(2) Transport roller 2
(3) Transport roller 3

Fig. 328: Position of the screws



- Take off two Allen screws using insert bit 6 and the **11** hexagonal screws using an open-end wrench size 13, and remove the bracket.

Fig. 329: Disconnecting patient table from the magnet



(1) Guide pin

- Carefully slide table away from the magnet and simultaneously pull out the cables.

Do not damage the cables.

Do not damage the covers.

9.3 Steps involved for separate transport

- Secure the disconnected cables of the patient table to ensure that the cables are not damaged during transport.
- Reattach the screws to the table electronics.

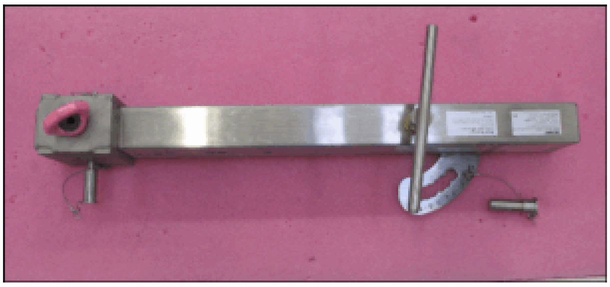
The patient table and the magnet are now transported separately.

9.3.1 Lifting

Only trained personnel are permitted to use the lifting gear.

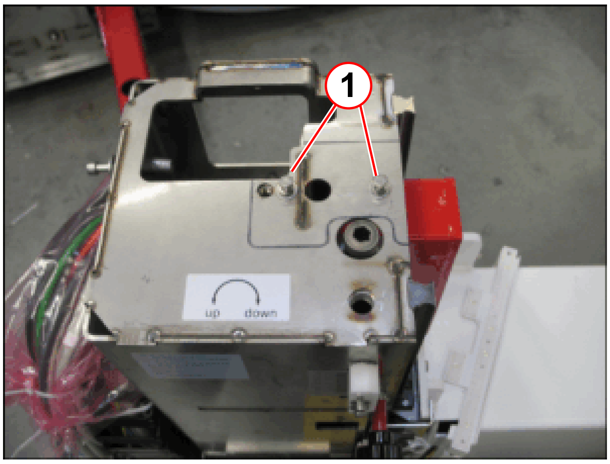
The lifting element for the patient table is designed for a maximum load of 350 kg.

Fig. 330: Patient table lifting element (Part no. 103 99 521)



If the patient table is removable, the removable tabletop must be removed from the baseplate before lifting; it must not be hoisted along with the rest of the components.

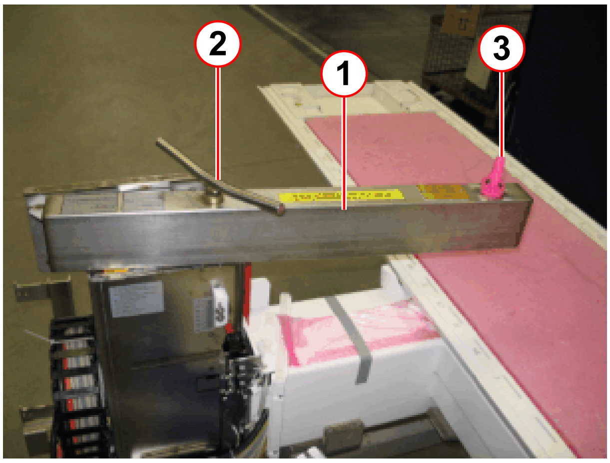
Fig. 331: Lifting column (example: Verio)



(1) Screws or brackets

- Remove the screws (1) so that the lifting element can be attached.

Fig. 332: Installing the lifting element

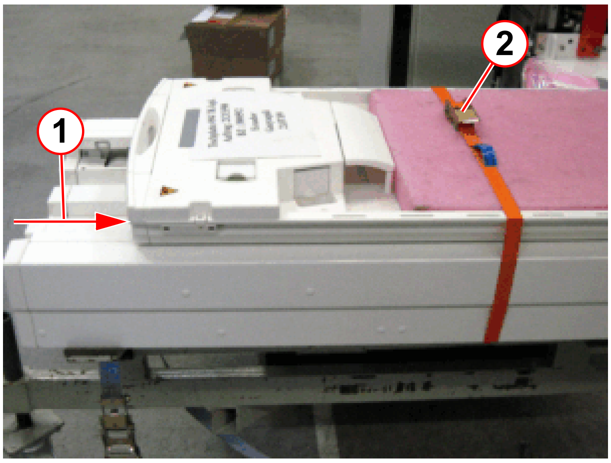


(1) Patient table lifting element
(2) Attachment screw
(3) Lifting lug

- Fasten the lifting element (1) to the lifting column with the screw (2).

> The tabletop must be secured with the tension belt to prevent slipping.

Fig. 333: Secure and level the patient table



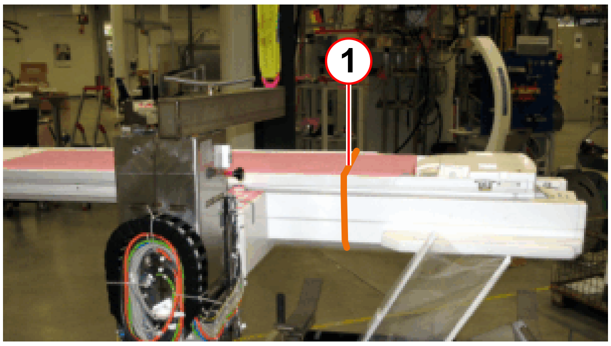
(1) Distance to leveling the patient table for lifting (approx. 20cm)
(2) Tension belt

So that the patient table is horizontal when raised, the tabletop needs to be moved approx. 20 cm.

- Hook the hoisting harness into the hoisting lug.
- Move the tabletop in the direction indicated by the arrow so far so that the patient table is horizontal when raised with the lifting member.

Carefully lift the patient table.

Fig. 334: Leveled patient table



(1) Tension belt

- Carefully lift the patient table.

9.4 Installing the patient table

9.4.1 Removing the covers

- Remove the service access cover.

9.4.2 Attaching the table support arm at the magnet

- Removing the screws at the table electronics and (➔ 1/Fig. 322 Page 226) swivel it.
- Slide table toward the magnet and flange it to the magnet (➔ 1/Fig. 329 Page 230) using a guide bolt.

NOTICE

Ensure that tabletop is pulled out up to the mechanical end stop.

» **Damage to the front funnel**

> Set the height using the transport rollers so that the guide pin can be inserted into the table.
>
> Do not damage the covers and cables!

- Install the table support arm with the 11 hexagonal screws and the two Allen screws using Loctite 221 (➔ Fig. 328 Page 230).
- Install the screws at the table electronics (➔ 1/Fig. 322 Page 226) and (➔ 1/Fig. 323 Page 226).
- Turn the transport rollers up and remove them.
- Install the lower cam switch (micro switch end stop) where marked (➔ 1/Fig. 326 Page 228).
- Removing the transport safety lock (➔ 1/Fig. 325 Page 227)
- Use the hand crank to raise the table up to the upper micro switch activation level.

> The tabletop has to be pulled out fully.

- Remove the hand crank (➔ 1/Fig. 324 Page 227).

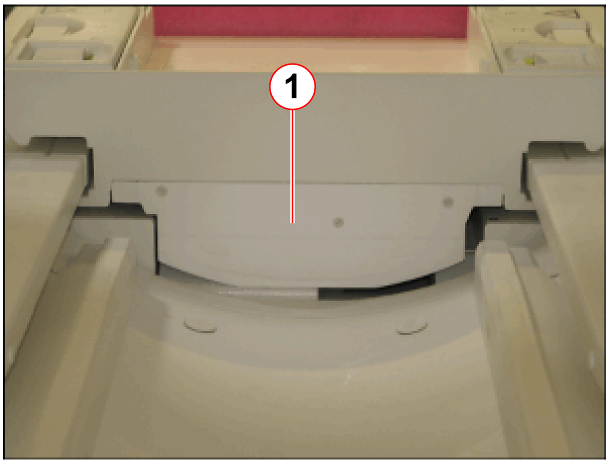
9.4.3 Connecting the cabling

- Connect the cables.
- Install the cover.

9.4.4 Connecting the cabling

- Install the cover (→ Fig. 319 Page 224) in back at the end of the table (two screws underneath the table).
- Attach the service access cover to the magnet.

Fig. 335: Cover of toothed belt mount (head end)



(1) Cover of toothed belt mount

- Reattach the cover (1) at the head end on the table top.

> Check to see whether the distance (3-8 mm) between the patient table and the front cover of the magnet suffices. (→ 3/Fig. 65 Page 69).

> Check the travel path of the table into the body coil. Adjust if necessary.

NOTICE

Instruct the project manager to inform the start-up technician (service technician) that the table was removed.

» **The service technician should check the functions of the patient table and the RF cabling (test tools, RCCS plug tests, RFIS tests).**

Version 02

- Chapter 04
  - (➔ Second remote Magnet Stop switch / Page 75) updated.
- Chapter 05
  - (➔ RCA cabinet / Page 81) updated.
- Chapter 06
  - (➔ Cabling / Page 100) additional note
  - (➔ MSUP/MPS cabling / Page 152) added.
  - (➔ Connecting the cold head to the He compressor F70 / Page 110) updated.
- Chapter 07
  - (➔ Line Voltage Adaptation, He Compressor F70 / Page 185) updated.
  - (➔ Connecting the He pressure lines: / Page 191) updated.
- Chapter 09
  - (➔ Fig. 335 Page 235) added.
- Editorial changes

Version 03

- Chapter 07
  - Adapting transformer added
- Chapter 09
  - (➔ Steps involved for separate transport / Page 231) updated.

Version 04

- Chapter 06
  - (➔ Connecting the gradient cables in the RF room / Page 129) picture added, torque changed and instruction to check the torque and polarity of the gradient cables by a third party added.
- Chapter 09
  - (➔ Lifting / Page 231) updated.

Version 05

- Chapter 02
  - New hint: The quench pipe may not be installed if the magnet is lifted.
- Chapter 06
  - Gradient cable connection (➔ Connecting the gradient cables / Page 118) updated.
    New hints and graphics:
    (➔ Fig. 156 Page 132)
    (➔ Tab. 4 Page 123)
    (➔ Fig. 158 Page 133)
  - Door contact (➔ Door Contact / Page 179) updated
  - Remote alarm (➔ Remote Alarm Contact / Page 178) updated
  - fMRI Trigger Converter (➔ fMRI Trigger Converter / Page 180) added

- New Connection (➔ Fig. 183 Page 150)

- Chapter 07
  - The Option “Adapter Transformer” is replaced by the Option “Stepping Transformer” (➔ Stepping Transformer / MR00-000.814.44).
- Chapter 08
  - (➔ Headphone Holder / Page 219) added.

Version 06

- Chapter 06
  - (➔ Fig. 185 Page 151) modification of the connector
  - (➔ Fig. 134 Page 122) added.
  - (➔ Cabling / Page 100) new note: "The cables must be kept more than 2 cm away from the ceiling."

Version 07

- Chapter 01
  - (➔ ShockBug / Page 19) added.
  - (➔ Gradient Supervision / Page 156) added.
  - (➔ Fig. 13 Page 27) updated.
- Chapter 06
  - Graphics (➔ Fig. 178 Page 146) updated.

Version 08:

- Chapter 06
  - (➔ Helium pressure line connections / Page 112) note added.
- All
  - Torque tolerances updated.
  - Torque wrench ideogram revised.

Version 09:

- Chapter 02
  - Lifting of the magnet system, warning added.

Version 10

Chapter 06

- Line voltages connection, changed torque 17.5 Nm into 3.0 Nm

Version 11

- Alignment of the patient table - work step added: Apply Loctite 221 to the counter nuts and alignment screws (defect 200876).

11 List of Hazard IDs

There are no Hazard IDs in this document.

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